First-Order Logic

A Concise Introduction

SECOND EDITION

John Heil

DERIVATION RULES

Rules of Inference

Modus Ponens (*MP*)
$p \supset q, p \vdash q$

Modus Tollens (*MT*)
$p \supset q, \neg q \vdash \neg p$

Hypothetical Syllogism (*HS*)
$p \supset q, q \supset r \vdash p \supset r$

Constructive Dilemma (*CD*)
$p \vee q, p \supset r, q \supset s \vdash r \vee s$

Conjunction Elimination ($\wedge E$)
$p \wedge q \vdash p$
$p \wedge q \vdash q$

Conjunction Insertion ($\wedge I$)
$p, q \vdash p \wedge q$

Disjunction elimination ($\vee E$)
$p \vee q, \neg p \vdash q$
$p \vee q, \neg q \vdash p$

Disjunction Insertion ($\vee I$)
$p \vdash p \vee q$

Conditional Proof (*CP*)

$$\begin{array}{|l} p \\ \vdots \\ q \\ \hline p \supset q \end{array}$$

Indirect Proof (*IP*)

$$\begin{array}{|l} \neg p \\ \vdots \\ q \wedge \neg q \\ \hline p \end{array}$$

Transformation Rules

Commutative Rule (*Com*)
$p \wedge q \dashv\vdash q \wedge p$
$p \vee q \dashv\vdash q \vee p$

Associative Rule (*Assoc*)
$p \wedge (q \wedge r) \dashv\vdash (p \wedge q) \wedge r$
$p \vee (q \vee r) \dashv\vdash (p \vee q) \vee r$

Principle of Tautology (*Taut*)
$p \dashv\vdash p \wedge p$
$p \dashv\vdash p \vee p$

DeMorgan's Law (*DeM*)
$p \wedge q \dashv\vdash \neg(\neg p \vee \neg q)$
$p \vee q \dashv\vdash \neg(\neg p \wedge \neg q)$

Distributive Rule (*Dist*)
$p \wedge (q \vee r) \dashv\vdash (p \wedge q) \vee (p \wedge r)$
$p \vee (q \wedge r) \dashv\vdash (p \vee q) \wedge (p \vee r)$

Exportation Rule (*Exp*)
$(p \wedge q) \supset r \dashv\vdash p \supset (q \supset r)$

Conditional Equivalence (*Cond*)
$p \supset q \dashv\vdash \neg p \vee q$

Biconditional Equivalence (*Bicond*)
$p \equiv q \dashv\vdash (p \supset q) \wedge (q \supset p)$

Contraposition (*Contra*)
$p \supset q \dashv\vdash \neg q \supset \neg p$

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Second Edition

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For

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Preface to the First Edition

Why another logic textbook? Why indeed. The market is flooded with textbooks, each of which fills, or purports to fill, a particular niche. Oddly, in spite of—or perhaps because of—the availability of scores of textbooks, many teachers of logic spurn commercial texts and teach from notes and handouts. This suggests that although there are many logic textbooks, there are not many good logic textbooks.

Logic texts fall into two categories. Some, like S. C. Kleene's *Mathematical Logic* (New York: John Wiley and Sons, 1967) and Benson Mates's *Elementary Logic* (New York: Oxford University Press, 1972), emphasize logic as a distinctive subject matter to be explicated by articulating, as elegantly as possible, the theory on which the subject matter rests. Others, too numerous to mention, focus on applications of logic, treating logic as a skill to be mastered, refined, and applied to arguments advanced by politicians, editorial writers, and talk show hosts. A few authors offer a middle ground, notably E. J. Lemmon in *Beginning Logic* (originally published in 1965, reissued by Hackett in 1978) and Paul Teller in his two-volume *A Modern Formal Logic Primer* (Englewood Cliffs, NJ: Prentice-Hall, 1989). Lemmon and Teller embed discussions of theory within a context that encourages the development of logical skills.

In what follows, I have elected to take this middle way. I focus on the construction of translations and derivations, but I locate these within a broader theoretical framework. The book assumes no prior contact with, or enthusiasm for, formal logic. My aim has been to introduce the elements of first-order logic gradually, in small steps, as clearly as possible. I have tried to write in a way that is congenial to students (and instructors) who might feel uncomfortable in symbolic domains. My approach to logic is not that of a card-carrying logician. This, I think, gives me something of an advantage in understanding what nonlogicians and symbolphobes find difficult or unintuitive. As a result, I spend more time explaining fundamental notions than other authors do. In my view, this pays dividends in the long run.

The volume covers elementary first-order logic with identity. I have not attempted to offer proofs for the soundness and completeness of the systems introduced. I have, however, offered sketches of what such proofs involve. These are included, with materials on the syntax and semantics of the systems, in sections on metalogic at the end of chapters 2 through 5. These sections could be skipped without loss of continuity. They are offered as springboards for more elaborate classroom discussions. For my own part, I think it important to include a dose of metalogic in an introductory course. Metalogic brings order to materials that are apt to seem arbitrary and ad hoc otherwise. Less obviously, an examination of the syntax, semantics, and metatheory of a formal system tells us something about ourselves. In mastering a formal system we come to terms with a domain that can be given a precise and elegant description. Any account of our psychology, then, must allow for our ability to understand and deploy systems with these formal characteristics.

The book began life in the summer of 1972. I had received support from the National Endowment for the Humanities to write a text that would combine logic with work in linguistic theory. My thought was that this was a case in which learning two things together was easier, more efficient, and more illuminating than learning either separately. The project culminated in a photocopied text inflicted on successive generations of students. In the ensuing years, linguists progressed from

transformational grammar, through generative semantics, and on to government binding theory. My interest in the linguistic theory waned, as did my enthusiasm for combining logic and linguistics in a single package. Meanwhile, the manuscript went through a series of photocopied incarnations. Although each version differed from its predecessor only a little, the cumulative effect has been massive.

Along the way, I received help from many people. The original project was inspired by John Corcoran, many of whose ideas I have filched shamelessly. I owe a large debt as well to David Zaret, who kindly read and offered advice on sections devoted to metalogic. Joseph DeMarco, Robert Ginsburg, Kathleen Smith, and two unnamed referees read an earlier version of the entire text and furnished countless criticisms and suggestions, all of which have improved the finished product. Many of my students provided suggestions, corrections, and advice. I am particularly grateful to Susan Peppers for reading and commenting on a large portion of the final draft. Barbara Hannan and Susan Moore caught numerous mistakes and infelicities in early versions put to use at Randolph-Macon Woman's College.

Angela Curran and Robert Stubbings provided indispensable help in ironing out technical difficulties at different stages of the project, and I am forever indebted to Piers Rawling for walking me through reformulations of the quantifier rules. My dean, Robert C. Williams, generously provided a computer, and Gary Jason a study guide designed to accompany this book. My greatest debt is to Harrison Hagan Heil, who brought me to appreciate the importance of saying clearly what can be said clearly.

Berkeley, January 1994

Preface to the Second Edition

The first edition of this book went out of print shortly after it was published. The publisher, Jones & Bartlett, sold its philosophy list to John Wiley & Sons, and the book eventually came to rest at Cengage Learning. Neither Wiley nor Cengage had an interest in keeping the book in print. As a result I have been using a PDF version in logic courses taught first at Davidson College, then at Washington University in St. Louis.

The files that issued in these PDFs were created so long ago that they were no longer editable. Over the intervening years, I was able to introduce minor corrections, but extensive changes were called for, changes that would require much more than simply tweaking PDFs. The situation resolved itself when I was lucky enough to be offered an opportunity to publish a new version of the book by Rick Todhunter, a senior editor at Hackett Publishing Company. The upshot is the volume you hold in your hand—or are viewing on a screen.

As in the case of the first edition, I have put a premium on readability. You do not really understand a topic until you can explain it to someone unfamiliar with it. In my own case, that meant making logic clear to myself first, then putting my understanding to work by declaiming the whys and wherefores of logic to students. In my experience, it is easier to learn something when you can see the point of it. With that in mind, I have sought to explain and motivate topics regularly taken for granted by logicians.

One example of this is the discussion of conditionals in chapter 2. Students are often left with the impression that truth conditions for logicians' conditionals are sharply at odds with conditionals we all use in everyday speech. Once you look more closely, you can see that the logician's conditional does a respectable job of capturing the logical core of everyday conditionals.

The book begins with an explication of a sentential logic, $\mathcal{L}_S$, followed by the presentation of a predicate logic, $\mathcal{L}_P$. Both $\mathcal{L}_S$ and $\mathcal{L}_P$ are natural deduction systems designed to approximate everyday deductive reasoning. The book differs from many introductory texts in including accounts of the syntax and semantics of both $\mathcal{L}_S$ and $\mathcal{L}_P$, as well as discussions of soundness and completeness. Some instructors might prefer to omit this material, but students often find it interesting, at least in outline.

In addition to countless changes, small and large, I have included in this edition a section on Prenex Normal Form, another topic some students find interesting and even useful. Like the material on soundness and completeness, this could be omitted without compromising the presentation of the nuts and bolts of first-order logic: the translation of sentences in natural language into $\mathcal{L}_S$ and $\mathcal{L}_P$, and the construction of derivations in both.

Throughout it all, I have tried to keep alive the idea that first-order logic has much to reveal about the languages we speak as we go about our lives and the thoughts we express in those languages. Although logic has important applications in many formal domains, I have chosen to highlight connections between logic and natural languages. This strikes a chord with students encountering the subject for the first time.

Readers aware of my antipathy toward philosophical reliance on talk of possible worlds might be surprised to see my invocation of possible worlds to explicate semantic features of $\mathcal{L}_S$ and $\mathcal{L}_P$. I have done so for two reasons. First, talk of alternative universes captures the imagination of many

students for whom thoughts of such things are perfectly natural. Second, students going on in philosophy will inevitably encounter endless references to possible worlds. Those students stand to benefit from being introduced to the jargon in a relatively benign environment.

This new edition of the book has benefited from questions and suggestions tendered by hundreds of students who have worked through the original and from undergraduate and postgraduate assistants who have been invaluable in moving the material from the printed page into undergraduates' heads. I have never lost my enthusiasm for teaching the subject. I hope some of this comes through in the book.

Some readers might wonder how I go about using the book in the classroom. I have only taught logic to students in North America, so my remarks here pertain to the North American model. They pertain, as well, to face-to-face teaching. As I type these words, I am preparing to teach for the first time at arms' length, so I shall need to adapt my usual practices to a slowly evolving postpandemic New Normal, at least for the foreseeable future.

I have found that students struggle when they fall behind and when they do not engage with the material. Being able to *follow* a translation or derivation when it is explained is one thing; being able to translate sentences and derive conclusions from premises is something else altogether. Learning to do logic is no different than learning any skilled activity. Success requires practice and repetition. The book is sprinkled with exercises aimed at encouraging students to *apply* what has been covered. I discuss these in class and send volunteers to the blackboard (yes, my university still uses blackboards) to write them out.

I do not mark students' work on these exercises but instead pair them with short, five-problem quizzes administered to students in a Logic Lab typically presided over by student assistants. (I have had excellent results with both undergraduate and postgraduate assistants.)

Logic Lab, which has traditionally convened for two hours, two evenings a week, serves two functions. First, it provides a venue for students taking quizzes. Second, it serves as a logic help desk. Students can show up and, if they are so inclined, get help on the material before taking the week's quiz and departing. (Because students take quizzes at different times over the course of the week, I do not return quizzes after they have been marked, but, once marked, students can look over their work in subsequent Logic Labs.) The quiz system keeps students from falling behind and helps them appreciate what they have mastered and what they have yet to master.

In addition to a dozen quizzes over the course of the term, I give two in-class tests and a cumulative final examination. The first test addresses sentential logic, $\mathcal{L}_S$, which is taken up in chapters 2 and 3. The second test concerns material in chapters 4 and 5, predicate logic, $\mathcal{L}_\mathcal{P}$. A final examination includes both $\mathcal{L}_S$ and $\mathcal{L}_\mathcal{P}$, along with material on reasoning under uncertainty (and the Hot Hand phenomenon) introduced in three term-ending sessions.

Not everyone using this book to teach logic to undergraduates will want to do things as I do them. The book is written to be self-standing, however, and in no way depends on the Logic Lab model. Indeed, I believe that anyone unfamiliar with the subject who sets out to learn formal logic could do so relying solely on the book. That, in any case, is what I set out to accomplish here.

Melbourne, July 2020

Acknowledgments

In reflecting on my debts to others for myriad corrections, suggestions, and advice on matters addressed in this book, I hardly know where to begin. The preface to the first edition acknowledges a number of students, colleagues, and others who were indispensable in the original project. Since then, many others have contributed in many ways, large and small, to its subsequent development. Earlier I mentioned Rick Todhunter, my editor at Hackett, who guided me through the publication process while exercising exceptional patience. I am grateful as well to two anonymous referees for helpful criticisms and suggestions.

Derek Braverman, Xiaoyu Ke, and Auke Montessori assisted me in implementing the first logic class taught with a version of this book. They assisted students in countless Zoom sessions (the course was taught remotely to a widely dispersed student cohort), and they discovered and corrected numerous textual infelicities, typographical and otherwise, in the manuscript. As noted in the preface, I was obliged to reenter much of the formal symbolic material, a process that inevitably led to slipups. Graham Renz and Derek Braverman graciously helped me identify the worst of these in the course of vetting solutions to the exercises. My copy editor, Lori Rider, made countless corrections and offered suggestions that improved the book immeasurably.

Students over many years helped me find better ways of doing what this book sets out to do. I am particularly grateful to the eighty-five students in my fall 2020 logic class at Washington University in St. Louis who not only suffered through a pandemic-induced asynchronous presentation of first-order logic but whose good sense, patience, and sharp eyes helped bring the book to life. Shelly Hykawy and the Hykawy family kindly granted permission for the reproduction of the drawing by C. B. Martin, *A Being in Search of a Variable for which to be a Value*, that appears opposite the title page.

Finally, I am indebted to my colleague, Bret Hyde, and his former student, Gus Heil, for advice on matters pertaining to linguistics, and to Harrison Hagan Heil, a center of calm in the midst of a storm, who left her mark on every page of this second editon of *First-Order Logic: A Concise Introduction*.

1. Introduction

1.00 Logic: What's Not to Like?

Why study logic? What is the point? What does logic have to offer? The best answer to these questions is that there is no simple answer. Why study economics? What is the point of history? What does biology have to offer? There are various reasons to study logic ranging from the mundane—'I need it to satisfy a degree requirement', 'I need an easy B'—to the sublime: 'Logic is fascinating, important, and I can't imagine life without it'.

Logic has certainly struck many as intrinsically fascinating, and its importance would be difficult to exaggerate. With luck, a measure of this fascination and importance will emerge in the chapters that follow. Even if the study of logic had little or no intrinsic significance, however, there would still be plenty of reasons for pursuing it.

Consider, first, the remarkable fact that human beings possess a capacity to learn and deploy languages. Aristotle (384–322 BCE) characterized human beings as rational animals. The brand of rationality that sets us off from other creatures is bound up with our linguistic endowment. Language is required for the expression—and perhaps even for the entertaining—of thoughts that make possible the kinds of coordinated social activity that give meaning to our lives. Your faithful dog Spot, it would seem, could have thoughts about goings-on around him—that you are at the front door, for instance. But how plausible is it to imagine Spot thinking—*now*—that you will be at the front door two days hence?

Without a language, Spot apparently lacks the wherewithal to express such a thought. It does not follow immediately that Spot could not, even so, entertain the thought. That seems right. But ask yourself what reason could you have for ascribing such a thought to Spot. Arguably, thoughts about a subject matter outside a thinker's immediate environment are, in one way or another, dependent on a linguistic background. Thoughts about spatially or temporally remote states of affairs, thoughts about abstract entities (sets and numbers, for instance), and thoughts about nonexistent things (square circles, four-sided triangles) seem, on the face of it, to require a linguistic medium for their expression, whether overt or covert.

One aim of this book is to convince you that logic provides insights into the structure of natural languages—the languages in which we converse. That language might or might not be English. For purposes of logic, it does not matter. The logical

Talk and Thought

Suppose, in your absence, Spot noses about quietly on Monday and Tuesday, then, on Wednesday, paces expectantly in the vicinity of the front door. Would this entitle someone observing Spot's behavior to conclude that, on Monday and Tuesday, Spot entertained the thought that you would return on Wednesday? Or does Spot's behavior suggest only that, on Wednesday, Spot thinks your arrival is imminent? What reasons could you have, here and now, for taking Spot to be entertaining a tensed thought, a thought about the future, one that might be expressed by means of the English sentence, '[Your name here] will return the day after tomorrow'?

forms that make up the subject matter of logic figure in the abstract logical forms underlying all natural languages. The study of logic, then, promises to provide insights into the character of language. Given the central place of language in human affairs, the study of logic can help illuminate human psychology.

The history of logic is usually taken to begin with Aristotle's classification of argument forms. Think of an argument as an ordered collection of sentences, one of which, a *conclusion*, is supported by one or more *premises*. Aristotle recognized that arguments exhibit repeatable *patterns*. Some of these patterns represent valid reasoning—their premises *support* or *imply* their conclusions—and some do not.

You will see later how to characterize the notion of implication more precisely. For the moment, you could think of premises as implying a conclusion when they *provide reasons to believe* that conclusion, in the following sense: if you accept the premises, you have reason to accept the conclusion.

Aristotle focused on simple *syllogistic* patterns of reasoning, patterns typically involving a pair of premises and a conclusion. He recognized that many of the arguments used by philosophers, politicians, scientists, and ordinary people could be understood as exhibiting combinations of these patterns. Aristotle reckoned that if an argument is *valid*, if its premises provide grounds for its conclusion, then any argument with the same pattern of sentences must be valid as well. This nicely illustrates the formal character of logic. Among other things, logic provides a way of studying and classifying repeatable forms or patterns of reasoning applicable to any subject matter.

As it happens, formal logic—the subject matter of this book—provides powerful techniques for assessing the validity and invalidity of arguments. This is bound to prove useful when you turn to the examination of arguments in ordinary life, the stuff of editorial pages and debates over public policy. It proves useful as well in the evaluation of theories about the universe and our place in it. Logic affords a framework for exhibiting the structure of lines of reasoning. To the extent that you can transform everyday reasoning into a standard logical form, you *discipline* that reasoning. In so doing, you might discover that some of your cherished beliefs are based on specious inferences—or that they are better supported than you had imagined.

Some philosophers have thought that elementary formal logic of the sort you will be encountering provides a *canonical notation* for the pursuit of knowledge. The idea is simple and elegant: anything you could intelligibly say or think about the world must be expressible in this favored idiom. Whatever is not so expressible would not be a candidate for truth or falsehood—or at any rate, *literal* truth and falsehood. Poetry and fiction, for instance, while not literally true, make legitimate claims on truth.

The thought, however, is that truths arising in poetry and fiction could be expressed more prosaically in the canonical language. If that is right, there is much to be learned about the structure of our conception of reality by studying logic. Even if you are suspicious of the concept of a canonical language, formal logic provides insight into one central aspect of ways we think about the universe and our place in it.

Whatever its standing in the broader scheme of things, logic has deep ties to mathematics and computer science. Anyone bent on pursuing serious work in either of these areas stands to benefit from an understanding of logic. Other disciplines, too, are connected to formal logic in fundamental ways. Quantum physics is sometimes held to mandate a nonstandard logic for the description of

quantum states. Such claims can be sensibly evaluated only against a background of a standard logic of the sort taken up here.

I have thus far omitted mention of one reason widely cited for embarking on the study of logic. By working at logic, you might expect to enhance your reasoning skills, thereby improving your performance on cognitive tasks generally. I have not emphasized this supposed benefit, however, because I am skeptical that there is much to it. Empirical studies cast doubts on the notion that training in logic leads to improvement in ordinary reasoning tasks of the sort you encounter every day outside the classroom. Formal logic, like most other learned disciplines, resists 'transference' across problem domains.

Take statistics. People trained in statistics typically fare no better than the rest of us at real-world tasks requiring reasoning under uncertainty. Similarly, a student who earns an A in logic can continue to make errors in everyday reasoning. The fault lies not with logic but with an ingrained tendency to compartmentalize what we learn. As a result we can fail to see that something we have learned in one context has straightforward applications in a different context.

Does logic, then, afford *nothing* in the way of overall improvement in reasoning? That too is doubtful. You would do well to scale down your expectations and recognize that, when you study logic, your improvement in reasoning prowess, if any, is likely to be incremental rather than dramatic. Still, in learning logic, you are more apt to become sensitive to species of bad reasoning. When problems are framed in ways that bring out their connections to the domain of formal logic, your training can serve you well.

Syllogistic Patterns of Reasoning

These two arguments exhibit similar forms, similar patterns of inference:

All whales are mammals.	All philosophers are logicians.
All mammals are warm-blooded.	All logicians are clever.
All whales are warm-blooded.	All philosophers are clever.

In each case, the sentence above the horizontal line, the *premise*, is taken to support the sentence below the line, the *conclusion*. These arguments are *valid*: their premises imply their conclusions. Must valid arguments have true premises? True conclusions? What of the following arguments?

All students are carnivores.	All kangaroos are marsupials.
All carnivores are marsupials.	All marsupials hop.
All students are marsupials.	All kangaroos hop.

These arguments are valid if their premises support their conclusions. *Are* they valid?

1.01 Practice Makes Less Imperfect

Aside from a variety of extrinsic reasons for taking up the study of logic, you might find the subject challenging and satisfying in itself, even exciting. I have sought to present materials in a way meant to draw you in and enable you to experience some of that excitement.

The success of the enterprise will require cooperation on your part. Mastery of formal logic comprises mastery of a range of *skills*. Logic encompasses a definite subject matter, but it includes as well techniques for the *perspicuous representation* of sentences in natural languages—that is, the translation of ordinary sentences into a formal notation that reveals something of their underlying logical structure—and for the construction of derivations. Both sentence representation and derivation construction require *practice*. Like practice at the piano, what is at first difficult, even alien, can, over time, evolve into something obvious and familiar.

The book is sprinkled with exercises designed to encourage the development of skills required for devising translations and constructing derivations. These exercises—and their components—vary in difficulty. Many are simple, some are more demanding, a few will push you to your limits. In every case, the value of exercises hinges on your *working through* them systematically. You will discover that there is a vast difference between, on the one hand, *following* someone's translating a sentence into a formal idiom or constructing a derivation and, on the other hand, translating the sentence or constructing the derivation yourself. Translation and derivation construction require the deployment of techniques and insights gained from practice, not simply the exercise of recognitional abilities.

The difference here is strictly analogous to that associated with any skillful activity. You can probably *hear* that a piece is played correctly on the piano, or *see* that a tennis serve is properly executed. From this, it scarcely follows that you could play the piece yourself or properly serve a tennis ball. To do so, you would need to practice, to repeat movements until they came smoothly and naturally.

Formal techniques used to devise translations and derivations include important *perceptual*—and not merely *cognitive*—components. Mastery of formal logic is by no means an exclusively cognitive or intellectual achievement. Some of us are said, like Spock, to think 'logically', while the rest of us (the author included) think in logically suspect ways. Perhaps logicians and mathematicians are predominantly 'left-brained', and the rest of us 'right-brained'. This picture is at least misleading and probably wrong.

In mastering techniques in formal logic, you must be prepared to repeat perceptual and cognitive maneuvers until they become routine. To that end, you will benefit from reworking—and re-reworking—sentence translations and derivations that strike you as troublesome. In so doing, you might imagine that you are learning nothing new. In one sense that is so. In practicing a piece on the piano, nothing new is learned; instead, a skill is enhanced. So it is with translations and derivations. Repetition improves and fine-tunes the pertinent skills.

You will discover that the individual steps required for the construction of derivations are mostly trivial moves you mastered in childhood. Seeing patterns among symbols calls on perceptual skills of the sort you deploy when you recognize species of birds or kinds of trees. You know, for instance, that if birds are feathered, and lorikeets are birds, it follows that lorikeets are feathered. Success in logic requires little more in the way of inferential sophistication. The challenge, rather, is largely perceptual. You must learn to *see* patterns within arrays of symbols, just as you *hear* them when you

say or think the earlier sentences about lorikeets. Once you can do this, it will be a simple matter to apply inferential strategies required for the construction of derivations.

Picture Henry, embarking on a leaf-gathering expedition for his eighth-grade biology class, armed with the *Audubon Field Guide to North American Trees*. At first Henry finds it difficult or impossible to identify leaves by comparing what he sees to pictures and descriptions in the guide. The guide displays a picture of an elm leaf. Does the leaf Henry is now examining match that picture? Well, it does in certain respects. It has a notch, however, where the pictured leaf does not. Is the notch damage done by an insect, or is it natural? If it is natural, is it uncharacteristic of elm leaves? Or do elm leaves perhaps vary in respect to such notches?

With practice, Henry will eventually acquire the knack of recognizing elm leaves. His success depends on his having come to appreciate what is and what is not relevant to a leaf's being an elm leaf. The knack is not one Henry could easily put into words. (Imagine trying to explain over the telephone how to identify a species of leaf to someone with no prior experience with leaves.)

In this regard, the perceptual skill Henry has acquired resembles the motor skills you master when you learn to walk, or tie a bow, or ride a bicycle, or ski. You learn such things by *doing* them, haltingly, at first, then, with practice, more fluidly, until, eventually, you perform them automatically. Early on, you deploy simple rules of thumb: *to turn, keep your weight on the downhill ski*. Later, you simply ski. The motor routines involved have been programmed to run without conscious direction.

So it is with the construction of translations and derivations. What is difficult at first becomes, with practice, fluid. Practice is essential and unavoidable. Some readers will have had a head start. You might be inclined to regard such people as smarter or 'more logical' than others. The abilities in question, however, are less intellectual than perceptual, less a matter of cogitation than of skillfully discerning patterns. If pattern recognition is right-brained, the skills required for success in logic are, perhaps surprisingly, right-brained skills.

Practice and repetition enable you to automatize conscious processes, a vital component in the mastery of any subject matter. Another, complementary, process, the process of bringing to conscious awareness what you already manage to do unselfconsciously, is no less important.

In speaking and understanding a language, you exercise a variety of *syntactic* and *semantic* skills. (Roughly, *syntax* designates the grammatical structure of sentences; *semantics* concerns their meanings.) When you set out to translate a particular English sentence into a formal idiom, you must first make explicit to yourself the semantics of the original sentence. Only then can you be confident that you have found a plausible formal counterpart. Your semantic knowledge, however, no less than your knowledge of walking, shoe tying, and skiing, is largely *implicit*: knowledge *how*, as distinct from knowledge *that*. You have it, but it is not easily recovered and made explicit.

In mastering logic, you are forced to convert your implicit linguistic know-how into an explicit appreciation of principles you unreflectively rely on in speaking and understanding. In the end, you stand to learn much about yourself and your fellow speakers.

1.02 $\mathcal{L}_S$ and $\mathcal{L}_P$

This book introduces two formal systems, two *languages*: $\mathcal{L}_S$ and $\mathcal{L}_P$. $\mathcal{L}_S$ is a simple *sentential* logic, a system the elements of which include 'atomic' sentences and sentential connectives that allow for the construction of complex 'molecular' sentences. Atomic sentences are analogous to simple English

sentences: 'Kangaroos are marsupials', 'Kangaroos hop'. A molecular sentence could be made up of these together with the connective *and*: 'Kangaroos are marsupials and kangaroos hop', or, more colloquially, 'Kangaroos are marsupials and hop'.

$\mathcal{L}_P$, in contrast, is a *predicate* logic, the sentences of which exhibit an internal, 'subatomic' structure. Because a sentential logic such as $\mathcal{L}_S$ is simpler than a predicate logic, $\mathcal{L}_S$ will be taken up first, opening the way to a consideration of $\mathcal{L}_P$, the primary target of this book.

Throughout it all, you would do well to bear in mind the importance of perceptual skills that, as I have insisted, are essential to your mastery of the material under discussion. You might bear in mind as well the importance of making clear to yourself what you, in one sense, already know about the language you speak. Thus prepared, you will be in a position to discover the austere beauty exhibited by formal systems such as $\mathcal{L}_S$ and $\mathcal{L}_P$ as well as their benefits and limitations as vehicles for the expression of thought and meaning.

2. The Language $\mathcal{L}_S$

2.00 A Formal Language

This chapter unveils a simple formal language, $\mathcal{L}_S$. Although $\mathcal{L}_S$ is indeed simple when compared to natural languages—English, German, Urdu—it exhibits a variety of interesting logical characteristics common to every language. You will be introduced first to the syntax and semantics of $\mathcal{L}_S$, then, in the next chapter, to its derivational structure. Throughout the discussion, features of $\mathcal{L}_S$ will be compared with those found in English. As it happens, you can learn a good deal about the logical character of familiar natural languages in the course of examining a simplified artefact.

As with any language, $\mathcal{L}_S$ incorporates an elementary *vocabulary* from which sentences are constructed. Sentences can be as complex as you like, so long as their complexity remains finite. There is no longest sentence in $\mathcal{L}_S$, but sentences in $\mathcal{L}_S$ (or those of English, for that matter) cannot be *infinitely* long. Sentences of $\mathcal{L}_S$, like their natural language counterparts, comprise *finite* strings of elements: sentences are *finitely constructible*. The significance of these points will become clear in the course of spelling out the syntax of $\mathcal{L}_S$ (§ 2.20).

Admittedly, ordinary human beings would be baffled by excessively complex sentences or those exceeding a certain length. (You might test yourself by reading Lucy Ellman's thousand-page novel, *Ducks, Newburyport* [Bloomsbury Publishing], most of which consists of a single sentence.) Such matters fall within the province of psychologists, however; they do not affect our characterization of a language.

Mathematicians are interested in defining and exploring characteristics of mathematical systems, but the *uses* of such systems and the cognitive difficulties that human beings might experience in dealing with them are not themselves of concern to mathematicians. The same holds for a logician aiming to characterize $\mathcal{L}_S$, or a linguist interested in characterizing Korean. In each case the goal is a description of an abstract object, a language. Linguistics is one thing, psycholinguistics something else altogether.

2.01 Sentential Constants and Variables

English sentences consist of nonempty but finite strings of words arranged in a particular order. Every sentence is made up of at least one word, and, as noted earlier, no sentence includes an infinite number of words. The building blocks of $\mathcal{L}_S$ sentences, in contrast, are not words, but elements that themselves function as sentences. These elements, the *sentential constants*, are represented by means of familiar uppercase letters, A, B, C, . . ., Z. Sentential constants in $\mathcal{L}_S$ function as simple declarative sentences do in English. They can occur individually or together with other sentential constants within complex sentences.

A simple atomic sentence, S, might play a role in $\mathcal{L}_S$ resembling the role played by the English sentence 'Socrates is wise'. Both sentences can be used to express a simple proposition. The $\mathcal{L}_S$ sentence,

like its English counterpart, can be combined with other simple sentences (in ways to be discussed) to produce sentences expressing complex propositions.

Elements of $\mathcal{L}_S$, like those occurring in natural languages, do not come with built-in meanings. *We* decide what a given element is to mean, how it is to be interpreted. In the case of natural languages, this decision might be conscious and deliberate, as when a scientist coins a term to designate a newly discovered particle, or it might be the result of a tacit social agreement shrouded in history. In learning a language we enter into an implicit agreement with others sharing the language, an agreement that allows us to use words in a way that reliably communicates what we intend to communicate.

Formal languages serve very different functions. $\mathcal{L}_S$ enables us to make clear the logical structure of complex sentences and logical relations among sentences in natural languages. As a result, it is possible to restrict the elements of $\mathcal{L}_S$ dramatically. Any sentence of $\mathcal{L}_S$ can be constructed from a small number of simple elements, because we can elect to interpret these simple elements differently on different occasions. (It is possible to supplement the elementary vocabulary of $\mathcal{L}_S$ by introducing subscripts: $A_1 \ldots, A_2, \ldots A_3, \ldots, A_{519}, \ldots$, but this is a complication that need not concern us here; see § 4.14.) Thus, on one occasion you might use the $\mathcal{L}_S$ expression, *S*, to mean

Socrates is wise.

On another occasion, you might use the very same symbol, *S*, to mean

Socrates is happy.

We can make life easier for one another by selecting letters that bear associative relations to corresponding English sentences. In the examples above, *S* was used to express propositions expressed, respectively, by the English sentences 'Socrates is wise' and 'Socrates is happy'. We might just as well have used *W* and *H*, or, for that matter, *R* and *L*. The choice of letters is constrained not by logic but by common sense: translate sentences in a way that will be easy for you to remember and for others to decipher. What you cannot do, however, is use the same letter in a single complex sentence as a stand-in for two *distinct* sentences.

Suppose you set out to represent in $\mathcal{L}_S$ something equivalent to the English sentence

Socrates is wise and Socrates is happy.

You cannot represent *both* simple constituent sentences—the sentence 'Socrates is wise' and the sentence 'Socrates is happy'—by means of an *S*. You must instead use distinct letters, *W* and *H*, for instance.

Occasionally we will need to talk in a general way about sentences in $\mathcal{L}_S$. We might, for instance, want to discuss kinds of sentences, or sentential structures generally, without restricting ourselves to particular sentences. Faced with a similar need, mathematicians resort to *variables*. In explaining the operation of multiplication, for instance, I could set out the following characterization:

Where *x* and *y* are integers, their product is equal to *x* added to itself *y* times.

Here *x* and *y* function as variables ranging over arbitrary numbers. In discussing $\mathcal{L}_S$, we deploy *sentential variables* that range over arbitrary $\mathcal{L}_S$ sentences. Sentential variables consist of the

*lower*case letters, *p, q, r, s, t*. The sentential variable, *p*, then, might be used to stand for any sentence of $\mathcal{L}_S$ you choose. The introduction of variables is more than a convenience. Without them, $\mathcal{L}_S$ and its characteristics could not be described in a suitably general way. Variables must not be mistaken for sentences, however. Just as *x* and *y* are not numbers, so *p* and *q* are not sentences of $\mathcal{L}_S$. Sentences in $\mathcal{L}_S$, then, include *only* uppercase letters.

Sentences and Propositions

Philosophers commonly speak of sentences as *expressing* propositions. Sentences vary from language to language. Propositions, in contrast, are taken to possess a kind of language-independent meaning.

Distinct sentences can be used to express the same proposition: the English sentence 'Snow is white' expresses the same proposition expressed by the French sentence 'La neige est blanche'. The same sentence could, on different occasions, be used to express different propositions: the English sentence 'They are flying planes' could be used to express two different propositions. Do you see what they are?

Although it is undeniably convenient to appeal to propositions in discussing language—you can say, for instance, that a sentence in one language translates a sentence in another language when both are used to express the same proposition—you should bear in mind that there is little agreement among philosophers as to what propositions are, or even whether such entities exist.

Exercises 2.01

Provide $\mathcal{L}_S$ translations of the English sentences below.

1. Socrates is brave.
2. Socrates loves Xantippi.
3. Xantippi loves Socrates.
4. The quick brown fox jumps over the lazy dog.
5. Every good boy does fine.

2.02 Truth-Functional Connectives

The richness of English and other natural languages arises from their capacity to yield an unlimited supply of sentences from a finite vocabulary. Mastering a language involves mastering techniques for producing and understanding sentences belonging to that infinite stock. One elementary technique for generating sentences consists of combining simple sentences into compounds. You might take the sentences 'Socrates is wise' and 'Socrates is happy' and conjoin them to form the compound sentence

Socrates is wise and Socrates is happy.

In the course of assembling simple sentences to produce compounds, we often modify the originals in a way that disguises their structure. Thus, although you could combine the sentences 'Socrates is wise' and 'Socrates is happy' to yield the sentence above, you are more likely produce something like this

Socrates, who is wise, is happy.

or, even more likely, this sentence:

Socrates is wise and happy.

In the first example, the sentence 'Socrates is wise' is converted to a relative clause, 'who is wise', and embedded inside the sentence 'Socrates is happy'. In the second example, elements in one sentence that are repeated in the other sentence have been dropped. These and many other such processes are common in natural languages.

In using English, you combine simple sentences to form larger, more informative sentences. This holds for $\mathcal{L}_S$ as well. There are, however, important differences. First, as the examples above illustrate, English sentences constructed from simpler English sentences typically include modifications of the original simple sentences. A sentence, combined with another, might be converted into a clause, or have its repeated elements dropped. In $\mathcal{L}_S$, simple sentences retain their identity. This is one reason a formal language such as $\mathcal{L}_S$ can be taken to reveal logical structure hidden or disguised in sentences used in natural languages.

A second difference between $\mathcal{L}_S$ and English is that the mechanism allowing for sentential combination in $\mathcal{L}_S$ is more restricted than the combinatory mechanisms typical of natural languages. In this chapter, you will be introduced to five *truth-functional sentential connectives* (also called *logical connectives*, *logical constants*, or *logical operators*). These serve to bind simple $\mathcal{L}_S$ sentences together to form more complex sentences. Truth-functional connectives, unlike their natural language counterparts, have no effect on the structure of the sentences they bind together. Complex $\mathcal{L}_S$ sentences are obvious compounds of simple $\mathcal{L}_S$ sentences.

As a reflection of this feature of $\mathcal{L}_S$, simple $\mathcal{L}_S$ sentences are called *atomic sentences*, and distinguished from compound *molecular sentences*. Molecules in nature are made up of atoms as parts. In making up a molecule, atoms retain their identities. Similarly, molecular sentences in $\mathcal{L}_S$ are made up of atoms that keep their sentential identities. As you have seen, natural languages are, in this respect, importantly different. In constructing a complex sentence, we typically transform the structure of its simple constituents, often beyond recognition. This feature of English, a feature each of us exploits constantly and unreflectively, can lead to difficulties when you set out to translate from

English into $\mathcal{L}_S$. In constructing a translation, in finding an $\mathcal{L}_S$ sentence that corresponds to some English sentence, you will often need to recover information no longer obviously present in the original sentence.

2.03 Negation: ¬

The symbol ¬ is used to represent negation in $\mathcal{L}_S$. By affixing ¬ to an $\mathcal{L}_S$ sentence, you negate it in much the same way you might negate an English sentence by appending the phrase 'It's not the case that . . .' Suppose you use the $\mathcal{L}_S$ sentence, *W*, to mean

Socrates is wise.

The sentence ¬*W* would mean

It's not the case that Socrates is wise.

or, more colloquially,

Socrates isn't wise.

Similarly, suppose you assign to *H* the meaning

Socrates is happy.

In that case, its negation, ¬*H*, would mean

It's not the case that Socrates is happy.

that is,

Socrates isn't happy.

Negation is the first of five *truth-functional sentential connectives* to be introduced. By comparing the operation of ¬ to the phrase 'It's not the case that . . .' in English, you acquire an informal grasp of its significance. That significance can be specified precisely by means of a *truth table*.

Sentences in $\mathcal{L}_S$ take on one of two values: true (T) or false (F). A truth table spells out the contribution a connective makes to the truth values of sentences in which it occurs. The table below characterizes the negation connective:

p	¬*p*
T	F
F	T

Truth-Functional Connectives

$\mathcal{L}_S$ makes use of five truth-functional connectives:

- ¬ negation (it's not the case that . . .)
- ∧ conjunction (. . . and . . .)
- ∨ disjunction (either . . . or . . .)
- ⊃ conditional (if . . . then . . . ; . . . only if . . .)
- ≡ biconditional (. . . if and only if . . .)

Notice that the table makes use of a sentential variable, p, rather than some particular sentence. This endows the definition with a level of generality it would otherwise lack. The truth table indicates that, given *any* $\mathcal{L}_S$ sentence, p, if p is true, then $\neg p$ is false, and if p is false, then $\neg p$ is true. Negation, then, *reverses the truth value* of sentences to which it is appended. In this regard, negation in $\mathcal{L}_S$ resembles negation in English. If the sentence 'Socrates is wise' is true, then 'It's not the case that Socrates is wise' (or 'Socrates isn't wise') is false; and if the original sentence is false, its negation is true.

$\mathcal{L}_S$ is a *truth-functional language*: the truth value of every sentence in $\mathcal{L}_S$ is a function of the truth values of its constituent sentences. In practice, this means that given any $\mathcal{L}_S$ sentence, p, you can precisely determine its truth value—determine whether it is true or false—if you know (i) the truth values of its constituent sentences, and (ii) the definitions of the truth-functional connectives. If you know that the sentence A is false, then you know (given the definition above) that $\neg A$ is true, and so on for every sentence in $\mathcal{L}_S$.

2.04 Conjunction: ∧

A second truth-functional connective represents the operation of *conjunction*, symbolized in $\mathcal{L}_S$ by an inverted wedge, ∧. As in the case of negation, conjunction in $\mathcal{L}_S$ mirrors conjunction in English. If you place a ∧ between two sentences, the result is a new compound sentence. Thus from W and H, you can construct the conjunction $W \wedge H$. In English, the conjunction

Socrates is wise and Socrates is happy.

results from using 'and' to conjoin the sentences 'Socrates is wise' and 'Socrates is happy'. You are free to combine negated with non-negated sentences or with other negated sentences to form more complex conjunctions:

$$\neg W \wedge H$$
$$W \wedge \neg H$$
$$\neg W \wedge \neg H$$

All of these are perfectly acceptable sentences, as are sentences built up from more than two atomic components:

$$(\neg W \wedge \neg H) \wedge A$$
$$(W \wedge \neg H) \wedge (A \wedge B)$$

In general, you can assemble conjunctions of any finite length. In each case the truth value of the resulting sentence will be a function of the truth values of its constituents.

A truth table characterization of the ∧ is set out below:

p q	$p \wedge q$
T T	T
T F	F
F T	F
F F	F

This truth table is more complicated than the truth table used to characterize negation. There, the aim was to specify the action of negation on single sentences: the negation of any sentence results in a reversal of the sentence's truth value—from true to false, or from false to true. This required only a *single* sentential variable. Every possible truth value of sentences over which that variable ranged could be specified by two rows in the table: a sentence to which a negation sign is appended can be true or false.

Because conjunction is used to conjoin *pairs* of sentences, a truth table characterization of ∧ requires additional rows to allow for the specification of every possible truth value combination of the conjoined sentences.

Sentences flanking the ∧ are called *conjuncts*. Every use of the ∧ involves a pair of conjuncts, so there are *four* possible combinations of truth values to be considered: (i) both conjuncts might be true; (ii) the first might be true, the second false; (iii) the first might be false, the second true; or (iv) both conjuncts might be false.

Truth Functions

$\mathcal{L}_S$ is a *truth-functional language*, a language in which the truth value of every sentence is a function of—is completely fixed by—the truth value of its constituent sentences. If you know the truth values of the simple sentences, you can determine the truth values of any complex sentence in which those simple sentences figure.

The connectives in $\mathcal{L}_S$ (¬, ∧, ∨, ⊃, ≡) are truth-functional connectives. This means that they are defined by reference to their contribution to the truth values of sentences in which they occur.

Think of ¬*p* or *p* ∧ *q* as functions—truth functions—in the sense in which x^2 is a mathematical function.

Truth tables resemble function tables, which depict in a tabular way the action of a particular function. The squaring function, for example, might be pictured by means of the following table:

x	x^2
0	0
1	1
2	4
3	9
4	16
5	25
⋮	⋮

The values appearing on the left side of the table represent the domain of the function; those to the right represent its range. Functions provide mappings from a domain to a range: they associate elements in the one with elements in the other. In the table above, elements in the set of positive integers are associated with elements of that same set. Truth tables associate truth values with truth values.

The truth table for ∧ exhibits the truth value of the resulting conjunction given each of these combinations of values for its conjuncts. A conjunction is true only when *both* of its conjuncts are true (the situation captured in the table's first row). In every other case, the resulting conjunction is false—as the remaining rows indicate.

This feature of conjunction in $\mathcal{L}_S$ tracks conjunction in English. In general, when English sentences are joined by 'and', the resulting compound sentence is true if *both* of its constituent sentences, both of its conjuncts, are true, and false otherwise. The sentence

Socrates is wise and happy.

is true when and only when the sentence 'Socrates is wise' and the sentence 'Socrates is happy' are both true. (Recall that the English sentence above is a stylistic variant of the sentence 'Socrates is wise and Socrates is happy'.)

2.05 Sentential Punctuation

Before delving further into the mysteries of $\mathcal{L}_S$, I invite you to reflect briefly on a problem of notational ambiguity. Suppose you are asked to find the value of the arithmetical expression

$$2 + 3 \times 5$$

In the absence of additional information, the expression is ambiguous, that is, it might mean

the sum of 2 and 3, times 5 (= 25)

or it might mean

2 added to the product of 3 and 5 (= 17)

The difference in the values of these readings of the original expression illustrates the reason mathematicians can ill afford ambiguity.

To avoid ambiguity, you could adopt various notational conventions. You might decide, for instance, always to perform operations in a left-to-right sequence. Were this convention followed in the example above, the expression would be interpreted in the first way. Alternatively, you might adopt a system of punctuation that made use of right and left parentheses so as to force one or another reading:

$$(2 + 3) \times 5$$

or

$$2 + (3 \times 5)$$

The rule, were anyone to take the trouble to formulate it, is that expressions occurring inside matching parentheses are to be replaced by the values of which they are functions. Thus, (2 + 3) would be replaced by 5, and (3 × 5) by 15.

In the case of $\mathcal{L}_S$, a similar technique will be adopted. Consider the sentence

$$\neg P \wedge Q$$

Is this expression to be read as the negation of P, $\neg P$, conjoined to Q? Or is it, rather, the negation of the *conjunction*, $P \wedge Q$? To minimize confusion $\mathcal{L}_S$ incorporates a convention whereby parentheses serve to make clear the *scope* of negation signs and other connectives. Thus, the negation of the conjunction

$$P \wedge Q$$

would be written as follows:

$$\neg(P \wedge Q)$$

If, in contrast, you intend the negation sign to apply exclusively to the first conjunct of a sentence, you need only omit the parentheses:

$$\neg P \wedge Q$$

All this can be summed up in a simple rule: *A negation sign applies only to the expression on its immediate right*. Consider the following sentences:

$$\neg(P \wedge Q) \wedge R$$
$$\neg((P \wedge Q) \wedge R)$$
$$P \wedge (Q \wedge \neg R)$$

In the first sentence, the scope of the negation sign includes the conjunction ($P \wedge Q$); it stops short of the right conjunct, R. In the second sentence, however, the entire complex expression is negated. In the last sentence, the scope of the negation sign includes only the rightmost conjunct, R.

Exercises 2.05

Provide $\mathcal{L}_S$ translations of the English sentences that follow using the $\neg$ and $\wedge$ connectives.
Let E = Elvis croons; F = Fenton investigates; G = George flees; H = Homer flees.

1. George and Homer flee.
2. Homer flees and George flees.
3. Fenton investigates and Homer doesn't flee.
4. It's not the case that both Fenton investigates and Homer doesn't flee.
5. Fenton investigates, Elvis croons, and George flees.
6. It's not the case that George and Homer flee.
7. Homer and George flee, and Fenton doesn't investigate.
8. Homer and George don't flee.
9. Fenton doesn't investigate, and George and Homer don't flee.
10. It's not the case that Homer and George flee, and Fenton doesn't investigate.

2.06 Disjunction: ∨

The third truth-functional connective to be introduced expresses *disjunction*, symbolized by a wedge, ∨. In English, disjunction is most familiarly expressed by the phrase 'either . . . or . . .', as in the sentence

Either it's raining or the sun is shining.

Disjunction in $\mathcal{L}_S$ differs in certain important respects from its English counterpart. The differences become clear once the ∨ is given a truth table characterization:

p q	$p \vee q$
T T	T
T F	T
F T	T
F F	F

As the truth table indicates, a disjunction in $\mathcal{L}_S$ is false when, and only when, *both* of its constituent sentences—its *disjuncts*—are false; otherwise the disjunction is true.

The second, third, and fourth rows of the truth table coincide nicely with our understanding of disjunction in English. Suppose I proclaim the disjunction above, 'Either it's raining or the sun is shining'. You would regard my utterance as true if one of the disjuncts is true: if it is raining but the sun isn't shining (the situation depicted in the second row of the truth table) or if the sun is shining and it is not raining (the third row). Similarly, you would take my utterance to be false if it turned out to be false that the sun is shining and false that it is raining (the table's fourth row).

It is harder to square the first row of the truth table characterization with typical English usage. According to the first row, if *both* of a disjunction's disjuncts are true, the disjunction as a whole is true. This might seem at odds with English usage. If, for instance, I said to you

Iola will arrive Monday or Tuesday.

you would expect to greet Iola on Monday *or* on Tuesday, but *not* on *both* days. This might well be the most common way of understanding 'either . . . or . . .' constructions in English.

Were we to spell out what we have in mind when we use a sentence like that above, we might put it like this:

Iola will arrive on Monday or on Tuesday, but not on both Monday and Tuesday.

Constructions of this sort express *exclusive* disjunction: 'either . . . or . . ., *and not both*'. As the truth table makes plain, a disjunction in $\mathcal{L}_S$ is true if *both* of its disjuncts are true: 'either . . . or . . ., *maybe both*'. While such constructions—*inclusive disjunctions*—occur less frequently in English than exclusive disjunctions, they do occur. Consider the sentence

Employees will be paid time-and-a-half for working on weekends or on holidays.

This sentence might appear on a contract you have made with your employer. It means, of course, that you will be paid extra for work done outside normal working hours: on weekends, holidays—or *both*. You would not be kindly disposed toward an employer who insisted on interpreting the disjunctive clause in an exclusive way, and refusing, for instance, to pay you time-and-a-half for the hours you put in last Saturday on the grounds that last Saturday was New Year's Day, and it is false that time-and-a-half need be paid for work on days that fall *both* on weekends and on holidays.

To capture the meaning of ordinary disjunctions, then, it is important to distinguish *exclusive* disjunction, *either . . . or, not both*, from $\mathcal{L}_S$-style *inclusive* disjunction, *either . . . or . . ., maybe both*. You might think it odd that logicians have elected to define disjunction in $\mathcal{L}_S$ in this inclusive vein, but the reason is simple: by using ∨ to represent inclusive disjunction, you can eat your cake and have it. With a little ingenuity, you can construct sentences that express exclusive disjunctions as well as sentences that express inclusive disjunctions.

Suppose the $\mathcal{L}_S$ sentence, W, is used to express the sentence 'Socrates is wise', H to express 'Socrates is happy', and B to express 'Socrates is bored'. Given these interpretations, together with our understanding of the truth-functional connectives ¬, ∧, and ∨, you are in a position to produce endless complex $\mathcal{L}_S$ sentences. Consider the following together with their English equivalents:

Ambiguous Sentences

Some of the sentences in the previous exercises are ambiguous; they have more than one meaning, hence more than one translation. Try to identify the sentences that are ambiguous. Notice whether distinct translations into $\mathcal{L}_S$ are required depending on which meaning is selected.

$$(W \wedge B) \vee H$$

Socrates is wise and bored, or he is happy.

$$(\neg W \vee B) \wedge \neg H$$

Socrates isn't wise or he's bored, and he isn't happy.

$$\neg W \vee \neg H$$

Socrates isn't wise or he isn't happy.

Provided you recognize that the third sentence leaves open the possibility that Socrates is *both* unwise and unhappy, these translations are straightforward.

A glance at the sentences above reveals a simple way of representing *exclusive disjunctions in* $\mathcal{L}_S$. Return for moment to the sentence

Iola will arrive Monday or Tuesday.

This sentence, as has been noted, would most naturally be used to assert that Iola will arrive on Monday or on Tuesday, but not on *both* Monday and Tuesday. Suppose you broke this asserted content down into components. First, the sentence indicates that Iola will arrive either on Monday or on Tuesday—and not, say, on Friday. This might be expressed as follows:

$$M \vee T$$

The English sentence also suggests that Iola will not arrive on *both* Monday and Tuesday. (At least it so informs us, *given* background information—for instance, that an *arrival* occurs at the onset of a visit, and that Iola's visit can begin on Monday only if it does not begin on Tuesday, and vice versa.) This aspect of the sentence's meaning could be captured in $\mathcal{L}_S$ via a negated conjunction:

$$\neg(M \wedge T)$$

This negated conjunction asserts that it is not the case that Iola will arrive both on Monday *and* on Tuesday, or more colloquially, *not both* Monday and Tuesday. The negation sign includes within its scope the *entire* conjunction.

A negated conjunction differs importantly from a conjunction of negations (the negation sign does not 'distribute'). The sentence above means something quite different from

$$\neg M \wedge \neg T$$

This sentence means that Iola will not arrive on Monday *and* not arrive on Tuesday, that she will arrive on *neither* day. This sentence is patently *inconsistent* both with the disjunctive sentence with which we began, 'Iola will arrive either Monday or Tuesday', and with the $\mathcal{L}_S$ sentence

$$M \vee T$$

which captures at least a part of the meaning of the English original. *No* disjunction is true if both of its disjuncts are false.

By now it should be clear that the exclusive disjunctive sense of the original English sentence can be represented in $\mathcal{L}_S$ by means of a conjunction of $\mathcal{L}_S$ sentences

$$M \vee T$$

which introduces the 'either . . . or . . .' component of the English original, and

$$\neg(M \wedge T)$$

which introduces the '. . . not both' component. When conjoined, these two segments become

$$(M \vee T) \wedge \neg(M \wedge T)$$

The first conjunct of this complex expression captures the *disjunctive* aspect of the English sentence, while its second conjunct captures its *exclusive* aspect.

In practice you can adopt the following principle in translating disjunctions from English into $\mathcal{L}_S$: translate disjunctions as *inclusive* disjunctions (that is, just using the $\vee$) unless they are *only* interpretable as exclusive

Ambiguity in $\mathcal{L}_S$

$\mathcal{L}_S$ makes use of the familiar mathematical convention of using parentheses to disambiguate sentences. (An ambiguous sentence is disambiguated when its intended meaning is made clear.) The expression

$$A \vee B \wedge C$$

is ambiguous as between

$$(A \vee B) \wedge C$$

and

$$A \vee (B \wedge C)$$

$\mathcal{L}_S$ can contain no ambiguous sentences, so the first expression is not a sentence of $\mathcal{L}_S$. It can be turned into a sentence by the addition of parentheses in either of the two ways set out above. Do these two sentences have the same meaning?

disjunctions. If a sentence could be read as expressing an inclusive disjunction, even if it might be read as expressing an exclusive disjunction as well, translate it using just the ∨.

Exercises 2.06

Provide $\mathcal{L}_S$ translations of the English sentences that follow using the ¬, ∧, and ∨ connectives. Let E = Elvis croons; F = Fenton investigates; G = George flees; H = Homer flees.

1. Elvis croons and Homer flees.
2. Homer flees or George doesn't.
3. Fenton investigates and either Homer doesn't flee or George flees.
4. It's not the case that either Fenton investigates or Homer doesn't flee.
5. Either it's not the case that Fenton investigates or Homer doesn't flee.
6. Either Homer or George flees, but not both.
7. Elvis croons and Homer and George flee, or Fenton doesn't investigate.
8. Homer or George doesn't flee.
9. Fenton doesn't investigate or Elvis doesn't croon, but not both.
10. Either Elvis croons or George or Homer flees.

2.07 The Conditional: ⊃

The fourth truth-functional $\mathcal{L}_S$ connective is the *conditional*. The horseshoe symbol, ⊃, as characterized in the truth table below, expresses in $\mathcal{L}_S$ roughly what is expressed in English by the phrases '. . . only if . . .' and 'if . . . then . . .' Conditional sentences have a central role in $\mathcal{L}_S$, one best appreciated if you compare them to 'if . . . then . . .' sentences in English. Consider, first, the truth table characterization of the ⊃:

p q	$p \supset q$
T T	T
T F	F
F T	T
F F	T

Reading $p \supset q$ as 'p is true *only if* q is true' or '*if* p is true, *then* q is true', you can begin to see how close conditionals in $\mathcal{L}_S$ come to those in English. The first two rows of the truth table fit nicely with our pre-$\mathcal{L}_S$ conception of conditionality. A conditional assertion is true if both its *antecedent*

(the sentence to the left of the ⊃) and its *consequent* (the sentence to the right) are true. Similarly, a conditional sentence with a true antecedent and a false consequent is clearly false.

Consider, for instance, the English sentences

It's raining only if the street is wet.

and

If it's raining, then the street is wet.

Both sentences could be translated into $\mathcal{L}_S$ as

$$R \supset W$$

If I assert this conditional, you are unlikely to object if you notice both that it is raining and that the street is wet, that is, if you notice that both its antecedent and its consequent are true. This corresponds to the first row of the truth table characterization. If you notice that it is raining (the antecedent of the conditional is true) and that the street is not wet (its consequent is false), you would declare my conditional false as the truth table's second row has it.

Now imagine a situation in which your observations correspond to the third and fourth rows of the truth table. Suppose, for instance, you observe that it is not raining but that the street is nevertheless wet (the truth table's third row), or that it is not raining and the street is not wet (the last row). Is it obvious that in those cases you ought to regard what I have said as *true*?

A short but unsatisfying answer to this question is that since these observations would not make the conditional sentence false, they make it true. Sentences in $\mathcal{L}_S$ must be either true or false, so an $\mathcal{L}_S$ sentence that is not false is thereby true.

Perhaps this is not so for English sentences, however. Your observing a cloudless sky and a dry street might not show that the sentence is false, but neither does your observation suggest that it is true. Perhaps English conditionals are *neither* true nor false when their antecedents are false; perhaps under those circumstances they lack a truth value.

Although this possibility cannot be ignored, there might be simpler explanations available for the apparent lack of fit between conditionals in English and those in $\mathcal{L}_S$.

You can get a feel for all this by reflecting on the logic of ordinary English conditionals. Return to the original sentence

If it's raining, then the street is wet.

Truth and Truth Conditions

Every sentence of $\mathcal{L}_S$ has a set of *truth conditions*: those circumstances under which it is true, and those circumstances under which it is false.

Every sentence of $\mathcal{L}_S$ has as well a truth *value*: it is either true or false.

The truth value of a sentence is fixed by

1. the sentence's truth conditions;
2. the state of the world at the time the sentence is uttered.

You can know the truth conditions of a sentence without knowing its truth value. You know the truth conditions of the sentence 'There is a pound of gold within one kilometer of the north pole of Venus'. You know what would make it true or false, even though you do not know its truth value—you do not know *whether* it is true or false.

which is, or so I have asserted, equivalent to 'It's raining only if the street is wet' and to the $\mathcal{L}_S$ sentence

$$R \supset W$$

One obvious feature of this conditional is that, in uttering it, I need not be interested in whether it is, at the time, raining, or whether the street happens to be wet. My aim, rather, is to assert that *it is false that it is both raining and the street is not wet*. More generally, 'if p, then q' indicates that you cannot have p be true without q being true as well. This is explicit in the second row of the truth table characterization of the ⊃.

Suppose you translate this gloss on the sentence into $\mathcal{L}_S$:

$$\neg(R \wedge \neg W)$$

and suppose you agree that this negated conjunction reasonably captures what I meant in asserting the original conditional sentence. Where does this leave us with respect to our original characterization of conditionals in $\mathcal{L}_S$? It would seem to follow that the English sentence

It's raining only if the street is wet.

means the same as the sentence

It's not the case that it is raining and the street isn't wet.

But what does it mean to say that two sentences have the same meaning?

Consider what you know when you know what the sentences above mean. Whatever else you might know, you know the conditions under which the sentence in question is true or false; you know the sentence's truth conditions. Again, this does not mean that you know *whether* the sentence is true or false. You know the truth conditions of the sentence 'At this moment, the number of pigeons on St. Peter's dome is even'. That is, you know what would make it true and what would make it false, despite having no idea whether the sentence is true or false.

This suggests that the meaning of a sentence is connected in some important way with its truth conditions. At any rate, two sentences with the same meaning might be thought to have the same truth conditions. In translating from one language to another, for instance, the goal is to find sentences with matching truth conditions.

In working out the truth conditions for ordinary English sentences, we are obliged to fall back on our tacit, intuitive knowledge of the language. In $\mathcal{L}_S$ matters are different. Because $\mathcal{L}_S$ is a truth-functional language, its truth-functional connectives have precise characterizations. As a result, you can easily specify the truth conditions of any $\mathcal{L}_S$ sentence. To do so, you need only construct a truth table for the sentence in question.

Thus far truth tables have been used exclusively to provide formal characterizations of connectives. *Given* these characterizations, however, you are in a position to devise truth table *analyses* of particular $\mathcal{L}_S$ sentences, analyses that provide a clear specification of the truth conditions of any sentence expressible in $\mathcal{L}_S$. Consider the $\mathcal{L}_S$ sentence

$$R \supset W$$

You can easily construct a truth table for this sentence that makes its truth conditions explicit. To do so, you need only call to mind the truth table characterization of the ⊃ connective:

$p\ q$	$p \supset q$
T T	T
T F	F
F T	T
F F	T

This truth table indicates, in essence, that a conditional sentence in $\mathcal{L}_S$ is true except when its antecedent (that is, its left-hand constituent) is true and its consequent (what is to the right) is false, the situation realized in the second row of the truth table above.

Applying this information to the $\mathcal{L}_S$ conditional yields the following truth table:

$R\ W$	$R \supset W$
T T	T
T F	F
F T	T
F F	T

The sentences *R* and *W* can each be true or false, and, in consequence, there are four possible truth value combinations of these sentences to consider. The truth table characterization of the ⊃ indicates that an $\mathcal{L}_S$ conditional is false when its antecedent is true and its consequent is false; it is true otherwise. Putting all this together yields the truth table above.

This truth table makes the truth conditions of the original sentence explicit. You could be said to *know* its truth conditions so long as you could reconstruct its truth table. As noted earlier, knowing the truth conditions of a given sentence is not the same as knowing its truth *value*. Whether or not a sentence is true depends on which row of the truth table corresponds to what is the case as a matter of fact.

You could think of each row of a truth table as describing a set of *possible worlds*, a possible world being *a way the world could be*. Knowing the truth conditions for a given sentence is a matter of knowing its truth value at *every possible world*, knowing what its truth value *would* be were the universe a particular way.

That might seem a daunting task, but the appearance is misleading. There is an infinitude of alternative universes, but only four possible combinations of truth values for the sentences *R* and *W*. What the truth table reveals is that at any world—that is, under any circumstances—in which *R* is true and *W* is false, the conditional $R \supset W$ is false; otherwise it is true. If you know the truth value of a given sentence in addition to its truth conditions, if you know whether or not it is true, then you know which of these sets of alternative worlds includes our universe.

'Possible Worlds'

Appeals to possible worlds to illuminate topics in philosophy originated with the philosopher G. W. Leibniz (1646–1716). Leibniz noted that the actual world might have been different in countless ways. Each of these ways it might have been is a possible world.

Leibniz is famous for arguing that you can account for the existence of the actual world only by supposing that the actual world is the best of all possible worlds, prompting a cynical response from Voltaire (1694–1778): 'If this is the best of all possible worlds, I should hate to see the others!'

Do possible worlds (other than the actual world) exist? Some philosophers have thought that they do, on the grounds that there are, objectively, ways the actual world might have been. Others regard talk about possible worlds as a convenient fiction.

Return now to the $\mathcal{L}_S$ sentence I suggested came close to capturing what we have in mind in asserting an ordinary English conditional. The starting point was the sentence

If it's raining, then the street is wet.

and the suggestion was that this sentence had the same truth conditions as the sentence

It's not the case that it is raining and the street isn't wet.

which is translated into $\mathcal{L}_S$ as

$$\neg(R \wedge \neg W)$$

What happens when you work up a truth table for this sentence and compare its truth conditions with the truth conditions of the original conditional sentence?

To construct a truth table for the sentence, you first break down the sentence into its constituent sentences. That is, before you determine the truth conditions for the sentence as a whole, you must determine the truth conditions for

$$R \wedge \neg W$$

the sentence contained inside the parentheses. Before working out the truth conditions for *this* sentence, however, you would first need to calculate the truth conditions for its right-hand conjunct

$$\neg W$$

Armed with that information, you could construct a truth table for the entire sentence.

This step-by-step procedure is set out in the truth table below:

R W	$\neg W$	$R \wedge \neg W$	$\neg(R \wedge \neg W)$
T T	F	F	T
T F	T	T	F
F T	F	F	T
F F	T	F	T

The two leftmost columns provide an inventory of the possible combinations of the truth values of R and W. The next column sets out the truth conditions for $\neg W$, the truth value of $\neg W$ given particular truth values for W. From the truth table definition of $\neg$, you know that $\neg W$ is false when W is true, and true when W is false.

Now, relying on the truth table characterization of $\wedge$, you can specify truth conditions for the conjunction $R \wedge \neg W$. A conjunction is true only when both of its conjuncts are true, and false otherwise. So the conjunction $R \wedge \neg W$ is true only in those rows of the truth table (only in those worlds) in which both R and $\neg W$ are true. The relevant columns are the R column and the $\neg W$ column.

Having established the truth conditions for the conjunction $R \wedge \neg W$, you need only determine the truth conditions for the negation of this conjunction. Recalling the truth table characterization of negation in $\mathcal{L}_S$, you know that negating a sentence reverses its truth value: a true sentence, negated, is false, and the negation of a false sentence is true. Thus, whenever the sentence $R \wedge \neg W$ is true, its negation $\neg(R \wedge \neg W)$ is false, and whenever $R \wedge \neg W$ is false, its negation is true. This is reflected in the rightmost column of the truth table.

The truth table just constructed provides an explicit representation of the truth conditions of the $\mathcal{L}_S$ sentence I suggested came close to capturing the sense of the original English conditional. The conditional 'If it's raining, then the street is wet' is equivalent to 'It's not the case that it's raining and the street isn't wet'. If you compare that truth table with the truth table constructed for the corresponding $\mathcal{L}_S$ conditional, you can see there is a perfect match:

R W	$\neg(R \wedge \neg W)$	$R \supset W$
T T	T	T
T F	F	F
F T	T	T
F F	T	T

The truth table shows clearly that the truth conditions for the two $\mathcal{L}_S$ sentences are identical. This means that, for our purposes, the sentences have the same meaning. Given that the one approximates our English original, the other must as well.

All this suggests that $\mathcal{L}_S$ conditionals are closer in meaning to English 'if . . . then . . .' sentences than is often supposed. It does not follow that conditional constructions in English invariably express conditionals in the manner of $\mathcal{L}_S$. English sentences containing 'if . . . then . . .' clauses can be used to express other relations. At first this might lead to intermittent fits of anxiety when you set out to translate English sentences into $\mathcal{L}_S$. Occasionally, English sentences that resemble $\mathcal{L}_S$-style conditionals will turn out not to be conditionals at all. Eventually, if you persist, these difficulties will sort themselves out and conditional translations into $\mathcal{L}_S$ will seem natural.

Conditionals Brought to Heel

Suppose you know that all the balls in a particular urn are the same color, but you do not know what that color is. In that case you know that the conditional 'if ball A is red, then ball B is red' is true even if 'ball A is red' and 'ball B is red' are both false—the situation captured in the last row in the truth table.

2.08 Conditionals, Dependence, and Sentential Punctuation

One additional feature of conditional sentences in $\mathcal{L}_S$ bears mention. In describing a sentence of the form 'If it's raining, then the street is wet' as a conditional sentence, it is tempting to regard the consequent, 'The street is wet', as *conditional* or *dependent* on the antecedent, 'It's raining'. That would be incorrect, however. What the conditional sentence asserts is that, if the antecedent is true, the consequent is true (otherwise the conditional is false). This does *not* amount to the assertion that the consequent is causally dependent on the antecedent.

The distinction is easy to miss because conditionals are often used to express causal relations. Consider the sentence

If you drink epoxy, you'll regret it (you'll regret drinking epoxy).

When you hear this sentence, you might naturally envisage a causal connection between your drinking epoxy and your subsequent regret. Your regret would be *brought about* by your drinking epoxy. Given that conditional 'if . . . then . . .' sentences are equivalent to '. . . only if . . .' sentences, however, the sentence above could be paraphrased as

You drink epoxy only if you'll regret it.

This paraphrase feels wrong. It appears to reverse the order of dependence: your regret now appears to lead to your drinking epoxy!

The appearance is misleading. It stems from a tendency to conflate logical and causal relations. The original sentence expresses, among other things, a particular logical relation between the conditional sentence and its constituent sentences 'You drink epoxy' and 'You'll regret it': if the first is true, then the second is true as well. Alternatively, it is not the case that the first is true and the second false. Doubtless what is responsible for the truth of the sentence is a causal relation between drinking epoxy and being sorry. But you can consider logical relations that sentences express independently of the state of the universe responsible for their truth or falsity. Conditional constructions compel us to do precisely that.

In practice, you can accustom yourself to distinguishing logical from causal relations by focusing explicitly on the truth conditions of sentences. Faced with the sentence 'If you drink epoxy, then you'll regret it', you first notice that the sentence comprises two component sentences, 'You drink epoxy' and 'You'll regret it'. The task then is to tease out the logical relation holding between these two sentences, that is, to think through the relation between the truth conditions of the original conditional sentence and those of the sentences making up that sentence's antecedent and consequent.

The conditional sentence is false (for whatever reason) when (i) the sentence 'You drink epoxy' is true *and* (ii) the sentence 'You'll regret it' is false: the first sentence is true *only if* the second is true. This is just what is captured in $\mathcal{L}_S$ by the $\supset$ connective. Someone's uttering the English sentence would doubtless convey *more* than this logical relation, but it expresses at least this much.

In translating English 'if . . . then . . .' conditionals into $\mathcal{L}_S$, begin by first locating the if-clause. The sentence associated with the if-clause belongs at the left of the conditional sign when the sentence is translated into $\mathcal{L}_S$. Although this might seem obvious, it is not so obvious when a sentence's if-clause is buried in the middle of the sentence, or when it follows a consequent then-clause. You can say

If it's raining then the street is wet.

or

The street is wet if it's raining.

Both sentences are translated into $\mathcal{L}_S$ as

$$R \supset W$$

In translating the second sentence, you would first spot the if-clause, 'It's raining', and identify this as the antecedent of the conditional.

If-clauses are one thing, *only-if*-clauses are another. A sentence of the form

$$p, \text{ if } q$$

is translated into $\mathcal{L}_S$ as

$$q \supset p$$

In contrast, a sentence of the form

$$p \text{ only if } q$$

goes into $\mathcal{L}_S$ as

$$p \supset q$$

Consider the English sentence

Gertrude will leave *if* Fenton investigates.

Assuming that G = 'Gertrude will leave' and F = 'Fenton investigates', this sentence can be translated into $\mathcal{L}_S$ as

$$F \supset G$$

This sentence is very different from the superficially similar sentence

Gertrude will leave *only if* Fenton investigates.

as its $\mathcal{L}_S$ translation reveals

$$G \supset F$$

In confronting an English conditional, then, first find the if-clause. This will be the antecedent of the conditional—*unless* the clause is an only-if-clause. In that case, the clause functions as the consequent of a conditional.

More complex conditionals follow suit. In § 2.05 parentheses were introduced to provide a simple way to disambiguate sentences. Negation signs obey the rule: *a negation sign applies only to the sentence to its immediate right*. Thus, in the sentence

$$\neg A \supset B$$

only the A is negated, while the negation sign in the sentence

$$\neg(A \supset B)$$

applies to the whole conditional, $A \supset B$.

Now consider the ambiguous expression

$$A \supset B \wedge C$$

This expression could be read as a conditional, the consequent of which is a conjunction:

$$A \supset (B \wedge C)$$

But the expression could just as easily be construed as a conjunction, the left conjunct of which is a conditional:

$$(A \supset B) \wedge C$$

A moment's reflection will reveal that the two sentences differ in their truth conditions. Potential ambiguities can be avoided by adhering to the principle: *no sentence can share connectives with more than one distinct sentence*. The principle is violated by the original ambiguous expression; B shares the connective $\supset$ with A, and the connective $\wedge$ with C. By adding parentheses, the ambiguity is dispelled. In the sentence

$$(A \supset B) \wedge C$$

A and B share the connective $\supset$, while $(A \supset B)$ and C share $\wedge$.

Exercises 2.08

Translate the following sentences into $\mathcal{L}_S$ using the $\neg$, $\wedge$, $\vee$, and $\supset$ connectives. If the English sentence is ambiguous, provide distinct $\mathcal{L}_S$ versions of it. But remember: no $\mathcal{L}_S$ sentence can be ambiguous. Let E = Elvis croons; F = Fenton investigates; G = George flees; H = Homer flees.

1. If Elvis croons, Homer flees.
2. Homer flees if Elvis croons.
3. Homer flees only if Elvis croons.
4. If George or Homer flees, Fenton investigates.
5. Fenton investigates if George or Homer flees.
6. If Elvis croons, then, if George flees, Fenton investigates.
7. If George and Homer flee, Elvis doesn't croon.
8. Homer or George flees if Elvis croons.
9. Fenton doesn't investigate if Elvis doesn't croon.
10. Fenton investigates only if either Elvis croons or George or Homer flees.

2.09 The Biconditional: ≡

Once you have conditional sentences under your belt, biconditionals are a piece of cake. A biconditional sentence amounts to a two-way conditional. English biconditionals are associated with the phrases '. . . if and only if . . .' and '. . . just in case . . .' Consider the English sentence

A solution is acid if and only if it turns litmus paper red.

This sentence incorporates the truth conditions not only of the conditional

If a solution is acid, then it turns litmus paper red.

but also of the complementary conditional

If a solution turns litmus paper red, then it is acid.

Biconditionals express bidirectional, 'back-to-back' conditionals in a single sentence. This means that biconditionals can be paraphrased by conjoining a pair of complementary conditionals.

Consider another English example:

A number is prime just in case it is divisible only by itself and one.

The sentence can be paraphrased by means of the conjoined pair

If a number is prime, then it is divisible only by itself and one.

and

If a number is divisible only by itself and one, then it is prime.

Ordinary conditionals are not in this way bidirectional. For instance, the English conditional

If it's Ferguson's bread, then it's wholesome.

does not imply

If it's wholesome, then it's Ferguson's bread.

The truth of the first sentence is compatible with the falsehood of the second sentence. You could assert with perfect consistency that if it is Ferguson's bread then it's wholesome, while denying that if it is wholesome, then it is Ferguson's bread: plenty of things other than Ferguson's bread are wholesome.

These remarks about English biconditionals apply straightforwardly to biconditionals in $\mathcal{L}_S$. First, consider the table below, which provides a characterization of the biconditional in $\mathcal{L}_S$.

p q	$p \equiv q$
T T	T
T F	F
F T	F
F F	T

Now recall the biconditional introduced earlier:

A solution is acid if and only if it turns litmus paper red.

You might represent this in $\mathcal{L}_S$ as a pair of conjoined, back-to-back conditionals:

$$(A \supset R) \wedge (R \supset A)$$

Compare the truth conditions of this sentence to those for the corresponding biconditional.

A R	$A \supset R$	$R \supset A$	$(A \supset R) \wedge (R \supset A)$	$A \equiv R$
T T	T	T	T	T
T F	F	T	F	F
F T	T	F	F	F
F F	T	T	T	T

The truth conditions of the biconditional sentence match those of the conjoined (back-to-back) conditionals. The sentences, despite their different appearances, have the same truth conditions; hence, for our purposes, they mean the same.

If you accept that (i) English biconditionals are nicely captured using conjoined, back-to-back conditionals in $\mathcal{L}_S$, and (ii) biconditionals in $\mathcal{L}_S$ expressed using the ≡ have the same truth conditions as back-to-back $\mathcal{L}_S$ conditionals, you can accept with a clear conscience (iii) the biconditional in $\mathcal{L}_S$ parallels biconditionality in English.

Exercises 2.09

Translate the following sentences into $\mathcal{L}_S$. If the English sentence is ambiguous, provide distinct $\mathcal{L}_S$ versions of it. But remember: no $\mathcal{L}_S$ sentence can be ambiguous. Let E = Elvis croons; F = Fenton investigates; G = George flees; H = Homer flees.

1. Homer flees if Elvis croons.
2. Homer flees only if Elvis croons.
3. Homer flees if and only if Elvis croons.
4. Homer flees if and only if either Elvis croons or Fenton investigates.

5. Elvis croons just in case George or Homer flees.
6. Elvis croons just in case George and Homer flee.
7. Elvis croons if and only if George and Homer don't flee or Fenton doesn't investigate.
8. Homer or George flees just in case Elvis croons.
9. If Fenton doesn't investigate, then Elvis doesn't croon if and only if George doesn't flee.
10. If Fenton investigates, then Elvis croons just in case George doesn't flee.

2.10 Complex Truth Tables

As the discussion above illustrates, truth tables are used both to define truth-functional connectives and to determine the truth conditions of $\mathcal{L}_S$ sentences. A truth table, in effect, exhibits the truth value of a sentence in every possible situation, or, as it is commonly put, at every possible world. These are represented by rows on the truth table.

Take the sentence

Snow is white.

This sentence is true at some worlds—those at which snow *is* white—and false at others.

S
T
F

Most sentences will be true at many possible worlds and false at many others, so each row in a truth table represents a *set* of possible worlds, the set, namely, in which the sentence has a particular assignment of truth values. Given an arbitrary sentence, p, you might wonder *how many* sets of possible worlds you ought to consider, how many rows we must include in p's truth table.

Happily, there is a simple way to calculate the number of rows required. First, count the number of *distinct atomic sentences* occurring in the target sentence. Every $\mathcal{L}_S$ sentence contains at least one atomic constituent, and it can contain many more. The number of distinct atomic sentences is arrived at by counting occurrences of distinct uppercase letters. Thus, the sentence

$$\neg A \supset B$$

contains *two* distinct uppercase letters, A and B, and the sentence

$$\neg A \supset (B \vee \neg C)$$

contains *three*, A, B, and C. Notice that the sentence

$$\neg A \supset (B \vee (\neg C \wedge A))$$

contains just *three* distinct uppercase letters. There are four uppercase letters, but only three *distinct* uppercase letters: *A*, *B*, and *C*.

Given this number, you can calculate, for a given sentence, how many sets of possible worlds are in play, how many rows to include in the sentence's truth table. You know that each atomic sentence has one of two values: true, false. A truth table provides an inventory of *all possible combinations* of these truth values. Because each atomic sentence can have one of two values, there will be 2^n possible combinations of values, where n is the number of distinct atomic sentences.

This means that truth tables for sentences containing a single atomic sentence require two rows ($2^1 = 2$), truth tables for sentences containing two distinct atomic sentences require four rows ($2^2 = 4$), those with three require eight rows ($2^3 = 8$), and so on.

In addition to making sure you have the correct number of rows, you must also be certain to include in a truth table every possible *combination* of truth values, every relevant set of possible worlds. A truth table might have enough rows but fail to do this because some rows repeat others, as in the truth table below.

A B	$A \supset B$
T T	T
F F	T
T T	T
F F	T

In this case the third row repeats the first, and the fourth row repeats the second. When that happens, one or more possible truth value combinations must have been left out. You can ensure that every possible combination of truth values is represented in a truth table by adopting a simple convention. Consider an $\mathcal{L}_S$ sentence introduced earlier:

$$\neg A \supset (B \vee (\neg C \wedge A))$$

This sentence contains three distinct atomic constituent sentences, so its truth table requires *eight* rows reflecting the eight possible combinations of truth values of *A*, *B*, and *C*. The convention for assigning truth values to truth table rows is this:

1. Set out at the left of the table the atomic constituents of the sentence for which the truth table is being constructed.

A B C	

2. Determine the number of rows required in the manner explained above. The truth table will require 2^n rows for n distinct atomic sentences.
3. In the rightmost column, in this example, the *C*-column, *alternate* *T*s and *F*s, beginning with *T*s.

4. In the next column to the left, here the *B*-column, again starting with *T*s, alternate *pairs* of *T*s and *F*s.

5. In the next column to the left, here the *A*-column, alternate *quadruples* of *T*s and *F*s, beginning with *T*s. Continue this procedure until the leftmost column is reached, each time doubling the length of strings of *T*s and *F*s, and alternating these groups beginning with *T*s. In the leftmost column, the first $2^n/2$ rows will contain *T*s and the remaining $2^n/2$ rows will hold *F*s.

A B C	
T T T	
T T F	
T F T	
T F F	
F T T	
F T F	
F F T	
F F F	

This technique, in addition to ensuring that you have an exhaustive listing of truth value combinations, also ensures that truth tables exhibit a *standard pattern* that makes them less tedious to construct than it would be otherwise.

Having established a framework, you can move on to calculate the truth conditions of the target sentence. In the case of complex sentences it is easy to make careless mistakes. Because a mistake at one point is inherited elsewhere, it is wise to proceed cautiously and systematically. Rather than attempting to ascertain the truth conditions for a complex sentence *all at once*, it is safer to work out the truth conditions for its components, moving 'inside out' from smaller to larger units until the level of the whole sentence is reached. In practice, the procedure works as follows:

1. Calculate the truth conditions for every *negated atomic sentence*. Here, the target sentence contains *two* negated atomic sentences, $\neg A$ and $\neg C$.

2. Calculate the truth conditions of sentences within parentheses working from the innermost parentheses out. In this case you would calculate the truth conditions for $(\neg C \wedge A)$, then those for $B \vee (\neg C \wedge A)$. Eventually you arrive at the truth conditions for the whole sentence.

This procedure is illustrated below. Truth conditions for $\neg A$ and $\neg C$ are set out at the left, and those for subsequent sentences make use of these calculations.

$A\ B\ C$	$\neg A$	$\neg C$	$\neg C \wedge A$	$B \vee (\neg C \wedge A)$	$\neg A \supset (B \vee (\neg C \wedge A))$
T T T	F	F	F	T	T
T T F	F	T	T	T	T
T F T	F	F	F	F	T
T F F	F	T	T	T	T
F T T	T	F	F	T	T
F T F	T	T	F	T	T
F F T	T	F	F	F	F
F F F	T	T	F	F	F

Confronted by a sprawling truth table, you might be tempted to take shortcuts, but shortcuts are ill advised. The more complex the sentence, the more ways you can go wrong, and the more caution you must exercise to avoid mistakes.

Once you absorb the principles of truth table construction, it is a simple, although occasionally grueling, matter to construct truth tables for any $\mathcal{L}_S$ sentence, and thereby to specify in a rigorous way that sentence's truth conditions. Were you to do this long enough and on enough sentences, you would begin to discover sentence pairs that, while differing syntactically, in their arrangement of elements, nevertheless exhibited *identical truth conditions*. In that case, you would be entitled to regard the sentences as having the same meaning.

None of this is surprising. Suppose you tell me that Iola procrastinates. When asked to explain, you say that Iola puts things off. In this case, the English sentences

Iola procrastinates.

and

Iola puts things off.

are *paraphrases* of one another and have the same truth conditions: they will be true or false at the same worlds. You can test this intuition by trying to envisage a possible world at which the first sentence is true and the second false (or vice versa).

When you discover paraphrase relations in a natural language like English, you discover, in effect, ways you might *eliminate* elements of the language without affecting our ability to describe our world. The sentences above illustrate the point. Given that those sentences are paraphrases, we could eliminate the word 'procrastinate' from English, thereby simplifying the language, without affecting its truth-stating capacity. The elimination of terms, of course, would affect other aspects of the language, for instance, its ability to express truths elegantly or poetically.

Such considerations need not deter us when we turn to $\mathcal{L}_S$. $\mathcal{L}_S$ is a formal idiom with no poetic pretentions. That being the case, you might wonder whether the vocabulary of $\mathcal{L}_S$ could be pared down without affecting its expressive power. $\mathcal{L}_S$ is pretty stark as it is. We need a stock of atomic sentences, so there would be no point in trying to limit these. But what about our treasured truth-functional

connectives? Might *these* somehow be reduced in number? Were their numbers diminished, what would be the effects on those left with the job of putting $\mathcal{L}_S$ through its paces?

Recall the earlier discussion of biconditionals. Every biconditional sentence corresponds to a pair of back-to-back conditionals. Thus, the sentence

$$A \equiv B$$

corresponds to the sentence

$$(A \supset B) \wedge (B \supset A)$$

These sentences are *logically equivalent*, that is, they have identical truth conditions—as you could easily prove by constructing a truth table for each sentence (as in § 2.09). The second sentence provides a paraphrase of the first. This suggests a way of *eliminating* the biconditional connective ≡ from $\mathcal{L}_S$ without affecting the language's expressive powers: whenever a biconditional is called for, you could instead use back-to-back conditionals.

So we could live without at least one element of $\mathcal{L}_S$, the biconditional connective. What of the others? Think back to the earlier discussion of conditionals in $\mathcal{L}_S$ and in English (§§ 2.07–2.08). I endeavored to motivate the usual truth table characterization of ⊃ by noting that ordinary English conditionals of the form

If *A* then *B*

could be regarded as having the same truth conditions as sentences of the form

It's not the case that *A* and not-*B*.

and in $\mathcal{L}_S$

$$\neg(A \wedge \neg B)$$

If you construct a truth table for this sentence, you discover that it has the same truth conditions as its conditional counterpart

$$A \supset B$$

At the time, this information was used to make a case for a particular characterization of the ⊃ connective. Now you can see that the logical equivalence we uncovered equips us with a technique for dispensing with the ⊃. Just as in the case of the biconditional, you could forego the ⊃, replacing it with a combination of ¬s and ∧s. Given that occurrences of ≡ could be replaced with back-to-back conditionals using just ∧s and ⊃s, and given the replaceability of the ⊃, it appears that *both* the ⊃ and the ≡ could be eliminated without affecting the expressive power of $\mathcal{L}_S$. Combinations of ⊃s and ∧s could stand in for occurrences of ≡s. Then, in the resulting sentences, occurrences of ⊃s could be replaced with ¬s and ∧s.

Might it be possible to go further? Might the ∧, the ¬, or the ∨ be eliminated? If so, *how far* might $\mathcal{L}_S$ be whittled down?

As it happens, you could get by in $\mathcal{L}_S$ with only negation ¬ and *one* of the connectives ⊃, ∧, ∨. For any sentence you might express using these, you could substitute a *logically equivalent* sentence, a sentence with identical truth conditions, that did not use the connective. You might, in a fit of logical

zeal, scrap all of the connectives other than the ¬ and, say, the ∨. The result would be a no-frills language of austere beauty—or, at any rate, an austere language.

Austere beauty has its price. Were $\mathcal{L}_S$ purged of expendable truth-functional connectives, the result would be a language much harder to use. Sentences that are easy to write in $\mathcal{L}_S$ as it is now constituted would become dreary strings of ¬s and ∨s. To illustrate, consider the sentence

$$A \equiv (B \wedge C)$$

Compare the following paraphrase of this sentence using only ¬s and ∨s:

$$\neg(\neg(\neg A \vee \neg(\neg B \vee \neg C)) \vee \neg((\neg B \vee \neg C) \vee A))$$

More complex sentences would require correspondingly more complex paraphrases. I leave it to you the reader to construct a truth table proof that the sentences above do indeed have the same truth conditions.

Exercises 2.10

Construct a truth table for each of the $\mathcal{L}_S$ sentences below.

1.	$\neg P \vee Q$	2.	$\neg P \wedge \neg Q$
3.	$\neg P \vee \neg Q$	4.	$\neg(\neg P \vee \neg Q)$
5.	$\neg(\neg P \wedge \neg Q)$	6.	$P \supset \neg Q$
7.	$P \supset (Q \wedge R)$	8.	$P \supset \neg(Q \wedge \neg R)$
9.	$P \wedge (Q \vee R)$	10.	$\neg(\neg P \vee (\neg Q \wedge \neg R))$

2.11 The Sheffer Stroke: |

Someone might take all this as a challenge. If you could manage with just two truth-functional connectives, might you get by with *even fewer*? Might it be possible somehow to preserve the expressive power of $\mathcal{L}_S$ wielding only a *single* connective?

The answer is no and yes. No, you could not get by with just one of the current stock of $\mathcal{L}_S$ connectives. The negation connective is required to make paraphrases work, but negation alone would not allow you to construct sentences containing more than a single atomic constituent. Suppose, however, a *new connective* were introduced, a connective that *combined* negation and some other logical operation. The result would be a language as expressive as $\mathcal{L}_S$, sublime after a fashion, but unrelentingly tedious to use.

This was in fact accomplished decades ago by H. M. Sheffer. Sheffer defined a stroke function, |, which subsequently came to be known as the Sheffer stroke, a truth table characterization of which is as follows:

$p\ q$	$p\|q$
T T	F
T F	T
F T	T
F F	T

Here $p|q$ are to be read as 'not both p and q'. Sheffer showed that all of the truth-functional connectives used in $\mathcal{L}_S$ could be replaced by his stroke connective. Imagine a Sheffer stroke equivalent of the placid $\mathcal{L}_S$ sentence

$$A \vee B$$

What would it look like?

Consider, first, that a simple negation, $\neg A$, expressed via Sheffer stroke notation, comes out as |

$$A|A$$

This can be shown by means of a truth table

$A\ B$	$A\|B$	$A\|A$	$\neg A$
T T	F	F	F
T F	T	F	F
F T	T	T	T
F F	T	T	T

Now consider the sentence

$$(A|A)|(B|B)$$

and its truth table

$A\ B$	$A\|A$	$B\|B$	$(A\|A)\ \|\ (B\|B)$	$A \vee B$
T T	F	F	T	T
T F	F	T	T	T
F T	T	F	T	T
F F	T	T	F	F

The truth conditions for this sentence match those for the original disjunction; the sentences are logically equivalent.

These truth tables reveal that negation and disjunction can be captured by means of Sheffer stroke notation. Given that every $\mathcal{L}_S$ sentence is logically equivalent to a sentence containing only ¬s and ∨s, it must be possible to construct a Sheffer stroke equivalent of *any* $\mathcal{L}_S$ sentence. You could, in this

way, pare down the logical vocabulary of $\mathcal{L}_S$ to a single connective |, although doing so would result in a loss of simplicity of expression. Short $\mathcal{L}_S$ sentences would become more ungainly and less winsome. The Sheffer stroke equivalent of

$$A$$

for instance, would be

$$(A|A)|(A|A)$$

For most of us, this is nothing if not confusing.

Alarmed at the prospect of facing protracted strings of symbols, you might consider *expanding* the set of truth-functional connectives—adding, for instance, a connective signifying exclusive disjunction ('either p or q, and not both p and q'). Were you to do so, you would shorten particular $\mathcal{L}_S$ sentences, just as you could shorten English sentences by adding terms to the language that replace whole phrases.

Adding connectives, however, like adding terms in a natural language, has a cost. An additional element—together with its effects on the truth conditions of sentences in which it figured—would need to be retained in memory. On the whole, it is not worth the bother.

Such issues concern not the expressive power of the language but the ease with which it can be deployed, its perceived simplicity. $\mathcal{L}_S$ was designed to strike a reasonable balance between notational austerity and ease of use. It reflects logical notions that in natural languages have evolved distinctive modes of expression.

We could, in English, get by without explicit conditional constructions, for instance, just as we could in $\mathcal{L}_S$, by replacing conditionals with combinations of other elements. English (and every other natural language) has evolved as it has because it fits us. Similarly, formal languages, including $\mathcal{L}_S$, have evolved as they have because they mirror logically central features of natural languages.

Exercises 2.11

For each of the $\mathcal{L}_S$ sentences below, devise a *logically equivalent* sentence, a sentence with the same truth conditions, but one that uses only the truth-functional connectives indicated. For each sentence construct a truth table that proves the equivalence.

1. $P \wedge Q$ {¬, ∨}
2. $P \supset Q$ {¬, ∧}
3. $P \vee Q$ {¬, ⊃}
4. $P \wedge Q$ { | }
5. $P \mid Q$ {¬, ∧}
6. $P \vee Q$ {¬, ∧}
7. $P \wedge Q$ {¬, ⊃}
8. $P \supset Q$ {¬, ∨}
9. $P \supset Q$ { | }
10. $P \mid Q$ {¬, ⊃}

2.12 Translating English into $\mathcal{L}_S$

Translating an English sentence into $\mathcal{L}_S$ is no different in principle from translating the sentence into another natural language. You recognize in advance that some of the sentence's features might prove untranslatable. Your goal is to find a sentence in the target language with truth conditions matching those of the original sentence. In the case of natural languages, you might also hope to preserve as well poetic qualities, connotations, and 'feel'. Owing to cultural differences, these often fail to survive translation, or they survive only in some transmuted form. Poetry and figurative speech aside, it seems likely that truth conditions can survive translation.

When it comes to translations into $\mathcal{L}_S$, there is no question of preserving anything other than truth conditions. $\mathcal{L}_S$, although charming in many respects, is stylistically undistinguished. More seriously, $\mathcal{L}_S$ lacks logical features common to all natural languages. As a result, some natural language sentences lack satisfying $\mathcal{L}_S$ counterparts. That will be so when no sentence in $\mathcal{L}_S$ has quite the same truth conditions as the English original. In such cases you must lower your sights and concoct an $\mathcal{L}_S$ sentence the truth conditions of which approximate those of the target sentence.

Despite its limitations, $\mathcal{L}_S$ meshes nicely with an indispensable segment of English (and, by extension, any natural language). Difficulties you will encounter in constructing translations stem less from the character of $\mathcal{L}_S$ than from a tendency to focus on English sentences in a way that obscures their truth conditions. This is not because you lack an appreciation of those truth conditions. On the contrary, you grasp them instantly and unselfconsciously. The trick is to make your implicit knowledge explicit.

You probably know how to ride a bicycle and how to tie a bow. But could you make this knowledge explicit? Imagine, for instance, trying to explain to someone how to ride a bicycle or tie a bow over the telephone. You learn a language by *speaking* it, just as you learn to ride a bicycle or tie a bow by *doing* it—badly, at first, then, with practice, with ease. Such learning is rarely a matter of your learning explicit principles, or if it is, the principles take hold only after they have become second nature.

In learning to ski, you learn that when you want to turn, you must put your weight on the downhill ski. You have mastered the technique, however, only when you are able to do it unthinkingly. In learning a second language, you might start with explicit rules and

Translatability

Translation is an art, not a science. This is due, in part, to our preference for translations that preserve the flavor of the original. But suppose that all you are interested in is the preservation of truth conditions, literal meanings of sentences. Is there any reason to think that some languages contain sentences the truth conditions of which could not be matched by some sentence or set of sentences in your language?

No one doubts that particular words in a language can lack exact counterparts in other languages. Presumably such words could be paraphrased, however.

Donald Davidson argues that languages, whatever their origins, must be intertranslatable (see his 'The Very Idea of a Conceptual Scheme', in *Inquiries into Truth and Interpretation* [Oxford: Clarendon Press, 1984]). Davidson asks: what evidence could you have that an utterance or inscription was a meaningful sentence if it could not be translated into a sentence of your language?

principles. Only when these become implicit, only when you can 'think in the language', could you be said to have learned it.

Speaking or understanding a language, like riding a bicycle, tying a bow, or skiing, is a skill. So is translation from English into $\mathcal{L}_S$. The more you do it, the easier it becomes. At first the process might feel awkward and unnatural, but, with persistence, it can become second nature. At the outset, you must sensitize yourself to logical features of English sentences that have significant truth-conditional roles. These features are familiar to all of us already—otherwise we would not understand one another. But it can sometimes be a struggle to see them for what they are. Most of the rest of this chapter will be devoted to hints and suggestions for bringing the truth conditions of English sentences to the surface. Once you can do that, translation into $\mathcal{L}_S$ is mostly painless. Mostly.

Notice, first, that word order, although important, is not an infallible guide to the logical character of English sentences. Consider the sentences

Spot bit Iola.

Iola bit Spot.

These sentences contain the same words, but in different arrangements. Word order is significant in such cases: it enables us to distinguish the actor and the recipient of the action, or, in classical terminology, the agent and patient. Do agents inevitably precede patients in sentences of this sort? No. Cases in which the order is reversed are common. Consider, for instance:

Iola was bitten by Spot.

Such examples bring to mind a useful distinction between the *surface structure* of a sentence and its *deep structure*. The sentences

Spot bit Iola.

Iola was bitten by Spot.

evidently have the same truth conditions despite differing in word order, and in other ways as well. Such sentences are in one sense the same and in another sense different. Their differences are obvious and outward, instantly detectable even by someone unfamiliar with English. Their sameness is visible only to someone who knows enough English to recognize that the sentences have identical truth conditions. Such sentences differ in their surface structure, but not in their deep structure.

Mastering a language involves acquiring the knack of recovering deep, logical forms from surface structures. Your skill at doing this is reflected in the ease with which you recognize that the sentences above have the same meaning. It is reflected, as well, in your recognition that, as in the examples below, a single sentence—a single surface structure—can be associated with distinct truth conditions—distinct deep structures.

Gertrude was frightened by the new methods.

The committee was looking for French and Italian teachers.

The lamb was too hot to eat.

They are flying planes.

Deep and Surface Structure

The distinction between deep and surface structure is due to the linguist Noam Chomsky [see, for instance, his *Syntactic Structures* (The Hague: Mouton, 1957)]. Chomsky argued that in learning a language, we learn the grammar of that language, a set of rules for generating sentences. Some rules generate deep structures, which make up the core of the language. Other rules transform deep structures into surface structures.

Surface structures are associated with sentences we speak and write. Deep structures are associated with sentences' truth conditions, sentences' *logical* structure.

Although a grasp of the truth conditions of a sentence requires a grasp of its deep structure, the notions are not equivalent. Deep structure includes abstract elements that combine to determine, *among other things*, a sentence's truth conditions.

The rules that generate surface structures from deep structures include options. Depending on which options are taken, a single deep structure can yield multiple surface structures. When this happens, the resulting sentences will have the same truth conditions. Ambiguous sentences occur when distinct deep structures yield identical surface structures.

Although Chomsky and other linguists have moved on, the distinction between deep and surface structure remains useful in discussions of translation from a natural language such as English into $\mathcal{L}_S$.

Each of these sentences is *ambiguous*; each could be used on different occasions to express different meanings.

The distinction between deep and surface structure can prove useful once you set about the construction of translations into $\mathcal{L}_S$. Because the goal is to devise $\mathcal{L}_S$ sentences that have the same truth conditions as particular English sentences, and because a sentence's truth conditions are determined by its deep structure, it is important not to be distracted by superficial characteristics of the sentences you set out to translate. The point will be amply and repeatedly illustrated in the course of examining familiar English constructions.

2.13 Conjunction

The identification of conjunctions in English, and their translation into $\mathcal{L}_S$, is not likely to raise special problems. In the sentences below, for instance, conjunction is easy to spot:

Fenton investigates and Iola procrastinates.

Fenton doesn't investigate and Iola doesn't procrastinate.

Their translation into $\mathcal{L}_S$ is gratifyingly straightforward:

$$F \wedge I$$

$$\neg F \wedge \neg I$$

Conjunctions are not invariably signaled in English by the appearance of 'and'. Each of the sentences below expresses a conjunction. So far as $\mathcal{L}_S$ is concerned, each possesses the same truth conditions.

Fenton investigates *and* Iola procrastinates.

Fenton investigates *but* Iola procrastinates.

Fenton investigates; *however*, Iola procrastinates.

Fenton investigates *yet* Iola procrastinates.

Fenton investigates *although* Iola procrastinates.

Although Fenton investigates, Iola procrastinates.

Each of these sentences goes over into $\mathcal{L}_S$ as

$$F \wedge I$$

The key in translating such sentences is to recognize that despite superficial differences, each expresses a conjunction: each is true just in case each of its conjuncts—'Fenton investigates' and 'Iola procrastinates'—is true.

Exercises 2.13

Translate the following sentences into $\mathcal{L}_S$. Let C = Callie escapes; F = Fenton investigates; G = Gertrude investigates; I = Iola finds the treasure; J = Joe is kidnapped.

1. Joe is kidnapped; however, Callie escapes.
2. Although Joe is kidnapped, Callie escapes.
3. Fenton investigates, but Gertrude doesn't.
4. Fenton and Gertrude don't both investigate.
5. Both Fenton and Gertrude investigate if Joe is kidnapped.
6. Fenton investigates if and only if Joe is kidnapped, but Gertrude doesn't investigate.
7. If Joe is kidnapped and if Callie escapes, then Gertrude investigates.
8. Gertrude investigates if Joe is kidnapped or Callie doesn't escape.
9. Either Fenton investigates and Callie escapes, or Joe is kidnapped and Gertrude investigates.
10. Although Gertrude investigates, Fenton doesn't, and Callie escapes.

2.14 Disjunction

Disjunction in $\mathcal{L}_S$ is no more difficult than conjunction provided you bear in mind the distinction between *inclusive* and *exclusive* disjunction (see § 2.6). Once this distinction is mastered, the remaining aspects of disjunction are within reach. In English, disjunction is typically, although not invariably, signaled by an occurrence of 'either . . . or . . .' The sentence

Either Fenton investigates or Iola procrastinates.

expresses a disjunction, and is translated into $\mathcal{L}_S$ as

$$F \vee I$$

Negated disjunctions are only slightly more complex. Consider the sentence

Neither Fenton nor Iola investigates.

The sentence is a good one on which to reflect in the manner suggested earlier. Suppose you set out to make the sentence's truth conditions explicit. In so doing, you can see that the sentence would be true when Fenton does not investigate and Iola does not investigate, and that it is false otherwise—false if Fenton investigates, or Iola investigates, or if they both do. This suggests the following $\mathcal{L}_S$ translation:

$$\neg F \wedge \neg I$$

This translation represents the English original not as a *dis*junction but as a *con*junction. If the aim is to find an $\mathcal{L}_S$ sentence with the right truth conditions, however, and if this sentence possesses the right truth conditions, then it is a perfectly satisfactory translation.

As it happens, the sentence could also be translated in a way that preserves the disjunctive character of its surface structure. Think of the sentence as saying

It's not the case that either Fenton or Iola investigates.

which is translated into $\mathcal{L}_S$ as

$$\neg(F \vee I)$$

If this is correct, both of these $\mathcal{L}_S$ translations should have the same truth conditions. You can easily check by constructing a truth table:

$F\ I$	$\neg F$	$\neg I$	$\neg F \wedge \neg I$	$F \vee I$	$\neg(F \vee I)$
T T	F	F	F	T	F
T F	F	T	F	T	F
F T	T	F	F	T	F
F F	T	T	T	F	T

The truth conditions for the two sentences match—as revealed by the fifth and seventh columns—so they have the same truth conditions. You have, then, two logically equivalent ways to translate 'neither . . . nor . . .' sentences.

Notice that the two $\mathcal{L}_S$ sentences

$$\neg(F \vee I)$$
$$\neg F \vee \neg I$$

are *not* equivalent in meaning—as the truth table below plainly shows:

$F\ I$	$\neg F$	$\neg I$	$\neg F \vee \neg I$	$F \vee I$	$\neg(F \vee I)$
T T	F	F	F	T	F
T F	F	T	T	T	F
F T	T	F	T	T	F
F F	T	T	T	F	T

The lack of an equivalence here (which mirrors an analogous lack of equivalence in the case of negated *conjunctions* encountered in § 2.6) is scarcely surprising. The first sentence says, 'It's not the case that either Fenton or Iola will investigate'; the second, 'Either Fenton won't investigate or Iola won't investigate'. The first sentence denies that either will investigate; the second is consistent with one, but not both, investigating.

Exercises 2.14

Translate the following sentences into $\mathcal{L}_S$. Let C = Chet loses the map; F = Frank calls home; G = Gertrude finds the treasure; I = Iola finds the treasure; J = Joe brings a shovel.

1. Either Iola or Gertrude finds the treasure.
2. Neither Iola nor Gertrude finds the treasure.
3. If Chet doesn't lose the map and Joe brings a shovel, then Iola or Gertrude finds the treasure.
4. If neither Iola nor Gertrude finds the treasure, Frank calls home.
5. Chet doesn't lose the map, or Joe doesn't bring a shovel, or Frank doesn't call home.
6. It's not the case that both Iola and Gertrude find the treasure.
7. If either Chet loses the map or Joe doesn't bring a shovel, Iola doesn't find the treasure.
8. Iola or Gertrude finds the treasure if Chet doesn't lose the map.
9. Either Chet loses the map or Iola and Gertrude don't find the treasure.
10. Iola or Gertrude finds the treasure if Chet doesn't lose the map and Joe brings a shovel.

2.15 Conditionals and Biconditionals

In introducing conditionals, I called your attention to a less-than-perfect fit between conditional surface structures in English and $\mathcal{L}_S$ conditionals. The appearance of a conditional 'if . . . then . . .' phrase in an English sentence is not an infallible sign that the sentence expresses a conditional. This feature of English makes the translation of conditionals particularly challenging. Moreover, the *order* of sentences in a conditional construction makes a difference in a way it does not in conjunctions or disjunctions. It is easy to become confused about what counts as the antecedent of a given conditional and what counts as its consequent. Here, as elsewhere, you will need to acquire the knack of *seeing through* the superficial structure of the target sentence to its deep structure.

In § 2.08, it became clear that English conditionals often occur in reverse order: consequents first, then antecedents. Instead of saying

If you put it into the oven, then the bread will rise.

you might just as well say

The bread will rise if you put it into the oven.

Both sentences are translated

$$B \supset R$$

In translating such sentences, the trick is to locate the if-clause and treat that as the antecedent. This does not mean that every occurrence of 'if' signals the antecedent of a conditional. Only-ifs and plain ifs make very different logical contributions to the sentences in which they occur. The sentence

The bread will rise only if you put it into the oven

is a straightforward '. . . only if . . .' conditional translated

$$R \supset B$$

Remember: when confusion threatens, you can always call to mind the truth conditions of the original sentence. In the present case, that might include asking whether the 'only-if' sentence would be obviously false if R were true and O false. If so, it expresses a conditional.

Although conditionals are associated with the phrases '. . . only if . . .' and 'if . . . then . . .', these are at most signs that a conditional *could* be in play, not guarantees that a conditional *is* in play. Consider the sentence

If you wash the dishes, then I'll give you a dollar.

This sentence contains a conditional component—reflected in the fact that you will expect to be given a dollar if you wash the dishes. What you probably do not expect, however, is to be given a dollar even when you do *not* wash the dishes. If you treat the sentence as a straightforward conditional, you would be taking it to be true under those very circumstances:

$W\,D$	$W \supset D$
T T	T
T F	F
F T	T
F F	T

The sentence is counted true in the third row of the truth table, the row in which I give you a dollar even though you do not wash the dishes. You might *hope* that I would do that, but the sentence by itself gives you no grounds for optimism in that regard. This suggests that the English original, despite its conditional appearance, could in fact express a biconditional:

I'll give you a dollar *if and only if* you wash the dishes.

As the truth table for the $\mathcal{L}_S$ version of this biconditional makes plain, you can expect to receive a dollar if you wash the dishes, but not otherwise:

$W\,D$	$W \equiv D$
T T	T
T F	F
F T	F
F F	T

So sentences that *appear* to be conditionals in English are sometimes used to express biconditionals instead. This is one source of the impression that conditionals in $\mathcal{L}_S$ differ radically from those in English. The fact that $\mathcal{L}_S$ conditionals do not always serve to translate English sentences containing '. . . only if . . .' and 'if . . . then . . .' constructions stems from a quirk of English: *conditional surface structures are not reliably paired with conditional deep structures*.

Having registered these subtleties, you will be relieved to learn that none of this is going to matter very much. The character of $\mathcal{L}_S$ is such that we can afford to allow ourselves to mistake unobvious English biconditionals for conditionals. The reason for this will become clear in the next chapter. For the present, recall that biconditionals can be thought of as *containing* conditionals. Biconditionals were introduced as expressing back-to-back conditional pairs. A mistaken translation of a biconditional into a conditional in $\mathcal{L}_S$ does not miss the mark entirely.

This does not mean that differences between conditionals and biconditionals can simply be ignored. Here is a safe policy: translate English sentences into $\mathcal{L}_S$ as biconditionals *only when they are obviously biconditionals*, only when they include some such phrase as '. . . if and only if . . .' or '. . . just in case . . .' The policy allows you to translate

If you wash the dishes, then I'll give you a dollar.

into $\mathcal{L}_S$ as a garden-variety conditional:

$$W \supset D$$

Biconditionals are reserved for sentences in which they are explicit:

You'll win if and only if you play hard.

A set is empty just in case it lacks members.

These sentences can be translated into $\mathcal{L}_S$ as

$$W \equiv P$$

$$E \equiv \neg M$$

Reserving biconditional constructions in $\mathcal{L}_S$ for obvious biconditionals reflects time-honored logical tradition. This makes perfect sense given the uses to which a formal language like $\mathcal{L}_S$ is typically put. You could always make your translations more subtle should the need arise.

Exercises 2.15

Translate the following sentences into $\mathcal{L}_S$. Let C = Callie spots the thief; F = Fenton makes an arrest; G = Gertrude falls asleep; J = Joe wrecks the roadster.

1. If Callie spots the thief, Fenton makes an arrest.
2. Joe doesn't wreck the roadster if Gertrude doesn't fall asleep.
3. If Gertrude doesn't fall asleep and Joe doesn't wreck the roadster, then Fenton makes an arrest.
4. Joe wrecks the roadster only if Gertrude falls asleep.
5. If Joe wrecks the roadster, Gertrude falls asleep.
6. Only if Gertrude falls asleep does Joe wreck the roadster.
7. Joe wrecks the roadster if and only if Gertrude falls asleep.
8. Fenton makes an arrest just in case Callie spots the thief and Joe doesn't wreck the roadster.
9. If either Joe wrecks the roadster or Callie doesn't spot the thief, Fenton doesn't make an arrest.
10. If Joe wrecks the roadster then Fenton makes an arrest if and only if Gertrude doesn't fall asleep.

2.16 Troublesome English Constructions

Some common English turns of phrase can cause special problems when you set out to capture them in $\mathcal{L}_S$. As always, successful translation depends on your making clear to yourself the truth conditions of the sentence to be translated. Errors occur when you do not take the time to think through the meaning of a particular sentence.

Consider the sentence

You'll gain weight unless you exercise.

The sentence says

If you don't exercise, then you'll gain weight.

This sentence can be translated into $\mathcal{L}_S$ as

$$\neg E \supset W$$

The translation illustrates a feature of 'unless' constructions generally. Typically, such constructions follow the pattern

p unless q

$$\neg q \supset p$$

The pattern is useful to remember. You might be tempted to represent such sentences *incorrectly* as

If you exercise, then you won't gain weight.

expressible in $\mathcal{L}_S$ as

$$E \supset \neg W$$

Although this sentence might be thought to be suggested by anyone asserting the original sentence, it differs dramatically from that sentence. The easiest way to see that this is so is to look at a parallel case:

You'll be caught unless you run.

This sentence asserts that

If you don't run, then you'll be caught.

Following the pattern above, it would be translated into $\mathcal{L}_S$ as

$$\neg R \supset C$$

The sentence does *not* tell us, however, that

If you are caught, then you don't run: $C \supset \neg R$

The original sentence leaves open the possibility that you are caught despite running; you might run into a trap, for instance, or be outrun, or trip and fall. A truth table comparison of the sentences shows they possess distinct truth conditions:

R C	$\neg R$	$\neg R \supset C$	$C \supset \neg R$
T T	F	T	F
T F	F	T	T
F T	T	T	T
F F	T	F	T

This truth table reveals something important about 'unless' constructions. Consider the fourth column, which presents the truth conditions of the correct translation of the original sentence, $\neg R \supset C$. Compare the truth conditions shown in this column with those for the disjunction, $C \vee R$, shown below:

R C	$\neg R$	$\neg R \supset C$	$C \vee R$
T T	F	T	T
T F	F	T	T
F T	T	T	T
F F	T	F	F

The columns match, so the truth conditions of $\neg R \supset C$ and $C \vee R$ are the same. So what? Well, you are now in a position to simplify the principle formulated earlier governing translation of 'unless' constructions:

$$p \text{ unless } q$$

$$p \vee q$$

To apply this principle, first locate the 'unless' clause, and rewrite the sentence in what could be called quasi-$\mathcal{L}_S$. Given the sentence

Unless it's stirred, the sauce will go bad.

let S represent 'The sauce is stirred', and let B stand for 'The sauce will go bad'. The sentence, in quasi-$\mathcal{L}_S$, is

$$B \text{ unless } S$$

which, following the principle, becomes

$$B \vee S$$

Intermediate quasi-$\mathcal{L}_S$ representations are often useful in translating otherwise confusing sentences.

A different kind of problem crops up in sentences such as the following:

If Fenton investigates, then, even though Callie doesn't, Homer will be caught.

Here, a conditional sentence

If Fenton investigates, then Homer will be caught.

incorporates an embedded sentence

Callie doesn't investigate.

The conditional seems clear enough, but what happens to this embedded sentence? Notice, first, that Callie's not investigating is not a *component* of the conditional, despite popping up in the midst of it. Suppose you translated the sentence

$$(F \wedge \neg C) \supset H$$

This translation would make Homer's being caught conditional on Fenton's investigating *and* on Callie's not investigating. In the English original, however, Homer's being caught is conditional solely on Fenton's investigating:

$$F \supset H$$

That Callie doesn't investigate is a piece of information added to, but not made a part of, the conditional.

You could translate the whole sentence simply by conjoining $\neg C$ with the original conditional:

$$(F \supset H) \wedge \neg C$$

This translation makes it clear that Homer's being caught is not conditional on Callie's not investigating.

In general, you can adopt this procedure when you encounter 'although', 'even though', and 'despite the fact' constructions. Such constructions typically signal the insertion of a piece of information best treated as a separate conjunct. Thus, the sentence

If Frank or Joe is kidnapped, then, despite the fact
that Iola escapes, Chet will not be happy.

would be translated as

$$((F \vee J) \supset \neg C) \wedge I$$

Although such translations can, on occasion, feel awkward, they capture the logical structure of their English counterparts, the most that could be hoped for in $\mathcal{L}_S$.

Exercises 2.16

Translate the following sentences into $\mathcal{L}_S$. Let C = Chet weighs anchor; F = Frank pilots *The Sleuth*; G = Gertrude is rescued; I = Iola signals; J = Joe wears a disguise.

1. Gertrude is rescued if Joe wears a disguise.
2. Iola signals unless Gertrude is rescued.
3. Gertrude isn't rescued unless Iola signals.
4. Unless Chet weighs anchor, Frank doesn't pilot *The Sleuth*.
5. Joe wears a disguise unless either Iola signals or Gertrude is rescued.
6. If Iola signals, then Joe wears a disguise unless Gertrude is rescued.
7. Joe wears a disguise although Iola doesn't signal.
8. Iola signals but Gertrude isn't rescued even though Joe wears a disguise.
9. If Frank pilots *The Sleuth*, then, even though Iola doesn't signal, Gertrude is rescued.
10. Frank pilots *The Sleuth*, despite the fact that Joe wears a disguise, if Chet weighs anchor.

2.17 Truth Table Analyses of $\mathcal{L}_S$ Sentences

Truth tables were introduced in the course of characterizing truth-functional connectives in $\mathcal{L}_S$. Those definitions can all be incorporated into a single grand table:

p q	$\neg p$	$p \wedge q$	$p \vee q$	$p \supset q$	$p \equiv q$
T T	F	T	T	T	T
T F	F	F	T	F	F
F T	T	F	T	T	F
F F	T	F	F	T	T

Given these characterizations, you can specify the truth conditions for any $\mathcal{L}_S$ sentence. This is a consequence of the fact that $\mathcal{L}_S$ is a truth-functional language: the truth conditions of every sentence are determined by the truth conditions of its constituent sentences.

Recall that knowing the truth conditions for a sentence amounts to knowing its truth value at all possible worlds. The rows in a truth table represent mutually exclusive and exhaustive sets of possible worlds. Such knowledge is less impressive than it sounds. All it takes to know the truth conditions of the sentence

The winning number in next week's state lottery is 566,792.

is to know what it means. You know this sentence is true at every world in which 566,792 is selected as the winning ticket, and false at all the rest. Were you to formulate a sentence of this sort mentioning every ticket for the state lottery, you would know the truth conditions for each of these sentences as well. What you would not know is which ticket will win, which of the myriad possible worlds you effortlessly consider is *your* world.

Picture God deliberating about which world to create, and settling on one of these (the best one possible, of course). In fixing the truth value of every sentence, God thereby fixes the world. You might imagine God scrutinizing an immense truth table, one with countless rows, each row representing a possible world. In choosing a world, God chooses one of these rows and, simultaneously, the truth value of every sentence.

Although no finite creature can pretend to approximate this level of knowledge, you are often in a position to narrow down the possibilities. You can say, with some justification, that your world belongs to a certain *set* of possible worlds, a set that might itself be infinite. This might be all you need to settle the truth values of particular sentences.

This idea is nicely illustrated in $\mathcal{L}_S$. Given a sentence, together with a specification of its truth conditions, you can determine its truth value provided you know the truth values of its atomic constituents. Suppose you encounter the sentence

It's not the case that either Iola and Callie escape or that Fenton investigates.

Translated into $\mathcal{L}_S$ this sentence becomes

$$\neg((I \wedge C) \vee F)$$

Suppose, further, that you know that C and F are true, and that I is false. This information, together with what you know about the truth-functional connectives $\neg$, $\wedge$, and $\vee$, enables you to calculate the truth value of the sentence easily.

$I\ C\ F$	$I \wedge C$	$(I \wedge C) \vee F$	$\neg((I \wedge C) \vee F)$
T T T	T	T	F
T T F	T	T	F
T F T	F	T	F
T F F	F	F	T
F T T	F	T	F
F T F	F	F	T
F F T	F	T	F
F F F	F	F	T

You know that C and F are true, and I false, so you know that the actual world belongs to the set of worlds picked out by the fifth row of the truth table. In that row, the sentence turns out to be *false*. The method is simple and, providing you are careful, foolproof. It is, however, unforgivably *tedious*. For sentences incorporating additional distinct atomic constituents, it would be decidedly more tedious. Recall that if a sentence contains n distinct atomic sentences, a truth table depicting its

truth conditions will have 2^n rows. A sentence containing six distinct atomic sentences, for instance, would require a sixty-four-row truth table.

Happily, there is a much simpler method of calculating the truth values of complex $\mathcal{L}_S$ sentences in cases in which you know the truth values of each of their atomic constituents. You can construct what amounts to a single row of the relevant truth table, the row that incorporates the truth values of the atomic constituents of the target sentence.

Consider the sentence

$$\neg((J \wedge \neg F) \supset (C \wedge J))$$

Suppose we know that F and J are true, and that C is false. We then

1. Write out the sentence and enter that sentence's truth value directly under each atomic sentence.

$$\begin{array}{c} \neg((J \wedge \neg F) \supset (C \wedge J) \\ \quad T \quad\quad T \quad\quad F \quad T \end{array}$$

2. Reverse the value of every negated atomic sentence.

$$\begin{array}{c} \neg((J \wedge \neg F) \supset (C \wedge J) \\ \quad T \quad\quad T \quad\quad F \quad T \\ \qquad\quad\; | \\ \qquad\quad\; F \end{array}$$

3. Working 'inside out', chart the values of molecular sentences.

$$\begin{array}{c} \neg((J \wedge \neg F) \supset (C \wedge J) \\ \quad T \quad\quad T \quad\quad F \quad T \\ \qquad\quad\; | \qquad\qquad | \\ \qquad\quad\; F \qquad\qquad | \\ \quad F \qquad\qquad\quad F \end{array}$$

4. You will eventually arrive at a determination for the sentence as a whole.

$$\begin{array}{c} \neg((J \wedge \neg F) \supset (C \wedge J) \\ \quad T \quad\quad T \quad\quad F \quad T \\ \qquad\quad\; | \qquad\qquad | \\ \qquad\quad\; F \qquad\qquad | \\ \quad F \qquad\qquad\quad F \\ \qquad\quad\; _________ \\ \qquad\qquad\; T \\ \quad F \end{array}$$

In this example, the truth value of the sentence is ultimately settled by the negation sign that includes the remainder of the sentence within its scope. Recall that the scope of a negation sign includes just the sentence to its immediate right. A negation sign to the left of an atomic sentence negates only that atomic sentence. A negation sign to the left of a matching pair of parentheses negates the sentence included within those parentheses.

Exercises 2.17

Determine the truth values of the $\mathcal{L}_S$ sentences below. Assume that A, B, and C are *true*, and that P, Q, R, and S are *false*.

1. $(Q \wedge B) \vee (\neg R \vee S)$
2. $(\neg Q \wedge (B \vee S)) \supset P$
3. $((P \wedge Q) \supset \neg A) \supset S$
4. $A \supset ((B \wedge \neg C) \vee (P \vee Q))$
5. $\neg(\neg(\neg P \wedge A) \supset ((B \vee Q) \wedge R))$
6. $((A \wedge B) \wedge P) \equiv (B \supset (C \vee S))$
7. $\neg((A \supset (P \vee C)) \wedge \neg(\neg B \vee \neg Q))$
8. $P \supset ((A \vee \neg B) \wedge (B \supset (C \vee Q)))$
9. $(S \vee (\neg Q \wedge \neg(R \vee S))) \equiv (\neg A \vee \neg(B \supset (\neg R \vee P)))$
10. $\neg((B \vee (P \wedge \neg Q)) \supset ((A \wedge \neg B) \vee (\neg P \supset (Q \vee R))))$

2.18 Contradictions and Logical Truths

Truth and falsity pertain to the *semantics* of sentences. A sentence's semantics is distinguished from the sentence's *syntax*. Every language has a syntax, principles specifying what does and what does not count as a sentence in that language. Most languages, indeed all natural languages, have a set of semantic principles that link the language to its subject matter as well. When you learn English, you learn the meanings of particular words. In the case of common nouns, for instance, you learn that 'tree' is used to designate trees, that 'star' is used to refer to stars, and that 'foot' is used to denote feet. But you also learn how words can be combined in sentences and used to make statements about actual and possible states of affairs. This requires your mastering semantic principles governing sentences.

Although $\mathcal{L}_S$ does not contain words, you might think of sentences in $\mathcal{L}_S$ as being assembled from one-word atomic sentences. Sentences are built from sentences by means of truth-functional connectives. A grasp of the semantics of $\mathcal{L}_S$ requires an understanding of the contribution these connectives make to the truth conditions of sentences in which they figure. Just as in English you recognize that the sentence

Snow is white and grass is green.

is true just in case both 'Snow is white' and 'Grass is green' are true, in $\mathcal{L}_S$

$$A \wedge B$$

is true if and only if A and B are both true. The truth-functional character of $\mathcal{L}_S$ allows for the truth conditions—the semantics—of any sentence to be specified mechanically by means of truth tables. Principles governing English and other natural languages are more complicated. Although we have all somehow mastered them, no one has yet succeeded in spelling them out in detail.

In $\mathcal{L}_S$, no less than in English, you can know what a sentence *means*, you can know its truth *conditions*, without knowing its truth *value*, without knowing *whether* the sentence is true or false. In most cases, knowing a sentence's truth value requires that you know something about the world. You know that the sentence

Snow is white.

is true because you know something about *snow*: its color in the world you inhabit.

Not all sentences are like this. In some cases, all it takes to know a sentence's truth value is a knowledge of the sentence's truth conditions. Consider the sentence

Snow is white or snow isn't white.

Anyone who understands this sentence knows that it could not *fail* to be true. Its truth value depends not on the way the world is but on the sentence's semantics. Given your understanding of conjunction and negation, you know that the sentence must be true. The point is clear if you translate the sentence into $\mathcal{L}_S$:

$$W \vee \neg W$$

and construct a truth table for it:

W	$\neg W$	$W \vee \neg W$
T	F	T
F	T	T

The sentence is true no matter what truth values are assigned to W, its lone atomic constituent. The sentence is true at all possible worlds, and a sentence true at all possible worlds is true of the actual world.

Sentences that are true at all possible worlds are called *logical truths*. Logical truths expressible in $\mathcal{L}_S$ are called *tautologies*, although, as it happens, not all logical truths are tautologies.

Sometimes, as in the case above, the tautologous character of a sentence is obvious. This is not always the case, however. Consider the English sentence

If Desdemona loves Cassio, then, if the Moon is made
of green cheese, Desdemona loves Cassio.

translated into $\mathcal{L}_S$ as

$$D \supset (M \supset D)$$

This sentence too is a logical truth, although that might not be obvious until you construct a truth table:

$D\ M$	$M \supset D$	$D \supset (M \supset D)$
T T	T	T
T F	T	T
F T	F	T
F F	T	T

The table reveals what intuition might not: the sentence is true regardless of the truth values of D and M; the sentence is true at all possible worlds.

Logically true sentences differ, on the one hand, from ordinary *contingent* sentences and, on the other hand, from *contradictions*. A contingent sentence in $\mathcal{L}_S$ is a sentence the truth value of which varies with assignments of truth values to its atomic constituents. More generally, a contingent sentence is true at some possible worlds and false at others. Most of the $\mathcal{L}_S$ sentences you have encountered thus far have been contingent. Had you bothered to construct truth tables for each of them, you would have discovered that they were true in at least one row of their respective truth tables and false in at least one row. An ordinary contingent sentence, 'Snow is white', is contingent because, although it is true here in our world, it is false at other worlds.

The truth table for a contradictory sentence, in contrast, shows that it is *false* in every row, and consequently false at all possible worlds.

Contradictions can be *explicit*, as in

Snow is white and snow is not white.

or merely *implicit*, as in

Desdemona loves Cassio and it's not the case that either the Moon isn't made of green cheese or Desdemona loves Cassio.

If you translate these sentences into $\mathcal{L}_S$ and construct a truth table for each, you can see that each is contradictory; each is false on every row of its truth table. The truth table for the first sentence looks like this:

W	$\neg W$	$W \wedge \neg W$
T	F	F
F	T	F

The second sentence's truth table looks like this:

$D\ M$	$\neg M$	$\neg M \vee D$	$\neg(\neg M \vee D)$	$D \wedge \neg(\neg M \vee D)$
T T	F	T	F	F
T F	T	T	F	F
F T	F	F	T	F
F F	T	T	F	F

The technique in $\mathcal{L}_S$ is straightforward. For any sentence, you can decide whether it is contingent, tautologous, or contradictory simply by constructing a truth table.

In the case of English sentences, matters are often less obvious. No comparable technique for working out the truth conditions of English sentences is available. As a result, you sometimes discover that sentences you had supposed were contingent are in fact contradictory or logically true.

A contradictory sentence cannot be true, so it cannot be used to impart information about the world. Logically true sentences, although true, also fail to impart information by failing to distinguish among possible worlds. If I tell you that passion fruit are purple, I provide you with information about the color of passion fruit. You know now that our world is one at which passion fruit are purple. If I say 'Passion fruit are purple or passion fruit are not purple', what I have said is true, but uninformative. You cannot tell, on the basis of what I have said, *anything* about passion fruit.

When your interest is in imparting information, it is not enough that what you say is true. Informative truths are those that *might have been false*. Science is sometimes said to be in the business of discovering truths. If that were all, however, the quickest route to a Nobel Prize would be to spend your days churning out logical truths.

A scientific hypothesis that is not 'falsifiable', one that could not have failed to be true, is empty. The informativeness of a hypothesis is proportional to the ratio of worlds at which it is false to those at which it is true. A *maximally informative* hypothesis would be one true at just one possible world. Logically true hypotheses are maximally *un*informative. Science aims not simply at the production of true hypotheses but at the production of informative hypotheses that are true—the more informative, the better.

At one time it was common to hear scientists and philosophers argue that psychoanalytic theories are unfalsifiable, hence uninformative. Suppose a diagnosis is pronounced correct when it elicits the patient's assent (the assent proving it true) *and* correct when it produces a denial (by being defensive the patient has proved the diagnosis correct). What could make it false? Unfalsifiable hypotheses, those true at every world, are informationally empty.

Exercises 2.18

Determine which of the $\mathcal{L}_S$ sentences that follow are *tautologous* (*logically true*), which are *contingent*, and which are *contradictory* by constructing a truth table for each.

1. $P \supset (\neg P \supset P)$
2. $(A \wedge \neg A) \supset Q$
3. $A \vee (B \supset (A \wedge C))$
4. $\neg(P \supset (\neg P \supset P))$
5. $(R \vee (P \wedge Q)) \equiv \neg(\neg R \wedge \neg(P \wedge Q))$
6. $((P \supset Q) \wedge (Q \supset R)) \supset (P \supset R)$
7. $((P \wedge \neg(Q \vee R)) \supset (R \wedge (P \supset S))) \wedge (S \supset (P \vee (Q \wedge R)))$
8. $((B \vee A) \wedge \neg B) \wedge \neg A$
9. $(A \supset (B \supset C)) \wedge (B \supset (A \supset \neg C))$
10. $\neg(T \supset \neg(S \wedge \neg P)) \supset ((T \wedge S) \vee P)$

2.19 Describing $\mathcal{L}_S$

An obvious method of offering a description of something is to provide an exhaustive inventory of its constituent parts together with an account of the relations these parts bear to one another. I might describe the chair in which I am sitting, for instance, by listing its parts—arms, legs, seat, and back—and indicating how each of these parts is related to the rest.

This technique is workable, however, only when the object of a description is not excessively complex. What is required for the description of sprawling abstract objects like English or $\mathcal{L}_S$? Both English and $\mathcal{L}_S$ comprise an infinite number of sentences, so there is no prospect of simply listing them all.

You might think of a description as a recipe that would enable anyone to decide, for any object whatever, whether it was or was not the object described. A description of Paris distinguishes Paris from every other city. A description of Socrates picks Socrates out of the set of philosophers. A description of English would provide a method for distinguishing English sentences from nonsentences.

In the case of $\mathcal{L}_S$, it is possible to provide such a distinguishing recipe that is both rigorous and elegant. The recipe is founded on a set of rules specifying the *syntax* of $\mathcal{L}_S$ and thereby enabling anyone privy to the rules to determine, for any object at all, whether it is, or is not, a sentence of $\mathcal{L}_S$. Such rules serve as recipes for the generation of all the sentences of $\mathcal{L}_S$ and only these: using finite resources, they give birth to an infinite object.

The syntactic rules of $\mathcal{L}_S$ amount to a *recursive definition* of 'sentence of $\mathcal{L}_S$'. Once formulated, the rules could then be put to work in an account of the semantics of $\mathcal{L}_S$. Think of the syntax and semantics of $\mathcal{L}_S$ as components of a description of $\mathcal{L}_S$.

The idea is perfectly general. You might set out to describe English or any other natural language by providing a generative syntactic component and a semantic component. To be sure, natural languages are many times more intricate than $\mathcal{L}_S$. Linguists and logicians have managed to provide syntactic and semantic analyses for narrowly circumscribed portions of English, for instance, but much remains uncharted. In contrast to English, $\mathcal{L}_S$ is completely charted. In that regard, the syntax and semantics of $\mathcal{L}_S$ could be seen as providing a simplified schematic model of what the syntax and semantics of a natural language would look like.

2.20 The Syntax of $\mathcal{L}_S$

An account of the syntax of $\mathcal{L}_S$ begins with a specification of $\mathcal{L}_S$'s vocabulary. Although $\mathcal{L}_S$ contains an infinite number of sentences, these are constructed from a finite, hence listable, vocabulary. Call the set of vocabulary elements $\boldsymbol{\mathcal{V}}$. $\boldsymbol{\mathcal{V}}$ is made up of three sets of elements:

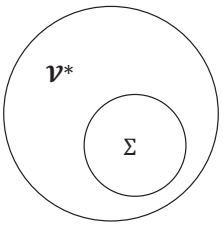
- $\boldsymbol{C}$ logical constants (truth-functional connectives) $\{\neg, \wedge, \vee, \supset, \equiv\}$
- $\boldsymbol{S}$ sentential constants $\{A, B, C, \ldots, Z\}$
- $\boldsymbol{P}$ left and right parentheses $\{(,)\}$

You can gain a sense of the task ahead by thinking of the endless ways in which the elements of $\boldsymbol{\mathcal{V}}$ could be combined to produce individual (finite) strings. Now imagine that each of these finite

sequences is a member of a set, $\boldsymbol{\mathcal{V}}^*$. Allowing for repetitions of elements, $\boldsymbol{\mathcal{V}}^*$ would contain infinitely many strings, among them:

$$A$$
$$\neg C$$
$$(P \supset Q) \vee M$$
$$)(G\neg \equiv$$
$$\supset \supset)Z$$

Plainly, some of the members of $\boldsymbol{\mathcal{V}}^*$ are $\mathcal{L}_S$ sentences, but many more are not. The goal now is to provide a way of specifying, for any member of $\boldsymbol{\mathcal{V}}^*$, whether it is or is not a sentence of $\mathcal{L}_S$. Think of $\boldsymbol{\mathcal{V}}^*$ as an infinite set of strings of symbols that includes a set of strings, Σ, comprising all and only the sentences of $\mathcal{L}_S$. The relation of Σ to $\boldsymbol{\mathcal{V}}^*$ is illustrated below.



Although Σ, the set of sentences of $\mathcal{L}_S$, is itself a proper subset of $\boldsymbol{\mathcal{V}}^*$ (that is, $\boldsymbol{\mathcal{V}}^*$ includes every member of Σ as well as strings not in Σ), it too is infinite.

Characterizing the syntax of $\mathcal{L}_S$ amounts to the construction of a finite collection of rules that would enable you to separate the sheep from the goats, to decide which strings of symbols in $\boldsymbol{\mathcal{V}}^*$ belong to Σ and which do not. This task can be simplified by the introduction of a technical notion, *boundedness*. A string of symbols enclosed between matching left and right parentheses is *bounded*. Examples of bounded strings include:

$$(P \supset B)$$
$$(G\neg\vee)$$
$$(\neg(\equiv))$$

Each of these strings is a member of $\boldsymbol{\mathcal{V}}^*$, but only the first qualifies for membership in Σ, hence as a sentence of $\mathcal{L}_S$.

Now it is possible to specify the requirements for sentencehood in $\mathcal{L}_S$, that is, to specify what it takes to be a member of Σ. All and only those members of $\boldsymbol{\mathcal{V}}^*$ that satisfy these rules are members of Σ, hence sentences of $\mathcal{L}_S$. (Recall that $\boldsymbol{S}$ is the set of sentential constants: A, B, C, . . ., Z.)

1. Every member of $\boldsymbol{S}$ is a sentence of $\mathcal{L}_S$.
2. If p is a sentence of $\mathcal{L}_S$, then $\neg p$ is a sentence of $\mathcal{L}_S$.
3. If p and q are sentences of $\mathcal{L}_S$, then $(p \wedge q)$ is a sentence of $\mathcal{L}_S$.
4. If p and q are sentences of $\mathcal{L}_S$, then $(p \vee q)$ is a sentence of $\mathcal{L}_S$.

Infinity

In ordinary speech, 'infinite' is treated as a synonym for 'unimaginably large'. Logicians and mathematicians have something else in mind when they speak of infinity. An infinite set is not merely a very large set; an infinite number is not just a very large number. Infinities are different in kind and not just degree from ordinary magnitudes.

One way to characterize an infinite set is as follows:

> φ is infinite if and only if the elements of φ stand in a one-one correspondence with a proper subset of the elements of φ.

(φ_p is a proper subset of φ just in case every element of φ_p is an element of φ, but there are elements of φ that are not elements of φ_p.)

The set of natural numbers, 0, 1, 2, 3, 4, . . ., is infinite. The natural numbers stand in a one-to-one relation of correspondence to a proper subset of its elements. There is, for instance, a one-to-one correspondence between the natural numbers (0, 1, 2, 3, 4, . . .) and the set of odd numbers (1, 3, 5, 7, 9, . . .). If that strikes you as impossible, you are probably trying to imagine how it would work with a finite set. It cannot work with a finite set, however, so the comparison is illegitimate.

The set of natural numbers is said to be denumerably infinite. Any set is denumerably infinite if it can be put into one-to-one correspondence with the set of natural numbers. Some infinite sets are not denumerable. The set of real numbers, for instance, is not denumerable. (The set of real numbers includes the rational numbers, those numbers expressible as ratios, or 'fractions' together with the irrational numbers, Π or √2, for instance, which cannot be expressed as ratios of two integers.)

5. If p and q are sentences of $\mathcal{L}_S$, then $(p \supset q)$ is a sentence of $\mathcal{L}_S$.
6. If p and q are sentences of $\mathcal{L}_S$, then $(p \equiv q)$ is a sentence of $\mathcal{L}_S$.
7. p is a sentence of $\mathcal{L}_S$ only if p is constructed in accord with rules 1-6.

Taken together, the rules spell out necessary (rule 7) and sufficient (rules 1–6) conditions for sentencehood in $\mathcal{L}_S$.

The rules provide a finite characterization of an infinite object, Σ, the set of sentences in $\mathcal{L}_S$. You can appreciate their elegance by applying them to a simple sentence:

$$(\neg(G \wedge L) \supset (\neg H \equiv K))$$

By rule 1, G, L, H, and K are all sentences of $\mathcal{L}_S$: all belong to $\boldsymbol{S}$, the set of sentential constants. Rule 2 specifies that $\neg H$ is a sentence of $\mathcal{L}_S$—a negated sentence is a sentence. The strings $(G \wedge L)$ and $(\neg H \equiv K)$ count as sentences of $\mathcal{L}_S$ by rules 3 and 6, respectively. $(G \wedge L)$ is a sentence of $\mathcal{L}_S$, so $\neg(G \wedge L)$ is an $\mathcal{L}_S$ sentence, again by rule 2. Finally, given that $\neg(G \wedge L)$ and $(\neg H \equiv K)$ are sentences of $\mathcal{L}_S$, $(\neg(G \wedge L) \supset (\neg H \equiv K))$ is an $\mathcal{L}_S$ sentence by rule 5.

The first six rules specify which strings of symbols count as sentences of $\mathcal{L}_S$. Rule 7 tells us that *only* these are sentences of $\mathcal{L}_S$. Rule 7 draws a boundary around the set of $\mathcal{L}_S$ sentences, excluding

members of $\boldsymbol{\mathcal{V}}^*$ that ought to be excluded. Taken together, the seven rules define 'sentence of $\mathcal{L}_S$': they are satisfied by, and only by, sentences of $\mathcal{L}_S$.

The *recursive* character of these rules is illustrated by rule 2. Given just the first two rules, you can generate an infinite set of $\mathcal{L}_S$ sentences. Thus, by rule 1 you know that A, for instance, is a sentence of $\mathcal{L}_S$. Rule 2 tells us that a negated sentence is a sentence, hence that $\neg A$ is a sentence of $\mathcal{L}_S$. Since $\neg A$ is a sentence, rule 2 tells us, again, that $\neg\neg A$ is a sentence; because *that* is a sentence, $\neg\neg\neg A$ is a sentence, as well, *and so on*. The '. . . and so on' here is warranted because the rule generates endless sentences by means of the simple device of piling up—concatenating—negation signs.

Recursive Definition

Σ, the set of sentences of $\mathcal{L}_S$, is infinite. Even so, that set can be exhaustively specified by means of a recursive definition.

A recursive definition proceeds by first characterizing an initial member (or finite collection of initial members) of the class. Second, rules are given for generating the remaining members of the set from these initial members.

A recursive rule is framed in a way that it can apply to its own output. If you begin with something stipulated to be a member of the set and apply an appropriate recursive rule to it, a new member of the set is generated. If you apply the rule to this generated member, the result is another member of the set—*and so on*.

Notice that, given the definition of 'sentence of $\mathcal{L}_S$' above, most of the expressions called sentences of $\mathcal{L}_S$ in this chapter would not in fact qualify as sentences! Rules 3 through 6 require that sentences containing truth-functional connectives (other than the negation sign) are *bounded*, that is, to be enclosed within parentheses. Thus the expression

$$R \supset B$$

would not count as a sentence. You could *turn it into* a sentence by adding parentheses:

$$(R \supset B)$$

Three responses to this shocking turn of events come to mind. First, you could bite the bullet and insist that alleged sentences be converted to genuine sentences by the addition of appropriate parentheses. Second, you could rewrite the rules, complicating them somewhat, so as to allow unbounded strings to count as sentences of $\mathcal{L}_S$. Third, you could adopt the informal convention of omitting 'needless' parentheses. I recommend the adoption—retroactively—of this third option.

2.21 The Semantics of $\mathcal{L}_S$

The semantics of $\mathcal{L}_S$ revolves around the notion of an *interpretation*. The syntactic rules of $\mathcal{L}_S$ yield an infinite collection of *uninterpreted* strings of symbols. In putting $\mathcal{L}_S$ to work, you assign meanings to sentential constants, members of the set $\boldsymbol{S}$, and make use of truth table characterizations of connectives to extend meanings to complex sentences. If I means that Iola is brave, for instance, and C means that Chet escapes, then $I \supset C$ means that Chet escapes if Iola is brave.

An assignment of meanings to the elements of $\boldsymbol{S}$ amounts to an assignment of truth values to each element. An interpretation of $\mathcal{L}_S$ is

(i) an assignment of truth values to sentential constants, and

(ii) the use of truth table characterizations of connectives to produce truth values for every sentence.

What is the connection between interpretation and truth? Reflect on what happens when you decide to let *C* represent

Callie is brave.

In so interpreting *C*, you in effect assign *C* a truth value. *C* will be true if and only if Callie *is* brave. Had you interpreted *C* differently, had you used *C* to mean that cassowaries are docile, for instance, *C* could well have had a different truth value. Whether *C* is true or false depends on what *C* means, how *C* is interpreted.

Given this background, it is possible to characterize the notion of an interpretation for $\mathcal{L}_S$:

> An interpretation, ***I***, of $\mathcal{L}_S$ is an assignment to each member of ***S*** (that is, to each atomic sentence) one and only one of the truth values {*true*, *false*} and an assignment to the members of ***S*** (that is, to the connectives) the truth-functional meanings set out below.

Think of an interpretation as a single row of a truth table in which every atomic sentence is assigned a truth value. Given *n* atomic sentences, the truth table would have2^n rows representing 2^n possible combinations of truth values of atomic constituents.

In setting out the syntax of $\mathcal{L}_S$, the atomic vocabulary was restricted to the uppercase letters *A* through *Z*. Thus characterized, a truth table representation of an interpretation of $\mathcal{L}_S$ would consist of one of 2^{26} or 67,108,864 rows of a truth table roughly the height of the Washington Monument. Had the vocabulary of $\mathcal{L}_S$ been expanded, perhaps by adding numerical subscripts to alphabetic characters, the number of truth value combinations could become literally uncountable. The aim, however, is not to construct a monstrous truth table but to provide a way to determine the truth conditions for any arbitrary sentence of $\mathcal{L}_S$, its truth value under any interpretation.

Now it is possible to define *true under an interpretation* for $\mathcal{L}_S$. Where ***I*** is an interpretation of $\mathcal{L}_S$, and p and q are sentences:

1. If p is a member of ***S***, then p is true under ***I*** if and only if ***I*** assigns the value true to p.
2. $\neg p$ is true under ***I*** if and only if p is not true under ***I***.
3. $(p \wedge q)$ is true under ***I*** if and only if p is true under ***I*** and q is true under ***I***.
4. $(p \vee q)$ is true under ***I*** if and only if p is true under ***I*** or q is true under ***I***, or both are true under ***I***.
5. $(p \supset q)$ is true under ***I*** if and only if q is true under ***I*** or p is not true under ***I***, or both.
6. $(p \equiv q)$ is true under ***I*** if and only if either p and q are both true under ***I*** or neither p nor q is true under ***I***.
7. p is false under ***I*** if and only if p is not true under ***I***.

Except in rule 1, p and q need not be atomic sentences.

Truth

Pilate asked, 'What is truth?' Philosophers have provided a number of answers in the form of definitions or 'theories' of truth. In general, our grasp of the notion of truth is better than our grasp of these theories.

The logician Alfred Tarski provided an important definition of truth in 'The Concept of Truth in Formalized Languages' (reprinted in many places, including *The Logic of Grammar*, Donald Davidson and Gilbert Harmon, eds. [Encino, CA: Dickenson Publishing Co., 1974]). Tarski's account of truth allows for a rigorous specification of the semantics of formal languages.

For our purposes, truth can be treated as a primitive, undefined notion, one that can be used to define other notions. Given this definition of truth under an interpretation, you can extend the semantics of $\mathcal{L}_S$. The first step is to define the notion of a *model*:

> An interpretation, $\boldsymbol{I}$, is a *model* of a sentence, p, if and only if $\boldsymbol{I}(p)$ is true. If Γ (*gamma*) is a set of sentences, then $\boldsymbol{I}$ is a model of Γ if and only if every sentence in Γ is true under $\boldsymbol{I}$.

An interpretation is a function from sentences to truth values, so the value of $\boldsymbol{I}(p)$ is the truth value an interpretation $\boldsymbol{I}$ assigns to a sentence p. If $\boldsymbol{I}$ is a model of a sentence, p, (or a set of sentences, Γ) then $\boldsymbol{I}$ assigns the value *true* to p (or to every member of Γ).

The notion of a model facilitates straightforward definitions of other semantic concepts:

> A sentence, p, is *logically true* if and only if every interpretation is a model of p (that is, p is true under every interpretation).

> A sentence, p, is *contradictory* if and only if p has no model (that is, p is false under every interpretation).

> A sentence, p, is *contingent* if and only if there is at least one model of p and at least one interpretation that is not a model of p (that is, p is true under some interpretations, and false under others).

The concept of an interpretation and the related concept of a model will prove useful in subsequent chapters. For the present, the lesson to be borne in mind is that every language, formal or natural, possesses both a syntax—a principled mechanism for distinguishing sentences from meaningless strings of elements of the language—and a semantics—a function associating meanings or truth conditions and sentences. Mastery of a language requires mastery of both. It requires something more as well. This something more is the subject of chapter 3.

3. Derivations in $\mathcal{L}_S$

3.00 Sentential Sequences

Your introduction to the deployment of sentences in $\mathcal{L}_S$ has only just begun. This might come as something of a shock, but consider: you credit others with understanding a word only if they can use that word appropriately in sentences. Sentences, like words, bear relations to one another. You recognize that the sentence

If Socrates is wise, then he is happy.

together with the sentence

Socrates is wise.

stand in a special relation to one another and to the sentence

Socrates is happy.

Anyone who failed to appreciate this relationship—anyone who did not see that the third sentence *follows from* the others—could not be said to have grasped the meaning of those sentences.

Chapter 2 focused on *intra*sentential logical relations. This chapter is devoted to an investigation of *inter*sentential logical relations, relations sentences bear to one another. The focus will no longer be on individual sentences but on *sequences* of sentences. Some sequences, *derivations*, exhibit definite structures. These structures are governed by rules just as sentences themselves are governed by rules. Learning to use these rules requires learning how to construct derivations in $\mathcal{L}_S$, and thus finally to be in a position to master the language.

3.01 Object Language and Metalanguage

Whenever you set out to discuss a language, you must do so by using some language or other. This situation sometimes gives rise to ambiguities and confusions. Philosophers are often accused of arguing about 'mere words'. Whether this is true or not, philosophers have from time to time proved vulnerable to a special class of linguistic confusion. St. Augustine relates the following conversation.

A: He was a terrible man. Manure came out of his mouth.

B: Is that possible?

A: Absolutely! He said 'Manure smells bad', and whatever he said must have come out of his mouth, so manure came out of his mouth.

Some theorists have devoted years to problems that, because they were based on a confusion of the sort illustrated by St. Augustine, were specious.

The White Knight's Song

In *Through the Looking Glass*, Lewis Carroll depicts a conversation between Alice and the White Knight.

> 'You are sad', the Knight said in an anxious tone: 'let me sing you a song to comfort you'.
>
> 'Is it very long?' Alice asked, for she had heard a good deal of poetry that day.
>
> 'It's long', said the Knight, 'but it's very, very beautiful. Everybody that hears me sing it—either it brings the tears into their eyes, or else . . .'
>
> 'Or else what?' said Alice, for the Knight had made a sudden pause.
>
> 'Or else it doesn't, you know. The name of the song is called *Haddocks Eyes*'.
>
> 'Oh, that's the name of the song, is it?' Alice said, trying to feel interested.
>
> 'No, you don't understand', the Knight said, looking a little vexed. 'That's what the name is *called*. The name really is *The Aged Aged Man*'.
>
> 'Then I ought to have said "That's what the song is called"?' Alice corrected herself.
>
> 'No, you oughtn't: that's quite another thing! The song is called *Ways and Means* but that's only what it's called, you know!'
>
> 'Well, what is the song, then?' said Alice, who was by this time completely bewildered.
>
> 'I was coming to that', the Knight said. 'The song really is *A-sitting On A Gate*: and the tune's my own invention'.
>
> Can you explain the White Knight's distinctions? Do you think he is entitled to them?

Peter Heath, *The Philosopher's Alice* (New York: St. Martin's, 1974), 218.

For this reason among others, it is important to distinguish between the use of language to talk about language and the use of language to talk about extralinguistic reality. In the conversation recounted by Augustine, *A* conflates these uses. That is, *A* runs together talk about a *word*, 'manure', and talk about that bit of nonlinguistic reality designated by the word 'manure': *manure*. *A* is guilty of a *use-mention* confusion.

The distinction between use and mention is an instance of a general distinction that comes into play whenever you pause to consider representations. In discussing cases in which some item—a word, a symbol, a picture, a thought—functions representationally, you are obliged to distinguish between representations or symbols themselves and *what* they represent or symbolize. When you represent representations, the sort of confusion illustrated by St. Augustine can trip up the unwary.

To stave off confusion, philosophers distinguish between an *object language*—the language under discussion—and a *metalanguage*—the language used to discuss some object language. The distinction is relative. No language is, in its own right, an object language or a metalanguage. A language could be either, depending on whether it is being *talked about*—in which case it is the object language—or being *used* to talk about another language—in which case it is the metalanguage. If you discuss $\mathcal{L}_S$ in English, then, for purposes of that discussion, $\mathcal{L}_S$ is the object language and English the metalanguage. Were you to formulate truths about English in $\mathcal{L}_S$, the roles would be reversed. $\mathcal{L}_S$ would be the metalanguage and English the object language.

Potential for confusion arises not so much in cases in which object language and metalanguage are clearly distinct, but in cases in which a language is used to talk about itself, cases in which a language serves *both* as a metalanguage and as an object language.

When this occurs, you can minimize confusion by adopting conventions that signal your intentions. Were your aim to discuss, in English, a particular English sentence, you might distinguish the sentence under discussion by the use of quotation marks, for instance, or by placing it on a line by itself. If I want to talk about the sentence 'Socrates is wise', I might put quotation marks around it, as I have just done, or put it on a separate line:

Socrates is wise.

In so doing, I make it clear that I am not *using* the sentence (to say that Socrates is wise) but talking about the sentence.

Locating an expression within quotation marks turns the expression into the *name of itself*. Suppose I tell you that the postal employee has seventeen letters, and compare this to my telling you that 'the postal employee' has seventeen letters. The first sentence says something about a postal employee, the second concerns *the expression* 'the postal employee'.

These sentences pretty obviously have different truth conditions, as do the following:

Bol is shorter than Muggsie.

'Bol' is shorter than 'Muggsie'.

By deploying quotation marks judiciously or by setting expressions off, you can avoid most use-mention confusions. Unfortunately, you cannot avoid them all. Consider the sentence

This sentence is false.

What are the truth conditions for such a sentence? The sentence is, or certainly seems to be, paradoxical: if false, it is true; if true, it is false.

Quotation marks would offer no help here. The fact that English, and for that matter any other natural language, includes such sentences has been taken as a sign that natural languages are themselves intrinsically flawed. You might or might not agree, but $\mathcal{L}_S$ is not similarly criticizable. The paradoxical sentence above cannot be translated into $\mathcal{L}_S$.

Exercises 3.01

Add quotation marks to the sentences below in such a way that, if possible, (i) they remain *grammatical* and (ii) they express plausible truths. (Thanks to John Corcoran.)

1. An acorn contains corn.
2. One plus one is not identical with two.
3. Without a cat there would be no catastrophe.
4. Sincerity involves sin.
5. French is not French.

6. One is not identical with one.
7. George said I am not a wimp.
8. Sentence 9 is true.
9. Sentence 8 is false.
10. I love the sound of a cellar door.

3.02 Derivations in $\mathcal{L}_s$

Derivations in $\mathcal{L}_S$ consist of sequences of sentences satisfying the following definition.

> A derivation is a finite, nonempty, ordered sequence of sentences, $\langle \Gamma, \phi \rangle$, in which (i) every member of Γ is either a premise or is introduced by means of an established rule, and (ii) ϕ (*phi*) is a single sentence.

Derivations are finite sequences of sentences in a particular order. A derivation consists of *premises*, accompanied by sentences *derived from* premises, followed by a *conclusion* consisting of a single sentence. The angle brackets, ⟨ and ⟩, enclose items that occur in a particular order: premises precede the conclusion.

In constructing a derivation, you make explicit the *connection* between premises and conclusion; you show that the conclusion *follows logically* from the premises. Derivations are often called *proofs* because, in constructing a derivation, you prove that a sequence is valid. The terms can be used interchangeably, although, as you will discover, proofs of validity need not involve derivations.

In English, a derivation might take the following form.

> If Socrates is wise, then he is happy.
>
> Socrates is wise.
>
> ---
>
> Socrates is happy.

The sentences above the horizontal line are premises. The sentence below the line is a conclusion. If the argument is a good one, the conclusion is a *logical consequence* of the premises—alternatively, the premises *logically imply* the conclusion.

Sentences are evaluated with respect to their truth or falsity. Arguments are not true or false, but *valid* or *invalid*. These notions are best characterized in stages, starting with a definition of validity in terms of logical implication (and the introduction of a new symbol).

> A derivation, $\langle \Gamma, \phi \rangle$, is valid if and only if Γ logically implies ϕ ($\Gamma \vDash \phi$).

The symbol $\vDash$ (lazy *pi*) belongs not to $\mathcal{L}_S$ but to the metalanguage. The symbol stands for *logically implies*, so $\Gamma \vDash \phi$ means that Γ logically implies ϕ. A definition of logical implication completes the characterization of validity.

> A set of sentences, Γ, logically implies a sentence, ϕ, ($\Gamma \vDash \phi$) if and only if ϕ cannot be false if Γ (each sentence in Γ) is true.

When Γ logically implies ϕ, then ϕ is said to be a *logical consequence* of Γ.

This is a lot to take in, so it is worth pausing briefly to reflect on these notions before venturing further.

When a sentence or set of sentences logically implies a sentence, the relation is what you have in mind when you recognize, even if only implicitly, that a particular sentence follows from another sentence or set of sentences. The relation is in play in the sequence mentioned earlier.

> If Socrates is wise, then he is happy.
>
> Socrates is wise.
>
> ---
>
> Socrates is happy.

Here, 'Socrates is happy' follows from 'If Socrates if wise, then he is happy' together with 'Socrates is wise'.

Logical implication can be characterized by referring to the truth conditions of sentences, their truth values across possible worlds. You can think of q as following from or being logically implied by p when at no possible world is p true and q false. If p and q make up a sequence in which p is a premise and q is the conclusion, then, when p logically implies q, the sequence is valid.

In the example above, there is no world at which the first two sentences are true and the third is false, no world at which it is true that if Socrates is wise, then he is happy, and true that Socrates is wise but false that Socrates is not happy.

Notice that this characterization does *not* require that the premises or the conclusion of a valid sequence be true. It requires only that *if* the premises are true, the conclusion must be true. If the conclusion of a valid sequence is false, one or more of the premises on which it rests must be false as well.

Validity is *truth-preserving*, then, but not *truth-guaranteeing*: if you begin with true premises and validly derive a conclusion from those premises, then the conclusion too must be true. If you know only that a conclusion follows validly from a set of premises, however, you do not thereby know whether the conclusion is true.

One way to show that a sequence is valid is to construct a derivation. Before discussing the mechanics of derivations, however, another, more flat-footed technique for proving validity is worth mentioning. The technique relies solely on truth tables. Consider the following sequence:

> If human beings are fish, then whales are fish.
>
> Human beings are fish.
>
> ---
>
> Whales are fish.

The reasoning in this sequence is perfectly valid. The conclusion, 'Whales are fish', follows from the premises: at no possible world are the premises true and the conclusion false. Yet the conclusion is false, as is one of the premises.

The sequence could be translated into $\mathcal{L}_S$ as

$$\begin{array}{l} H \supset W \\ H \\ \hline W \end{array}$$

Once you have put the sequence into $\mathcal{L}_S$, you can construct a truth table to prove validity:

H W	$H \supset W$
T T	T
T F	F
F T	T
F F	T

The truth table contains no row in which the premises, H and $H \supset W$, are both true and the conclusion, W, is false. In two rows, the second and fourth, the conclusion is false, but in those rows one of the premises is false as well.

Now consider another sequence.

If Socrates drinks the hemlock, then he dies.

Socrates doesn't drink the hemlock.

Socrates doesn't die.

Is the reasoning in this sequence valid? Does the conclusion follow from—is it logically implied by—the premises? Is there a possible world at which the premises are true and the conclusion false? You might waver—maybe, maybe not. Once you put the sequence into $\mathcal{L}_S$, however, the answer becomes clear.

H D	$H \supset D$	$\neg H$	$\neg D$
T T	T	F	F
T F	F	F	T
F T	T	T	F
F F	T	T	T

The third row depicts a possible world at which the premises of the original sequence are true and its conclusion false. That world might be one at which hemlock is lethal (as it is in the actual world) and in which Socrates does not drink the hemlock but dies anyway—from some other cause.

An argument of this sort in fact incorporates a well-known fallacy: *denying the antecedent*. From a conditional sentence, together with the denial of its antecedent, you cannot infer the denial of the consequent. In everyday life it is easy to miss fallacious—that is, invalid—arguments because you are distracted by your knowledge of the truth or falsity of their premises and conclusions. Yet it is perfectly possible for all of the premises and the conclusion of an argument to be true and for the argument still to be invalid. The last row of the truth table above illustrates precisely this possibility.

When you evaluate arguments in English, then, it is important to attend to the logical relations holding between premises and conclusion and learn to ignore the isolated truth values of their constituent sentences.

Conditionals and Logical Implication

Logical implication, the relation captured by the $\models$, is sometimes confused with the conditional relation captured by the $\supset$. You might, for instance, be tempted to read sentences of the form

$p \supset q$

as '*p* implies *q*'. If 'implies' is taken to mean 'logically implies', however, this reading is seriously off base. Consider a simple conditional sentence in English:

If the sky is blue, then Socrates is wise.

This sentence concerns the color of the sky and a particular individual, Socrates. The conditional is, as a matter of fact, true—its antecedent and consequent are both true. In contrast, the sentence

'The sky is blue' implies 'Socrates is wise'.

concerns not Socrates and the color of the sky, but, as the quotation marks indicate, a pair of *sentences*. It is, in addition, *false*. The sentence

The sky is blue.

does not imply—that is, logically imply—the sentence

Socrates is wise.

You can easily envisage a world at which the sky is blue, but Socrates is not wise.

Exercises 3.02

For each of the $\mathcal{L}_S$ sequences below, construct a truth table that demonstrates its validity or invalidity, one that shows whether its premises do or do not logically imply its conclusion. If the sequence is invalid, indicate the row or rows that establish this fact.

1. $P \supset Q$, $R \supset P$, R / Q
2. $P \supset Q$, $P \supset R$ / $Q \supset R$
3. $P \supset Q$, $\neg(Q \wedge R)$ / $P \supset \neg R$
4. $P \vee \neg Q$, $\neg(Q \wedge \neg R) \supset \neg S$, $\neg P$ / $\neg Q$
5. $P \wedge Q$, $Q \supset (P \vee R)$ / R
6. $P \supset (Q \supset P)$ / P
7. $(P \wedge Q) \supset R$, $P \wedge \neg R$ / $\neg Q$
8. $P \supset Q$, $R \supset Q$, R / P
9. $P \supset Q$, Q / P
10. $P \vee Q$, P / $\neg Q$
11. $P \wedge Q$, $P \supset (R \vee \neg Q)$ / R
12. $P \supset Q$, $\neg Q$ / $\neg P$
13. $P \supset Q$, $R \supset S$, $P \vee R$ / $Q \vee S$
14. $P \supset Q$, $\neg Q$ / $P \supset R$
15. $(P \vee Q) \supset R$, $P \vee S$, $\neg S$ / R

3.03 The Principle of Form

Logic is silent as to the truth or plausibility of premises and conclusions. Its province is validity. A valid argument need not have a true conclusion. But if an argument incorporates true premises and if its conclusion follows validly from those premises, then that conclusion must be true. An ideal knower needs both true sentences and valid inferential procedures.

Philosophers since Descartes (1596–1650) have dreamed of basing all knowledge on a few indisputable truths. Those truths, together with appropriate inferences, would yield a complete system of knowledge. Cartesian foundationalism is out of fashion, but the need for valid inferences is universal and timeless.

Truth tables provide a mechanical, algorithmic technique for proving the validity of sequences in $\mathcal{L}_S$. As the previous exercises clearly illustrate, however, truth tables are unwieldy and tedious, at least for human reasoners. Although they afford exhaustive analyses of $\mathcal{L}_S$ sequences, truth tables

Algorithms

An algorithm is a procedure that guarantees a particular result in a finite number of steps.

Think of an algorithm as prescribing a step-by-step method for achieving a previously specified goal. Each step is entirely mechanical, calling for no insight or intelligence. In this sense, algorithmic solutions to problems are 'mindless'—although considerable intelligence is generally required to invent them!

Imagine that you want to find your way through a maze. The maze is finite, simply connected (its walls are connected to one another or to the maze's outer wall), and you are indifferent to the time it takes to locate the exit. You can guarantee success by following a simple algorithm: *turn left at every corner*. Although the procedure will lead you up blind alleys, it does not require that you remember which paths you have taken. You need only be able to identify corners, to execute the instruction to turn left—and to recognize the exit when you eventually reach it.

Truth tables afford algorithmic procedures for determining validity or invalidity in $\mathcal{L}_S$. You can mechanically establish that a sequence is valid by constructing a truth table for the sequence and noting whether, on any row of the table, the premises are true and the conclusion is false. If no such row appears, the sequence is valid; it is invalid otherwise.

lend themselves to notational errors. Worse, truth table proofs of validity are applicable only to sequences that can be smoothly translated into $\mathcal{L}_S$. Much, perhaps most, of the reasoning we use every day outstrips the meager resources of $\mathcal{L}_S$.

A more intelligent, interesting, and natural way to demonstrate validity is to construct a derivation. Derivations are founded on inferential principles that extend well beyond $\mathcal{L}_S$. A derivation comprises a sequence of sentences, including premises and a conclusion, in which the reasoning from premises to conclusion is spelled out explicitly. The derivational component of $\mathcal{L}_S$ has been designed to reflect familiar informal patterns of reasoning. $\mathcal{L}_S$ constitutes what logicians call a *natural deduction system*, one in which derivational proofs of validity are founded on natural patterns of reasoning.

Derivations in $\mathcal{L}_S$ depend on a *principle of form*:

> If a sequence, $\langle \Gamma, \phi \rangle$, is valid, then any sequence *with the same form* is valid.

The key to understanding the principle is to understand what it means for two sentences, or sentence sequences, to have *the same form*. You can acquire an intuitive grip on this notion by considering examples.

Consider a valid sequence discussed earlier:

> If human beings are fish, then whales are fish.
>
> Human beings are fish.
>
> ---
>
> Whales are fish.

and in $\mathcal{L}_S$:

$$\begin{array}{l} H \supset W \\ H \\ \hline W \end{array}$$

The principle of form states that, if this sequence is valid, any sequence with the *same form* is valid. What, then, is the form of the sequence? Suppose you set out the sequence using variables:

$$\begin{array}{l} p \supset q \\ p \\ \hline q \end{array}$$

The sequence consists of a pair of premises, one of which is a conditional sentence, the other is the antecedent of that conditional, and the conclusion is the consequent of the conditional. This is the form of the sequence. The following sequences have the same form as the original sequences:

If the sky is blue, then Socrates is wise.
The sky is blue.

Socrates is wise.

In $\mathcal{L}_S$:

$$\begin{array}{l} B \supset W \\ B \\ \hline W \end{array}$$

Although the sameness of pattern is obvious in these examples, that is not always the case. The $\mathcal{L}_S$ sequence below has the same form as those above:

$$\begin{array}{l} \neg(A \vee (C \supset D)) \supset (D \wedge \neg C) \\ \neg(A \vee (C \supset D) \\ \hline D \wedge \neg C \end{array}$$

This might surprise you. At first glance, this complex sequence looks nothing like the original sequence. Closer examination reveals that the sequence, like the earlier one, consists of (i) a conditional sentence, (ii) the antecedent of that conditional, followed by (iii) its consequent. The antecedents and consequents are themselves nonatomic, but that is irrelevant when it comes to the sequence's form. The sequence matches the pattern

$$\begin{array}{l} p \supset q \\ p \\ \hline q \end{array}$$

You will be in a position to take on derivations in $\mathcal{L}_S$ once you learn to distinguish such patterns. The construction of derivations is a perceptual task far more than it is an intellectual one, something akin to bird-watching or solving picture puzzles (see chapter 1). Patterns, hidden at first, can, with practice, come to seem obvious.

3.04 Inference Rules: *MP*, *MT*

Earlier, the sequence below was shown to be valid:

$$\begin{array}{l} H \supset W \\ H \\ \hline W \end{array}$$

Generalizing the pattern yielded a generalized sequence of the form

$$\begin{array}{l} p \supset q \\ p \\ \hline q \end{array}$$

The principle of form tells us that every sequence with this form is valid. The pattern of inference exhibited in this sequence is both familiar and natural and, for that reason, deserves to be accorded to the status of a *rule*. The traditional Latin name for this rule is *modus ponendo ponens*—for short, *modus ponens*, or *MP*.

$$(MP) \qquad p \supset q, p \vdash q$$

This formulation of *MP* indicates that $p \supset q$, together with p, *deductively yield* q. That is, *MP* permits you to derive q from $p \supset q$ and p. The variables p and q are understood to range over any $\mathcal{L}_S$ sentences. The turnstile, $\vdash$, indicates that the element on its right is derivable from the elements on its left: if you have the elements on the left, you are entitled to assert the one on the right. This rule, in concert with other rules, is used in the derivation of conclusions from premises.

The mechanics of derivations are best learned by example. Suppose you set out to show that the conclusion of the sequence mentioned earlier is derivable from its premises.

$$\begin{array}{l} \neg(A \vee (C \supset D)) \supset (D \wedge \neg C) \\ \neg(A \vee (C \supset D) \\ \hline D \wedge \neg C \end{array}$$

Modus ponens enables you to support the move from the two premises—the sentences above the horizontal line—to the conclusion. To facilitate the procedure, each line in the derivation is given a number. When you need to refer to a particular sentence in a derivation, you can do so by way of the number of the line on which the sentence occurs.

Line numbers contribute to the intelligibility of derivations. Intelligibility is aided by the use of other devices as well. A plus sign to the left of a sentence indicates that the sentence is a premise.

A question mark is placed to the left of the conclusion to be derived, indicating that the sentence is taken to follow deductively from the premises.

In the course of constructing derivations, you will often find it necessary to add sentences that have been derived by means of a rule from other sentences en route to the conclusion. To the right of every sentence added in this way you place (i) the name of the rule that supports its insertion, and (ii) the line number or numbers of sentences to which the rule was applied. Putting this all together, a derivation for the preceding sequence could be constructed as follows:

1.	$+ \neg(A \vee (C \supset D)) \supset (D \wedge \neg C)$	
2.	$+ \neg(A \vee (C \supset D)$	
3.	$? (D \wedge \neg C)$	
4.	$D \wedge \neg C$	1, 2 *MP*

The sentences in lines 1 and 2 are premises, as the + to their left indicates. Line 3 contains the conclusion you intend to derive. A conclusion, signaled by a ?, is never itself used in a derivation. It is inserted following the premises to indicate the *goal* of the derivation. The sentence on line 4 is derived, so its introduction must be supported with the name of the rule that entitles you to enter it, *MP*, together with the line numbers of sentences to which the rule has been applied. The notation in line 4 indicates that $(D \wedge \neg C)$ was derived from sentences in lines 1 and 2 via *MP*.

Derivation will be introduced in two stages. Rule *MP* is one example of an inference rule. Inference rules mirror principles of reasoning we all employ unselfconsciously in our everyday thinking. This is less so in the case of transformation rules, which will be discussed in due course (§§ 3.11–3.15).

Rules are formulated not in $\mathcal{L}_S$ but in the metalanguage. Rules tell you when it is permissible to derive sentences from sentences. For this reason, rules are formulated using *variables* (lowercase *p*'s and *q*'s). As the derivation above makes clear, these variables do not stand for atomic $\mathcal{L}_S$ sentences but for any $\mathcal{L}_S$ sentence whatever.

Inference and Implication

Implication is a relation among sentences. Inference is a mental act. 'If *A* then *B*', coupled with '*A*', logically implies '*B*'. This means that if both 'If *A* then *B*' and '*A*' are true, '*B*' must be true: there is no world at which the first two sentences are true and the third is not true.

Your accepting the first two sentences above does not thereby rationally oblige you to infer or accept the third. At most, reason demands your acceptance of '*B*' or your rejection of either 'If *A* then *B*' or '*A*'. It is standard practice in the sciences and in everyday life to reject a hypothesis when we learn that it implies something false.

Ordinary sentences, singly and in combination, imply endless other sentences. We take the trouble to infer these other sentences only when we have some reason to do so.

Whole Lines

Modus ponens (*MP*), in common with every other inference rule, applies only to *whole lines* of derivations. The following application of *MP* violates this restriction.

1. $+ B \vee (S \supset W)$
2. $+ S$
3. W 1, 2 *MP* (*invalid, violates the restriction*)

Differently put, *MP* cannot be applied to sentences that are themselves parts of sentences. The reason should be clear: the conditional sentence in line 1 above is part of a disjunction. Suppose I tell you

> Either the sky is blue or, if I sell you a lottery ticket for $5, you will win the lottery.

You would be foolish to infer that, were I to sell you a ticket, you would win the lottery, even if you had every reason to believe that my disjunctive utterance was truthful. The moral is that in $\mathcal{L}_S$ rules of inference apply only to whole sentences, not to sentences that are themselves parts of sentences.

A rule similar to *modus ponens*, *modus tollendo tollens*—*modus tollens* (*MT*)—enables you to derive the denial of the antecedent of a conditional sentence given (i) the conditional sentence and (ii) the denial of its consequent.

$$(MT) \qquad p \supset q, \neg q \vdash \neg p$$

Consider a parallel sequence in English:

> If Socrates is wise, then he is happy.
> Socrates isn't happy.
> ---
> Socrates isn't wise.

The pattern of reasoning is straightforward and familiar. Consider its deployment in $\mathcal{L}_S$:

$W \supset H$
$\neg H$

$\neg W$

You can construct a derivation of this sequence, marking each premise with a +, using a ? to indicate the conclusion to be derived, and showing that the conclusion does follow from the premises by *MT*:

1. $+ W \supset H$
2. $+ \neg H$
3. $? \neg W$
4. $\neg W$ 1, 2 *MT*

Derivations are not always so straightforward. Consider the English sequence

If it's not a holiday, then the motorway is jammed.

The motorway isn't jammed.

It's a holiday.

The sequence consists of a conditional sentence, 'If it's not a holiday, then the motorway is jammed', followed by the denial of the consequent of that conditional, 'The motorway isn't jammed'. The conclusion, 'It's a holiday', permitted in accord with *modus tollens*, is the denial of the antecedent. Why not 'It's not the case that it isn't a holiday?' Negated negations are equivalent to unnegated sentences. 'It's not the case that it isn't a holiday' is logically equivalent to 'It's a holiday', as the truth table below shows:

H	$\neg H$	$\neg\neg H$
T	F	T
F	T	F

The truth conditions of $\neg\neg H$ are the same as those for H.

3.05 Sentence Valence

How might this feature of negation be captured in $\mathcal{L}_S$? One option would be to introduce a rule—double negation (*DN*)—that would allow p to be derived from $\neg\neg p$. The derivation below reflects the application of such a rule, together with *MT*, to an $\mathcal{L}_S$ version of the sequence above.

1.	$+ \neg H \supset J$	
2.	$+ \neg J$	
3.	$? H$	
4.	$\neg\neg H$	1, 2 *MT*
5.	H	4 *DN*

Although *DN* could be included in the list of derivation rules for $\mathcal{L}_S$, a less cumbersome mechanism is available, one that will streamline many derivations.

Every sentence has a *valence*, positive or negative. A negated sentence has a *negative valence*. A sentence not preceded by a negation sign has a *positive valence*. The sentences below have a positive valence:

$$P$$

$$P \vee Q$$

$$P \supset (Q \vee \neg R)$$

The valence of the following sentences is negative:

$$\neg P$$

$$\neg(P \vee Q)$$

$$\neg(P \supset (Q \vee \neg R))$$

In $\mathcal{L}_S$, negation signs appearing in rules are to be read as indications of the *relative valence* of expressions to which they are affixed. *MT* applies when you have a conditional sentence together with a sentence identical to its consequent but with the opposite valence. From this you can infer the negation of its antecedent—that is, a sentence identical to the antecedent but with the opposite valence. Reversing the valence of a negated sentence yields an *unnegated* sentence. This means that in the derivation above, you can infer *H directly* from lines 1 and 2 via *MT*.

1.	$+ \neg H \supset J$	
2.	$+ \neg J$	
3.	$? H$	
4.	H	1, 2 *MT*

The same point applies to negated consequents, as illustrated by the derivation below:

1.	$+ A \supset \neg B$	
2.	$+ B$	
3.	$? \neg A$	
4.	$\neg A$	1, 2 *MT*

MT permits the derivation of the negation of the antecedent of a conditional, given the negation of its consequent. The consequent of the conditional $A \supset \neg B$ is $\neg B$. Reversing the valence of $\neg B$ yields B.

3.06 Hypothetical Syllogism: *HS*

MP and *MT* mirror everyday forms of reasoning with conditional sentences. A third rule, *hypothetical syllogism* (*HS*), involving conditionals, governs inferences of the sort illustrated below.

If Socrates is wise, then he is happy.

If Socrates is happy, then he is satisfied.

If Socrates is wise, then he is satisfied.

The reasoning here has the form

$$(HS) \qquad p \supset q, q \supset r \vdash p \supset r$$

Such sequences consist of a pair of conditionals in which the consequent of one matches the antecedent of the other, from which a third conditional can be extracted. Conditionals are *transitive*: if r is conditional on q, and q is conditional on p, then r is conditional on p.

A derivation of the sequence above would look like this:

1. $+ W \supset H$
2. $+ H \supset S$
3. $? W \supset S$
4. $W \supset S$ 1, 2 *HS*

Inference rules, including *HS*, apply whether the conditionals involved are simple or complex, and regardless of their order of occurrence in a derivation. The derivation below illustrates both possibilities:

1. $+ \neg(A \vee \neg B) \supset (C \supset D)$
2. $+ (R \wedge S) \supset \neg(A \vee \neg B)$
3. $? (R \wedge S) \supset (C \supset D)$
4. $(R \wedge S) \supset (C \supset D)$ 1, 2 *HS*

Here, as elsewhere, the trick is to begin cultivating the knack of recognizing *patterns* amidst sometimes bewildering arrays of symbols.

Exercises 3.06

Construct derivations for the $\mathcal{L}_S$ sequences below using *MP*, *MT*, and *HS*, and interpreting occurrences of ¬ in *MT* as an instruction to *reverse the valence* of the expression to the right of the ¬.

1. $+ P \supset (Q \vee R)$
   $+ \neg(Q \vee R)$
   $? \neg P$

2. $+ \neg P \supset \neg Q$
   $+ \neg P$
   $? \neg Q$

3. $+ \neg P \supset Q$
   $+ \neg Q$
   $? P$

4. $+ P \supset Q$
   $+ \neg S$
   $+ \neg(Q \supset R) \supset S$
   $? P \supset R$

5. $+ P \supset \neg Q$
   $+ Q$
   $+ \neg S \supset P$
   $? S$

6. $+ P \supset (Q \supset R)$
   $+ P$
   $+ Q$
   $? R$

7. $+ \neg(P \supset Q) \supset R$
   $+ \neg R$
   $+ P$
   $? Q$

8. $+ P \supset Q$
   $+ Q \supset R$
   $+ P$
   $? R$

9. $+ P \supset \neg Q$
   $+ Q$
   $+ R \supset P$
   $? \neg R$

10. $+ \neg(P \supset R) \supset \neg Q$
    $+ P$
    $+ Q$
    $? R$

11. $+ \neg P \supset \neg Q$
    $+ Q$
    $+ P \supset (S \wedge T)$
    $? S \wedge T$

12. $+ P \supset Q$
    $+ Q \supset \neg R$
    $+ R$
    $? \neg P$

13. $+ \neg(P \supset Q) \supset R$
    $+ R \supset S$
    $+ \neg S$
    $? P \supset Q$

14. $+ \neg(P \wedge \neg S) \supset (Q \vee R)$
    $+ (Q \vee R) \supset \neg T$
    $+ T$
    $? P \wedge \neg S$

15. $+ (P \supset S) \supset T$
    $+ T \supset \neg(Q \vee \neg R)$
    $+ ((P \supset S) \supset \neg(Q \vee \neg R)) \supset U$
    $? U$

Derivation Heuristics

A *heuristic* is a rule of thumb, a principle that can prove useful in the solution of a problem but does not guarantee a solution. Heuristics are distinguished from algorithms, solution-guaranteeing procedures.

Truth tables serve as algorithms for the assessment of validity or invalidity in $\mathcal{L}_S$. By constructing a truth table, you can 'mechanically' establish that a sequence is valid or invalid.

Matters are different when you set out to prove validity by means of a derivation. There is no mechanical technique for applying rules of inference to premises in such a way as to guarantee the derivation of the conclusion of every valid sequence. Your failure to derive a conclusion could mean either that the sequence is invalid or that, although it is valid, you have not yet discovered a sequence of rule applications that would demonstrate its validity.

This does not mean that the construction of derivations is a matter of blind luck. Once you have the knack, you can often *see* how a derivation might go. Acquiring the knack involves the acquisition of pattern recognition skills: you learn to see—in the literal sense—familiar patterns in the midst of unfamiliar arrays of symbols.

One rule of thumb or heuristic commonly thought to help in the construction of derivations is this:

> Focus on the conclusion and 'work backward' to the premises.

In so doing, you can narrow down the range of options in the application of rules.

3.07 Rules for Conjunction: ∧*I*, ∧*E*

Rules *MP*, *MT*, and *HS* operate on conditionals. Now consider a pair of rules involving conjunction. The first rule, *conjunction introduction* (∧*I*), allows you to put two sentences together to make a conjunction.

$$(\wedge I) \qquad p, q \vdash p \wedge q$$

The rule reflects the pattern of reasoning in the sequence below:

Socrates is wise.

Socrates is happy.

Socrates is wise and happy.

The sequence, translated into $\mathcal{L}_S$, incorporates an application of $\wedge I$.

1.	W	
2.	H	
3.	$W \wedge H$	1, 2 $\wedge I$

A second rule, *conjunction elimination* ($\wedge E$), goes in the reverse direction. From

Socrates is wise and happy.

you are permitted to infer either or both of the sentences

Socrates is wise.

and

Socrates is happy.

The rule is formulated in two parts, making this option explicit.

$$(\wedge E) \qquad p \wedge q \vdash p$$
$$p \wedge q \vdash q$$

Applying $\wedge E$ to an $\mathcal{L}_S$ version of the English sequence above enables you to 'bring down' each conjunct individually.

1.	$+\ W \wedge H$	
2.	W	1 $\wedge E$
3.	H	1 $\wedge E$

As in the case of rules of inference generally, conjunction rules apply only to whole lines in derivations. From

$$\neg(W \wedge H)$$

you could derive neither W nor H. Such an inference ignores the negation sign and is clearly invalid as the English sequence below illustrates.

It's not the case that Socrates is wise and happy.

Socrates is wise.

You could prove the invalidity of the $\mathcal{L}_S$ sequence by constructing a truth table. This would show that there is at least one row in which the premise $\neg(W \wedge H)$ is true and W is false and a row in which $\neg(W \wedge H)$ is true and H is false.

Exercises 3.07

Construct derivations for the $\mathcal{L}_S$ sequences below using *MP*, *MT*, *HS*, ∧*I*, and ∧*E*.

1. $+ (P \supset Q) \wedge \neg R$
   $+ P$
   $? Q$

2. $+ P \wedge (\neg Q \wedge \neg R)$
   $? \neg R$

3. $+ \neg(P \wedge \neg Q) \supset R$
   $+ \neg R$
   $? \neg Q$

4. $+ \neg P$
   $+ Q$
   $+ (\neg P \wedge Q) \supset R$
   $? R$

5. $+ P \wedge Q$
   $? Q \wedge P$

6. $+ P \supset (Q \wedge \neg R)$
   $+ P$
   $? \neg R$

7. $+ P \supset (Q \wedge \neg R)$
   $+ P$
   $? \neg R \wedge Q$

8. $+ (P \wedge Q) \supset (R \wedge S)$
   $+ Q$
   $+ P$
   $? R$

9. $+ \neg(P \wedge Q) \supset (P \vee Q)$
   $+ \neg(P \vee Q)$
   $? Q$

10. $+ S \wedge ((P \equiv Q) \supset R)$
    $+ P \equiv Q$
    $? R$

11. $+ P \supset Q$
    $+ R \supset S$
    $+ P \wedge R$
    $? Q \wedge S$

12. $+ P \supset Q$
    $+ Q \supset (R \wedge S)$
    $+ P \wedge T$
    $? S \wedge T$

13. $+ (P \wedge Q) \supset \neg(S \wedge T)$
    $+ \neg S \supset \neg T$
    $+ T$
    $? \neg(P \wedge Q)$

14. $+ P \supset (Q \supset \neg R)$
    $+ P \wedge Q$
    $? \neg R$

15. $+ S \wedge \neg Q$
    $+ \neg(R \wedge (S \supset T)) \supset Q$
    $? T$

3.08 Rules for Disjunction: $\vee I$, $\vee E$

Disjunction rules differ from those affecting the $\wedge$ connective in a way that mirrors the truth-functional difference between disjunction and conjunction. *Disjunction introduction* ($\vee I$) permits you to append any sentence you please to a given sentence as a disjunct. The added sentence need not be present earlier in the derivation; it need not be imported from some other line. The idea is straightforward. Suppose you know that

Socrates is happy.

You can infer that

Socrates is happy or wise.

or, for that matter,

Socrates is happy or whales are fish.

These disjunctions add no information to the sentences from which they were derived. They do not say, for instance, that, in addition to being happy, Socrates is wise or that whales are fish. 'Socrates is wise' and 'Whales are fish' are present only as components of disjunctions. The rule, expressed in $\mathcal{L}_S$, has the form

$$(\vee I) \qquad p \vdash p \vee q$$

In practice, $\vee I$ most often comes into play when you need to add something to a derivation that is not already present. If, in looking over a sequence, you notice that an atomic sentence occurs in the conclusion that is absent from the premises, the chances are excellent that you would need to call on $\vee I$ should you set out to construct a derivation.

Rule $\vee I$ permits the addition of atomic sentences to atomic sentences

1.	$+ H$	
2.	$H \vee W$	1 $\vee I$

as well as the addition of nonatomic, molecular sentences.

1.	$+ H$	
2.	$H \vee \neg(W \supset \neg P)$	1 $\vee I$

The rule thus permits the introduction of *any* sentence to a sentence already present, provided you restrict its application to whole lines, as the sequence below illustrates.

1.	$+ H \vee W$	
2.	$(H \vee W) \vee \neg(W \supset \neg P)$	1 $\vee I$

When ¬(*W* ⊃ ¬*P*) is added to *H* ∨ *W*, it is added to the whole sentence, not to *W* alone. The derivation below is defective.

1. + *H* ∨ *W*
2. *H* ∨ (W ∨ *P*) 1 ∨*I* (*incorrect*)

Disjunction introduction's companion rule, *disjunction elimination* (∨*E*), differs from its conjunction counterpart, ∧*E*, in requiring two sentences for its application. Given a disjunction, together with the denial of one of its disjuncts (that is, a sentence identical to one of its disjuncts but possessing the opposite valence), ∨*E* permits you to derive the remaining disjunct. Thus, in English,

Either Socrates is wise or Socrates is happy.

Socrates isn't wise.

Socrates is happy.

The pattern of reasoning invoked exhibits the following form:

W ∨ *H*

¬*W*

H

The sequence below, which differs from the one above only in featuring the negation of the second disjunct, is equally sound:

Either Socrates is wise or Socrates is happy.

Socrates isn't happy.

Socrates is wise.

As with ∧*E*, the formulation of ∨*E* includes two parts.

(∨*E*) $p \vee q, \neg p \vdash q$

$p \vee q, \neg q \vdash p$

Applying ∨*E* to the sequence above yields

1. + *W* ∨ *H*
2. + ¬*H*
3. ? *W*
4. *W* 1, 2 ∨*E*

Here, as elsewhere, negation signs occurring in the formulation of rules pertain to the valences of sentences, and not to the occurrence of actual negation signs in sentences under consideration. Given a disjunction,

$$\neg R \vee \neg(S \wedge P)$$

together with a sentence identical with one of its disjuncts but with the opposite valence, R, you could derive the remaining disjunct:

$$\neg(S \wedge P)$$

More formally:

1.	$+ \neg R \vee \neg(S \wedge P)$	
2.	$+ R$	
3.	$? \neg(S \wedge P)$	
4.	$\neg(S \wedge P)$	1, 2 $\vee E$

You could not, however, apply the rule to a sentence that is itself contained within another sentence.

1.	$+ P \wedge (Q \vee \neg R)$	
2.	$+ R$	
3.	Q	1, 2 $\vee E$ (*incorrect*)

The restriction applies to $\vee E$ just as it applies to every rule of inference.

Exercises 3.08

Construct derivations for the $\mathcal{L}_S$ sequences below using *MP*, *MT*, *HS*, ∧*I*, ∧*E*, ∨*I*, and ∨*E*.

1. $+ P$
   $+ (P \vee \neg R) \supset Q$
   $? Q$

2. $+ (P \vee Q) \supset (R \wedge S)$
   $+ P$
   $? S$

3. $+ P \supset Q$
   $+ P \vee \neg R$
   $+ R$
   $? Q$

4. $+ P \supset (Q \vee R)$
   $+ \neg(Q \vee R) \vee S$
   $+ \neg S$
   $? \neg P$

5. $+ (P \wedge Q) \vee \neg(R \wedge S)$
   $+ S$
   $+ R$
   $? Q$

6. $+ P \supset \neg(Q \wedge R)$
   $+ (Q \wedge R) \vee S$
   $+ \neg S$
   $? \neg P$

7. $+ P \supset Q$
   $+ P$
   $? P \wedge (Q \vee R)$

8. $+ P \supset (Q \vee \neg S)$
   $+ P \wedge S$
   $? Q$

9. $+ P$
   $+ \neg Q$
   $+ \neg(Q \vee R) \supset \neg P$
   $? R$

10. $+ P \supset (Q \wedge R)$
    $+ S \vee \neg T$
    $+ S \supset P$
    $+ T$
    $? Q$

11. $+ P \supset (Q \vee R)$
    $+ S \supset (P \wedge \neg R)$
    $+ S \wedge T$
    $? Q$

12. $+ P \supset Q$
    $+ (Q \vee (R \supset S)) \supset (S \vee T)$
    $+ \neg S \wedge P$
    $? T$

13. $+ (P \vee \neg Q) \supset R$
    $+ P$
    $? R \vee (S \equiv \neg T)$

14. $+ P \supset Q$
    $+ (Q \vee R) \supset (R \vee \neg S)$
    $+ P \wedge \neg R$
    $? \neg S$

15. $+ P \supset \neg(Q \vee R)$
    $+ Q \wedge S$
    $+ T \vee P$
    $? T$

3.09 Conditional Proof: *CP*

The conditional sentence 'If Socrates is a philosopher, then he is happy' evidently follows from the sentence 'Philosophers are happy'. How might you go about showing this? A natural way to reason would be as follows:

> Philosophers are happy. If Socrates is a philosopher, he must be happy. Given that philosophers are happy, then, if Socrates is a philosopher, he is happy.

In so reasoning, you would be deriving a conditional sentence by (i) supposing its antecedent true 'for the sake of argument' and (ii) showing that, given your other premise, if this antecedent is true, the consequent is true. This amounts to establishing that the conditional sentence follows from the premises with which you began.

The pattern of reasoning is captured in $\mathcal{L}_S$ by a *conditional proof* rule (*CP*). The rule allows you to enter a sentence, p, as a *supposition*. If q could be shown to follow from this supposition (together with any other available premises), you would be entitled to affirm the conditional sentence $p \supset q$. The rule is set out as follows:

(*CP*)
$\ulcorner p$
$\vdots$
$q \lrcorner$
$p \supset q$

The $\ulcorner$ marks a *suppositional premise*. It functions in derivations just as 'suppose' functions in English sequences. The $\lrcorner$ indicates that a supposition has been *discharged*.

Conditional proof is useful (and often indispensable) in the derivation of conditional sentences. *CP* permits you to introduce, at any point in a derivation, the antecedent of a conditional you want to derive. The suppositional sentence, p, can be used in the derivation just as though it were a premise, its suppositional character p signaled by the presence of the $\ulcorner$.

In deriving q, the consequent of the pertinent conditional, you discharge this supposition and *close off* the application of *CP* by placing a $\lrcorner$ around q. You then enter the conditional, $p \supset q$, the antecedent of which is the sentence introduced as the supposition, and the consequent the last sentence derived, the sentence to the right of the $\lrcorner$. To indicate the scope of your supposition, connect the $\ulcorner$ and $\lrcorner$ as shown in the example below:

1.	$+ S \supset M$	
2.	$? S \supset (S \wedge M)$	
3.	$\ulcorner S$	
4.	$? S \wedge M$	
5.	M	1, 3 *MP*
6.	$S \wedge M \lrcorner$	3, 5 $\wedge I$
7.	$S \supset (S \wedge M)$	3–6 *CP*

Here, line 3 introduces S as a supposition. A *subgoal*, $S \wedge M$, is announced in line 4 to mark the sentence you intend to derive with the help of the supposition. Once this subgoal is derived, the supposition, S, is discharged and the conditional proved on the basis of S is set out in line 7. This derivation typifies the role of conditional proof in derivations.

CP can be used to derive *any* conditional sentence needed to complete a derivation. That sentence could, but need not, be the conclusion. Once a supposition is entered, attention shifts from the original conclusion to the subgoal, the consequent of the conditional you intend eventually to derive. When that consequent is derived, however, and the supposition is bracketed—signaling that it has been discharged—sentences contained within the brackets become *inactive* and cannot be used in subsequent lines.

CP is a powerful and versatile rule. This becomes evident when you embed applications of *CP* inside one another. Embedded conditional proofs are useful for deriving conditional sentences that themselves contain conditional consequents. Imagine setting out to derive the sentence below:

$$(S \supset P) \supset (S \supset R)$$

You might enter as a supposition

$$S \supset P$$

and seek, as a subgoal the sentence

$$S \supset R$$

This subgoal is *itself* a conditional, so you might establish it using an embedded *CP* strategy. The derivation below illustrates this technique:

1.	$+ P \supset (Q \vee R)$	
2.	$+ (S \wedge P) \supset \neg Q$	
3.	$? (S \supset P) \supset (S \supset R)$	
4.	$S \supset P$	
5.	$? S \supset R$	
6.	S	
7.	$? R$	
8.	P	4, 6 *MP*
9.	$Q \vee R$	1, 8 *MP*
10.	$S \wedge P$	6, 8 $\wedge I$
11.	$\neg Q$	2, 10 *MP*
12.	R	9, 11 $\vee E$
13.	$S \supset R$	6–12 *CP*
14.	$(S \supset P) \supset (S \supset R)$	4–13 *CP*

Such derivations might at first seem daunting. With practice, they become natural allies in the construction of derivations. You need only recognize that when a conditional sentence is needed in a

derivation, you can often obtain the sentence by first introducing its antecedent as a supposition and then deriving its consequent. The strategy is applied in the derivation below to obtain a sentence that is then used to derive the conclusion.

1.	$+ P \supset (\neg Q \vee R)$	
2.	$+ Q$	
3.	$+ (P \supset R) \supset S$	
4.	$? S$	
5.	$\boxed{P}$	
6.	$\quad ? R$	
7.	$\quad \neg Q \vee R$	1, 5 *MP*
8.	$\quad R$	2, 7 $\vee E$
9.	$P \supset R$	5–8 *CP*
10.	S	3, 9 *MP*

Here, rule *CP* is used to derive not the conclusion of the sequence, S, but a sentence used in the course of deriving the conclusion, $P \supset R$.

Exercises 3.09

Construct derivations for the $\mathcal{L}_S$ sequences below using rules discussed thus far, including *CP*.

1. $+ P \supset Q$
   $+ P \supset (Q \supset R)$
   $+ Q \supset (R \supset S)$
   $? P \supset S$

2. $+ P \supset (S \supset (Q \wedge R))$
   $+ (Q \wedge R) \supset \neg P$
   $+ T \supset S$
   $? P \supset \neg T$

3. $+ P \supset Q$
   $+ R \supset S$
   $? (P \wedge R) \supset (Q \wedge S)$

4. $+ (P \wedge Q) \supset R$
   $+ P$
   $? Q \supset R$

5. $+ P \supset ((Q \vee R) \supset S)$
   $+ (S \vee T) \supset W$
   $? P \supset (Q \supset W)$

6. $+ P \supset (Q \vee R)$
   $+ P \supset \neg Q$
   $? P \supset R$

7. $+ (P \supset R) \supset (Q \supset S)$
   $+ Q$
   $? (P \supset R) \supset S$

8. $+ P \supset S$
   $+ R \supset S$
   $? P \supset (R \supset S)$

9. $+ (P \wedge R) \vee \neg S$
   $+ Q \vee T$
   $+ P \supset \neg Q$
   $? (S \supset T) \vee R$

10. $+ Q \supset (T \vee S)$
    $+ \neg R \wedge \neg T$
    $+ P$
    $? P \wedge (Q \supset S)$

11. $+ \neg P \vee (S \supset Q)$
$+ S \wedge T$
$? P \supset (Q \vee R)$

12. $+ (P \vee \neg T) \supset ((S \vee T) \supset Q)$
$+ \neg P \vee S$
$? (P \supset Q) \vee (S \supset T)$

13. $+ T \supset \neg P$
$+ T \vee (S \vee R)$
$+ \neg R$
$? P \supset S$

14. $+ P \supset (S \vee T)$
$+ (S \vee T) \supset (Q \supset (R \vee \neg S))$
$+ S$
$? P \supset (Q \supset R)$

15. $+ (Q \supset R) \supset \neg S$
$+ T \supset (\neg(Q \supset R) \supset P)$
$? S \supset (T \supset (P \vee Q))$

3.10 Indirect Proof: *IP*

A common strategy for establishing the truth of an assertion is to show that, were the assertion false, it would lead to an absurdity, in which case the assertion must be true. Reasoning of this sort—*reductio ad absurdum*, or plain *reductio*, or, as here, *indirect proof*—is widespread in ordinary life and in mathematics and the sciences.

Holmes argues that the suspect, Smith, could not have been in London at the time of the murder by showing that the supposition that Smith was in London leads to an impossibility: had Smith been in London at the time, she could not have witnessed the accident in Berkshire. A mathematical proof that there is no largest prime number begins with the supposition that there *is* a largest prime and shows that this supposition has paradoxical consequences.

In $\mathcal{L}_S$ the format deployed in indirect proofs resembles that used for *CP*:

$$(IP)\ \begin{array}{|l} \overline{\neg p} \\ \vdots \\ q \wedge \neg q \\ \hline p \end{array}$$

Suppose that you want to prove A. Using *IP*, you introduce A's negation, $\neg A$, as a supposition, and then endeavor to show that this supposition leads to a contradiction: any sentence, simple or complex, conjoined with its negation.

$$q \wedge \neg q$$

Once you derive a contradiction, you can *discharge* the supposition and enter the sentence you set out to prove. The sentences below are contradictory, hence, on this characterization, paradoxical.

$$A \wedge \neg A$$

$$(P \vee \neg Q) \wedge \neg(P \vee \neg Q)$$

A Greatest Prime?

A prime number is a number divisible only by itself and 1. Is there a greatest prime, a prime number greater than any other prime number?

The Greek mathematician Euclid (c. 300 BCE) offered a famous *reductio* argument, an indirect proof, that there is no greatest prime number.

1. Suppose there were a greatest prime, *x*.
2. Let *y* be the product of all primes less than or equal to *x* plus 1: $y = (2 \times 3 \times 5 \times 7 \times \ldots \times x) + 1$.
3. If *y is* prime, then *x* is not the greatest prime because *y* is greater than *x*.
4. If *y* is *not* prime, *y* has a prime divisor, *z*, and *z* is different from each of the primes 2, 3, 5, 7, . . ., *x* less than or equal to *x*. Thus, *z* must be a prime greater than *x*.
5. But *y* is either prime or not prime.
6. Therefore, *x* is not the greatest prime.
7. Therefore, there is no greatest prime.

See E. Nagel and J. R. Newman, *Gödel's Proof* (New York: New York University Press, 1967), 37–38.

As was the case in *CP*, a supposition in *IP* is accompanied by a subgoal. In the case of *CP*, this subgoal is a sentence making up the consequent of a conditional sentence you hope to derive. In deploying *IP*, the subgoal is the derivation of some sentence—*any* sentence—together with its negation The indefinite character of the subgoal is signaled by an ×. Consider an application of *IP*:

1.	$+ P \vee R$	
2.	$+ R \supset S$	
3.	$+ \neg S \vee \neg R$	
4.	$? P$	
5.	$\neg P$	
6.	$? \times$	
7.	R	1, 5 $\vee E$
8.	S	2, 7 *MP*
9.	$\neg S$	3, 7 $\vee E$
10.	$S \wedge \neg S$	8, 9 $\wedge I$
11.	P	5–10 *IP*

You can derive S and $\neg S$ as shown in lines 8 and 9, respectively. These sentences contradict one another. In line 10, the contradiction is made explicit and the suppositional premise discharged.

Bear in mind that the contradictory sentence could be *any* $\mathcal{L}_S$ sentence conjoined with its negation, including the sentence you might be hoping to derive—as in the example below.

1.	$+ P \supset (Q \vee R)$	
2.	$+ \neg P \supset R$	
3.	$+ \neg Q$	
4.	$? R$	
5.	$\neg R$	
6.	$? \times$	
7.	P	2, 5 *MT*
8.	$Q \vee R$	1, 7 *MP*
9.	R	3, 8 $\vee E$
10.	$R \wedge \neg R$	5, 9 $\wedge I$
11.	R	5–10 *IP*

IP and *CP* can be used together, one embedded inside the other, as the derivation below illustrates.

1.	$+ \neg Q \vee (P \supset (R \vee S))$	
2.	$+ \neg S \wedge \neg R$	
3.	$? P \supset \neg Q$	
4.	P	
5.	$? \neg Q$	
6.	Q	
7.	$? \times$	
8.	$P \supset (R \vee S)$	1, 6 $\vee E$
9.	$R \vee S$	4, 8 *MP*
10.	$\neg S$	2 $\wedge E$
11.	R	9, 10 $\vee E$
12.	$\neg R$	2 $\wedge E$
13.	$R \wedge \neg R$	11, 12 $\wedge I$
14.	$\neg Q$	6–13 *IP*
15.	$P \supset \neg Q$	4–13 *CP*

Here, the derivation of a conditional sentence is accomplished by (i) entering the antecedent of the conditional as a supposition, then (ii) showing that the negation of the conditional's consequent leads to a contradiction. In line 6, Q is taken to be the negation of $\neg Q$. You need not enter $\neg\neg Q$ as the supposition, and then appeal to a principle of double negation, converting $\neg\neg Q$ to Q. Negation signs in applications of *IP*, just as in every other rule, serve as indicators of valence. *IP* should be read

as permitting the supposition of a sentence, the valence of which is the reverse of the sentence you intend to derive.

Exercises 3.10

Construct derivations for the $\mathcal{L}_S$ sequences below using rules discussed thus far, including *IP*.

1. $+ P \supset Q$
   $+ R \supset P$
   $+ R \vee (Q \wedge S)$
   $? Q$

2. $+ P \vee Q$
   $+ P \supset (R \wedge S)$
   $+ (R \wedge S) \supset Q$
   $? Q$

3. $+ (P \vee Q) \supset R$
   $+ \neg R$
   $? \neg P$

4. $+ \neg R \supset \neg(\neg P \vee Q)$
   $+ \neg R$
   $? P$

5. $+ (P \vee Q) \supset (R \supset \neg S)$
   $+ (S \vee T) \supset (P \wedge R)$
   $? \neg S$

6. $+ P \supset Q$
   $+ S \supset T$
   $? (P \vee S) \supset \neg(\neg Q \wedge \neg T)$

7. $+ P \supset ((Q \wedge R) \vee S)$
   $+ (Q \wedge R) \supset \neg P$
   $+ S \supset (T \supset \neg P)$
   $? P \supset \neg T$

8. $+ (P \vee Q) \supset (R \supset S)$
   $+ \neg P \supset T$
   $+ R \wedge \neg S$
   $? T$

9. $+ P \supset Q$
   $+ (Q \vee R) \supset S$
   $+ \neg S$
   $? \neg(P \vee S)$

10. $+ P \supset (\neg Q \wedge R)$
    $+ S \vee \neg T$
    $+ P \vee T$
    $? Q \supset S$

11. $+ S \supset (Q \vee R)$
    $+ S \vee (P \supset S)$
    $? P \supset (Q \vee R)$

12. $+ \neg(S \wedge T) \supset (Q \supset R)$
    $+ P \supset \neg T$
    $? P \supset (Q \supset R)$

13. $+ P \vee Q$
    $+ (P \vee R) \supset (S \supset T)$
    $+ S \wedge (T \supset Q)$
    $? Q$

14. $+ P \vee S$
    $+ S \supset (R \supset T)$
    $+ R \wedge (T \supset P)$
    $? P \vee Q$

15. $+ P \supset S$
    $+ P \vee (S \wedge T)$
    $? S \vee \neg T$

3.11 Transformation Rules: *Com, Assoc, Taut*

The rules discussed thus far, rules of inference, facilitate the derivation of sentences from sentences in the construction of derivations. Are these rules sufficient to establish the validity of any valid sequence in $\mathcal{L}_S$?

Consider the sequence below:

1. $+ P \supset Q$
2. $+ (R \vee Q) \supset S$
3. $+ \neg S$
4. $? \neg(P \vee S)$

Suppose you set out to derive the conclusion using *IP*.

1.	$+ P \supset Q$		
2.	$+ (R \vee Q) \supset S$		
3.	$+ \neg S$		
4.	$? \neg(P \vee S)$		
5.	$\overline{P \vee S}$		
6.	$? \times$		
7.	P	3, 5 $\vee E$	
8.	Q	1, 7 *MP*	
9.	$Q \vee R$	8 $\vee I$	

You can see that S, together with $\neg S$ in line 3, is a contradiction. Might you use lines 2 and 9 to obtain S via *MP*, and thus the contradiction?

That would be a promising strategy were it not for the fact that *MP* cannot apply to lines 2 and 9. *MP* requires a conditional sentence together with the conditional sentence's antecedent. The sentence on line 9, $Q \vee R$, is not the antecedent of the conditional on line 2, $(R \vee Q) \supset S$. $Q \vee R$ and $R \vee Q$ are distinct $\mathcal{L}_S$ sentences.

In English, you can reverse the order of the disjuncts in a disjunctive sentence without affecting the sentence's truth conditions. The sentences below mean the same:

Socrates is wise or Socrates is happy.

Socrates is happy or Socrates is wise.

The same *commutative* principle holds for disjunctions in $\mathcal{L}_S$. You can reverse the order of the disjuncts in a disjunctive $\mathcal{L}_S$ sentence, thereby producing a new sentence with the same truth conditions as the original:

$$W \vee H$$

$$H \vee W$$

Conjunction in English and in $\mathcal{L}_S$ is commutative as well. In both English and $\mathcal{L}_S$, reversing the order of a conjunction's conjuncts has no effect on the truth conditions of the sentence as a whole.

Socrates is wise and Socrates is happy.

Socrates is happy and Socrates is wise.

The $\mathcal{L}_S$ counterparts of these sentences are also logically equivalent: they have the same truth conditions.

$$W \wedge H$$

$$H \wedge W$$

These remarks lie behind a transformation rule that permits the reversal of conjuncts and disjuncts, the *commutative rule*.

(*Com*) $\quad p \wedge q \dashv\vdash q \wedge p$

$\quad p \vee q \dashv\vdash q \vee p$

Com permits the reversal of conjunctive and disjunctive pairs, atomic or nonatomic. Applying *Com* to the following sentences:

$$\neg(A \wedge B) \vee (P \supset Q)$$

$$(A \vee (P \supset Q)) \wedge \neg(\neg A \wedge (Q \vee R))$$

yields, respectively,

$$(P \supset Q) \vee \neg(A \wedge B)$$

$$\neg(\neg A \wedge (Q \vee R)) \wedge (A \vee (P \supset Q))$$

The addition of *Com* makes it possible to complete the derivation begun earlier.

1.	$+ P \supset Q$	
2.	$+ (R \vee Q) \supset S$	
3.	$+ \neg S$	
4.	$? \neg(P \vee S)$	
5.	$\boxed{P \vee S}$	
6.	$\times$	
7.	P	3, 5 $\vee E$
8.	Q	1, 7 *MP*
9.	$Q \vee R$	8 $\vee I$
10.	$R \vee Q$	9 *Com*
11.	S	2, 10 *MP*
12.	$S \wedge \neg S$	3, 11 $\wedge I$
13.	$\neg(P \vee S)$	5–12 *IP*

Transformation rules, unlike rules of inference, are *bidirectional*. A sentence matching the form on the left-hand side can be transformed into one matching the form on the right, and vice versa. Bidirectionality is signaled by back-to-back turnstiles, ⊣⊢, which indicate that the expression on the left deductively yields the expression on the right, and the expression on the right deductively yields the expression on the left.

Transformation rules are distinguished from rules of inference in another way. A transformation rule can be applied to sentences that are parts of other sentences. An inference rule such as *MP* applies only to whole lines. *MP* could not be used to derive *R* in the sequence below:

1. $+ P \vee (Q \supset R)$
2. $+ Q$
3. $? R$
4. ⋮ ⋮

The restriction to whole sentences does not apply to transformation rules. Transformation rules license the replacement of sentences with logically equivalent sentences that have matching truth conditions. The truth-functional architecture of $\mathcal{L}_S$ means that substituting a logically equivalent sentence for one that is part of a larger sentence has no effect on the larger sentence's truth conditions. Thus, although the application of *Com* in the sequence below transforms only a part of a sentence, it is perfectly acceptable.

1. $P \supset (Q \vee R)$
2. $P \supset (R \vee Q)$ 1 *Com*

The transformation goes through even if the disjunctive antecedent to which *Com* was applied had been negated:

1. $P \supset \neg(Q \vee R)$
2. $P \supset \neg(R \vee Q)$ 1 *Com*

The rule is applied to the sentence inside the parentheses. Because $\mathcal{L}_S$ is truth-functional, if $Q \vee R$ and $R \vee Q$ have identical truth conditions, $P \supset \neg(Q \vee R)$ and $P \supset \neg(R \vee Q)$ have the same truth conditions.

Transformation Rules

In practice, transformation rules facilitate the manipulation of $\mathcal{L}_S$ sentences so as to achieve a pattern suitable for the application of one or more inference rules. Although the logical relationships exemplified by transformation rules are commonplace, the rules themselves might not seem to resemble natural patterns of reasoning.

Think of transformation rules as pattern-synthesizing devices that enable you to bend $\mathcal{L}_S$ sentences to your will so as to produce patterns amenable to manipulation by rules of inference.

A second transformation rule, the *associative rule* (*Assoc*), permits you to slide parentheses back and forth across sequences of ∧s and ∨s.

(*Assoc*) $p \wedge (q \wedge r) \dashv\vdash (p \wedge q) \wedge r$

$p \vee (q \vee r) \dashv\vdash (p \vee q) \vee r$

Using *Assoc*, you could, for instance, transform the sentence

$$\neg A \vee (B \vee \neg C)$$

into the sentence

$$(\neg A \vee B) \vee \neg C$$

As in the case of transformation rules generally, *Assoc* can be applied to sentences that are themselves parts of sentences. Consider an application of *Assoc* in the sequence below:

1.	$(M \wedge (A \wedge \neg C)) \wedge (\neg P \supset Q)$	
2.	$((M \wedge A) \wedge \neg C) \wedge (\neg P \supset Q)$	1 *Assoc*

The rule could have been applied differently in the case of the second sentence. Thus $A \wedge \neg C$ could have been treated as an element the q element in the formulation of *Assoc* and obtained

3.	$M \wedge ((A \wedge \neg C) \wedge (\neg P \supset Q))$	1 *Assoc*

How you apply a rule on an occasion depends on the sentence you want to derive, which in turn depends on the derivation.

Although transformation rules, unlike rules of inference, are applicable to parts of sentences, they share with inference rules a general restriction: no line can result from the application of more than one rule. Both walking and doing derivations require you to proceed one step at a time. The application of more than one rule, or the application of a single rule more than once, is not permitted in the derivation of a line. This means that a sequence like that below requires two steps, two applications of *Com*:

1.	$+(P \vee Q) \wedge R$	
2.	$? R \wedge (Q \vee P)$	
3.	$R \wedge (P \vee Q)$	1 *Com*
4.	$R \wedge (Q \vee P)$	3 *Com*

Com does not permit the derivation of line 4 from line 1 in a single step. Every line in a derivation results from a single application of a single rule.

A third transformation rule bears mention in this context. The rule, the *principle of tautology* (*Taut*), is formulated below.

(*Taut*) $p \dashv\vdash p \wedge p$

$p \dashv\vdash p \vee p$

Taut expresses the principle that a sentence is logically equivalent to a conjunction or disjunction of itself.

The label 'principle of tautology' is potentially misleading. Appropriately formulated, *every* derivation rule in $\mathcal{L}_S$ expresses a tautology. The use of the label in this case could be defended by noting that the relation depicted in *Taut* is *patently* tautological, whereas those encompassed by other rules are less obvious. If you utter the same sentence twice, then, pretty clearly, you have said no more than you would have said had you uttered it just once.

Although the principle of tautology has little application in English, it can figure centrally in $\mathcal{L}_S$. As with any derivation rule, it applies to any sentence of the appropriate form, simple or complex, as the sequence below illustrates:

1.	$+ A \vee A$	
2.	$+ \neg(P \supset Q)$	
3.	$+ \neg(A \wedge \neg B) \vee \neg(A \wedge \neg B)$	
4.	A	1 *Taut*
5.	$\neg(P \supset Q) \wedge \neg(P \supset Q)$	2 *Taut*
6.	$\neg(A \wedge \neg B)$	3 *Taut*

Taut can be applied from left to right, as it is in line 5 above, or right to left, as in lines 4 and 6.

Derivation Heuristics

Single-premise derivations:

> In derivations consisting of a single premise, observe the difference between this premise and the desired conclusion, and apply rules so as to diminish this difference gradually.

Start with large differences, and move in the direction of smaller differences, comparing the current line with the conclusion until you achieve a match.

In such cases, 'working backward' from the conclusion to the premise can be especially helpful.

Exercises 3.11

Construct derivations for the $\mathcal{L}_S$ sequences below using rules discussed thus far, including *Com*, *Assoc*, and *Taut*. Remember: only *one* application of a rule per line.

1. $+ P \vee (Q \wedge \neg R)$
   $? (\neg R \wedge Q) \vee P$

2. $+ (P \supset Q) \vee (R \vee S)$
   $? (S \vee R) \vee (P \supset Q)$

3. $+ (P \wedge P) \vee (Q \vee R)$
   $? R \vee (P \vee Q)$

4. $+ (P \vee Q) \wedge (R \wedge S)$
   $? (R \wedge (P \vee Q)) \wedge S$

5. $+ (P \vee (Q \supset R)) \vee (S \vee T)$
   $? ((P \vee S) \vee T) \vee (Q \supset R)$

6. $+ P \supset ((R \vee Q) \supset S)$
   $+ (T \vee S) \supset W$
   $? P \supset (Q \supset W)$

7. $+ \neg(R \wedge T) \supset \neg S$
   $+ (P \vee (Q \wedge Q)) \supset S$
   $? Q \supset R$

8. $+ (P \vee (Q \vee R)) \supset T$
   $+ (S \vee \neg T) \supset R$
   $? T$

9. $+ P \supset Q$
   $+ (R \vee S) \supset \neg Q$
   $? \neg(P \wedge S)$

10. $+ P \supset ((Q \wedge R) \vee S)$
    $+ (R \wedge Q) \supset \neg P$
    $+ T \supset \neg S$
    $? P \supset \neg T$

11. $+ (P \vee Q) \supset (R \wedge S)$
    $+ Q$
    $? T \vee R$

12. $+ P \supset R$
    $+ P \vee (R \wedge S)$
    $? Q \vee R$

13. $+ P \supset (S \supset T)$
    $+ S \wedge (Q \vee (\neg T \vee \neg P))$
    $+ Q \supset (S \supset R)$
    $? P \supset (Q \wedge R)$

14. $+ P \vee S$
    $+ S \supset (T \supset P)$
    $+ S \supset T$
    $? R \vee (Q \vee P)$

15. $+ S \supset Q$
    $+ Q \vee (S \wedge T)$
    $? P \vee (Q \wedge Q)$

3.12 Transformation Rules: *DeM*

The next transformation rule, *DeMorgan's Law* (*DeM*) (after the British logician Augustus DeMorgan [1808–1871]), lays bare an important logical feature of conjunctions and disjunctions. Consider the English conjunction

Socrates is wise and Socrates is happy.

If you think about it, you can see that this conjunction is equivalent to the disjunction

It's not the case that either Socrates isn't wise or Socrates isn't happy.

Similarly, a disjunction

Socrates is wise or Socrates is happy.

is equivalent to an appropriately negated conjunction

It's not the case that both Socrates isn't wise and Socrates isn't happy.

The negated sentences are stilted and clumsy, but a moment's reflection reveals that they mean the same as their more elegant counterparts. This sameness of meaning is expressed by DeMorgan's Law:

$$(DeM) \qquad p \wedge q \dashv\vdash \neg(\neg p \vee \neg q)$$
$$p \vee q \dashv\vdash \neg(\neg p \wedge \neg q)$$

In derivations, *DeM* enables you to move between ∨s and ∧s. Applying *DeM* to $\mathcal{L}_S$ versions of the English sentences above, you obtain the transformational pairs

1.	$+ W \wedge H$	
2.	$+ W \vee H$	
3.	$\neg(\neg W \vee \neg H)$	1 *DeM*
4.	$\neg(\neg W \wedge \neg H)$	2 *DeM*

Bearing in mind that the occurrence of a negation sign in a rule is to be interpreted as an instruction to *reverse the valence* of the negated expression, *DeM* permits the conversion of a conjunction into a disjunction, or a disjunction into a conjunction, provided you reverse the valence of (i) each conjunct (or disjunct) and (ii) the expression as a whole. Given the disjunction

$$A \vee \neg B$$

DeM licenses its transformation into the conjunction

$$\neg(\neg A \wedge B)$$

The valence of each disjunct is reversed, as is the valence of the whole expression. Applying *DeM* to

$$\neg(P \wedge \neg(Q \vee R))$$

yields the logically equivalent sentence

$$\neg P \vee (Q \vee R)$$

Here, the valence of the conjuncts is reversed. P becomes $\neg P$, and $\neg(Q \vee R)$ becomes $(Q \vee R)$. The valence of the entire expression, which is negative in the original sentence, is positive in its transformed counterpart. *DeM* could be applied a second time to the portion of the sentence within the parentheses to obtain

$$\neg P \vee \neg(\neg Q \wedge \neg R)$$

Remember, however, that a derivation of this sentence from the original requires *two* applications of *DeM*.

Derivation Heuristics

Negation signs:

> Do not be distracted by negation signs in looking for appropriate rules; *negation signs will take care of themselves*.

Focus on patterns of elementary sentences and connectives. In the course of a derivation, negation signs inevitably come and go. Chances are good that unless you make a careless mistake, the derivation of a sentence with the right connectives and the right atomic constituents in the right order will result in a sentence with the right distribution of negation signs.

Exercises 3.12

Construct derivations for the $\mathcal{L}_S$ sequences below using all of the rules discussed thus far, including *DeM*. Remember: only one application of a rule per line.

1. $+ P \vee (Q \wedge R)$
   $? \neg(\neg P \wedge (\neg Q \vee \neg R))$

2. $+ \neg(\neg P \vee (\neg Q \wedge \neg R))$
   $? P \wedge (Q \vee R)$

3. $+ (\neg P \vee \neg Q) \wedge (R \wedge S)$
   $? \neg((\neg R \vee \neg S) \vee (P \wedge Q))$

4. $+ P \supset \neg(Q \wedge (R \vee \neg S))$
   $? P \supset (\neg Q \vee (\neg R \wedge S))$

5. $+ \neg((\neg P \wedge \neg Q) \vee (\neg R \vee S))$
   $? (P \vee Q) \wedge (R \wedge \neg S)$

6. $+ Q \supset S$
   $+ S \supset P$
   $? P \vee \neg Q$

7. $+ P \supset (Q \supset \neg(R \vee S))$
   $+ Q$
   $? P \supset \neg S$

8. $+ P \supset (Q \vee S)$
   $+ \neg S$
   $? \neg P \vee Q$

9. $+ P \supset (Q \vee S)$
   $+ P \supset (R \vee \neg S)$
   $? P \supset (Q \vee R)$

10. $+ P \vee (Q \vee R)$
    $+ Q \supset (R \wedge S)$
    $? P \vee R$

11. $+ P \wedge (S \supset \neg Q)$
    $+ P \supset S$
    $? \neg(\neg P \vee Q)$

12. $+ S \vee (Q \supset R)$
    $+ P \vee (Q \wedge (T \vee \neg R))$
    $? S \vee (T \vee P)$

13. $+ R \supset \neg(\neg P \wedge Q)$
    $+ R \vee T$
    $+ Q \wedge (P \supset \neg S)$
    $? S \supset T$

14. $+ S \vee (P \supset (R \supset Q))$
    $+ S \vee P$
    $+ R$
    $? Q \vee S$

15. $+ S \supset (P \wedge \neg Q)$
    $+ S \vee (P \wedge R)$
    $? P$

3.13 Transformation Rules: *Dist*, *Exp*

Transformation rules invoke relations of logical equivalence, licensing the replacement of *any* sentence with another sentence provided the sentences have the same truth conditions. As noted earlier, this is possible because, unlike English, $\mathcal{L}_S$ is a truth-functional language, so the truth conditions of sentences remain unaffected when sentences they contain are replaced with logically equivalent sentences.

Although some transformation rules have obvious parallels in English (*Com* and *DeM* come to mind), others do not. The *distributive rule* (*Dist*), for instance, lacks graceful English applications.

$$(Dist) \qquad p \wedge (q \vee r) \dashv\vdash (p \wedge q) \vee (p \wedge r)$$
$$p \vee (q \wedge r) \dashv\vdash (p \vee q) \wedge (p \vee r)$$

What would be an example of a plausible transformation in English corresponding to *Dist*? Consider the English sentence

Socrates is wise, and he is either happy or brave.

The sentence is equivalent to

Socrates is either wise and happy or wise and brave.

The second sentence is awkward, but when you think carefully about it, you can see that it possesses the same truth conditions as the original sentence.

Dist licenses moves from conjunctions, in which one conjunct is itself a disjunction, to a disjunction of conjunctions—and vice versa; and from a disjunction in which one disjunct is itself a conjunction, to a conjunction of disjunctions—and vice versa. Applying *Dist* (right to left) to the sentence

$$(\neg A \vee C) \wedge (\neg A \vee \neg D)$$

yields

$$\neg A \vee (C \wedge \neg D)$$

Less obviously, perhaps, *Dist* applied (left to right) to the sentence

$$(\neg P \supset Q) \wedge (\neg(R \wedge S) \vee \neg T)$$

results in the sentence

$$((\neg P \supset Q) \wedge \neg(R \wedge S)) \vee ((\neg P \supset Q) \wedge \neg T)$$

This example dramatizes the need for practice in the recognition of patterns in $\mathcal{L}_S$. Once you have the knack, the application of the rule becomes routine. Until that happens, patterns might seem unobvious—or worse, there might seem to be no pattern.

Dist effects the expansion or contraction of sentences containing ∧s and ∨s. The *exportation rule* (*Exp*) allows for the manipulation of conditional sentences:

$$(Exp) \qquad (p \wedge q) \supset r \dashv\vdash p \supset (q \supset r)$$

Derivation Heuristics

Using *Dist* in derivations

> Consider applying *Dist* when you encounter complex sentences containing repeated elements.

Dist transforms sentences into longer sentences—or into shorter sentences, depending on the direction of application. The sentence

(1) $(F \wedge \neg G) \vee (\neg G \wedge (E \supset H))$

with an intermediate application of *Com*

(2) $(\neg G \wedge F) \vee (\neg G \wedge (E \supset H))$

yields (or, taken in the opposite direction, is yielded by) the sentence

(3) $\neg G \wedge (F \vee (E \supset H))$

Notice the presence of shared elements (the $\neg G$) in sentence (1), and compare the length of sentence (3) to sentences (1) and (2).

Exp permits the substitution of a conditional with a conjunctive antecedent for a conditional with a conditional consequent, and vice versa. This principle is at work in your recognition that the English sentences below have the same truth conditions.

If Socrates is wise and happy, then he is brave.

If Socrates is wise, then, if he is happy, he is brave.

Translated into $\mathcal{L}_S$:

$$(W \wedge H) \supset \mathrm{B}$$
$$W \supset (H \supset \mathrm{B})$$

More complex applications of *Exp* are illustrated in the sequence below:

1.	$+ (\neg A \wedge B) \supset (\mathrm{C} \supset \neg \mathrm{D})$	
2.	$+ (P \wedge \neg(Q \vee R)) \supset S$	
3.	$((\neg A \wedge B) \wedge \mathrm{C}) \supset \neg \mathrm{D}$	1 *Exp*
4.	$P \supset (\neg(Q \vee R) \supset S)$	2 *Exp*

The application of *Exp* in line 3 moves from right to left, while the application in line 4 moves left to right. If the patterns are not clear to you, try circling components of the complex sentences that correspond to elements in the formulation of the rule.

Exercises 3.13

Construct derivations for the $\mathcal{L}_S$ sequences below using rules discussed thus far, including *Dist* and *Exp*. Remember: *only one application of any rule per line.*

1. $+ \neg P \vee (\neg Q \wedge \neg R)$
   $? \neg((P \wedge R) \vee (P \wedge Q))$

2. $+ \neg(P \vee Q) \supset R$
   $? \neg P \supset (\neg Q \supset R)$

3. $+ (P \vee Q) \supset (R \supset S)$
   $? ((P \wedge R) \vee (Q \wedge R)) \supset S$

4. $+ P \vee (Q \wedge \neg R)$
   $? (P \vee Q) \wedge \neg(\neg P \wedge R)$

5. $+ \neg(\neg P \vee (\neg Q \wedge \neg R)) \supset S$
   $? P \supset ((Q \vee R) \supset S)$

6. $+ P \supset (Q \wedge R)$
   $+ R \supset (Q \supset S)$
   $? P \supset S$

7. $+ P \supset (S \wedge Q)$
   $+ S \supset R$
   $? (\neg P \vee Q) \wedge (\neg P \vee R)$

8. $+ (P \wedge R) \supset Q$
   $+ P \supset R$
   $? P \supset Q$

9. $+ (P \wedge Q) \supset R$
   $+ (Q \wedge R) \supset S$
   $? (P \wedge Q) \supset S$

10. $+ (S \vee T) \supset (\neg P \vee \neg R)$
    $+ S \vee (Q \wedge T)$
    $? P \supset \neg R$

11. $+ S \supset (P \vee (Q \wedge R))$
    $+ (P \vee R) \supset ((P \vee Q) \supset R)$
    $? S \supset R$

12. $+ S$
    $+ \neg R \supset T$
    $? (R \vee S) \wedge (R \vee T)$

13. $+ Q \supset (T \supset (S \supset P))$
    $+ (S \supset P) \supset R$
    $+ S \supset (Q \wedge T)$
    $? R$

14. $+ P \supset (Q \supset R)$
    $+ S \supset (P \wedge Q)$
    $+ (S \supset R) \supset T$
    $? P \supset T$

15. $+ ((P \wedge Q) \supset R) \supset S$
    $+ T \supset (P \supset (Q \supset R))$
    $? T \supset ((S \vee P) \wedge (S \vee R))$

3.14 Rules for Conditionals: *Contra*, *Cond*

Conditional sentences in both English and $\mathcal{L}_S$ can be *contraposed* without affecting their truth conditions. The contrapositive of a conditional sentence is a sentence in which the conditional's antecedent and consequent position exchange places, and the valence of each is reversed. The English conditional sentence below is paired with its contrapositive:

If Socrates is wise, then he is happy.

If Socrates is not happy, then he is not wise.

The same principle is at work in the rule for *Contraposition* (*Contra*):

$$(Contra) \qquad p \supset q \dashv\vdash \neg q \supset \neg p$$

Applying *Contra* to $\mathcal{L}_S$ versions of the sentences above you obtain:

$$W \supset H$$

$$\neg H \supset \neg W$$

The sequence below features applications of *Contra* to complex sentences:

1.	$+ \neg A \supset (B \wedge \neg C)$	
2.	$+ \neg(P \supset Q) \supset \neg(R \vee \neg S)$	
3.	$\neg(B \wedge \neg C) \supset A$	1 *Contra*
4.	$(R \vee \neg S) \supset (P \supset Q)$	2 *Contra*

Conditional equivalence (*Cond*) is another rule useful for transforming and manipulating conditional sentences.

$$(Cond) \qquad p \supset q \dashv\vdash \neg p \vee q$$

Cond licenses move between ⊃s and ∨s. The rule allows for the substitution of a ⊃ for a ∨, or vice versa, provided the valence of the sentence to the left of the connective is reversed.

Everyday applications of *Cond* are not ready to hand because, in English, 'if . . . then . . .' is so often used to express something more than simple conditionality. If the English sentences below do not appear equivalent, that is probably because you are giving 'if . . . then . . .' a sense different from the sense given to the ⊃.

If it's raining, then the street is wet.

Either it's not raining or the street is wet.

Differences between conditionals in English and in $\mathcal{L}_S$ can be overplayed (a point discussed in § 2.07). In this case, if you accept the original conditional sentence, you will, at least on your better days, accept the claim that either it is not raining or the street is wet. True, the disjunctive sentence leaves open the possibility that it's not raining and the street is wet anyway, but that possibility is left open by the original conditional sentence as well. Three applications of *Cond* are illustrated in the sequence below:

1.	$+ A \supset R$	
2.	$+ \neg A \supset (B \wedge \neg C)$	
3.	$+ R \vee \neg S$	
4.	$\neg A \vee R$	1 *Cond* (applied left to right)
5.	$A \vee (B \wedge \neg C)$	2 *Cond* (applied left to right)
6.	$\neg R \supset \neg S$	3 *Cond* (applied right to left)

The sequence provides examples of applications of *Cond* to both simple and complex sentences and examples of its use in each direction.

Derivation Heuristics

Conditional Proof (*CP*)

Proofs featuring conditional conclusions are obvious candidates for *CP*.

But

Derivations need not have a conditional conclusion for *CP* to be useful. *CP* can be used to derive conditional sentences that are transformable into other sentences.

Suppose, for instance, you are faced with deriving

$\neg S \vee T$

You might suppose S and derive T, yielding $S \supset T$, then use *Cond* to convert this conditional to $\neg S \vee T$.

Exercises 3.14

Construct derivations for the $\mathcal{L}_S$ sequences below using rules discussed thus far, including *Contra* and *Cond*. Remember: *only one application of any rule per line.*

1. $+ P \supset Q$
   $? Q \vee \neg P$

2. $+ \neg P \supset (Q \wedge R)$
   $? \neg(Q \wedge R) \supset P$

3. $+ (P \vee Q) \supset R$
   $? (\neg P \vee R) \wedge (\neg Q \vee R)$

4. $+ \neg P \supset (\neg Q \supset R)$
   $? (P \vee Q) \vee R$

5. $+ (P \vee Q) \supset (R \vee S)$
   $? (R \vee S) \vee (\neg P \wedge \neg Q)$

6. $+ (\neg S \vee R) \supset (T \supset P)$
   $+ S \supset R$
   $? \neg T \vee P$

7. $+ P \supset Q$
   $+ S \supset T$
   $? \neg(P \vee S) \vee (Q \vee T)$

8. $+ ((P \supset Q) \supset R) \supset S$
   $+ R$
   $? S$

9. $+ (\neg P \supset Q) \supset R$
   $+ S \supset (\neg Q \supset P)$
   $+ S \vee T$
   $? \neg T \supset R$

10. $+ \neg P \vee R$
    $+ (P \wedge \neg R) \vee S$
    $+ (R \wedge S) \supset Q$
    $? P \supset Q$

11. $+ (P \wedge \neg Q) \supset S$
    $+ (P \supset Q) \supset \neg T$
    $? S \vee \neg T$

12. $+ S \supset R$
    $+ R \supset \neg(T \supset Q)$
    $? S \supset T$

13. $+ P \vee Q$
    $+ Q \supset S$
    $+ P \supset S$
    $? S \wedge (\neg P \supset Q)$

14. $+ \neg P \supset Q$
    $+ S \supset \neg(P \vee Q)$
    $+ R \supset S$
    $? R \supset T$

15. $+ P \supset (Q \wedge R)$
    $+ (P \supset R) \supset S$
    $? S$

3.15 Biconditional Sentences: *Bicond*

The rules discussed thus far do not provide a means for coping with derivations featuring biconditionals. Consider the English sequence below:

The substance is acid if and only if it turns litmus paper red.

The substance turns litmus paper red.

The substance is acid.

The reasoning is clearly valid. Suppose the sequence is translated into $\mathcal{L}_S$.

1. $+ A \equiv R$
2. $+ R$
3. $? A$

How would you go about proving the sequence valid? The sticking point is the biconditional in the first sentence. Biconditionals were introduced in chapter 2 as equivalent to conjunctions of back-to-back conditional sentences. This logical equivalence is exploited in a *biconditional equivalence* rule (*Bicond*).

(*Bicond*) $\quad p \equiv q \dashv\vdash (p \supset q) \wedge (q \supset p)$

Applying the rule to the sequence above, the derivation can now be completed.

1.	$+ A \equiv R$	
2.	$+ R$	
3.	$? A$	
4.	$(A \supset R) \wedge (R \supset A)$	1 *Bicond*
5.	$R \supset A$	4 $\wedge E$
6.	A	2, 5 *MP*

By permitting the conversion of biconditional sentences to conjoined conditionals, *Bicond* allows for the elimination of biconditionals in derivations. This reflects our treatment of biconditionals in everyday reasoning. In most contexts, biconditionals (like the biconditional in the English sequence above) are heard as back-to-back conditionals and treated accordingly.

Exercises 3.15

Construct derivations for the $\mathcal{L}_S$ sequences below using rules discussed thus far, including *Bicond*.

1. $+ P \equiv Q$
   $? (P \supset Q) \wedge (\neg P \supset \neg Q)$

2. $+ P \equiv Q$
   $? (P \vee \neg Q) \wedge (\neg P \vee Q)$

3. $+ \neg((P \supset Q) \supset \neg(Q \supset P))$
   $? P \equiv Q$

4. $+ (P \supset Q) \wedge \neg(\neg P \wedge Q)$
   $? P \equiv Q$

5. $+ P \equiv Q$
   $? Q \equiv P$

6. $+ (P \vee Q) \supset (R \equiv \neg S)$
   $+ (S \vee T) \supset (P \wedge R)$
   $? \neg S$

7. $+ P \vee Q$
   $+ Q \equiv (R \wedge S)$
   $+ (R \vee P) \supset T$
   $? T$

8. $+ P \supset (Q \equiv R)$
   $+ \neg S \supset (P \vee R)$
   $+ P \equiv Q$
   $? S \vee R$

9. $+ P \equiv Q$
   $+ (P \supset R) \supset (P \wedge S)$
   $? \neg(Q \wedge S) \supset \neg(P \wedge R)$

10. $+ S \equiv T$
    $+ S \supset (P \vee Q)$
    $? \neg Q \supset (T \supset P)$

11. $+ P \equiv Q$
    $? (P \wedge Q) \vee (\neg P \wedge \neg Q)$

12. $+ P \supset (Q \equiv R)$
    $+ (\neg Q \vee R) \supset T$
    $? P \supset T$

13. $+ P \equiv (Q \vee R)$
    $+ \neg R \vee \neg S$
    $+ S \supset P$
    $? S \supset Q$

14. $+ P \equiv (\neg Q \vee R)$
    $+ (Q \supset R) \supset S$
    $+ S \supset \neg P$
    $? \neg P$

15. $+ P \vee (Q \wedge R)$
    $+ Q \equiv S$
    $? \neg S \supset P$

3.16 Constructive Dilemma: *CD*

Consider the English sequence below:

> The liquid in the beaker is either an acid or a base.
>
> If it's an acid, it turns litmus paper red.
>
> If it's a base, it turns litmus paper cobalt blue.
>
> ---
>
> The liquid in the beaker either turns litmus paper red or turns it blue.

The sequence is valid. You could express it in $\mathcal{L}_S$ as follows (letting A = the liquid is an acid, B = the liquid is a base, R = the liquid turns litmus paper red, and C = the liquid turns litmus paper blue).

1. $+ A \vee B$
2. $+ A \supset R$
3. $+ B \supset C$
4. $? R \vee C$

The sequence can be proved valid using *IP* and rules already introduced.

1.	$+ A \vee B$	
2.	$+ A \supset R$	
3.	$+ B \supset C$	
4.	$? R \vee C$	
5.	$\neg(R \vee C)$	
6.	$? \times$	
7.	$\neg R \wedge \neg C$	5 *DeM*
8.	$\neg R$	7 $\wedge E$
9.	$\neg A$	2, 8 *MT*
10.	$\neg C$	7 $\wedge E$
11.	B	1, 9 $\vee E$
12.	$\neg B$	3, 10 *MT*
13.	$B \wedge \neg B$	11, 13 $\wedge I$
14.	$R \vee C$	5–13 *IP*

The pattern of reasoning occurring in the sequence, together with variations on that pattern, is so common that it could be incorporated into a rule, *constructive dilemma* (*CD*), which permits an inference directly from the premises of the sequence to its conclusion.

(*CD*) $p \vee q, p \supset r, q \supset s \dashv\vdash r \vee s$

The rule expresses the idea that, given two conditionals and the disjunction of their antecedents, you can infer a disjunction of their consequents.

Applying the rule to the sequence above results in a simple, one-step derivation.

1.	$+ A \vee B$	
2.	$+ A \supset R$	
3.	$+ B \supset C$	
4.	$? R \vee C$	
5.	$R \vee C$	1, 2, 3 *CD*

CD is a rule of inference, not a transformation rule, so it cannot be applied to sentences within sentences, but only to whole sentences. Judicious application of rule *CD* can result in shorter, less complicated derivations. The derivation above is nine steps shorter than the previous derivation in which *CD* was not used.

The advantage afforded by rule *CD* is not merely that it enables some derivations to be shortened. It allows the construction of derivations that are easier to execute because they are closer to the ways we ordinarily reason. Consider the sequence below:

Either Elvis or Fenton sneezed.

If Elvis sneezed, Gertrude and Joe are mistaken.

If Fenton sneezed, Callie and Gertrude are mistaken.

Gertrude is mistaken.

The sequence is valid. If the premises are true, the conclusion must be true as well. How would you go about proving that the sequence is valid in $\mathcal{L}_S$?

The conclusion is not a disjunction, so it might not occur to you to try rule *CD*, but think of it this way. Suppose a sequence includes a disjunction, $p \vee q$. Suppose, further, that the sequence includes *or implies* conditional sentences, the antecedents of which consist of the respective disjuncts p and q, and the consequents of which are the same sentence, r. Differently put, suppose the sequence includes a disjunction, $p \vee q$, and includes as well $p \supset r$ and $q \supset r$, the two disjuncts, p and q, matching the antecedents of the two conditionals. In that case, using rule *CD*, the sentence $r \vee r$ can be derived, which is equivalent, by *Taut*, to r.

This strategy is applicable to the sequence above. (Let E = Elvis sneezed, F = Fenton sneezed, G = Gertrude is mistaken, J = Joe is mistaken, and C = Callie is mistaken.)

1.	$+ E \vee F$	
2.	$+ E \supset (G \wedge J)$	
3.	$+ F \supset (C \wedge G)$	
4.	$? G$	
5.	$\boxed{E}$	
6.	$? G$	
7.	$G \wedge J$	2, 5 *MP*
8.	G	7 $\wedge E$
9.	$E \supset G$	5–8 *CP*
10.	$\boxed{F}$	
11.	$?G$	
12.	$C \wedge G$	3, 10 *MP*
13.	G	12 $\wedge E$
14.	$F \supset G$	10–13 *CP*
15.	$G \vee G$	1, 9, 14 *CD*
16.	G	15 *Taut*

The use of *CP* together with *CD* can save time and agony in derivations of sequences that might otherwise resist solution.

3.17 Acquiring a Feel for Derivations

Done! You have now encountered all the rules needed for the construction of derivations in $\mathcal{L}_S$. This is an excellent place to pause and reflect on some of the more general lessons that have emerged in this chapter.

Derivation rules belong to the metalanguage, not to $\mathcal{L}_S$, and include both rules of inference and transformation rules. Transformation rules govern the replacement of sentences with logically equivalent counterparts. Inference rules differ from transformation rules by virtue of applying only to whole lines of derivations.

Negation signs appearing in rule formulations indicate not negation signs affixed to sentences but the sentences' relative valences. Transformations and inferences permitted by the derivation rules sometimes require the reversal of valences. Conditional equivalence (*Cond*), for instance, permits the replacement of a conditional sentence, $A \supset B$, with a disjunction, $\neg A \vee B$, provided the valence of the conditional's antecedent is reversed. Nonnegated antecedents, antecedents with a positive valence, take on negation signs, and negated antecedents lose them. This is old hat. I mention it now in order to make clear the point of excluding from our list of rules an explicit rule for double negation.

Transformation rules can also be useful in establishing subgoals. When faced with a derivation in which the route to the conclusion is not obvious, it is often helpful to think of ways in which the transformation rules could be used to transform the conclusion. Play with the conclusion separately, apart from the derivation by applying transformation rules to it. It is entirely possible that one of these transformed sentences would be easier to derive. When that is so, you need only derive the transformed sentence, then reverse the process to arrive at the conclusion.

This strategy is a version of the more general strategy of working backward from the conclusion. You can look over the conclusion and, in your head, work backward to the premises. In a difficult derivation, this might not take you far, but the aim is, as always, to narrow the gap between where you are and where you want to be.

As you familiarize yourself with the rules and their applications in derivations, familiar patterns will begin to stand out. After practice, the solution of a derivation that once seemed formidable will, often enough, be obvious at a glance—even though the derivation that springs from a flash of insight might turn out to have many steps.

At first, insights might be hard to come by, even for relatively simple derivations. When that is so, you might find it useful to apply rules—*MP*, *MT*, $\wedge E$, $\vee E$, for instance—that result in simplifying or breaking down more complex sentences. Simpler sentences are often easier to manipulate.

I have harped on the importance of practice, but the role played by practice cannot be overemphasized. The skills required for the construction of derivations are largely perceptual. They can be honed only through repetition. You might find it helpful, for instance, to take a derivation you have already constructed and *reconstruct* it on a fresh sheet of paper.

Derivation Heuristics

Indirect Proof (*IP*):

Sometimes *IP* affords simpler derivations of conditional sentences than *CP*.

Suppose you want to derive

$A \supset B$

but you get nowhere using *CP*. You might try introducing $\neg(A \supset B)$ as a supposition, then deriving a contradiction.

Note that $\neg(A \supset B)$ can be transformed into $\neg(\neg A \vee B)$ via *Cond*. Negated disjunctions are transformable using *DeM* into useful conjunctions, in this case, $A \wedge \neg B$. Now each conjunct can be 'brought down' by $\wedge E$ and used separately in the derivation.

Breaking sentences down in this way is sometimes helpful when you are stuck in the midst of a derivation.

When in doubt, break complex sentences into simpler sentences.

Look for conjunctions that can be separated into elements, and for applications of *MP*, *MT*, $\wedge E$, and $\vee E$ rules that facilitate the breaking down of complex sentences.

This is akin to playing scales on the piano: it might feel mindlessly repetitious, but it is part of what it takes to acquire the requisite skill. Eventually, the light dawns, and arrays of symbols that once appeared shapeless form themselves into units marching inevitably toward a conclusion.

Exercises 3.17

Prove the validity of each sequence below, incorporating an application of *CD*.

1. $+ \neg(P \wedge Q)$
   $+ S \supset P$
   $+ S \supset Q$
   $? \neg S$

2. $+ S \supset P$
   $+ Q \supset P$
   $+ \neg Q \supset S$
   $? P$

3. $+ P \supset (R \vee S)$
   $+ Q \supset (R \vee S)$
   $? (P \vee Q) \supset (R \vee S)$

4. $+ P \supset (R \wedge T)$
   $+ Q \supset (S \wedge T)$
   $? (P \vee Q) \supset (R \vee S)$

5. $+ \neg P \supset S$
   $+ \neg R \supset (S \vee T)$
   $+ \neg(P \wedge R)$
   $? S \vee T$

6. $+ P \vee R$
   $+ P \supset (Q \wedge \neg S)$
   $+ (\neg R \vee T) \wedge \neg S$
   $? Q \vee T$

7. $+ \neg P \vee Q$
   $+ R \supset S$
   $? (P \vee R) \supset (Q \vee S)$

8. $+ \neg P$
   $+ \neg Q$
   $? (P \vee Q) \supset (R \vee S)$

9. $+ P \supset (S \vee T)$
   $+ Q \supset (S \vee T)$
   $+ \neg T$
   $? (P \vee Q) \supset S$

10. $+ P \supset (Q \vee R)$
    $+ S \supset (R \vee T)$
    $+ \neg R$
    $? (P \vee S) \supset (Q \vee T)$

11. $+ S \vee \neg T$
    $+ P \vee \neg Q$
    $? (Q \vee T) \supset (S \vee P)$

12. $+ P \vee Q$
    $+ R \vee S$
    $? \neg(Q \wedge S) \supset (P \vee R)$

13. $+ P \vee Q$
    $+ P \supset (R \vee S)$
    $+ \neg Q \vee (S \vee R)$
    $? R \vee S$

14. $+ S \supset T$
    $+ R \supset (T \vee Q)$
    $+ (T \vee Q) \supset P$
    $? (S \vee R) \supset (T \vee P)$

15. $+ P \supset S$
    $+ Q \supset S$
    $? ((P \wedge R) \vee (Q \wedge R)) \supset S$

Because each of these sequences could be proved valid without the use of *CD*, you can use them to practice applications of the other rules by proving them *without* resorting to *CD*.

3.18 Proving Invalidity

Instances of reasoning encountered in everyday life are often flawed in one way or another. Sometimes unwarranted assumptions contaminate an argument so that, even when its conclusion is validly supported, you have little reason to accept that conclusion. Sometimes arguments are out-and-out invalid. The premises of an argument could be plausible, and the conclusion might be something you would like to believe, yet the conclusion fails to follow from the premises.

Derivations establish the validity of sequences. What of invalid sequences? If you fail to prove a sequence valid, you do not thereby prove it invalid. Maybe you have simply overlooked something. Suppose you are stymied by a particular derivation. You suspect that the sequence is invalid. How might you prove that it is?

One way to establish that a sequence is invalid is by means of a truth table (as in § 3.02). Bearing this in mind, you might construct a truth table and see whether, on any row of that truth table, the premises of the suspicious sequence are true and its conclusion is false. The strategy lacks appeal, however. Truth tables, although reliable, are unwieldy and potentially confusing when they require many rows. As it happens, an alternative is available!

Consider the sequence below:

1. $+ P \supset Q$
2. $+ R \supset S$
3. $+ R \vee Q$
4. $? P \vee S$

Suppose you have tried without success to construct a derivation to prove this sequence valid, and you suspect that it is invalid. You could construct a truth table to confirm your suspicions.

$P\,Q\,R\,S$	$P \supset Q$	$R \supset S$	$R \vee Q$	$P \vee S$
T T T T	T	T	T	T
T T T F	T	F	T	T
T T F T	T	T	T	T
T T F F	T	T	T	T
T F T T	F	T	T	T
T F T F	F	F	T	T
T F F T	F	T	F	T
T F F F	F	T	F	T
F T T T	T	T	T	T
F T T F	T	F	T	F
F T F T	T	T	T	T
F T F F	T	T	T	F
F F T T	T	T	T	T
F F T F	T	F	T	F
F F F T	T	T	F	T
F F F F	T	T	F	F

The shaded row of the truth table reveals that it is possible for the sequence's premises to be true and its conclusion false.

Earlier I hinted at a streamlined technique for achieving the same goal. The technique enables you to construct what amounts to a single row of a truth table that establishes the invalidity of an invalid sequence.

A sequence is invalid if there is an interpretation—an assignment of truth values to the atomic constituents—that makes the premises true and the conclusion false. Given a sequence $\langle \Gamma, \phi \rangle$, the question is whether it is possible to assign values to the atomic constituents of Γ and ϕ that result in the sentences in Γ's being true and ϕ's being false. The discovery of such an assignment would constitute a proof that the sequence is invalid. Best of all, the proof would have been accomplished without the construction of a truth table.

How would this work? Start with an assignment of truth values to the conclusion that makes it false. If this assignment can be consistently extended to the premises so as to make them true, you will have established that the sequence is invalid. The procedure is illustrated below for the sequence above. First, set out the sequence horizontally, with premises to the left and the conclusion to the right, separated from the premises by a vertical line.

$$P \supset Q \quad R \supset S \quad R \vee Q \mid P \vee S$$

Next, assign values to the atomic sentences in the conclusion so as to make the conclusion false. In this case the conclusion consists of a disjunction, so there is only one way to do this: by assigning the value false to both constituents, P and S.

$$\begin{array}{llll} P \supset Q & R \supset S & R \vee Q & \mid P \vee S \\ & & & \begin{array}{cc} \mathrm{F} & \mathrm{F} \\ \cup & \\ \mathrm{F} & \end{array} \end{array}$$

Assignments of truth values must be consistent, so these values must be carried over to the premises.

$$\begin{array}{llll} P \supset Q & R \supset S & R \vee Q & \mid P \vee S \\ \mathrm{F} & \mathrm{F} & \mathrm{F} \; \mathrm{F} & \\ & & \cup & \\ & & \mathrm{F} & \end{array}$$

Now, look for truth value assignments to the remaining premises that result in their being true. In this case, R must be false—otherwise the second premise would be false. If R is false, Q must be true if the third premise is to be true. These assignments result in the premises being true.

$$\begin{array}{llll} P \supset Q & R \supset S & R \vee Q & \mid P \vee S \\ \mathrm{F} \; \mathrm{T} & \mathrm{F} \; \mathrm{F} & \mathrm{F} \; \mathrm{T} & \mathrm{F} \; \mathrm{F} \\ \cup & \cup & \cup & \cup \\ \mathrm{T} & \mathrm{T} & \mathrm{T} & \mathrm{F} \end{array}$$

In this way you can uncover a consistent assignment of truth values, an interpretation under which the premises of the sequence are true and its conclusion is false.

$$\mathrm{I}: \{P = \mathrm{F}, Q = \mathrm{T}, R = \mathrm{F}, S = \mathrm{F}\}$$

Compare this interpretation to the shaded row in the original truth table. There, the values of P, Q, R, and S match those set out above.

This technique can be extended to any $\mathcal{L}_S$ sequence. When you can find a consistent assignment of truth values that makes the conclusion of a sequence false and its premises true, you have a tidy proof of invalidity. Failure to find such an assignment, of course, does not establish the validity of a sequence. Just as you could fail to come up with a proof for a perfectly valid sequence, so you might overlook an interpretation for an invalid sequence that would establish its invalidity.

A complication remains. There is often more than one way to assign values to a conclusion to make it false and to premises to make them true. It might be necessary to test several interpretations before abandoning an attempt to prove invalidity and embarking on a proof for validity.

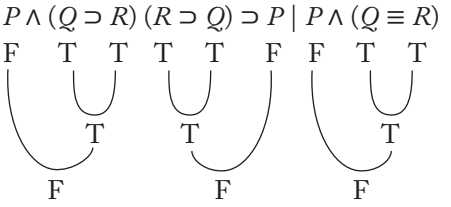
Consider the sequence below:

1. $+ P \wedge (Q \supset R)$
2. $+ (R \supset Q) \supset P$
3. $? P \wedge (Q \equiv R)$

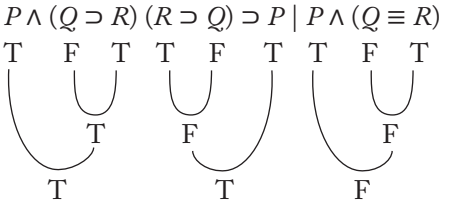
which could be represented horizontally:

$$P \wedge (Q \supset R)\ (R \supset Q) \supset P \mid P \wedge (Q \equiv R)$$

Several assignments of truth values to the conclusion of this sequence would render it false. Some of these assignments would, when carried over into the premises, result in false premises as well. You could, for instance, make the conclusion false by assigning the value false to *P*, and true to *both* *Q* and *R*.



That assignment would result in both premises being false, however. Were you to stop there and assume that because this assignment failed to establish invalidity, no assignment would do so, you would be in error. As the diagram below illustrates, other assignments must be considered.



Here is an assignment of truth values, an interpretation under which the conclusion of the sequence is false and its premises are true.

$$\text{I: } \{P = \text{T}, Q = \text{F}, R = \text{T}\}$$

Exercises 3.18

For each of the sequences below (i) construct a derivation proving its validity, or (ii) provide an interpretation demonstrating its invalidity and a chart showing the application of this interpretation. (Hint: you could waste a lot of time trying to derive the conclusion of an invalid sequence, so, in most cases, you would do well to try to prove invalidity first. If that proves impossible, you could move on to a proof of validity for the sequence.)

1. $+ \neg(P \wedge Q) \supset ((R \supset S) \supset T)$
   $+ R \wedge (P \vee Q)$
   $? S \supset T$

2. $+ P \equiv (Q \vee R)$
   $+ \neg Q$
   $? \neg P \supset R$

3. $+ P \equiv \neg Q$
   $+ (Q \wedge R) \supset (S \vee T)$
   $+ \neg(P \wedge R)$
   $? (R \supset \neg S) \supset T$

4. $+ (P \vee Q) \supset (R \equiv S)$
   $+ \neg(\neg S \wedge P)$
   $+ R \supset T$
   $? P \supset (T \wedge R)$

5. $+ P \supset (Q \supset R)$
   $+ R \supset S$
   $+ (T \wedge U) \supset P$
   $+ \neg(\neg U \wedge \neg Q)$
   $? T \supset (Q \wedge S)$

6. $+ \neg P \supset (Q \supset R)$
   $+ (P \vee S) \supset T$
   $+ R \supset (P \vee S)$
   $+ \neg T$
   $? \neg Q$

7. $+ \neg(P \wedge Q) \supset R$
   $+ R \supset S$
   $+ (P \wedge S) \supset Q$
   $? \neg R \vee Q$

8. $+ P \supset (Q \equiv R)$
   $+ \neg S \supset (P \vee R)$
   $+ P \equiv Q$
   $? S \wedge R$

9. $+ \neg(P \wedge Q)$
   $+ (P \supset R) \supset (P \wedge S)$
   $? (P \wedge R) \supset (Q \wedge S)$

10. $+ P \supset (Q \supset R)$
    $+ \neg R$
    $? P \supset \neg Q$

11. $+ P \supset (Q \vee R)$
    $+ S \supset (T \vee R)$
    $? (P \vee S) \supset ((T \vee Q) \vee R)$

12. $+ P \supset (Q \vee R)$
    $+ S \supset (T \vee R)$
    $? (P \vee S) \supset R$

13. $+ R \supset (P \vee Q)$
    $+ S \supset (P \vee Q)$
    $+ P \vee Q$
    $? R \wedge S$

14. $+ (P \vee Q) \supset R$
    $+ (P \vee Q) \supset S$
    $+ \neg S$
    $? \neg P$

15. $+ R \supset Q$
    $+ P \supset (R \vee S)$
    $? P \wedge Q$

3.19 Theorems

Theorems are expressions that follow exclusively from axioms of a formal system without additional assumptions. In proving theorems in Euclidean geometry, you derive geometrical truths from Euclidean axioms. Once a theorem is shown to follow from an axiom, you are free to use it as a shortcut in derivations of theorems because you could always trace the missing steps back to the axioms on which the system is based. A theorem such as the Pythagorean theorem

The square of the hypotenuse of a right triangle =
the sums of the squares of the two remaining sides

requires no assumptions other than those concerning points, lines, and angles spelled out in the Euclidean axioms.

Suppose, in contrast, you set out to determine the length of the hypotenuse of a right triangle formed by the intersection of three highways in Nebraska. In this case, you would require, in addition to the axioms of geometry, facts concerning the angle of each intersection and distances between intersections. The answer you obtain is not a theorem of geometry but a purportedly fact-stating sentence about a triangle formed by roads in Nebraska. Geometers prove theorems. Carpenters and land surveyors use geometry to establish actual areas and boundaries.

The axioms of a system determine the character of the system. Euclidean geometry is commonly taken to have five axioms from which every Euclidean theorem is derivable. The axioms are geometrical sentences that occupy a privileged position within the system: they are presumed not to require proof.

For centuries geometers assumed that the Euclidean axioms were definitive of the structure of space that gave them a natural basis. The geometers were wrong. Space, as we now believe, is not Euclidean. Even if space were Euclidian, however, that would have no bearing on the standing of the axioms of Euclidian geometry. Geometers no longer suppose that the axioms of a formal system need worldly support. If the axioms are properly formulated, they define a self-contained system that might or might not have an application to the universe.

Arithmetic, like geometry, constitutes a formal system. Familiar arithmetical truths such as

$$7 + 5 = 12$$

$$295 \times 172 = 50{,}740$$

express arithmetical theorems. These can be derived from the axioms of arithmetic.

What are the axioms of arithmetic? That question received various answers in the twentieth century. The most familiar axiomatizations of both arithmetic and Euclidean geometry are founded on five axioms.

There is nothing magical about the number five, however. A formal system might feature any number of axioms. Indeed, $\mathcal{L}_S$ represents a limiting case: $\mathcal{L}_S$ possesses no axioms at all. Although it is possible to devise an equivalent, axiom-based system, $\mathcal{L}_S$ is a *natural deduction system*. Because $\mathcal{L}_S$ is not founded on axioms, it resembles *natural reasoning*. When we reason in ordinary life and in the

sciences, we appeal to premises, but almost never to sentences that have the character of axioms. So it is with $\mathcal{L}_S$.

Despite lacking axioms, $\mathcal{L}_S$ does have theorems. How is that possible? Theorems are sentences belonging to a formal system that are derivable from the axioms of the system alone. How, then, could $\mathcal{L}_S$ possibly yield theorems? If a theorem follows from the axioms of a system and $\mathcal{L}_S$ has no axioms, then a theorem of $\mathcal{L}_S$ would be a sentence derivable from the empty set of sentences, derivable from thin air.

You might recall the discussion of contingent, contradictory, and logically true sentences in § 2.18. Logically true sentences are sentences that cannot fail to be true. A logically true sentence is true under every interpretation. Deriving a logically true sentence requires no outside assistance, no appeal to premises.

Magic? Consider the sentence below:

$$P \vee \neg P$$

Now, consider a derivation of this sentence using *IP*:

1.	$\neg(P \vee \neg P)$	
2.	? ×	
3.	$\neg P \wedge P$	1 *DeM*
4.	$P \vee \neg P$	3 *IP*

The derivation leads off with a suppositional premise: suppose that $P \vee \neg P$ is false. This leads to a contradiction, so $P \vee \neg P$ must be true. A turnstile, $\vdash$, with nothing to its left indicates that a particular sentence is a theorem. Thus

$$\vdash P \vee \neg P$$

marks off the sentence, $P \vee \neg P$, as a theorem of $\mathcal{L}_S$.

The $\vdash$ is used to express the derivability relation. Given a sequence, $\langle \Gamma, \phi \rangle$, if $\Gamma \vdash \phi$, then ϕ is derivable from Γ, that is, a derivation of ϕ from Γ is possible. If a sentence, p, is a theorem, if $\vdash p$, then p is derivable from the empty set of sentences. This means that the first step in the derivation of a theorem must be a supposition. As the sequence above illustrates, derivations of theorems revolve around *IP* or *CP*, or some *IP*/*CP* hybrid.

How might this work for a more complex theorem?

$$\vdash (P \supset Q) \equiv (\neg P \vee Q)$$

This theorem is a biconditional. You could derive it by treating it as a two-way conditional. The sentence $(P \supset Q) \supset (\neg P \vee Q)$ could be derived first, then $(\neg P \vee Q) \supset (P \supset Q)$. These are conjoined, and *Bicond* is applied to the conjunction to yield the biconditional.

1.	$P \supset Q$	
2.	? $\neg P \vee Q$	
3.	$\neg P \vee Q$	1 *Cond*
4.	$(P \supset Q) \supset (\neg P \vee Q)$	1-3 *CP*
5.	$\neg P \vee Q$	
6.	? $P \supset Q$	
7.	$P \supset Q$	5 *Cond*
8.	$(\neg P \vee Q) \supset (P \supset Q)$	5–7 *CP*
9.	$((P \supset Q) \supset (\neg P \vee Q)) \wedge ((\neg P \vee Q) \supset (P \supset Q))$	4, 8 $\wedge I$
10.	$(P \supset Q) \equiv (\neg P \vee Q)$	9 *Bicond*

This completes your introduction to the elements of $\mathcal{L}_S$. The next section turns to a discussion of proofs *about* $\mathcal{L}_S$. Before venturing into that discussion, you might find it useful to review §§ 2.19–2.21.

Exercises 3.19

Construct derivations for the sentences below proving that they are theorems of $\mathcal{L}_S$.

1. $\vdash \neg(P \wedge \neg P)$
2. $\vdash P \supset (\neg P \supset P)$
3. $\vdash (P \wedge \neg P) \supset Q$
4. $\vdash ((P \supset Q) \wedge \neg Q) \supset \neg P$
5. $\vdash ((P \supset Q) \wedge P) \supset Q$
6. $\vdash (P \vee Q) \equiv \neg(\neg P \wedge \neg Q)$
7. $\vdash P \supset P$
8. $\vdash (P \supset Q) \vee (Q \supset P)$
9. $\vdash P \supset (\neg Q \supset P)$
10. $\vdash \neg(P \supset Q) \equiv (P \wedge \neg Q)$
11. $\vdash (P \equiv Q) \equiv (\neg P \equiv \neg Q)$
12. $\vdash \neg P \supset (P \supset Q)$
13. $\vdash P \supset ((P \wedge Q) \vee (P \wedge \neg Q))$
14. $\vdash (P \supset Q) \supset (P \supset (Q \equiv P))$
15. $\vdash ((P \supset Q) \supset R) \supset ((P \supset Q) \supset (P \supset R))$

Derivation Heuristics

Derivations of theorems involving conditionals are sometimes simpler using *IP* rather than *CP*.

Consider the theorem below:

$\vdash (P \supset (Q \supset P))$

It might occur to you to derive the theorem using an embedded *CP*:

1.	$\boxed{P}$	
2.	$? Q \supset P$	
3.	$\quad\boxed{Q}$	
4.	$\quad ? P$	

The theorem can be proved in this way—for instance, by obtaining P via *IP*—but a simpler *IP* derivation is available:

1.	$\boxed{\neg(P \supset (Q \supset P))}$	
2.	$? \times$	
3.	$\neg(\neg P \vee (Q \supset P))$	1 *Cond*
4.	$\neg(\neg P \vee (\neg Q \vee P))$	3 *Cond*
5.	$P \wedge \neg(\neg Q \vee P)$	4 *DeM*
6.	$\neg(\neg Q \vee P)$	5 $\wedge E$
7.	$Q \wedge \neg P$	6 *DeM*
8.	$\neg P$	7 $\wedge E$
9.	P	5 $\wedge E$
10.	$P \wedge \neg P$	8, 9 $\wedge I$
11.	$P \supset (Q \supset P)$	1–10 *IP*

3.20 Soundness and Completeness of $\mathcal{L}_S$

In discussing derivations, I have moved back and forth between talk of derivability and talk of validity. An unstated assumption has been that whenever you construct a derivation, you have established the validity of the sequence; you have shown that its premises logically imply its conclusion. Now, ask yourself whether you have any reason, beyond your trust in the author, to think that this assumption is warranted. What assurance do you have that every derivable sequence in $\mathcal{L}_S$ really is a valid sequence?

This might seem an odd question, but think about it. When you *derive* a sequence, you apply rules that concern only configurations of uninterpreted strings of symbols. When you prove that a sequence is valid using a truth table, in contrast, you focus on *interpretations* of those strings of symbols, assignments of truth values to the symbols making up the strings. Why think these are connected?

This question concerns the *soundness* of $\mathcal{L}_S$. Soundness is defined as follows:

> $\mathcal{L}_S$ is *sound*, if and only if, for every set of sentences, Γ, and any sentence, ϕ, if Γ *deductively yields* ϕ ($\Gamma \vdash \phi$), then Γ *logically implies* ϕ ($\Gamma \vDash \phi$).

To say that Γ deductively yields ϕ ($\Gamma \vdash \phi$) is to say that there is a derivation from Γ to ϕ that uses the derivation rules included in $\mathcal{L}_S$. If Γ logically implies ϕ ($\Gamma \vDash \phi$), then there is no interpretation under which Γ is true and ϕ false. If $\mathcal{L}_S$ is sound, then every derivable sequence is a valid sequence.

What of the complementary question: is every valid sequence derivable? This question concerns the *completeness* of $\mathcal{L}_S$:

> $\mathcal{L}_S$ is *complete* if and only if, for every set of sentences, Γ, and any sentence, ϕ, if $\Gamma \vDash \phi$, then $\Gamma \vdash \phi$.

If $\mathcal{L}_S$ is complete, then whenever a sequence is valid, there is a derivation establishing its validity that relies only on the rules set out in this chapter. This is not to say that you could always find such a derivation, only that if you are unable to devise a derivation, the fault lies with you, not with $\mathcal{L}_S$.

You can achieve an intuitive understanding of soundness and completeness by considering the relation between the concepts of validity and logical implication on the one hand, and, on the other hand, the concept of derivability. A derivation in $\mathcal{L}_S$ is a finite ordered sequence of sentences, $\langle \Gamma, \phi \rangle$, in which one sentence, ϕ (the conclusion), is derivable from the other sentences, Γ (the premises), by means of one or more derivation rules. ϕ's derivability from Γ (alternatively, Γ's yielding ϕ) is represented by the turnstile: $\Gamma \vdash \phi$. Theorems are special cases of derivations. A theorem, τ (*tau*), is a sentence derivable from the empty set of sentences: $\vdash \tau$.

Derivability is a *syntactic* concept. Derivation rules permit you to add sentences to a sequence given syntactic features of other sentences present in the sequence regardless of their interpretation. When you set out to construct a derivation, you pay attention only to combinations of shapes. You are indifferent to the meanings of these shapes.

Validity, in contrast, is a *semantic* concept. Validity is defined by reference to *truth* and the derivative notion of an interpretation. A sequence is valid when its premises logically imply its conclusion. The premises of a sequence logically imply its conclusion when there is no interpretation under which the premises are true and the conclusion is false. Establishing the soundness and completeness of $\mathcal{L}_S$ amounts to establishing a correspondence between syntactic and semantic categories, between derivability and validity, between $\vdash$ and $\vDash$.

In the course of producing derivations, you might have noticed that some of the rules you were using provide convenient shortcuts. They could have been omitted, although omitting them would require adding steps to derivations.

Consider $\vee E$:

$$(\vee E) \qquad p \vee q, \neg p \vdash q$$

You could dispense with this rule and still come up with derivations for all derivable sequences. Suppose, for instance, you were confronted with the sequence below:

1. $+ A \vee B$
2. $+ \neg A$
3. $? B$

Here it would be natural to use $\vee E$, but in its absence, you could convert the disjunction to a conditional, and apply *MP*:

1.	$+ A \vee B$	
2.	$+ \neg A$	
3.	$? B$	
4.	$\neg A \supset B$	1 *Cond*
5.	B	2, 4 *MP*

Indispensable rules are distinguished from those that you could, at the cost of some inconvenience, do without. Think of an indispensable rule as a *primitive* rule. The remaining rules are *derived rules*. Which of the rules set out for $\mathcal{L}_S$ are primitive, and which are derived? In part, this is a matter of choice. The sequence above shows that $\vee E$ is dispensable: applications of *Cond* and *MP* could be substituted for applications of $\vee E$. The order could have been reversed. Applications of *Cond* and $\vee E$ could have been replaced applications of *MP*.

You could settle on a set of primitive rules by selecting rules none of which is replaceable by, that is, *derivable from*, other rules (or combinations of rules). The objective here is not to whittle down $\mathcal{L}_S$ but to simplify proofs for completeness and soundness. If you could prove that a version of $\mathcal{L}_S$ featuring only primitive rules is sound and complete, you would have shown that $\mathcal{L}_S$ is sound and complete.

You would have established that $\mathcal{L}_S$ is sound if you could show that the primitive rules are *truth-preserving*: their application to true sentences always yields true sentences. Derived rules constitute shortcuts that are themselves derivable via applications of primitive rules. If the primitive rules are truth-preserving, then $\mathcal{L}_S$ is

Primitive and Derived Rules

A derived rule is replaceable by applications of one or more primitive rules. You have some latitude in choosing which rules are to be primitive, and which derived, but how *much* latitude?

For $\mathcal{L}_S$ to be *complete*, its rules must be adequate to yield derivations of every valid sequence expressible in $\mathcal{L}_S$. To simplify proofs for *soundness*, the set of primitive rules should be the smallest possible. The target, then, is a set of rules that contains *all and only* rules required to yield derivations for every valid $\mathcal{L}_S$ sequence.

The following rules could be taken as primitive:

$\wedge I$, $\wedge E$, $\vee I$, $\vee E$, *MP*, *IP*, *Bicond*

The remaining rules, including all the transformation rules, are derivable from these, provided only that negation signs in the formulation of rules are read as indicators of the relative valences of sentences. (Without this proviso, a rule for double negation would need to be included among the primitives: $\neg\neg p \vdash p$.)

sound. $\mathcal{L}_S$ is complete, provided that every valid sequence is provably valid by means of a derivation that uses only primitive derivation rules.

I shall not set out proofs for the soundness and completeness of $\mathcal{L}_S$ here, but the concepts are so important that it is worth reflecting on what the proofs would involve. Note first that proofs for soundness and completeness are constructed in the metalanguage: they are not proofs *in* $\mathcal{L}_S$, they are proofs *about* $\mathcal{L}_S$. This is no accident.

What would a proof for soundness look like? Recall that $\mathcal{L}_S$ is sound if and only if every derivable sequence is valid. A sequence is valid if its premises logically imply its conclusion. If a set of sentences, Γ, logically implies a sentence, ϕ, then if every sentence in Γ is true, ϕ must be true. Suppose the premises of an arbitrary sequence are true. If you could show that

> if a sentence is true, then any sentence derived from that sentence by means of a primitive derivation rule is true

you would have shown that $\mathcal{L}_S$ is sound.

This would provide an *inductive proof* for the soundness of $\mathcal{L}_S$ (not to be confused with ordinary inductive reasoning). If a rule permits the derivation of only true sentences from true sentences, it is *truth-preserving*. Showing that $\mathcal{L}_S$ is sound is a matter of showing that each of the derivation rules is truth-preserving in this sense.

With two exceptions, *CP* and *IP*, you can use truth tables to establish that the primitive derivation rules of $\mathcal{L}_S$ are truth-preserving. Because of their form, *CP* and *IP* require special treatment. This complicates the proof but does not change it in any fundamental way.

Rule $\wedge I$ (p, $q \vdash p \wedge q$), for instance, could be shown to be truth-preserving via a truth table.

p q	$p \wedge q$
T T	T
T F	F
F T	F
F F	F

The truth table includes no row in which both p and q are true and $p \wedge q$ is false, so applications of $\wedge I$ are truth-preserving.

A proof that $\mathcal{L}_S$ is complete is more complicated, even in outline. The aim would be to establish that every valid sequence is derivable; that is, a derivation could be constructed for every sequence in which the premises logically imply the conclusion.

The first step would be to show that every logical truth expressible in $\mathcal{L}_S$ is derivable as a theorem. The second step would be to prove that every valid sequence is derivable. A simple two-step plan, but how would it work? Hold on to your hat!

Consider the truth table for an arbitrary sentence expressible in $\mathcal{L}_S$, $\neg P \wedge Q$.

Mathematical Induction

Suppose you want to show that every natural number has a particular property, Ψ (*psi*). (A natural number is one of the numbers 0, 1, 2, 3, 4, . . .)

You could do so if you could show that

(i) 0 has Ψ

(ii) if a number has Ψ, then its successor (the number following it in the series) has Ψ.

Thus: if 0 has Ψ, then the successor of 0, 1, has Ψ; if 1 has Ψ, then 2, its successor, has Ψ, and so on.

This technique, proof by *mathematical induction*, affords a way of establishing that every member of an unlimited or infinite series of objects has a particular property if the first member of the series has the property, and if any member of the series has it, the next member has it.

Mathematical induction should not be mistaken for what is commonly called *inductive reasoning*. Ordinary inductive reasoning is distinguished from *deductive reasoning*. When you reason deductively, the truth of your premises guarantees the truth of your conclusion. From 'All people are mortal' and 'Socrates is a person', you can infer 'Socrates is mortal'.

In contrast, when you reason inductively, your premises provide only *probabilistic* support for your conclusion. From '80 percent of logicians are right-handed' and 'Karen is a logician', you can infer that 'Karen is probably right-handed'. In so doing, you are reasoning inductively.

Mathematical induction is a species of deductive reasoning!

P Q	$\neg P$	$\neg P \wedge Q$
T T	F	F
T F	F	F
F T	T	T
F F	T	F

This sentence *or its negation*, $\neg(\neg P \wedge Q)$, is derivable from premises consisting of its atomic constituents, negated or not, depending on whether those constituents are false or true on a given row of the truth table.

$$P, Q \vdash \neg(\neg P \wedge Q)$$
$$P, \neg Q \vdash \neg(\neg P \wedge Q)$$
$$\neg P, Q \vdash (\neg P \wedge Q)$$
$$\neg P, \neg Q \vdash \neg(\neg P \wedge Q)$$

What is true for the sentence in this example turns out to be provably true for *every* sentence of $\mathcal{L}_S$: every $\mathcal{L}_S$ sentence, or its negation, is derivable from premises consisting of the sentence's atomic constituents, negated or not.

Now consider the logical truths, sentences true under every interpretation. A logically true sentence, $P \supset (Q \supset P)$, for instance, has the value *true* in every row of its truth table. It can be shown that if this is so, then $P \supset (Q \supset P)$ is derivable whatever the values of P and Q.

$$P, Q \vdash P \supset (Q \supset P)$$
$$P, \neg Q \vdash P \supset (Q \supset P)$$
$$\neg P, Q \vdash P \supset (Q \supset P)$$
$$\neg P, \neg Q \vdash P \supset (Q \supset P)$$

If $P \supset (Q \supset P)$ is so derivable, $P \supset (Q \supset P)$ is derivable as a theorem of $\mathcal{L}_S$. More generally, every logical truth expressible in $\mathcal{L}_S$ is derivable as a theorem: if τ is implied by any sequence, including the empty sequence ($\models \tau$), then τ is derivable from any sequence including the empty sequence ($\vdash \tau$).

The details are intricate, but, given this result, it is possible to prove that if $\Gamma \models \phi$, then $\Gamma \vdash \phi$, every valid sequence is derivable. The idea is straightforward. Every sequence has a 'corresponding conditional'. Consider the sequence below:

1. $+ \neg P \supset Q$
2. $+ \neg Q$
3. $? P$

The sequence is valid, so

$$\neg P \supset Q, \neg Q \models P$$

You can construct a corresponding conditional sentence from this sequence, replacing the $\models$ with a $\supset$, and conjoining the premises:

$$((\neg P \supset Q) \wedge \neg Q) \supset P$$

If the original sequence is valid, then its corresponding conditional must be logically true. Why? Well, a conditional is false only if its antecedent—here the conjoined premises of the original sequence, $((\neg P \supset Q) \wedge \neg Q)$—is true and its consequent—the conclusion of the original sequence, P—is false. If the sequence is valid, this would be impossible. If every valid sequence corresponds to a logically true conditional, then, and if every logically true sentence is derivable, the conditional corresponding to every valid sequence is derivable. And if that is so, the sequence is derivable.

These brief remarks on soundness and completeness barely scratch the surface. Perhaps they are enough, however, to give you a feel for what would be involved in a proof that $\mathcal{L}_S$ is both sound and complete. More detailed discussions of the proofs can be found in Geoffrey Hunter's *Metalogic: An Introduction to the Metatheory of Standard First Order Logic* (Berkeley: University of California Press, 1971); S. C. Kleene's *Mathematical Logic* (New York: John Wiley & Sons, 1967); and Paul Teller's *A Modern Formal Logic Primer: Predicate Logic and Metatheory* (Englewood Cliffs, NJ: Prentice-Hall, 1989).

4. The Language $\mathcal{L}_P$

4.00 Frege's Legacy

Despite its many charms, as formal languages go, $\mathcal{L}_S$ is a blunt instrument. $\mathcal{L}_S$ captures an important range of logical relations among sentences, its limitations quickly become evident when you consider a sequence mentioned in chapter 3 as an example of an obviously valid pattern of reasoning.

All people are mortal.

Socrates is a person.

Therefore, Socrates is mortal.

The sequence is plainly valid: if its premises are true, its conclusion must be true. When you represent the sequence in $\mathcal{L}_S$, however, its validity is masked (P = 'All people are mortal', S = 'Socrates is a person', and M = 'Socrates is mortal').

1. $+ P$
2. $+ S$
3. $? M$

To find an interpretation under which the conclusion, M, is false and the premises, P and S, are true, you need only assign the value *true* to P and S, and *false* to M. The example illustrates one respect in which $\mathcal{L}_S$ is logically thin. $\mathcal{L}_S$ cannot represent logical features *internal* to atomic sentences.

The class of valid derivations in $\mathcal{L}_S$ includes only those the validity of which is determined by the truth-functional structure of individual sentences. As the sequence above makes clear, not all logical relations are like this. $\mathcal{L}_S$ captures *inter*sentential logical relations among atomic sentences but is oblivious to *intra*sentential relations. This had been appreciated since the time of Aristotle, but not until the late nineteenth century did Gottlob Frege (1848–1925) devise a notational system capable of expressing the relevant logical relations.

Virtually the whole of modern logic rests on Frege's work. This chapter and the next concern a Fregean language, $\mathcal{L}_P$, that provides a framework for representing and exploiting a range of logical relations unrepresentable in $\mathcal{L}_S$. You might (or might not) be relieved to know that $\mathcal{L}_S$ will not be left behind but simply absorbed into $\mathcal{L}_P$.

4.01 Terms

Capturing the validity of the English sequence with which the chapter opened requires digging into the structure of the sentences that make it up. In $\mathcal{L}_S$, these are unstructured black boxes.

Consider another English sequence:

All philosophers are clever.

Some Newfoundlanders are philosophers.

Therefore, some Newfoundlanders are clever.

This sequence too appears valid: if the premises are true, the conclusion could not be false. Suppose nouns and adjectives in the sequence are replaced by letters. The sequence might then be represented schematically:

All *P* are *Q*.

Some *R* are *P*.

Therefore, some *R* are *Q*.

You could substitute any nouns or adjectives you please for *P*, *Q*, and *R*, and the resulting sequence is valid. For instance:

All students are conscientious.

Some southpaws are students.

Therefore, some southpaws are conscientious.

P, *Q*, and *R* are not sentences, however, but parts of sentences. The validity of the sequence depends not on the meanings of *P*, *Q*, and *R* but on formal relations among the sentences of which they are elements.

Consider a third sequence:

Some students are conscientious.

Some southpaws are students.

Therefore, some southpaws are conscientious.

This resembles the previous sequence with respect to the terms substituted for *P*, *Q*, and *R*.

Some *P* are *Q*.

Some *R* are *P*.

Therefore, some *R* are *Q*.

The second sequence differs from the first only in the substitution of 'some' for 'all' in its first sentence. Does this matter? The previous sequence was valid; this one is not. You can easily imagine circumstances under which the premises are true and the conclusion is false. Some students might be conscientious and some southpaws might be students without there being any conscientious southpaws. The truth of the premises is consistent with the falsehood of the conclusion, so the sequence is invalid.

Finally, consider the sequence

All students are conscientious.

Some southpaws are conscientious.

Therefore, some southpaws are students.

This sequence returns to the 'all'/'some' pattern of the original sequence, but the *P*, *Q*, and *R* elements are differently distributed.

All *P* are *Q*.

Some *R* are *Q*.

Therefore, some *R* are *P*.

What is the effect on the sequence's validity? You can again envisage circumstances under which students and some southpaws are conscientious even though no southpaws are students. The sequence is invalid.

The validity and invalidity of such sequences evidently depends on

(i) the arrangement of terms associated with *P*, *Q*, and *R*, and

(ii) the 'all'/'some' pattern.

Some arrangements of *P*s, *Q*s, *R*s, 'all', and 'some' result in valid sequences, and others do not.

Focus, first, on the *P*s, *Q*s, and *R*s. These express *general terms*, not sentences. General terms include common nouns (student, southpaw), adjectives (conscientious, wise), and intransitive verbs (leaps, sits). Sentences are true or false; general terms are *true of* (or fail to be true of) individual entities. 'Student' holds true of each student, 'conscientious' is true of whatever is conscientious, and 'leaps' is true of anything that leaps. This is not to say that a general term must be true of something. 'Mermaid' and 'phlogiston' are true of nothing.

Terms exhibit superficial grammatical differences. You would say 'Socrates is a philosopher', but 'Socrates is wise', not 'Socrates is *a* wise', and 'Socrates sits' but not 'Socrates philosophers'. There is less to these differences than meets the eye. In the interest of uniformity, 'Socrates is wise' and 'Socrates sits' could be paraphrased as 'Socrates is a wise thing' and 'Socrates is a sitting thing'.

In addition to common nouns, adjectives, and intransitive verbs, general terms include ordinary transitive verbs—'sees', 'admires', 'lifts'—and comparative constructions having the form of complex transitive verbs—'is wiser than', 'is taller than', 'is between'. These terms are true not of objects considered singly but of ordered pairs (or triples, or, more generally, of ordered *n*-tuples) of objects.

An ordered pair of objects is a collection of two objects taken in a particular order. 'Is taller than' is true of every ordered pair of individuals the first member of which is taller than its second. 'Is between' is true of every ordered triple of objects the first member of which is between the remaining two members.

A measure of order can be brought to all this by treating all general terms as verbs, some of which are commonly expressed as unaccompanied adjectives. 'Wise' and 'red' are adjectives, but you could regard them as components of the complex intransitive verbs 'is wise' and 'is red'. Returning to the list above, you can see that 'is conscientious', 'is a southpaw', 'is a philosopher', 'is a sitting thing' provide serviceable alternatives to the originals.

Not every general term has a single word as an English equivalent. 'Is a conscientious student' is, from the point of view of English, a complex general term true of conscientious students; 'is a black bowlegged swan' is a general term true of black bowlegged swans.

The set or class of objects of which a general term is true is the term's *extension*. The extension of the term 'is a conscientious student' is the set of conscientious students; the extension of the term 'is a mermaid' is the empty set. 'Is a mermaid', 'is a griffin', and 'is a square circle', despite differing in meaning, all have the same extension. These terms are, as far as we know, true of nothing at all.

General terms are distinguished from *singular terms*. A singular term purports to designate a unique individual. Singular terms include proper names such as 'Socrates', 'New South Wales', and 'Mt. Everest'. They include, as well, descriptions, at least those intended to designate unique individuals, *definite descriptions*, 'the teacher of Plato', for instance, 'the state in which Sydney is located', 'the tallest mountain'.

Combining a general term and a singular term yields a *predication*, a simple sentence in which a general term purports to be true of an object (or an ordered n-tuple of objects). The English sentences below express simple predications:

Socrates is a philosopher.

The teacher of Plato leaps.

Socrates climbs Mt. Everest.

New South Wales is rectangular.

Although both proper names ('Socrates', 'Mt. Everest') and definite descriptions ('the teacher of Plato', 'the state in which Sydney is located') function as singular terms, they differ in other ways. How they differ is a topic better approached after a consideration of terms in $\mathcal{L}_p$.

Exercises 4.01

Circle each singular term and draw a box around each general term in the sentences below. Indicate with a number (1, 2, 3, . . .) whether a general term is true of individual objects taken singly, ordered pairs of individuals, or ordered triples, etc.

1. Callie is tall.
2. Joe is taller than Iola.
3. Callie is taller than Joe.
4. If Callie is taller than Joe and Joe is taller than Iola, then Callie is taller than Iola.
5. Callie is taller than Joe or Iola.
6. Gertrude sits between Frank and Joe.
7. Callie admires Fenton.
8. Fenton admires himself.
9. Chet dislikes Fenton only if Fenton admires himself.

10. Iola is shorter than Callie or Joe, but taller than Fenton.
11. Gertrude gives *The Sleuth* to Frank and Joe.
12. Callie and Iola live in Bayport.
13. Fenton is a detective but Gertrude isn't.
14. If Gertrude is a detective, she admires Frank and Joe.
15. Frank and Joe are brothers.

4.02 Terms in $\mathcal{L}_P$

Formal languages, including $\mathcal{L}_S$ and $\mathcal{L}_P$, provide a perspective on linguistic structure that can illuminate natural languages, including English. Your coming to appreciate the role of terms in $\mathcal{L}_P$ will enable you to see more clearly their role in English, which is often disguised or hidden in thickets of grammatical complexity. This section and the next concern the elements of $\mathcal{L}_P$.

In $\mathcal{L}_P$ the uppercase letters, *A*–*Z*, serve as *predicates* or *predicate constants*. These function in $\mathcal{L}_P$ not as sentences, as they did in $\mathcal{L}_S$, but as general terms. You could represent the general term 'is wise' with the predicate *W*, for instance, the general term 'is a philosopher' with the predicate *P*, and so on.

Lowercase letters, *a*–*t*, the *individual constants*, function as proper names do in English. You might use the individual constant *s* as you would the name 'Socrates', the individual constant *n* as the name 'New South Wales', and the individual constant *m* as the name 'Mars'. The lowercase letters *u*–*z* are *individual variables*. Variables have a distinctive role that will be revealed in the next section.

The simplest sentences in $\mathcal{L}_P$ are *predications*: individual constants paired with predicates. Consider the English predication

Socrates is a philosopher.

Suppose *s* stands in for the name 'Socrates', and *P* expresses the general term 'is a philosopher'. With these assignments, you could represent the sentence in $\mathcal{L}_P$ as follows:

Ps

The general term 'is a philosopher' expresses a predicate true of, or not true of, individual objects. The term is true of the individual named by 'Socrates', and not true of the individual named by 'Mt. Everest'. Socrates, but not Mt. Everest, belongs to the extension of 'is a philosopher'. General terms are introduced paired with variables. Just as in $\mathcal{L}_S$ you might use *P* to mean 'Socrates is a philosopher', in $\mathcal{L}_P$ you might stipulate that *Px* means '*x* is a philosopher'. The variable, *x*, is a placeholder to be filled by a name. If *s* is 'Socrates' and *e* is 'Everest', then *Ps* expresses in $\mathcal{L}_P$ what the English sentence 'Socrates is a philosopher' expresses, and *Pe* expresses in $\mathcal{L}_P$ what is expressed by the English sentence 'Everest is a philosopher'.

You could think of general terms in English—'is wise', 'is a philosopher'—as incomplete symbols with slots that must be filled to yield a self-standing symbol. These slots are *argument places*, and the

items filling the slots, *arguments*. The terms 'is wise' and 'is a philosopher' each have a single argument place that correspond to one-place predicates in $\mathcal{L}_p$.

A *one-place predicate*, a predicate with a single argument place, is a *monadic predicate*, so *Px* is a monadic predicate. Here the variable *x* serves as a metalinguistic placeholder, a dummy symbol occupying the argument place in much the way the space in '_ is a philosopher' might. *Px* can be turned into a sentence by replacing the *x* with an individual constant, as in *Ps* and *Pe*.

Now consider the English sentence

Socrates climbs Mt. Everest.

This sentence includes two proper names, 'Socrates' and 'Mt. Everest', and a general term, 'climbs'. 'Climb' is a transitive verb expressing a general term true not of objects taken singly but of ordered pairs of objects. This is reflected in $\mathcal{L}_p$ by attaching to the predicate not one but two singular terms:

Cse

In this case the $\mathcal{L}_p$ predicate corresponding to 'climbs' is a *two-place predicate*, a predicate with two slots or argument places corresponding to the two spaces in '_climbs_'. A predicate of this sort is a *dyadic predicate*. Two-place predicates are introduced in a special way. You might say, for instance, that *C*①② represents '① climbs ②'. This would mean that, given the previous interpretations of *s* and *e*, if you place *s* in the ① slot and *e* in the ② slot, the result is a predication true just in case Socrates climbs Mt. Everest. Conversely, if you put *e* in the ① slot and replace ② with *s*, the result is a sentence true if and only if Everest climbs Socrates.

General terms, then, can be monadic ('is wise', 'is conscientious') and dyadic ('climbs', 'is taller than'). A monadic general term is true of objects; a dyadic term is true of ordered pairs of objects and represented in $\mathcal{L}_p$ by one- and two-place predicates, respectively. Some terms are true (or not) of ordered *triples* of objects. The English general term 'is between' might be represented in $\mathcal{L}_p$ by a *three-place* or *triadic predicate* *B*①②③ meaning '① is between ② and ③'. Given this interpretation, you could translate the English sentence

Clio is between Euterpe and Melpomene.

into $\mathcal{L}_p$ as

Bcem

Once an interpretation is specified for *B*①②③, and once it is settled that *c* = 'Clio', *e* = 'Euterpe', and *m* = 'Melpomene', you could not change the order of the individual constants without changing the truth conditions of the sentence. The sentence

Bmce

is true just in case the English sentence

Melpomene is between Clio and Euterpe.

is true.

Monadic, dyadic, and triadic predicates in $\mathcal{L}_p$ correspond to monadic, dyadic, and triadic general terms in English. How far could you go in this direction? Sentences must be finite. This limits the

Words and World

What do singular terms and general terms designate or 'correspond to' in the world?

One traditional answer is that singular terms correspond to particular things, and general terms correspond to properties of things and relations they bear to one another. The singular term 'Socrates' designates the particular individual, Socrates, the general term 'is wise' designates the property had by things that are wise, and 'is taller than' designates the relation one thing has to another when it is taller than the other.

The distinction between particulars, on the one hand, and properties and relations, on the other hand, has struck many philosophers as fundamental. Particulars are said to be unique, dated, individual things. Properties, in contrast, are taken to be repeatable. Properties, unlike particular entities, can be shared (or 'instantiated') by many distinct particulars. Properties, on such a conception, are *universals*.

Plato thought of universals as existing independently of their instances. Other philosophers have taken universals to be general entities capable of being wholly present in many distinct places and times. If the balls on a billiard table *share* the property of sphericality, they have *the same* property; sphericality is wholly present in each ball.

$\mathcal{L}_\mathcal{P}$ is neutral on such questions. What $\mathcal{L}_\mathcal{P}$ offers is a way of regimenting talk about the world so that it is possible to be clear about what sorts of entity we are committing ourselves to when we say what we do. (See § 4.15 below.)

number of argument places a predicate could have. This leaves open the possibility of predicates and corresponding general terms, with any finite number of argument places. With sufficient patience, you might concoct a predicate true of, and only of, ordered sets containing a dozen objects. The phrase 'n-place predicate' affords a way of designating a predicate while leaving open whether it is monadic, dyadic, triadic, or something more.

Armed with a supply of predicates and individual constants, you would be in a position to represent simple English sentences in $\mathcal{L}_\mathcal{P}$. If these are supplemented by our old friends the connectives ($\neg$, $\wedge$, $\vee$, $\supset$, $\equiv$), the scope of translation can be broadened to include truth functions of these simple sentences. Take the sentence

If Euterpe is wise, she is wiser than Clio.

Supposing Wx = 'x is wise', and $W①②$ = '① is wiser than ②', you could translate the sentence into $\mathcal{L}_\mathcal{P}$ as follows:

$$We \supset Wec$$

Consider the English sentence

If Euterpe is wiser than Clio and Melpomene, then she is wiser than Terpsichore.

Translation of this sentence into $\mathcal{L}_\mathcal{P}$ requires only that you construct appropriate predications and combine these with connectives just as you would in $\mathcal{L}_\mathcal{S}$:

$$(Wec \wedge Wem) \supset Wet$$

The English sentence

If neither Clio nor Melpomene is wiser than
Euterpe, then Terpsichore is not wiser than Euterpe.

could be translated into $\mathcal{L}_P$ as

$$\neg(Wce \vee Wme) \supset \neg Wte$$

The technique for translating English sentences into $\mathcal{L}_S$ extends smoothly to the representation of simple predications and truth functions of predications in $\mathcal{L}_P$. Matters become more complicated—and more interesting—when you move beyond predications.

Exercises 4.02

Construct $\mathcal{L}_P$ equivalents for the sentences in the previous exercise (4.01).

4.03 Quantifiers and Variables

Predications in $\mathcal{L}_P$ resemble atomic sentences in $\mathcal{L}_S$. Were $\mathcal{L}_P$ limited to predications and truth functions of predications, it would be logically on a par with $\mathcal{L}_S$. The power of $\mathcal{L}_P$ arises not from its capacity to express predications but from its capacity to represent predications in a way that expresses *generality* of the sort exhibited by the syllogism at the outset of this chapter.

All people are mortal.

Socrates is a person.

Therefore, Socrates is mortal.

Names are useful ways of designating individuals and properties that are important to us: Socrates, Mt. Everest, water, oxygen. Most individual things lack names. These can be identified *ostensively*—by exhibiting them, or pointing to them—or by means of *descriptions*. I might, on a whim, name the desk in my office, but I am more likely to identify it simply as *that desk over there* or *the desk in my office*.

This is unsurprising. Were we obliged always to refer to objects by name, communication would falter. Were I to decide to call my desk *Clyde*, for instance, you would be at a loss to understand my request to retrieve an object left on Clyde. You would have no way of knowing that I was referring to my desk unless you knew its name in advance. The world contains too many things for us to designate them all by names: a language that relied exclusively on names to designate objects would be dramatically limited in scope.

Singular terms in $\mathcal{L}_P$ include, in addition to names, *descriptions*, linguistic devices purporting to designate unique objects. Thus far the focus has been exclusively on predications incorporating proper names. Once you move beyond names, you can begin to appreciate the real power of $\mathcal{L}_P$.

Much of our talk about the world, and virtually all of our scientific talk, is framed in general terms. Consider, for instance, the English sentences

The man in the corner is a spy.

Whales are mammals.

All people are mortal.

Some planets have more than one moon.

Each of these sentences comments on the world. None does so by means of a name. $\mathcal{L}_\mathcal{P}$ is capable of expressing such sentences in a way that makes their logical features salient.

Why is generality important? You might attempt to express generality in a language by *listing* individuals. You might, for instance, try to paraphrase the English sentence

Whales are mammals.

by first giving a name to each whale, and then saying of each named individual that it is both a whale and a mammal. This strategy, however, besides being impossibly cumbersome, would neglect an essential feature of the original sentence. In referring to the class of whales, we take the class to be *open-ended*: producing an exhaustive list of its members would be not merely tedious but impossible. In allowing for reference to unnamed individuals and for open-endedness, generality differs essentially from lists, even exhaustive lists, of individuals.

Generality is introduced into $\mathcal{L}_\mathcal{P}$ by means of *quantifiers* and *variables*. The lowercase letters *a* through *t* are used in $\mathcal{L}_\mathcal{P}$ as names. The remaining lowercase letters, *u*, *v*, *w*, *x*, *y*, and *z*, function as

Names

In $\mathcal{L}_\mathcal{P}$, lowercase letters *a*–*t* function as proper names. Names are used to designate or refer to individual objects.

What of names used to designate more than one object? And what of names that designate nonexistent objects?

You can use 'Aristotle' to designate the Greek philosopher Aristotle, the teacher of Alexander. But many individuals have been called 'Aristotle'. What connects our use of this name on a particular occasion to just one of these?

There is nothing in the name itself that attaches it to one individual rather than another. Because the same name can be used by different people in different circumstances to designate different individuals, using a name informatively requires fixing its reference. Most often we rely on context. 'Aristotle' deployed in a philosophy lecture is fixed to a particular Athenian philosopher. If you named your dog Aristotle, my asking you 'How is Aristotle today?' fixes the name to a particular dog.

What of names that lack bearers, 'Harry Potter', for instance, or 'Camelot'? The existence of a name in a language does not guarantee the existence of an object corresponding to the name. A linguistic sign counts as a name when it is used to designate a unique object. If the object does not exist, the sign fails to perform its job, but through no fault of its own. When you use a sign as a name, you commit yourself to the existence of an individual corresponding to the name.

Maybe names of fictional individuals are not being used as names but serve a different function. What do you think?

variables ranging over individuals. The idea is familiar from mathematics. The numerals 1, 2, 3, 4 are names of numbers. When your interest lies in making a general claim about numbers, you deploy variables, *x*'s and *y*'s, $x^2 + y^2 = z^2$, for instance. In this case, the *x*'s and *y*'s are understood as standing for numbers, just not for any particular numbers.

Variables came into play in $\mathcal{L}_S$, but only in the metalanguage, only when it was necessary to refer to sentences generally. Variables were used setting out derivation rules, for instance:

$$(MP) \qquad p \supset q, p \vdash q$$

Variables have an analogous role in $\mathcal{L}_\mathcal{P}$. Suppose the English sentence 'Mars is spherical' is represented as *Sm*. Then the expression *Sx* could be interpreted as ascribing sphericality to an arbitrary individual, *x*.

In fact, as noted in § 4.02, the expression *Sx* is not a proper sentence of $\mathcal{L}_\mathcal{P}$. It resembles an English expression of the form

_ is spherical

This expression is not an English sentence, although it could easily be turned into a sentence in more than one way. You could replace '_' with a *name*:

Venus is spherical.

or you could replace the '_' with a *description*

The second planet from the Sun is spherical.

Another option would be to replace the '_' with the English counterpart of a *quantifier*:

Something is spherical.

Everything is spherical.

Translation of these sentences into $\mathcal{L}_\mathcal{P}$ requires the use of variables in the sentences themselves, and not simply in the metalanguage. When variables occur in sentences of $\mathcal{L}_\mathcal{P}$, they cannot stand on their own. Variables are always accompanied by *quantifiers*.

$\mathcal{L}_\mathcal{P}$ includes two quantifiers:

universal quantifier: $\forall\alpha$

existential quantifier: $\exists\alpha$

A quantifier consists of a symbol—an inverted *A* or a backward *E*—together with a variable. In the examples above, the Greek letter α (alpha) is used as a metalinguistic variable ranging over individual variables in $\mathcal{L}_\mathcal{P}$. (Greek letters are needed because $\mathcal{L}_\mathcal{P}$ itself exhausts the Roman alphabet.) In putting quantifiers to use in sentences, the α's above would be replaced by some individual variable, *u*, *v*, *w*, *x*, *y*, *z*, as in $\forall x$, $\forall y$, $\exists x$, $\exists y$. . .

Universal quantifiers express what would be expressed in English by the phrases

all α

every α

any α

Existential quantifiers can be read as

some α

at least one α

Quantifiers in $\mathcal{L}_p$, in common with their natural-language cousins, never occur in isolation. A quantifier is always attached to some other expression.

The simplest quantified sentences are made up of a single quantifier and a single monadic predicate. Using Sx to mean 'x is spherical'

$$\forall x Sx$$

can be read

For all x, x is spherical (that is, everything is spherical).

An existentially quantified counterpart

$$\exists x Sx$$

could be read as

There is at least one x, such that x is spherical (something is spherical).

The metalanguage variable, α, has been replaced by an authentic $\mathcal{L}_p$ variable, x, in each quantifier occurrence above. An x has been appended to the predicate S so that the variable occurring in the quantifier *matches* the variable occurring in the argument place of the predicate. This ensures that the quantifier 'picks up' the variable.

Think of variables occurring with predicates as pronouns, grammatical elements the significance of which is determined by relations they bear to other elements in the sentence. In the sentence 'Socrates forgot his hat when he left the Agora', occurrences of the pronouns 'his' and 'he' refer to Socrates. If you replace 'Socrates' in the sentence with 'Glaucon', 'his' and 'he' would refer to Glaucon.

In $\mathcal{L}_p$, the significance of a variable/pronoun is fixed by the quantifier with which it is associated and by the predicate (or predicates) to which it is affixed. This aspect of variables will become clearer once you see how it works in practice.

Differences in the logical structure of universally and existentially quantified expressions surface when you consider more complex sentences. One difference can be illustrated by means of examples. Consider the English sentence

Quasi- $\mathcal{L}_p$

In moving between English and $\mathcal{L}_p$, it is frequently useful to employ 'quasi-$\mathcal{L}_p$', a mixture of English and $\mathcal{L}_p$. Quantifiers in quasi-$\mathcal{L}_p$ can be read as

$\forall x$: For all x . . .

$\exists x$: There is an x . . .

Complex expressions and sentences can be read in quasi-$\mathcal{L}_p$ as well. The standard forms of quantified sentences, for instance, can be read in quasi-$\mathcal{L}_p$ as follows:

$\forall x(Px \supset Qx)$: For all x, if x is P, then x is Q.

$\exists x(Px \wedge Qx)$: There is an x such that x is P and x is Q.

All plants are green.

Letting Px = 'x is a plant' and Gx = 'x is green', the sentence could be translated into $\mathcal{L}_\mathcal{P}$ as

$$\forall x(Px \supset Gx)$$

In quasi-$\mathcal{L}_\mathcal{P}$

For all x, if x is a plant, x is green.

Notice the occurrence of the variable x both in the quantifier and in the argument places of the predicate letters. The quantifier ties together these occurrences of x in a way that is brought out nicely by the quasi-$\mathcal{L}_\mathcal{P}$ gloss accompanying the sentence.

Compare this universally quantified sentence with an existentially quantified counterpart:

Some plants are green.

The sentence can be translated into $\mathcal{L}_\mathcal{P}$ as

$$\exists x(Px \wedge Gx)$$

In quasi-$\mathcal{L}_\mathcal{P}$

There is at least one x such that x is a plant and x is green.

Think of these sentences as *paradigms*, typifying examples of universally and existentially quantified sentences. For the most part, universally quantified sentences are expressed as conditionals, and existentially quantified sentences are conjunctions. In this regard, $\mathcal{L}_\mathcal{P}$ illuminates the logic of sentences featuring 'all' and 'some', a logic disguised in the English originals.

Consider again the English sentence and its $\mathcal{L}_\mathcal{P}$ translation:

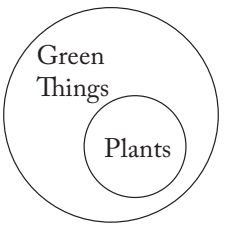
All plants are green.

$$\forall x(Px \supset Gx)$$

Predicates function as general terms that are true of, or not true of, individual objects. (If a predicate is not monadic, it is true of, or not true of, ordered n-tuples of objects, a qualification that can be omitted here.) Recall that the class or set of things of which a general term is true is the *extension* of the term. A term's extension consists of the set of individuals answering to or *satisfying* the term. You could interpret the sentence above as saying something about the relation classes of objects bear to one another.

If anything is in the class of plants, then it is in the class of green things.

The class relation expressed by the sentence can be depicted by means of a Venn diagram:



The class of plants is *included* in the class of green things. The *inclusion* relation is nicely captured by a universally quantified conditional sentence that says

Take any object at all; call that object x; if x is in the class of plants, then x is in the class of green things.

Does it follow that if something is green it is a plant? No. The sentence leaves open the possibility that the class of green things has members that are not plants.

Compare this with the sentence

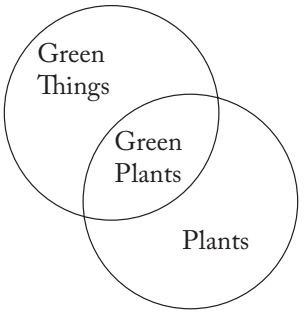
Some plants are green.

$$\exists x(Px \wedge Gx)$$

This sentence might be taken to mean

There is at least one thing, x, that is a member of both the class of plants and the class of green things.

Diagrammatically represented:



Universally quantified sentences express the class *inclusion* relation. Existentially quantified sentences express class *intersection*. They indicate that the intersection of the class of plants and the class of green things is not empty; at least one individual is a member of *both*.

These simple relationships, class inclusion and class intersection, are carried through in the use of universal and existential quantifiers, even in complex sentences. With few exceptions, universally quantified sentences and sentence parts are built around a conditional core; existentially quantified sentences and sentence parts are built around conjunctions.

Exercises 4.03

Translate the English sentences below into $\mathcal{L}_\mathcal{P}$. Let Sx = 'x is a sleuth'; Ax = 'x is an aunt'; Cx = 'x is cautious'. Use appropriate lowercase letters for names.

1. Frank and Joe are sleuths.
2. Gertrude isn't a sleuth, she's an aunt.
3. Some aunts are sleuths.
4. If some aunts are sleuths, then some sleuths are aunts.
5. If Gertrude is a sleuth, then some aunts are sleuths.
6. Fenton is both cautious and a sleuth.
7. All sleuths are cautious.
8. Every aunt is cautious.
9. Iola is neither an aunt nor cautious.
10. If all aunts are cautious, then Gertrude is cautious.
11. Gertrude is a sleuth only if some aunts are sleuths.
12. Some sleuths are both cautious and aunts.
13. Frank and Joe are cautious if and only if every sleuth is cautious.
14. Every aunt, if she is cautious, is a sleuth.
15. If Gertrude is a sleuth, then at least one aunt is a sleuth.

4.04 Bound and Free Variables

In $\mathcal{L}_\mathcal{P}$, every quantifier incorporates a variable that captures and *binds* the expression to which it is affixed. Variables occurring in the argument places of predicates resemble pronouns in English, terms that share a designation within a sentence. The mechanism in $\mathcal{L}_\mathcal{P}$ is uncomplicated. For a quantifier to capture a variable in a predicate, the variable in the quantifier must *match* the variable in the predicate. The $\mathcal{L}_\mathcal{P}$ sentence

$$\forall x(Fx \supset Gx)$$

could be read as

All Fs are Gs.

This reading is made possible by the matching pattern of variables. 'All' refers both to Fs and to Gs because the variable contained in the universal quantifier, x, matches the variable filling the

argument places in both predicate expressions. The matching variables ensure that the objects said to be *F*s are the same as those said to be *G*s.

A quantifier *picks up* any matching variable falling within its *scope*. The scope of a quantifier can be understood exactly as the scope of a negation sign is understood. Consider the $\mathcal{L}_{\mathcal{P}}$ sentences below:

$$\neg Fa \supset Ga$$

$$\neg(Fa \supset Ga)$$

In the first sentence, the scope of the negation sign includes only *Fa*. The scope of the negation sign in the second sentence includes everything falling within the matching parentheses to its right, $Fa \supset Ga$.

In the same way, the scope of the universal quantifier, $\forall x$, in the two expressions below includes the expression to its immediate right:

$$\forall xFx \supset Gx$$

$$\forall x(Fx \supset Gx)$$

In the first case, only *Fx* falls within the scope of the quantifier, but *Gx* does not. In the second case, the entire expression within parentheses, $Fx \supset Gx$, falls within its scope.

A variable is *bound* when it both

(i) falls within the scope of a quantifier, and

(ii) matches the variable contained in the quantifier.

If a variable is not bound, it is *free*. Occurrences of variables included in quantifiers are neither bound nor free. Their job is to *do* the binding.

In the first example above, the *x* in *Fx* is bound by the universal quantifier to its immediate left, but the *x* in *Gx* is free. In both cases the variable *x* matches the variable in the universal quantifier, but only the *x* in *Fx* falls within its scope.

In $\mathcal{L}_{\mathcal{P}}$, no sentence can contain free variables. Variables in sentences containing quantifiers can function as pronouns only when those variables are bound by quantifiers. The occurrence of a free variable in an $\mathcal{L}_{\mathcal{P}}$ expression is comparable to the occurrence of a _ in the midst of what otherwise would be an ordinary English sentence. Suppose Fx = '*x* is friendly' and Gx = '*x* is good'. Then the first expression above might be represented in English as

If everything is friendly, then _ is good.

and this is not an English sentence.

Exercises 4.04

Examine the expressions below and indicate (i) the scope of each quantifier and (ii) which variables, if any, occur freely. Indicate quantifier scope by means of a line, and circle free variables.

Example: $\forall x(Fx \supset Gx) \wedge Hx$

1. $\forall xFy \supset Gx$
2. $\exists x(Fx \wedge Gx)$
3. $\forall x(Fx \supset Gx) \supset \exists x(Fx \wedge Gx)$
4. $\exists y(Fx \wedge Gx) \wedge \forall x(Fy \supset Gy)$
5. $\forall x((Fx \supset Gx) \supset (Hx \wedge Ix))$
6. $\exists y((Fy \wedge Gy) \wedge (Hy \wedge Iy))$
7. $(\exists y(Fy \wedge Gy) \wedge (Hy \wedge Iy))$
8. $((\exists yFy \wedge Gy) \wedge (Hy \wedge Iy))$
9. $Fa \supset Ga$
10. $\forall x(Fx \supset Gx) \supset Ha$

4.05 Negation

The occurrence of negation in quantified sentences poses no special challenges. Quantifiers can fall within the scope of negation signs, and vice versa. Compare the English sentences

Not all sleuths are married.

All sleuths are unmarried.

Both sentences are variants of the core sentence

All sleuths are married.

Letting Sx = 'x is a sleuth' and Mx = 'x is married', this core sentence can be translated into $\mathcal{L}_p$ as

$$\forall x(Sx \supset Mx)$$

How would you translate 'Not all sleuths are married'? The sentence is the negation of the original—'Not all . . .', so you would translate the sentence as

$$\neg\forall x(Sx \supset Mx)$$

What about 'All sleuths are unmarried'? In this case you would negate only the predicate expression, 'x is married', Mx, not the quantifier

$$\forall x(Sx \supset \neg Mx)$$

If you are like most of us, you will find it helpful to turn to quasi-$\mathcal{L}_p$, both in the course of translating English sentences into $\mathcal{L}_p$ and in figuring out the meanings of $\mathcal{L}_p$ sentences. The sentence above in quasi-$\mathcal{L}_p$ would be

For all x, if x is a sleuth, then it is not the case that x is married.

The same principles apply to existentially quantified sentences. Suppose

$$\exists x(Sx \wedge Mx)$$

represents the English sentence

Some sleuths are married.

How would you translate the English sentences below into $\mathcal{L}_p$?

No sleuths are married.

Some sleuths are unmarried.

The first sentence tells you that it is not the case that at least one sleuth is married.

$$\neg\exists x(Sx \wedge Mx)$$

(Remember that an existential quantifier means 'some' in the sense of 'at least one'.) If it is not the case that at least one sleuth is married, then no sleuths are married. If some sleuths are unmarried, then at least one individual is both a sleuth and is not married:

$$\exists x(Sx \wedge \neg Mx)$$

In quasi-$\mathcal{L}_p$

There is an x such that x is a sleuth and it is not the case that x is married.

In mastering quantified sentences and sentence parts, it is important not to lose sight of sentences containing names. Consider the sentence

Some Greek is a philosopher.

Letting Gx = 'x is a Greek' and Px = 'x is a philosopher', this sentence would be translated into $\mathcal{L}_p$ as

$$\exists x(Gx \wedge Px)$$

What about the sentence

If Socrates is a philosopher, then some Greek is a philosopher.

The antecedent of this conditional contains a name, an individual constant, not a variable:

$$Ps \supset \exists x(Gx \wedge Px)$$

Another example:

If Socrates is admired by some philosopher, he is admired by every philosopher.

Letting $A①②$ = '① admires ②', you could translate the sentence into $\mathcal{L}_p$ as

$$\exists x(Px \wedge Axs) \supset \forall x(Px \supset Axs)$$

or in quasi-$\mathcal{L}_p$

If there is at least one *x* such that *x* is a philosopher, and *x* admires Socrates, then for all *x*, if *x* is a philosopher, *x* admires Socrates.

Variables take on significance only when matched with a quantifier, but individual constants stand on their own.

Exercises 4.05

Translate the English sentences below into $\mathcal{L}_p$. Let Sx = '*x* is a sleuth'; 'Wx = *x* is wily'; 'Kx = *x* is kidnapped'. Use appropriate lowercase letters for names. Remember that it is often helpful to think through a sentence in quasi-$\mathcal{L}_p$ first.

1. Every sleuth is wily.
2. If Chet is kidnapped, then some sleuths aren't wily.
3. If some sleuths aren't wily, then not all sleuths are wily.
4. If all sleuths are wily, then if Fenton isn't wily, he isn't a sleuth.
5. If Gertrude and Callie are kidnapped, then no sleuth is wily.
6. No sleuth fails to be wily.
7. Not every sleuth fails to be wily.
8. If some sleuths are kidnapped, then not all sleuths are wily.
9. If some sleuths aren't kidnapped, then not all sleuths fail to be wily.
10. If neither Gertrude nor Callie is kidnapped, then Iola is wily.
11. If no sleuths are wily, then some sleuth is kidnapped.
12. Some sleuths are kidnapped even though every sleuth is wily.
13. If not all sleuths are kidnapped, then Frank or Joe isn't kidnapped.
14. Gertrude is kidnapped only if some sleuth isn't wily.
15. If a sleuth is wily, she isn't kidnapped.

4.06 Complex Terms

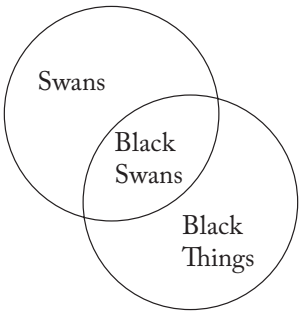
Predicates in $\mathcal{L}_p$ correspond to general terms in English. Individual objects in the extension of a general term are members of a set or class of objects, each of which is such that the general term is true of it. English includes a plentiful supply of words expressing general terms. When the need

arises, we make up new words and add them to our language. Meteorologists have an expanded vocabulary for the description of clouds, entomologists for the description of insects, physicists for the description of particles.

Words can fall into disuse when we lose interest in distinctions they mark. Eighteenth-century English, for instance, contained words for emotions that do not survive in twentieth-century English. The distinctions remain, although our interest in making them has, for whatever reason, waned.

Were general terms expressible only as individual words, and singular terms limited to names, language use would be a dreary affair, confined to situations in, or connected to, the immediate surroundings of speakers and hearers. As it is, we find it possible to speak of absent and unnamed objects by using descriptions: 'the desk in my office', 'the jacaranda tree in the quad', 'the goddess of wisdom'.

Another device is available for the manufacture of general terms on the spot. English contains the general terms 'is a swan' and 'is black'. Putting these together you can fabricate a new general term, 'is a black swan'. The extension of this term includes objects belonging to the *intersection* of the class of swans and the class of black things. It includes everything that is *both* a swan and black:



Similarly, you can assemble a new general term, 'is a black bowlegged swan'. The term's extension is the intersection of the class of black things, the class of swans, and the class of bowlegged things. The technique can be extended ad lib.

Terms manufactured in this way function as *complex terms*. Sentences containing complex terms are straightforwardly translatable into $\mathcal{L}_\mathcal{P}$. Consider, for instance, the sentence

Every black swan is graceful.

In this case 'black swan' expresses a complex term. The sentence could be translated as follows (assuming that Sx = 'x is a swan', Bx = 'x is black', and Gx = 'x is graceful'):

$$\forall x((Sx \wedge Bx) \supset Gx)$$

In quasi-$\mathcal{L}_\mathcal{P}$

For all x, if x is a swan and x is black (if x is a member of both the class of swans and the class of black things), then x is graceful (x is included in the class of graceful things).

Notice that the sentence follows the paradigm for universally quantified expressions illustrated by

All swans are graceful.

In $\mathcal{L}_\mathcal{P}$

$$\forall x(Sx \supset Gx)$$

The antecedents of the conditionals differ—the first sentence features a complex term in the form of a conjunction in its antecedent, and the second does not—but the pattern is the same. In each case, one class is said to be included in another.

What of existentially quantified expressions containing complex terms as in the English sentence below:

Some black swans are graceful.

This would be translated into $\mathcal{L}_\mathcal{P}$ as

$$\exists x((Sx \wedge Bx) \wedge Gx)$$

In both the universally quantified sentence above and in this sentence, $Sx \wedge Bx$ expresses the complex general term 'is a black swan'. The term is one for which English has no single word, so a complex term must be improvised. Ignoring that complication, the sentences exhibit the standard forms of universally and existentially quantified sentences, respectively.

Exercises 4.06

Translate the English sentences below into $\mathcal{L}_\mathcal{P}$. Let Sx = 'x is a sleuth'; Cx = 'x is clever'; $C①②$ = '① is more clever than ②'; Ex = 'x escapes'. Use appropriate lowercase letters for names.

1. Every clever sleuth escapes.
2. Some sleuth is more clever than Fenton.
3. No sleuth is more clever than Gertrude.
4. Every sleuth is more clever than Chet.
5. If Fenton escapes, then some clever sleuth escapes.
6. If Gertrude doesn't escape, then no clever sleuth escapes.
7. If neither Gertrude nor Fenton escapes, then some clever sleuth doesn't escape.
8. Not every clever sleuth escapes.
9. Every clever sleuth fails to escape.
10. If some clever sleuth escapes, then some clever sleuth is more clever than Gertrude.
11. If Frank and Joe escape, then Gertrude is more clever than Callie or Iola.

12. Frank escapes only if some clever sleuth is more clever than Gertrude.
13. Not every clever sleuth fails to escape.
14. If some clever sleuth escapes, then every clever sleuth escapes.
15. Not every clever sleuth is more clever than Gertrude.

4.07 Mixed Quantification

The translation of many sentences into $\mathcal{L}_\mathcal{P}$ requires the use of more than one quantifier. When a sentence contains multiple quantifiers, the quantifiers and the variables they bind must be kept properly sorted out. Consider the English sentence

Cats like fish.

Taken out of context, this sentence is *quadruply* ambiguous. It could be used to mean any of the following:

All cats like all fish.

All cats like some fish.

Some cats like all fish.

Some cats like some fish.

These sentences introduce another collection of paradigms for recurring patterns in English. Suppose Cx = 'x is a cat', Fx = 'x is a fish', and $L①②$ = '① likes ②'.

First,

All cats like all fish.

This sentence can be translated into $\mathcal{L}_\mathcal{P}$ as

$$\forall x(Cx \supset \forall y(Fy \supset Lxy))$$

In quasi-$\mathcal{L}_\mathcal{P}$

For all x, if x is a cat, then, for all y, if y is a fish, x likes y.

Four features of the translation deserve mention:

i. Every variable is bound by a quantifier. All occurrences of the variable x fall within the scope of the $\forall x$ quantifier, and all occurrences of the variable y fall within the scope of the $\forall y$.

ii. The second quantifier, $\forall y$, falls within the scope of the first quantifier, $\forall x$. When this happens, each quantifier must incorporate a distinct variable; otherwise the sentence would be ambiguous.

iii. The connectives associated with each quantifier follow the paradigm. Universally

quantified expressions are built around conditionals, and so it is in this case. The $\forall x$ applies to a conditional sentence that happens to have a conditional consequent; the second quantifier, $\forall y$, applies to that conditional consequent.

iv. The variables, *x* and *y*, function as pronouns. The pattern of variables, quantifiers, and predicate letters establishes that *x*'s are cats and *y*'s are fish. In the context of the sentence, the two-place predicate, *Lxy*, means that cats like fish. Had it been *Lyx*, the liking relation would be reversed: because *x*'s are cats and *y*'s are fish, the sentence would say that fish like cats.

Translations of the remaining sentences in the original list follow suit. In each case, the four points discussed above—suitably amended—apply.

All cats like some fish.

In $\mathcal{L}_\mathcal{P}$

$$\forall x(Cx \supset \exists y(Fy \wedge Lxy))$$

And in quasi-$\mathcal{L}_\mathcal{P}$

For all *x*, if *x* is a cat, then there is a *y* such that *y* is a fish, and *x* likes *y*.

Here an *existential* quantifier, $\exists y$, is affixed to the consequent of a universally quantified conditional (reflecting its reference to *some* fish), and the connective is adjusted accordingly: a $\supset$ is paired with a universal quantifier, $\forall x$, and $\wedge$ with an existential quantifier.

An analogous pattern can be observed in translating

Some cats like all fish.

In $\mathcal{L}_\mathcal{P}$

$$\exists x(Cx \wedge \forall y(Fy \supset Lxy))$$

In quasi-$\mathcal{L}_\mathcal{P}$

There is an *x* such that *x* is a cat, and for all *y*, if *y* is a fish, *x* likes *y*.

Again, the pattern of quantifiers and connectives honors the paradigms. The sentence is an existentially quantified conjunction, the second conjunct of which is a universally quantified conditional.

Finally,

Some cats like some fish.

In $\mathcal{L}_\mathcal{P}$

$$\exists x(Cx \wedge \exists y(Fy \wedge Lxy))$$

and in quasi-$\mathcal{L}_\mathcal{P}$

There is an *x* such that *x* is a cat, and there is a *y* such that *y* is a fish, and *x* likes *y*.

The sentence consists of an existentially quantified conjunction included as a conjunct within an existentially quantified conjunction.

Attention to sentences featuring mixed quantifiers leads inevitably to an observation about the significance of quantifier *order* within sentences. Consider the English sentence

Every sailor fancies some port.

The sentence, like many (most? all?) English sentences taken out of context, is ambiguous. The sentence could be used to mean that every sailor fancies some port or other, a port that might vary across sailors: some fancy Sydney, others fancy San Diego, still others fancy Reykjavik.

Alternatively, the sentence could be used to mean that some particular port—Sydney, say—is fancied by every sailor. These different interpretations of the English original are reflected in the order in which quantifiers are introduced. Thus, letting Sx = 'x is a sailor', Px = 'x is a port', and $F①②$ = '① fancies ②', the $\mathcal{L}_\mathcal{P}$ sentence

$$\forall x(Sx \supset \exists y(Py \wedge Fxy)$$

says, in quasi-$\mathcal{L}_\mathcal{P}$

For all x, if x is a sailor, then there is a y such that y is a port, and x fancies y.

In English:

Every sailor fancies some port (*some port or other*).

Compare this sentence with

$$\exists x(Px \wedge \forall y(Sy \supset Fyx)$$

In quasi-$\mathcal{L}_\mathcal{P}$

There is an x such that x is a port, and for all y, if y is a sailor, y fancies x.

which, in ordinary English would be

There is some port fancied by every sailor.

The order of occurrence of the variables used in the argument places of the predicate $F①②$ (① fancies ②) is significant. Their order reflects the pronominal character of variables generally. In the first sentence, x is taken to be a member of the class of sailors, hence x is the *subject* of the fancying relation; in the second sentence, x is specified as a *port*, hence it is the *object* of fancy.

Meta- and Object Language Variables

You will have noticed that variables appearing in the argument places in sentences need not match the variables used in specifying an interpretation of the $\mathcal{L}_\mathcal{P}$ predicate. Sx was said to mean 'x is a sailor', and Px meant 'x is a port'. Variables so used are mere placeholders.

The interpretation indicates that *whatever* goes in the argument place occupied by x in Sx is being said to be a sailor. If you subsequently write 'Sy', you are saying 'y is a sailor', and 'Ss' would mean 'Socrates is a sailor'.

Exercises 4.07

Translate the English sentences below into $\mathcal{L}_\mathcal{P}$, letting Sx = 'x is a sleuth'; Cx = 'x is a criminal'; Mx = 'x is married'; $A①②$ = '① arrests ②'. Use appropriate lowercase letters for names.

1. Every sleuth arrests some criminal.
2. No sleuth arrests any criminal.
3. Every criminal is arrested by some sleuth.
4. Some sleuth arrests every criminal.
5. Some unmarried sleuth arrests every married criminal.
6. Fenton arrests every unmarried criminal.
7. If every married sleuth arrests some unmarried criminal, then some criminal is unmarried.
8. Not every sleuth arrests some criminal.
9. Every sleuth fails to arrest some criminal.
10. Not every married sleuth fails to arrest some unmarried criminal.
11. All married criminals are arrested by some sleuth.
12. Every sleuth fails to arrest some unmarried criminal.
13. No sleuth arrests every married criminal.
14. If Fenton arrests every criminal, then he arrests every married criminal.
15. If not every married criminal is arrested by some sleuth, then some criminals aren't arrested by any sleuth.

4.08 Translational Odds and Ends

Consider the sentence

Some dog is missing.

Translated into $\mathcal{L}_\mathcal{P}$ (letting Dx = 'x is a dog' and Mx = 'x is missing') results in the sentence

$$\exists x(Dx \wedge Mx)$$

In quasi-$\mathcal{L}_\mathcal{P}$

There is an x such that x is a dog, and x is missing.

Now consider a superficially similar sentence:

Something is missing.

You might be tempted to read this sentence in quasi-$\mathcal{L}_\mathcal{P}$ as

There is an x such that x is a thing, and x is missing.

and so to translate it

$$\exists x(Tx \wedge Mx)$$

The temptation is to be resisted. Saying that 'x is a thing' is not to predicate anything of x. It is simply to indicate a something to which predicates can be applied. To say, in $\mathcal{L}_\mathcal{P}$, that something is missing, you need only

$$\exists x Mx$$

This says

There is an x (i.e., there is something, x) such that x is missing.

The point extends to sentences incorporating mixed quantifiers. First, consider the English sentence

Some sailor is taller than any landlubber.

Assuming that Sx = 'x is a sailor', Lx = 'x is a landlubber', and $T①②$ = '① is taller than ②', the sentence goes into $\mathcal{L}_\mathcal{P}$ as

$$\exists x(Sx \wedge \forall y(Ly \supset Txy))$$

Now, reflect on the sentence

Something is taller than any landlubber.

Once again, 'thing' is not functioning as a predicate, so the sentence in $\mathcal{L}_\mathcal{P}$ would be

$$\exists x \forall y(Ly \supset Txy)$$

In quasi-$\mathcal{L}_\mathcal{P}$

There is an x such that, for all y, if y is a landlubber, x is taller than y.

Finally, consider the English sentence

Everything is taller than something.

The sentence leaves utterly open the *sorts* of thing being compared, so the only predicate in play is 'taller than'.

$$\forall x \exists y Txy$$

In quasi-$\mathcal{L}_\mathcal{P}$

For all x, there is some y such that x is taller than y.

English words sometimes appear to function as terms when they do not. There are, as well, cases in which terms are hidden or disguised. Consider the English sentence

Everyone scorns someone.

The word 'everyone' here could be paraphrased 'every *person*'; 'someone' could be paraphrased 'some (or at least one) *person*'. 'Everyone' and 'someone' differ fundamentally from 'everything' and 'something', despite a superficial similarity. Letting Px = 'x is a person' and $S①②$ = '① scorns ②', the sentence above is translated into $\mathcal{L}_\mathcal{P}$ as

$$\forall x(Px \supset \exists y(Py \wedge Sxy))$$

The English indefinite article, 'a' (or 'an'), can prove irksome. The sentence below provides one illustration:

A computer is a machine.

Although, at first glance, the sentence might appear to refer to a particular computer (*a* computer . . .), a little reflection reveals that it is intended to say something about computers generally, the class of computers, namely, that if it is true of something that it is a computer, then it is true of that something that it is a machine. So the sentence is equivalent to

All computers are machines.

or simply

Computers are machines.

Given that Cx = 'x is a computer' and Mx = 'x is a machine',

$$\forall x(Cx \supset Mx)$$

When it comes to translation, the aim is always to come up with a sentence the truth conditions of which approximate those of the original. In many cases you will need to burrow beneath the surface structure of sentences you are translating so as not to trip over syntactic forms that can be misleading when taken out of context.

You might expect 'any', 'all', and 'every' to signal the presence of a universal quantifier. Often they do. In the sentence below, 'any' picks out every member of the class of hamsters:

Any hamster that bites is dangerous.

Assuming that Hx = 'x is a hamster', Bx = 'x bites', and Dx = 'x is dangerous', this sentence can be translated into $\mathcal{L}_\mathcal{P}$ as

$$\forall x((Hx \wedge Bx) \supset Dx)$$

But consider the superficially similar sentence

If any hamster bites, they all do (that is, every hamster bites).

This sentence does not mean 'If all (or every) hamster bites . . .' but 'If some (at least one) hamster bites . . .', and, in consequence, it goes into $\mathcal{L}_\mathcal{P}$ as

$$\exists x(Hx \wedge Bx) \supset \forall x(Hx \supset Bx)$$

Notice that the sentence has been translated using x's in both quantifiers. This is permissible because the quantifiers do not overlap in scope. The sentence could have been translated as

$$\exists x(Hx \wedge Bx) \supset \forall y(Hy \supset By)$$

Translation Applied

Consider the English sentence

> If horses are animals, then all heads of horses are heads of animals.

How might this be translated into $\mathcal{L}_\mathcal{P}$? (Let Hx = 'x is a horse'; Ax = 'x is an animal'; and $H①②$ = '① is the head of ②'.) The sentence has a conditional antecedent

> If horses are animals, then . . .

Thus

$$\forall x(Hx \supset Ax) \supset$$

The consequent is more challenging.

> . . . all heads of horses are heads of animals.

This goes into $\mathcal{L}_\mathcal{P}$ as

$$\forall x(\exists y(Hy \wedge Hxy) \supset \exists z(Az \wedge Hxz))$$

In quasi-$\mathcal{L}_\mathcal{P}$

> For all x, if there is a y such that y is a horse and x is the head of y, then there is a z such that z is an animal and x is the head of z.

The whole sentence in $\mathcal{L}_\mathcal{P}$ looks like this:

$$\forall x(Hx \supset Ax) \supset \forall x(\exists y(Hy \wedge Hxy) \supset \exists z(Az \wedge Hxz))$$

Note that, strictly speaking, a 'y' could have been used in place of the 'z'. Can you see why?

[The sentence is discussed in W. V. Quine, *Methods of Logic*, 3rd ed. (New York: Holt, Rinehart & Winston, 1972), 142–43.]

The two sentences are logically equivalent. Either is an acceptable translation.

In working out translations, it is all too easy to focus on occurrences of particular words and take these as infallible signs of logical structure. The same word in different sentential contexts can express different meanings, however. You learn words by learning to use them in sentences. When words are taken out of context, their contribution to the meanings of sentences in which they occur is lost.

Consider the English sentence

> Euterpe is bolder than Clio or Melpomene.

An 'or' signals a disjunction, right? Well, is 'or' used in this sentence to express a disjunction? Would the sentence below be an acceptable paraphrase?

Euterpe is bolder than Clio or Euterpe is bolder than Melpomene

The original sentence says that Euterpe is bolder than *either*. Thus

Euterpe is bolder than Clio *and* Euterpe is bolder than Melpomene.

Once you see this, translation into $\mathcal{L}_\mathcal{P}$ is a snap. Letting $B①②$ = '① is bolder than ②',

$$Bec \wedge Bem$$

The basic unit of meaning is the sentence. You could think of the meaning of a word as the contribution that word makes to the truth conditions of sentences in which it occurs. Learning the meanings of words is a matter of learning the roles they play in sentences. Providing a definition of an individual word requires *abstracting out* the contribution the word makes to the meanings of sentences in which it occurs. As the foregoing illustrates, the same word can affect the truth conditions of sentences differently in different sentential contexts. This is reflected in the fact that dictionaries often provide multiple meanings for individual words, and this is just one more reason translation can prove tricky—and interesting.

Exercises 4.08

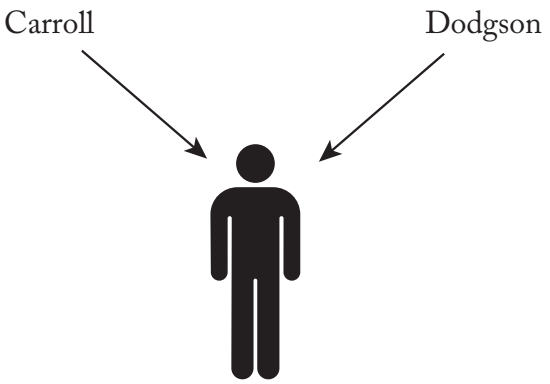
Translate the English sentences that follow into $\mathcal{L}_\mathcal{P}$. Let Sx = 'x is a sleuth'; Px = 'x is a person'; Wx = 'x is wiley'; $A①②$ = '① admires ②'. Use appropriate lowercase letters for names.

1. A sleuth is a wily person.
2. Someone is admired by every sleuth.
3. Every sleuth admires something.
4. No sleuth is admired by everyone.
5. If Fenton is a wily sleuth, he is admired by someone.
6. If Fenton isn't a wily sleuth, he is admired by no one.
7. Callie admires a wily sleuth.
8. If Callie admires Fenton, she admires a wily sleuth.
9. If a sleuth isn't wily, nothing is.
10. A wily sleuth admires nothing.
11. A person who fails to be wily is not admired by anyone.
12. Not every wily sleuth is admired by Fenton.
13. Every wily sleuth is admired by someone.
14. Some wily sleuth is admired by everyone.
15. A wily sleuth is admired by everyone.

4.09 Identity

Identity is a relation everything bears to itself and to nothing else. The concept of identity is the concept of *self-sameness*. If *x* is identical with *y*, then *x* and *y* are *one and the same* individual. This way of putting it has a faintly paradoxical ring. How can *two* things be *one and the same* thing? The paradox is only apparent. If *x* and *y* are identical, then there are not *two* things, *x* and *y*, but only a single thing twice designated.

We require a concept of identity precisely because things can be variously named and described. The identity relation enables us to indicate that two names or descriptions designate one and the same individual. Lewis Carroll and Charles Dodgson are one and the same person.



The identity relation in $\mathcal{L}_p$ holds among the objects designated by individual terms. If *a* and *b* are identical, the terms '*a*' and '*b*' designate one and the same individual. Cicero *is—is identical with*—Tully, Hesperus (the Evening Star) *is* Phosphorus (the Morning Star), Scott *is* the author of *Waverly*, the masked bandit *is* the dashing prince.

The 'is' of identity differs importantly from the 'is' of predication. The sentences below exhibit the 'is' of identity:

Hesperus is Phosphorus.

Lewis Carroll is Charles Dodgson.

This 'is' differs from the 'is' of predication illustrated by these sentences:

Hesperus is bright.

Carroll is English.

Here 'is' appears as an undetachable component of a pair of general terms: 'is bright' and 'is English'.

An 'is' of identity can typically be replaced by the phrase 'is nothing but' without affecting the truth conditions of the sentence in which it occurs. Phosphorus is nothing but Hesperus, and Carroll is nothing but Dodgson. If Phosphorus is bright and Carroll is English, however, it does not follow that Phosphorus is nothing but bright, or that Carroll is nothing but English.

Identity and Resemblance

Identity is a relation between something and itself. If Cicero is identical with Tully, Cicero and Tully are one and the same individual.

Identity is distinguished from exact similarity or resemblance. Twins Judy and Trudy are identical, not in the sense that there is just one of them with two names, but in the sense that each exactly resembles the other.

English affords a way of distinguishing cases in which 'identical' is used to designate the identity relation from those in which it is used to express resemblance.

> Cicero is identical *with* Tully.
>
> Judy is identical *to* (not *with*) Trudy.

If Cicero is identical *with* Tully, then, because everything resembles itself, Cicero would be identical *to* Tully. Judy's being identical *to* her twin Trudy does not, however, mean that Judy is identical *with* Trudy.

Sadly, philosophers (who should know better) often use 'identical to' when they mean 'identical with', thereby squandering a useful linguistic resource. In some cases, it is unclear whether they mean to be invoking identity or exact similarity.

The basis of this distinction is clear. Identity is a relation every object bears to itself and to no other object. Phosphorus is Hesperus just in case the object named by 'Phosphorus' is identical with the object named by 'Hesperus'. Phosphorus is bright, however, just in case the object named 'Phosphorus' is a member of the class of bright objects.

Identity is a two-place relation that could be represented just as you might represent any other two-place relation: $I①②$. The significance of the identity relation is profound, however, so logicians prefer to represent the identity relation with a dedicated symbol, the identity sign, $=$. The identity sign does not express equality but identity: '$a = b$' does not mean 'a and b are *equal*'; '$a = b$' means *a is b*.

Just as the argument places in $I①②$ must be filled by individual names or variables, the identity sign is always flanked by individual names or variables. You could have

$$a = b$$

$$x = y$$

but not

$$Pa = Pb$$

$$\exists xFx = \exists xGx$$

Because the identity sign is always flanked by individual names or variables, you can omit parentheses without producing ambiguity. This is simply a reflection of the fact that $a = b$ is a stand-in for Iab, and just as you would not translate 'If Carroll is an author, then Carroll is Dodgson' as

$$Ac \supset (Icd)$$

but as

$$Ac \supset Icd$$

hence

$$Ac \supset c = d$$

The identity sign locks together the *c* and *d*, just as *Icd* would, so parentheses would be redundant.

The negation of an identity relation is indicated by a ≠. Rather than using

$$\neg(a = b)$$

or maybe

$$\neg a = b$$

to deny that '*a*' and '*b*' designate one and the same individual, you would use

$$a \neq b$$

The ≠ further reduces the need for redundant parentheses, and its familiarity makes its use uncomplicated.

You are now in a position to translate English sentences incorporating identity relations into $\mathcal{L}_\mathcal{P}$. Consider the English sentence

If Twain is an author and Clemens isn't,

then Twain isn't Clemens.

Do not be confused by occurrences in the sentences of both the *is* of predication and the *is* of identity. The sentence can be translated into $\mathcal{L}_\mathcal{P}$ without recourse to quantifiers (letting Ax = '*x* is an author'):

$$(At \wedge \neg Ac) \supset t \neq c$$

Now reflect on a more complex sentence involving quantification:

Clemens is the only author.

Although it is not immediately obvious, this sentence harbors an occurrence of the identity relation. The sentence tells us both that Clemens is an author and that he is the *only* author. How might that be expressed in $\mathcal{L}_\mathcal{P}$?

If Clemens is the only author, then anything that is an author, anything of which 'is an author' is true, must be Clemens. Putting all this together yields

$$Ac \wedge \forall x(Ax \supset x = c)$$

In quasi-$\mathcal{L}_\mathcal{P}$

Clemens is an author, and, for all *x*, if *x* is an author, *x* is Clemens.

The sentence illustrates the extent to which the identity relation can play a central but unobvious role in the logic of everyday talk. More generally: in looking carefully at the truth conditions of apparently unremarkable sentences, it is often possible to discern a formidable logical structure hidden in plain sight.

Identity

Everything is identical with itself and nothing else. It seems to follow that statements of identity must be either trivially true:

Lewis Carroll = Lewis Carroll

or patently false:

Lewis Carroll = Arnold Schwarzenegger

Still,

Lewis Carroll = Charles Dodgson

is an identity statement that is both true and informative.

One reason identity is an indispensable concept is that names and objects are not perfectly correlated, and because it is possible to refer to a single object in many ways. We need some way of indicating that objects variously designated are one and the same:

Lewis Carroll = the author of *Jabberwocky*

the author of *Jabberwocky* = the lecturer in mathematics

The foregoing illustrates one of the benefits of studying logic. Each of us deploys the concept of identity continuously, but without an explicit awareness that we are doing so. Were you to lack an implicit grasp of the concept of identity, you would be severely limited in what you could think or say. In studying $\mathcal{L}_p$, your implicit knowledge is brought to the surface and made explicit: you learn something important about yourself.

Exercises 4.09

Translate the English sentences below into $\mathcal{L}_P$. Let Sx = 'x is a sleuth'; $A①②$ = '① is the aunt of ②'. Use appropriate lowercase letters for names.

1. Gertrude is Joe's aunt.
2. Gertrude is Miss Hardy.
3. Gertrude is Joe's only aunt.
4. Joe's only aunt isn't a sleuth.
5. Frank's only aunt is Joe's only aunt.
6. Joe isn't Frank.
7. If Frank isn't Joe's aunt, then Frank isn't Joe's only aunt.
8. If Gertrude is Miss Hardy, then, if Gertrude is a sleuth, Miss Hardy is a sleuth.
9. If Gertrude is a sleuth and Miss Hardy isn't a sleuth, then Gertrude isn't Miss Hardy.
10. Either Miss Hardy is Joe's aunt or Miss Hardy isn't Gertrude.
11. Fenton is the only sleuth.
12. If there are any sleuths, Fenton is the only sleuth.
13. No sleuth is Frank's aunt.
14. If Gertrude is Joe's aunt, she isn't a sleuth only if she isn't Miss Hardy.
15. Not every sleuth is Joe's aunt.

4.10 At Least, at Most, Exactly

Suppose Px is used to mean 'x is a philosopher'. In that case, the $\mathcal{L}_P$ sentence

$$\exists x Px$$

might naturally be taken to express what would be expressed in English as

There is at least one philosopher.

Now consider the English sentence

There are at least two philosophers.

You might think that this sentence would be expressible in $\mathcal{L}_P$ as

$$\exists x Px \wedge \exists y Py$$

This translation fails to measure up, however. The sentence says, twice over, that there is at least one philosopher. But saying twice that there is at least one philosopher is not the same as saying that there are at least *two* philosophers; it is simply to repeat yourself. The $\mathcal{L}_\mathcal{P}$ sentence (and its English counterpart) would be true at worlds containing a single philosopher.

You can ensure that worlds of the latter sort are excluded by adding that the philosophers in question are distinct:

$$x \neq y$$

Putting these elements together, you obtain

$$\exists x(Px \wedge \exists y(Py \wedge x \neq y))$$

In quasi-$\mathcal{L}_\mathcal{P}$

There is at least one x such that x is a philosopher and there is a y such that y is a philosopher and x and y are distinct.

Similarly, you could translate

There are at least *three* philosophers.

as

$$\exists x(Px \wedge \exists y((Py \wedge x \neq y) \wedge \exists z(Pz \wedge (x \neq z \wedge y \neq z))))$$

Now reflect the English sentence

There is *at most* one philosopher.

The truth conditions for this sentence make it true at worlds containing no more than one philosopher, including those benighted worlds containing no philosophers at all. You can cook up an $\mathcal{L}_\mathcal{P}$ equivalent by making use of universal quantifiers:

$$\forall x(Px \supset \forall y(Py \supset x = y))$$

In quasi-$\mathcal{L}_\mathcal{P}$

For all x, if x is a philosopher, then for all y, if y is a philosopher, then x and y are one and the same.

Compare this sentence with the $\mathcal{L}_\mathcal{P}$ translation of 'There is at least one philosopher' set out above. Quantifiers and their associated major connectives differ systematically—existentials taking $\wedge$'s and $\supset$'s accompanying universals.

By now you might be in a position to work out a translation for

There are at most *two* philosophers.

The sentence is true at worlds containing no more than two philosophers (including those dismal worlds bereft of philosophers), and false at all other worlds. The $\mathcal{L}_\mathcal{P}$ sentence

$$\forall x(Px \supset \forall y((Py \wedge x \neq y) \supset \forall z(Pz \supset (z = x \vee z = y))))$$

How Many Gods?

People disagree on whether there are any gods and, if there are, how many there are. You now have the resources to represent these differences in $\mathcal{L}_\mathcal{P}$ (Gx = x is a god).

Theism: there is at least one god
$\exists x Gx$

Atheism: there are no gods
$\neg\exists x Gx$

Agnosticism: there is at most one god
$\forall x(Gx \supset \forall y(Gy \supset x = y))$

Monotheism: there is exactly one god
$\exists x(Gx \wedge \forall y(Gy \supset x = y))$

Gnosticism: there are exactly two gods
$\exists x(Gx \wedge \exists y((Gy \wedge x \neq y) \wedge \forall z(Gz \supset (z = x \vee z = y))))$

If nothing else, you should now be in a position to state your theological position in a precise way.

fills the bill. In quasi-$\mathcal{L}_\mathcal{P}$

For all x, if x is a philosopher, then, for all y, if y is a philosopher other than x, then, if anything, z, is a philosopher, z is identical with x or with y.

I leave it to you to extrapolate from this sentence to sentences mentioning three or more philosophers.

Having worked through translations of 'at least one' and 'at most one', you might wonder about translations of 'exactly one'. Take the sentence

There is exactly one philosopher.

In saying that there is exactly one member of a certain class, in this case the class of philosophers, you would be saying that there is *at least one* member of the class, and *at most one* member. To translate 'exactly n' sentences into $\mathcal{L}_\mathcal{P}$, you need only combine these elements. 'There is at least one philosopher' is expressible in $\mathcal{L}_\mathcal{P}$ as:

$$\exists x Px$$

'and there is at most one philosopher' is captured by:

$$\wedge \forall y(Py \supset x = y)$$

Putting these elements together:

$$\exists x(Px \wedge \forall y(Py \supset x = y))$$

(Notice the pairing of connectives and quantifiers.) In quasi-$\mathcal{L}_P$

> There is at least one philosopher, *x*, and for all *y*, if *y* is a philosopher, then *x* and *y* are one and the same.

Moving from a sentence of this sort to sentences mentioning *exactly two* (or, more generally, exactly *n*) philosophers is unsurprising.

$$\exists x(Px \wedge \exists y((Py \wedge x \neq y) \wedge \forall z(Pz \supset (z = x \vee z = y))))$$

The sentence combines 'at least two' with 'at most two' to yield 'exactly two'. In quasi-$\mathcal{L}_P$

> There is at least one philosopher, *x*, and a philosopher, *y*, and *x* and *y* are distinct; and if anything, *z*, is a philosopher, *z* is identical with *x* or *y*.

You might regard all this as interesting but doubt its significance. The next section focuses on an application of identity that is at the heart of linguistic communication.

4.11 Definite Descriptions

Quantification makes it possible to refer to sets or classes of individuals without enumerating their members. The identity relation makes it possible to refer to individual class members without using names. You might refer to a certain philosopher by means of a name, 'Aristotle', or by means of a *definite description*: 'the teacher of Alexander'.

Definite descriptions are ubiquitous. A language that did not permit the construction of descriptions would be communicatively unwieldy. Mutual understanding would require a shared stock of names for anything you wanted to talk about. Names can be learned ostensively. Pointing to a passer-by, I say to you, 'That's Natasha', thereby equipping you to know who I am referring to when I later say, 'Tomorrow is Natasha's birthday'.

Identity and Indiscernibility

The philosopher Leibniz (*Discourse on Metaphysics*: § 9) is usually credited with formulating a controversial principle of identity:

> *Identity of indiscernibles*: if every predicate true of *x* is true of *y* and vice versa, then $x = y$.

According to this principle, if *x* and *y* are exactly alike in every respect (if *x* and *y* are *indiscernible*), then *x* and *y* are the selfsame object, $x = y$. Really? Why couldn't two things share all of their properties—two electrons, for instance?

A distinct but closely related principle is less controversial.

> *Indiscernibility of identicals*: if $x = y$, then every predicate true of *x* is true of *y* and vice versa.

The principles differ importantly. Can you see why? (You might start by looking again at the earlier discussion of identify and resemblance.)

More often, names are introduced by descriptions. You are taught that Athena is the Greek goddess of wisdom, that Canberra is the capital of Australia, that Marshall was the first chief justice. Lacking descriptions, the names you learned would be limited to labels for individuals present and salient to both you and whoever introduced you to the name. Under those circumstances, communication would be precarious at best.

The linguistic role of definite descriptions mirrors the role of names. A definite description purports to designate a particular, that is definite, object. As in the case of names, an object can be designated by more than one description. In general, definite descriptions and names are interchangeable in sentential contexts:

Erato is clever.

The muse is clever.

Each sentence can be used to ascribe cleverness to one and the same individual. Letting Cx = 'x is clever', the first sentence can be translated into $\mathcal{L}_p$ as

$$Ce$$

What of the second? Following Bertrand Russell (1872–1970), a description such as 'the muse', used in a particular context, aims to designate exactly one object. Were this not so, were there, for instance, no muse in the vicinity, or were there more than one, the description (and any sentence containing it) would be defective. An ordinary existentially quantified sentence in $\mathcal{L}_p$ captures only the first of these conditions:

$$\exists x Mx$$

This sentence says that there is *at least one* muse, leaving open the possibility that there is more than one muse.

A definite description, in contrast, excludes this possibility. What would it take to narrow the focus? Russell argued that the sentence 'There is at least one muse' should be conjoined with the sentence 'There is *at most one* muse'.

$$\exists x(Mx \wedge \forall y(My \supset x = y))$$

That is, there is *at least one* muse and *at most one*, which amounts to *exactly one*. If this is right, if the definite description 'the muse' is equivalent to 'there is at least one muse, and at most one' ('there is exactly one'), you could translate 'The muse is clever' as

$$\exists x((Mx \wedge \forall y(My \supset x = y)) \wedge Cx)$$

The name 'Erato' designates a particular individual. The definite description in a different way ensures that x (at least in the context in which it is used) designates a particular individual. All that remains is an attribution of cleverness to the object of that designation.

Consider a slightly more complicated sentence that incorporates a definite description:

The young philosopher respects Socrates.

The sentence features a definite description:

The young philosopher . . .

The description implies that, in the context in which the sentence is uttered, there is one, and only one—exactly one—young philosopher. Suppose Px = 'x is a philosopher', Yx = 'x is young', and $R①②$ = '① respects ②'. To say that there is exactly one young philosopher is to say that there is *at least one*

$$\exists x(Px \wedge Yx) \ldots$$

and *at most one*.

$$\ldots \forall y((Py \wedge Yy) \supset x = y) \ldots$$

Conjoining these yields *exactly one*.

$$\exists x((Px \wedge Yx) \wedge \forall y((Py \wedge Yy) \supset x = y))$$

To this you need only add that this young philosopher, x, admires Socrates.

$$\ldots Axs$$

Putting this together with the definite description yields

$$\exists x(((Px \wedge Yx) \wedge \forall y((Py \wedge Yy) \supset x = y)) \wedge Axs)$$

Identity and Relations

The identity relation is *transitive* and *symmetric*. It is transitive because

if $x = y$ and $y = z$, then $x = z$

In this respect identity resembles the *greater than* relation

if $x > y$ and $y > z$, then $x > z$

The greater than relation is not symmetric, however. Although

if $x = y$, then $y = x$

it is *not* true that

if $x > y$, then $y > x$

Not all relations are transitive. Some, like the *is-next-to* relation, are *nontransitive*:

if x is next to y and y is next to z,
x *might* or *might not be* next to z

Other relations, for instance the *is the mother of* relation, are *intransitive*.

if x is the mother of y, and y is the
mother of z, x is *not* the mother of z

(Notice the pattern of parentheses.) The sentence has the form of a complex conjunction, one conjunct of which is itself a conjunction.

$$((p \wedge q) \wedge r)$$

The p conjunct is itself a conjunction, $(Px \wedge Yx)$, and the q conjunct is a conditional with a conjunctive antecedent, $\forall y((Py \wedge Yy) \supset x = y)$. These relations are marked by nested parentheses.

Russell's account of definite descriptions is plausible only so long as descriptions are used in contexts that narrow the range of objects over which variables range. You can speak sensibly of 'the young philosopher' only in a context that makes it clear that you intend something like 'the young philosopher in that group of philosophers over there'. In this context, 'the young philosopher' would be the one and only young philosopher in the group.

On Russell's view, then, 'the young philosopher' implies that a salient collection of objects includes exactly one young philosopher. It does not imply that the whole universe includes exactly one young philosopher. (Russell introduced the theory of descriptions in 'On Denoting', *Mind* 14 [1905]: 479–93.)

If nothing else, this should serve as a reminder of the central role of context in the semantics of natural languages. You might ask yourself whether it might be possible to eliminate this role, perhaps by replacing contexts with descriptions of contexts. Such descriptions would, if successful, move what is ordinarily in the background to the foreground.

Wittgenstein on Language and Thought

Language disguises thought—so much so that from the outward form of the clothing it is impossible to infer the form of the thought beneath it, because the outward form of the clothing is not designed to reveal the form of the body, but for entirely different purposes.

The tacit conventions on which the understanding of everyday language depends are enormously complicated.

(Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*, trans. D. F. Pears and B. F. McGuinness [London: Routledge & Kegan Paul, 1961], § 4.002.)

You are doubtless itching to apply what you have learned about identity to translations of sentences into $\mathcal{L}_\mathcal{P}$. Before tackling exercises 4.11, however, you might reflect on the nuts and bolts of definite descriptions.

Definite descriptions of the form 'the F. . .' are analyzable into conjunctions: 'there is at least one F' and '. . . at most one F'. These can be translated into $\mathcal{L}_\mathcal{P}$ as follows:

there is at least one F: $\exists xFx \ldots$

there is at most one F: $\ldots \wedge \forall y(Fy \supset x = y)$

which, when conjoined, yield

there is exactly one F: $\exists x(Fx \wedge \forall y(Fy \supset x = y))$

The middle expression above is not a sentence of $\mathcal{L}_\mathcal{P}$. It contains, in addition to a dangling connective, a free variable, an occurrence of x that is not picked up by a quantifier and thus acquires significance only when the expression is included in the sentence at the bottom of the list.

Exercises 4.11

Translate the English sentences below into $\mathcal{L}_p$. Let Sx = 'x is a sleuth'; Wx = 'x is wily'; Px = 'x is a person'; $A①②$ = '① admires ②'. Use appropriate lowercase letters for names.

1. The sleuth is wily.
2. There is at least one wily sleuth.
3. At most there is one wily sleuth.
4. The wily sleuth admires someone.
5. Everyone admires the wily sleuth.
6. Gertrude admires the wily sleuth.
7. There are exactly two wily sleuths.
8. There are no more than two wily sleuths.
9. If Fenton and Gertrude are sleuths, then there are at least two sleuths.
10. Fenton admires the two sleuths.
11. No one admires every wily sleuth.
12. Someone admires the wily sleuth.
13. At least two people admire Fenton.
14. The sleuth admires Fenton if he is wily.
15. If Fenton is admired by the wily sleuth, then Fenton is admired by at least one person.

4.12 Comparatives, Superlatives, Exceptives

By now the importance of the concept of identity should be evident. Were you, or anyone else, to lack the concept, you would find it impossible to say or think much of what you now say and think. These observations can be extended by investigating the logic of comparatives, superlatives, and (what might be called) exceptives.

Consider a simple comparative in English:

Euterpe is wiser than Clio.

Letting $W①②$ = '① is wiser than ②', the sentence is translated into $\mathcal{L}_p$ as

$$Wec$$

Now reflect on the sentence

Euterpe is the wisest Muse.

Note, first, that this sentence is not equivalent to the sentence

Euterpe is wiser than any Muse.

In $\mathcal{L}_\mathcal{P}$ (using Mx to mean 'x is a Muse')

$$\forall x(Mx \supset Wex)$$

This would be correct only if Euterpe were not a Muse. If Euterpe is a Muse, however, and she is, the translation could not be correct. Euterpe cannot be wiser than *any* Muse. Euterpe cannot be wiser than *herself*! If Euterpe is the wisest Muse, then

Euterpe is wiser than *any other* Muse (that is, any Muse *other than Euterpe*).

Thus put, you should be able to see your way to an $\mathcal{L}_\mathcal{P}$ translation.

$$Me \wedge \forall x((Mx \wedge x \neq e) \supset Wex)$$

In quasi-$\mathcal{L}_\mathcal{P}$

Euterpe is a Muse, and, for all x, if x is a Muse *other than Euterpe*, Euterpe is wiser than x.

By itself, the sentence both in English and in $\mathcal{L}_\mathcal{P}$ does not imply that there are any Muses other than Euterpe. It would be true even if Euterpe were the only Muse. Indeed, were Euterpe the only Muse, the sentence would be trivially true.

When you say that some object possesses more of a given trait than some other object or collection of objects, you employ a *comparative* construction—as in

Euterpe is wiser than Clio.

Superlatives come into play when you say that an object possesses more of that trait than anything else. A superlative is a species of comparative. An object belonging to some class is compared to every other member of that class. The wisest Muse is the Muse wiser than any other Muse; the tallest building is the building taller than any other building. The wisest Muse need not be the wisest thing, and the tallest building need not be the tallest thing.

Exceptives compare an object belonging to some class of objects to other members of the class *other than* one or more specified members. An example:

Clio is the wisest Muse except for Euterpe.

According to this sentence, Clio is the wisest member of the class consisting of every Muse *minus* Clio and minus Euterpe. The sentence does not imply that Euterpe is wiser than Clio, although she might be. The sentence implies only that Clio is no wiser than Euterpe. The possibility that Clio and Euterpe are tied with respect to wisdom is left open.

Translating the sentence into $\mathcal{L}_\mathcal{P}$ is straightforward so long as you bear these points in mind. As in the case of the superlative sentence above, the sentence indicates, first, that both Clio and Euterpe are Muses:

$$Mc \wedge Me$$

You are not entitled to add that Clio and Euterpe are not one and the same individual: $c \neq e$. The sentence might strongly *suggest* that this is so, but it does not say so explicitly. Instead, a class consisting of all Muses *except* Clio and Euterpe is designated. Clio is wiser than any member of *that* class:

$$\forall x((Mx \wedge (x \neq c \wedge x \neq e)) \supset Wcx)$$

Once again, the class of Muses must exclude Clio as well as Euterpe; otherwise the sentence would imply that Clio is wiser than herself. Excluding both Euterpe and Clio requires the use of a conjunction, $x \neq c \wedge x \neq e$. Combining these elements yields

$$(Mc \wedge Me) \wedge \forall x((Mx \wedge (x \neq c \wedge x \neq e)) \supset Wcx)$$

In quasi-$\mathcal{L}_\mathcal{P}$

> Clio is a Muse and Euterpe is a Muse, and, for all x, if x is a Muse other than Clio and other than Euterpe, Clio is wiser than x.

Two more related kinds of sentence bear mention here. An example of the first occurs in the English sentence

> Only Euterpe is wise

The sentence expresses the thought that the class of wise things includes just one member, Euterpe. If anything is a member of this class, it must be Euterpe. In $\mathcal{L}_\mathcal{P}$, and assuming that Wx = 'x is wise',

$$We \wedge \forall x(Wx \supset x = e)$$

The same translational pattern applies to sentences featuring complex terms:

> Euterpe is the only wise Muse.

In $\mathcal{L}_\mathcal{P}$

$$(Me \wedge We) \wedge \forall x((Mx \wedge Wx) \supset x = e)$$

The translation makes the sentence's truth conditions explicit: Euterpe is a wise Muse, and anything that is a wise Muse is Euterpe.

Exercises 4.12

Translate the English sentences below into $\mathcal{L}_\mathcal{P}$. Let Sx = 'x is a sleuth'; Bx = 'x is brave'; $T①②$ = '① is taller than ②'. Use appropriate lowercase letters for names.

1. Fenton is taller than Chet.
2. Fenton is not taller than some sleuth.
3. Callie is the tallest sleuth.
4. No sleuth is taller than Callie.
5. The tallest sleuth is Callie.
6. Except for Callie, Fenton is the tallest sleuth.
7. The tallest sleuth is brave.
8. The tallest sleuth is taller than Chet.
9. If Callie is the tallest sleuth, then she is brave.
10. If Fenton is a sleuth, he's not the tallest sleuth.
11. Only Callie is a brave sleuth.
12. Only Callie and Fenton are brave sleuths.
13. The brave sleuth is taller than Fenton or Chet.
14. If any sleuth is brave, then the tallest sleuth is brave.
15. The tallest sleuth is the brave sleuth.

4.13 Times and Places

Sentences featuring the pronouns 'someone', 'everyone', 'no one' were discussed in § 4.08. In using these pronouns, you are quantifying over *people*. Recall the parings of English sentences and their $\mathcal{L}_\mathcal{P}$ translations below:

$$\text{Someone is mortal} \rightarrow \exists x(Px \wedge Mx)$$

$$\text{Everyone is mortal} \rightarrow \forall x(Px \supset Mx)$$

$$\text{No one is mortal} \rightarrow \neg\exists x(Px \wedge Mx)$$

These examples generalize to sentences that mention times—'sometimes', 'always', 'never'—and places—'somewhere', 'everywhere', 'nowhere'. In deploying such sentences you are quantifying over times and places, respectively. Letting Tx = 'x is a time' and $H①②$ = '① is happy at ②',

Iola is sometimes happy → $\exists x(Tx \wedge Hix)$

Chet is always happy → $\forall x(Tx \supset Hcx)$

Gertrude is never happy → $\neg\exists x(Tx \wedge Hgx)$

Letting Px = 'x is a place' and $A①②$ = '① is admired at ②',

Compton is admired somewhere → $\exists x(Px \wedge Acx)$

Elvis is admired everywhere → $\forall x(Px \supset Aex)$

Beetlebum isn't admired anywhere → $\neg\exists x(Px \wedge Abx)$

These paradigms can be extended to more complex cases (letting Px = 'x is a person', Tx = 'x is a time' and $H①②$ = '① is happy at ②').

Everyone is sometimes happy → $\forall x(Px \supset \exists y(Ty \wedge Hxy))$

Someone is always happy → $\exists x(Px \wedge \forall y(Ty \supset Hxy))$

Someone (or other) is always happy → $\forall x(Tx \supset \exists y(Py \wedge Hyx))$

The detective is never happy → $\exists x((Dx \wedge \forall y(Dy \supset x = y)) \wedge \neg\exists z(Tz \wedge Hxz))$

(Notice the difference between the second and third sentences above.)

Finally, letting Px = 'x is a person', Lx = 'x is a place (a location)', and $A①②$ = '① is admired at ②',

Everyone is admired somewhere → $\forall x(Px \supset \exists y(Ly \wedge Axy))$

Someone is admired everywhere → $\exists x(Px \wedge \forall y(Ly \supset Axy))$

Everywhere someone (or other) is admired → $\forall x(Lx \supset \exists y(Py \wedge Ayx))$

The detective isn't admired anywhere → $\exists x((Dx \wedge \forall y(Dy \supset x = y)) \wedge \neg\exists z(Lz \wedge Axz))$

4.14 The Domain of Discourse

Variables in sentences in $\mathcal{L}_p$ are taken to range indiscriminately over everything. Take the English sentence below (uttered in desperation):

Everything is lost.

How would you translate this sentence into $\mathcal{L}_p$ (letting Lx = 'x is lost')? What could be easier?

$\forall xLx$

Thus translated, the sentence states that everything—literally *everything*: people, stars, prime numbers, grains of sand on the surface of Mars—is lost. Ridiculous, maybe, but in the absence of further information, this is exactly how you should translate the sentence. The messy specter of context rears its head again.

The set of individuals over which variables in a sentence (or a collection of sentences, or a language) range is called the *domain of discourse* for that sentence (collection of sentences, language).

The domain of discourse for sentences in $\mathcal{L}_{\mathcal{P}}$, unless otherwise specified, is unrestricted: it includes everything.

You can always narrow the domain explicitly. If, for instance, in saying that

Everything is lost.

you mean to be saying that

Everything in Mallacoota is lost.

(perhaps as the result of a bushfire), you could express this in $\mathcal{L}_{\mathcal{P}}$ (letting $I①② = ①$ is in ②) as

$$\forall x(Ixm \supset Lx)$$

Natural languages are not used in a vacuum. In conversing with others, you rely heavily on contextual information in deciding how to express your thoughts and how to interpret what others mean by their utterances. The significance of context put in an appearance in the discussion of definite descriptions in § 4.10. If 'the philosopher is wise' is analyzed along the lines suggested by Russell ('There is *exactly one* philosopher, *x*, and *x* is wise'), then most uses of definite descriptions depend on an implicit narrowing of the domain—to those present in the speaker's vicinity, for instance.

You might try to build these contextual restrictions into the sentence: 'The philosopher in the kitchen on 20 January 2021'. It is far from clear, however, that contextual information could be ever be made entirely explicit. (You will have noticed that 'the kitchen' is itself a definite description.)

Suppose, as seems likely, the implicit narrowing we use in interpreting language could never be fully spelled out. Were that so, an explanation of the semantics of English and other natural languages would require an appeal to something in addition to the semantics of individual sentences. How might that work?

Suppose I set out to explain your understanding of the sentence 'The philosopher is wise' used on a particular occasion. I recognize that your way of understanding the sentence is colored by your grasp of the context in which the sentence is used. Perhaps you use your contextual knowledge to 'add to' the sentence behind the scenes. The beefed-up sentence would incorporate an explicit representation of what you mean that does not require an appeal to context.

Philosophers have long dreamed of doing just this, creating a perfect, self-contained language, the sentences of which do their job without reliance on context. The results have not been encouraging. No matter how much is added to a sentence, a grasp of its meaning on a particular occasion would apparently require an understanding of the context. Background, contextual knowledge appears to differ *in kind* from explicit knowledge.

When you say 'Everything is lost' on a particular occasion, you do not take yourself to be making a claim about absolutely everything. When you say that everyone is invited to a party at your house, you do not mean, nor does your audience understand you to mean, literally *everyone*.

Even supposing that the role of context is ineliminable, you can and do take explicit steps to narrow the class of individuals over which you take your discourse to range. One narrowing technique is mirrored in $\mathcal{L}_{\mathcal{P}}$. You can narrow a claim by specifying that it concerns only a particular class of individuals. You invite *everyone in the office* to a party, for instance.

The sentence about Mallacoota above is another example of this kind of explicit narrowing. So long as the goal is to reproduce in $\mathcal{L}_{\mathcal{P}}$ sentences the explicit truth conditions of which *approximate* the truth conditions of ordinary English utterances, this device helps, although, inevitably,

it falls short. Were the aim of logic to expunge all uncertainty, all indeterminacy in language, the result would only serve to increase the distance between logic and ordinary speech. A more modest approach is preferable.

Context is not going away, but, in using $\mathcal{L}_\mathcal{P}$, you can allow the same latitude in the interpretation of sentences in $\mathcal{L}_\mathcal{P}$ that you would allow in the interpretation of English sentences. As a result, the significance of a sentence in $\mathcal{L}_\mathcal{P}$ could vary from occasion to occasion. That is so in any case, however, and it is certainly so for utterances of English sentences.

Just as you recognize that 'everything' in 'everything is lost' can denote different everythings on different occasions, you recognize that the $\mathcal{L}_\mathcal{P}$ sentence

$$\forall x Lx$$

can range over different individuals at different times. Allowing a role for context, you would not be lowering your sights, merely conceding the inevitable. The moral: in using $\mathcal{L}_\mathcal{P}$, what variables are taken to range over depends both on the predicates contained in a sentence and on contextual factors, stated and unstated.

Why not make a virtue of necessity? Imagine that you are setting out to formulate axioms of arithmetic and to prove theorems from those axioms. Your interest would lie only with numbers, their properties, and relations. With that in mind, you might make this explicit in each individual sentence. Each sentence would explicitly limit the range of its variables to numbers, just as in the sentence above, the range of variables was limited to things in Mallacoota. After a while, however, that might begin to seem pointlessly repetitive. For

There is an even prime number.

you would write something like

$$\exists x(Nx \wedge (Ex \wedge Px))$$

(supposing Nx = 'x is a natural number', Ex = 'x is even', and Px = 'x is prime'). The sentence

Every number has a successor.

would be expressed in $\mathcal{L}_\mathcal{P}$ (letting $S①② =$ '① is the successor of ②')

$$\forall x(Nx \supset \exists y((Ny \wedge x \neq y) \wedge Syx))$$

In each case the sentence explicitly restricts the domain to the class of natural numbers, Nx. (The natural numbers are the whole numbers 0, 1, 2, 3, 4, . . .)

Taking a different tack, you might take a step back and impose a restriction in the metalanguage on the class of objects over which variables in the object language are to be taken to range. You might, for instance, announce that the domain of discourse is the class of natural numbers. In so doing, you would avoid having to include in every sentence an indication that you are talking only about members of the class of numbers, and not cabbages or kings. The sentence

There is an even prime number.

could be translated into $\mathcal{L}_\mathcal{P}$ as

$$\exists x(Ex \wedge Px)$$

Compare this translation to the earlier translation that presumed an unrestricted domain. You are concerned only with individuals belonging to the class of natural numbers, so, in saying that something is both even and prime, you are saying that some *number* is both even and prime.

By restricting the domain in this way, a comparable economy could be realized for any sentence concerned with numbers.

Every number has a successor.

is translated into $\mathcal{L}_p$, assuming a domain restricted to numbers, as

$$\forall x \exists y\, (x \neq y \wedge Syx)$$

Again, compare this translation to the translation of the same sentence above written without benefit of a restricted domain.

Introducing a restricted domain of discourse is not something to be taken lightly. Restricting a domain requires stage setting. You must supply the context by making it clear to your intended audience what you have in mind. Domain restricting is useful when you are faced with the prospect of formulating a large number of complex sentences, all of which share a subject matter. Otherwise, it is best to assume an unrestricted domain and rely on context to pick up the slack.

Exercises 4.14

Translate the English sentences below into $\mathcal{L}_p$, first assuming an unrestricted domain of discourse, then assuming a domain restricted to the class of sleuths. Let Sx = 'x is a sleuth'; Bx = 'x is brave'; A①② = ① admires ②'. Use appropriate lowercase letters for names.

1. Every sleuth admires some sleuth.
2. Every sleuth admires some brave sleuth.
3. Some sleuth admires every sleuth.
4. Some brave sleuth admires every sleuth.
5. There is exactly one brave sleuth.
6. There is at most one brave sleuth.
7. The brave sleuth admires Fenton.
8. There are at most two sleuths.
9. Every sleuth admires the brave sleuth.
10. No sleuth admires any sleuth who isn't brave.

4.15 The Syntax of $\mathcal{L}_\mathcal{P}$

A description of the syntax of $\mathcal{L}_\mathcal{P}$ parallels the description of the syntax of $\mathcal{L}_\mathcal{S}$ in § 2.20. The quantificational structure of $\mathcal{L}_\mathcal{P}$ adds a measure of complexity to the task, but its overall character is the same. You begin by specifying classes of symbols from which sentences of $\mathcal{L}_\mathcal{P}$ are to be constructed. The class of logical connectives, ***C***, and the class of parentheses, ***P***, are the same in $\mathcal{L}_\mathcal{P}$ and $\mathcal{L}_\mathcal{S}$:

C truth-functional connectives {¬, ∧, ∨, ⊃, ≡}

P left and right parentheses {(,)}

$\mathcal{L}_\mathcal{S}$ includes a collection of sentential constants, themselves (atomic) sentences of $\mathcal{L}_\mathcal{S}$. In $\mathcal{L}_\mathcal{P}$ these are replaced with individual terms, predicate letters, and quantifiers designated by Greek letters: α designates the set of individual variables; β designates the set of individual constants; Φ designates the set of predicate letters; and Υ (upsilon) designates the set of quantifiers.

α $\{u, v, w, x, y, z\}$

β $\{a, b, c, \ldots, t\}$

Φ $\{A, B, C, \ldots, Z\}$

Υ $\{\forall u, \forall v, \ldots, \forall z, \exists u, \exists v, \ldots, \exists z\}$

Next, the set of unquantified atomic sentences, Π (pi), is specified. Π comprises finite strings of symbols consisting of some member of ϕ followed by one or more members of β (that is, a predicate letter followed by one or more individual constants):

$$\Pi\ \{\phi_0\beta_0,\ \phi_0\beta_1,\ \phi_0\beta_2,\ \ldots,\ \phi_1\beta_0,\ \ldots,\ \phi_0\beta_0\beta_1,\ \ldots,\ \phi_n\beta_0\ldots\beta_n\}$$

In Π, $\phi_0, \phi_1, \ldots, \phi_n$ are members of Φ (*A*, *B*, *C*, . . ., *Z*) and $\beta_0, \beta_1, \ldots, \beta_n$ are members—not necessarily distinct—of β (*a*, *b*, *c*, . . ., *t*). Expressions $\phi_n\beta_0\ldots\beta_{nn}$ must be finite. Π includes, then, *Aa*, *Aab*, *Faa*, *Gfh*, and so on. Π does *not* include expressions containing variables—*Gxa*, *Fx*, . . . Members of Π are simple, self-standing, $\mathcal{L}_\mathcal{S}$-like unquantified sentences.

What of sentences that express identity relations: '$a = b$', '$c \neq d$', for instance? Identity is a dyadic relation that could be expressed by a two-place predicate, $I①②$. The two identity statements above could be represented as *Iab*, ¬*Icd*. As explained in § 4.09, $\mathcal{L}_\mathcal{P}$ honors identity with its own distinctive symbol, =, owing to the central role of identity in translations. As far as the syntax of $\mathcal{L}_\mathcal{P}$ is concerned, however, identity is just one more two-place relation, $I①②$. Treating identity in this way allows the syntax to be simplified.

Now it is possible to provide a *recursive* specification of the set of sentences of $\mathcal{L}_\mathcal{P}$ analogous to rules used for constructing sentences in $\mathcal{L}_\mathcal{S}$. This set, Σ (uppercase sigma), is the set of sentences lacking quantifiers, those built solely from members of ***C*** (the logical constants), ***P*** (parentheses), and Π (*Aa*, *Aab*, *Cnm*, . . .).

Σ can then be extended to include sentences incorporating quantifiers and variables. As in the syntax of $\mathcal{L}_\mathcal{S}$, a *bounded string* is a string of symbols enclosed between left and right parentheses. Lowercase Greek *sigmas* (σ_i, σ_j) designate arbitrary members of Σ.

Rules reminiscent of $\mathcal{L}_\mathcal{S}$ get the ball rolling.

1. Every member of Π is a member of Σ.
2. If σ_i is a member of Σ, then $\neg\sigma_i$ is a member of Σ.
3. If σ_i and σ_j are members of Σ, then $(\sigma_i \wedge \sigma_j)$ is a member of Σ.
4. If σ_i and σ_j are members of Σ, then $(\sigma_i \vee \sigma_j)$ is a member of Σ.
5. If σ_i and σ_j are members of Σ, then $(\sigma_i \supset \sigma_j)$ is a member of Σ.
6. If σ_i and σ_j are members of Σ, then $(\sigma_i \equiv \sigma_j)$ is a member of Σ.

These rules, just as those used to produce the class of sentences of $\mathcal{L}_S$, are recursive. They begin with a base clause (rule 1), stipulating that members of Π are members of Σ. The remaining rules provide recipes for generating the remaining members of Σ from this stipulated basis. The recursive character of these generative rules is expressed in their taking as inputs members of Σ and producing as outputs new members of Σ.

Thus far, Σ includes $\mathcal{L}_{\mathcal{P}}$ sentences of the form

$$Fa$$

$$(Fa \supset Ga)$$

$$((Gab \wedge Fmn) \vee \neg Ps)$$

As with $\mathcal{L}_S$, the outermost parentheses can, in practice, be left off when they enclose a self-standing sentence.

Rules 1–6 yield only sentences of $\mathcal{L}_{\mathcal{P}}$ containing neither quantifiers nor variables. Sentences with quantifiers are obtained by operating on these sentences. This is accomplished by means of a pair of additional rules, one for each quantifier.

7. Suppose $\phi\beta_0 \ldots \beta_n$ is a member of Σ containing β_i, a member of β, and that α_i is a variable, not in $\phi\beta_0 \ldots \beta_n$. Let $\phi\alpha_i$ be the string obtained from $\phi\beta_0 \ldots \beta_n$ by replacing at least one occurrence of β_i by α_i. Then $\forall\alpha_i(\phi\alpha_i)$ is a member of Σ.
8. Suppose $\phi\beta_0 \ldots \beta_n$ is a member of Σ containing β_i, a member of β, and that α_i is a variable, not in $\phi\beta_0 \ldots \beta_n$. Let $\phi\alpha_i$ be the string obtained from $\phi\beta_0 \ldots \beta_n$ by replacing at least one occurrence of β_i by α_i. Then $\exists\alpha_i(\phi\alpha_i)$ is a member of Σ.

Rules 1–8 set out a recipe for constructing sentences of $\mathcal{L}_{\mathcal{P}}$. A final rule makes it explicit that *only* objects satisfying these conditions count as sentences of $\mathcal{L}_{\mathcal{P}}$:

9. All and only members of Σ are sentences of $\mathcal{L}_{\mathcal{P}}$.

Although the overall form of the definition is familiar from the recursive characterization of *sentence of* $\mathcal{L}_S$ in § 2.20, rules 7 and 8 call for comment. These rules are best understood through examples. Consider the $\mathcal{L}_{\mathcal{P}}$ sentence

$$\forall x(Fx \supset Gx)$$

Neither *Fx* nor *Gx* counts as a member of Σ; each contains a free variable, *x*, so neither counts as a sentence of $\mathcal{L}_\mathcal{P}$. For the same reason, the expression

$$(Fx \supset Gx)$$

does not count as a sentence. In contrast,

$$(Fa \supset Ga)$$

is a member of Σ, hence a sentence. Both *Fa* and *Ga* are members of Π, and so, by rule 1, count as members of Σ. According to rule 5, a ⊃ flanked by members of Σ is a member of Σ, so the expression above is a member of Σ. Rule 7 allows for the replacement of one or more occurrences of the individual constant *a* in this expression with a variable. Suppose both occurrences of *a* were replaced by *x*, thereby obtaining

$$(Fx \supset Gx)$$

which, with the addition of a universal quantifier, (∀*x*), yields a quantified sentence of $\mathcal{L}_\mathcal{P}$, a respectable member of Σ.

$$\forall x(Fx \supset Gx)$$

Rule 7 permits the replacement of *one or more* occurrences of *a* with a variable, so the very same input expression could yield

$$\forall x(Fx \supset Ga)$$

$$\forall x(Fa \supset Gx)$$

$$\forall y(Fy \supset Gy)$$

$$\forall y(Fa \supset Gy)$$

Rule 7 includes a clause specifying that the variable α_i does not already occur in $\phi\beta_0 \ldots \beta_n$. The prohibition blocks expressions of the form

$$\forall x \exists x Fxx$$

By rule 1, the expression *Faa* is a member of Σ, hence a sentence. An application of rule 8 yields a new member of Σ, the quantified sentence

$$\exists x Fxa$$

So far, so good. Suppose, now, rule 7 is applied to this sentence. In so doing, the prohibition requires that the remaining occurrence of *a* be replaced by some variable *not already present in the sentence*. As a result, rule 7 permits

$$\forall y \forall x Fxy$$

but not

$$\forall x \exists x Fxx$$

The same prohibition blocks expressions of the form

$$\forall x(Fx \supset \forall x Gx)$$

while allowing

$$\forall x Fx \supset \forall x Gx$$

and

$$\forall x Fx \supset \exists y Gy$$

Rules 7 and 8, in light of this clause, cannot be used to generate expressions that include quantifiers the variables of which match and the scope of which overlaps.

Rules 7 and 8 exclude, as well, expressions of the form

$$\forall x(Fa \supset Ga)$$

Such expressions incorporate *vacuous* quantifiers, quantifiers that pick up no variables.

Finally, rule 9 excludes from the set of sentences of $\mathcal{L}_\mathcal{P}$ expressions not constructed in accord with rules 1–9. Taken together, the nine rules and their accompanying definitions provide necessary and sufficient conditions for sentencehood in $\mathcal{L}_\mathcal{P}$. The conditions are *sufficient* because anything satisfying them counts as a sentence of $\mathcal{L}_\mathcal{P}$; the conditions are *necessary* because *only* items satisfying them count as sentences of $\mathcal{L}_\mathcal{P}$.

4.16 The Semantics of $\mathcal{L}_\mathcal{P}$

The syntax of $\mathcal{L}_\mathcal{P}$ builds on the syntax of $\mathcal{L}_\mathcal{S}$. $\mathcal{L}_\mathcal{P}$'s semantics extends the semantics of $\mathcal{L}_\mathcal{S}$ in a comparable fashion. As in $\mathcal{L}_\mathcal{S}$, the semantics of $\mathcal{L}_\mathcal{P}$ begins with an account of what an *interpretation* of a sentence involves, then, exploiting the recursive account of sentences in $\mathcal{L}_\mathcal{P}$, proceeds to spell out what it is for a sentence in $\mathcal{L}_\mathcal{P}$ to be *true under an interpretation*.

An interpretation, $\mathcal{I}$, of $\mathcal{L}_\mathcal{P}$ consists in

(i) a nonempty, finite or infinite, domain of objects, D;

(ii) an assignment to each individual constant (a, b, c, . . ., t) a member of D;

(iii) an assignment to each monadic predicate (A, B, C, . . ., Z) a set of objects, and to each n-adic predicate a set of ordered n-tuples of objects.

Individual constants and predicates can be said to designate objects and classes of objects assigned to them.

Note that clause (ii) allows for the assignment of an object to more than one individual constant, but no individual constant can be ambiguous; no constant can be assigned more than one object. The same object could be assigned to a and to b, for instance, but distinct objects cannot be assigned to a. This reflects the fact that a single individual can be designated by more than one name (recall Phosphorus and Hesperus in § 4.09). Clause (iii) places an analogous restriction on predicates.

A domain might consist of students studying logic, a collection of stars, or the natural numbers. A domain could just as easily consist of a wildly gerrymandered collection of objects: Socrates, the number two, and this book. The only restriction on collections of objects that make up domains is that the sets must contain at least one object.

Given an interpretation, $\mathcal{I}$, it is possible to spell out what it is for a sentence true under $\mathcal{I}$, first for unquantified sentences of $\mathcal{L}_\mathcal{P}$, then for sentences containing quantifiers.

1. If σ_i is a member of Π, then σ_i is true under $\mathcal{I}$, if and only if the objects $\mathcal{I}$ assigns to the individual constants of σ_i are members of the set of ordered *n*-tuples assigned to the predicate contained in σ_i.

Recall that σ_i is any member of Π, the set of unquantified simple sentences of $\mathcal{L}_\mathcal{P}$, those lacking both variables and connectives (*Fa*, *Gb*, *Hcd*, etc.). On rule 1, σ_i is true under an interpretation, $\mathcal{I}$, just in case the object or ordered *n*-tuple of objects assigned by $\mathcal{I}$ to the individual constant or constants of σ_i are members of the set of objects or ordered *n*-tuples of objects assigned to its predicate. (Think of an object by itself as a *one*-tuple.)

A sentence such as

Fab

is true under $\mathcal{I}$, just in case the objects assigned to *a* and *b* are members of the set of pairs or ordered pairs assigned to *F*, such that *a* is the first member, *b* the second.

2. $\neg\sigma_i$ is true under $\mathcal{I}$, if and only if σ_i is not true under *I*.
3. $(\sigma_i \wedge \sigma_j)$ is true under $\mathcal{I}$, if and only if σ_i is true under $\mathcal{I}$ and σ_j is true under $\mathcal{I}$.
4. $(\sigma_i \vee \sigma_j)$ is true under $\mathcal{I}$, if and only if σ_i is true under $\mathcal{I}$ or σ_j is true under $\mathcal{I}$, or both.
5. $(\sigma_i \supset \sigma_j)$ is true under $\mathcal{I}$, if and only if σ_i is not true under $\mathcal{I}$ and σ_j is true under $\mathcal{I}$, or both.
6. $(\sigma_i \equiv \sigma_j)$ is true under $\mathcal{I}$, if and only if both σ_i and σ_j are true under $\mathcal{I}$ or neither is true under $\mathcal{I}$.

These rules are counterparts of those used in § 2.21 to characterize the notion of truth under an interpretation for $\mathcal{L}_\mathcal{S}$. This is unsurprising. This segment of $\mathcal{L}_\mathcal{P}$ consists of unquantified atomic sentences and truth functions of atomic sentences.

A reminder: to simplify the presentation, sentences of the form $a = b$ will be treated as predications of the form *Iab*. The only difference between $I①②$ and an ordinary two-place predicate is that the interpretation of $I①②$ remains fixed: on *every* interpretation, *Iab* is true just in case *a* and *b* co-designate, that is, the object assigned to *a* is one and the same as the object assigned to *b*.

Consider an application of rules 1–6 in a particular case: an interpretation of $\mathcal{L}_\mathcal{P}$ having as its domain the planets. Constants designate planets and predicates designate sets of *n*-tuples as follows:

R:	the set of ringed planets
A:	the set of planets having an atmosphere
G:	the set of ordered pairs of planets in which the first planet is greater in mass than the second
a:	Mars
b:	Earth
c:	Uranus

Call this interpretation $\mathcal{I}^*$. Now consider the sentence

$$Gbc$$

According to rule 1, this sentence is true under $\mathcal{I}^*$ just in case the objects assigned by $\mathcal{I}^*$ to b and c are such that $\langle b,c\rangle$ is a member of the set of ordered pairs assigned by $\mathcal{I}^*$ to G. In English: the sentence is true under $\mathcal{I}^*$ just in case Earth is more massive than Uranus. (Earth's mass does not exceed Uranus's, so, under $\mathcal{I}^*$, Gbc is false.)

Now reflect on a more complex sentence:

$$(Aa \wedge Rc) \supset \neg Gac$$

By rule 1, you know that Aa is true under $\mathcal{I}^*$ if and only if Mars has an atmosphere, and Rc is true if and only if Uranus is ringed. Rule 3 tells us that $(Aa \wedge Rc)$ is true under $\mathcal{I}^*$ just in case both Aa and Rc are true. (Both *are* true under the interpretation, so the conjunction is true.) Again, by rule 1, you know that Gac is true if and only if Mars is greater in mass than Uranus. According to rule 2, $\neg Gac$ is true under $\mathcal{I}^*$ just in case Gac is not true. The mass of Uranus is greater than that of Mars, so Gac is not true under the interpretation, and $\neg Gac$ is true. Finally, by rule 5, a conditional sentence is true under an interpretation if and only if either its consequent is true under that interpretation, or its antecedent is false, or both. In the present case, the consequent, $\neg Gac$, is true under $\mathcal{I}^*$, so the sentence as a whole is true under that interpretation.

Although this might strike you as impossibly complicated, if you take a few minutes to work through the example, you will see that it simply makes explicit the implicit principles you have been deploying all along in translating sentences into $\mathcal{L}_\mathcal{P}$. The same is true for quantified sentences.

The semantics of quantification begins with the notion of β-variant interpretations. (Here I am following Benson Mates, *Elementary Logic*, 2nd ed. [New York: Oxford University Press, 1972], chap. 4.) Suppose that $\mathcal{I}_i$ and $\mathcal{I}_j$ are interpretations of $\mathcal{L}_\mathcal{P}$, and β is an individual constant (a, b, c, . . ., t). Then $\mathcal{I}_i$ is a β-variant of $\mathcal{I}_j$ just in case $\mathcal{I}_i$ and $\mathcal{I}_j$ differ at most with respect to the object they assign to β. Note that, if $\mathcal{I}_i$ and $\mathcal{I}_j$ differ at most with respect to the object they assign to β, then $\mathcal{I}_i$ and $\mathcal{I}_j$ have the same domain. Further, on this characterization $\mathcal{I}_i$ is a β-variant of itself.

Returning to the example used earlier, a β-variant interpretation would be one, the domain of which is the planets, but in which b designates Uranus, not Earth, and A designates not the set of planets having an atmosphere but the set of ringed planets.

Now suppose that $\forall\alpha\phi\alpha$ and $\exists\alpha\phi\alpha$ are sentences of $\mathcal{L}_\mathcal{P}$ consisting of a quantifier followed by an expression containing no free variable other than α. Were some variable other than α free in either, neither $\forall\alpha\phi\alpha$ nor $\exists\alpha\phi\alpha$ would be sentences of $\mathcal{L}_\mathcal{P}$ because they would contain a free variable not captured by $\forall\alpha$ or by $\exists\alpha$. Finally, let $\phi\alpha/\beta$ be the result of replacing every free occurrence of α in $\phi\alpha$ with an individual constant, β; that is, it is the result of replacing every instance of the variable α freed when $\forall\alpha$ or $\exists\alpha$ are removed with a constant, β.

You can see how this works by supposing that $\forall\alpha\phi\alpha$ is the sentence

$$\forall x(Ax \supset \exists yRy)$$

In this sentence, α is x, and $\phi\alpha$ is the expression minus the universal quantifier, $\forall x$:

$$Ax \supset \exists yRy$$

The expression is not a sentence because it contains a free variable, x. Now, $\phi\alpha/\beta$ is the result of replacing this x with some individual constant, β.

$$Aa \supset \exists yRy$$

This is a sentence of $\mathcal{L}_p$, and so, given an interpretation, it has a truth value.

When a variable freed in this way is replaced by an individual constant, it is replaced by the first individual constant not already present in $\phi\alpha$. You could take the class of individual constants to be ordered, a being the first constant, b being the second, c being the third, and so on.

Now a complete characterization of 'true under $\mathcal{I}$' for $\mathcal{L}_p$ is within range.

7. $\forall\alpha\phi\alpha$ is true under $\mathcal{I}$, if and only if $\phi\alpha/\beta$ is true under every β-variant of $\mathcal{I}$.
8. $\exists\alpha\phi\alpha$ is true under $\mathcal{I}$, if and only if $\phi\alpha/\beta$ is true under at least one β-variant of $\mathcal{I}$.
9. σ_i is false under $\mathcal{I}$, if and only if σ_i is not true under $\mathcal{I}$.

How are rules 7 and 8 to be understood? First, consider a simple case, given the interpretation, $\mathcal{I}^*$, set out earlier.

$$\forall xRx$$

According to rule 7, this sentence is true under $\mathcal{I}^*$ just in case Ra is true under every a-variant of $\mathcal{I}^*$. An a-variant of $\mathcal{I}^*$ is an interpretation differing from $\mathcal{I}^*$ at most with respect to the individual assigned to the individual constant a. $\mathcal{I}^*$ assigns Mars to a. One a-variant of $\mathcal{I}^*$ is the interpretation that assigns Earth to a. Another a-variant assigns Uranus to a. Applying rule 7, $\forall xRx$ is true under $\mathcal{I}^*$ if and only if Ra is true under *every* a-variant of $\mathcal{I}^*$. Thus, $\forall xRx$ is true under $\mathcal{I}^*$ just in case Mars, Earth, and Uranus are each ringed. (Mars and Earth lack rings, so $\forall xRx$ is not true under $\mathcal{I}^*$.)

Now consider the sentence discussed earlier:

$$\forall x(Ax \supset \exists yRy)$$

This sentence, on rule 7, is true under $\mathcal{I}^*$ if and only if the sentence

$$Aa \supset \exists yRy$$

is true under every a-variant of $\mathcal{I}^*$, so the sentence is true under $\mathcal{I}^*$ only if

$$Ab \supset \exists yRy$$

$$Ac \supset \exists yRy$$

are true as well.

A conditional sentence is true under an interpretation just in case either its antecedent is false or its consequent is true, or both. The consequent of the conditional under consideration is itself a quantified sentence:

$$\exists yRy$$

By rule 8, this sentence is true under $\mathcal{I}^*$ just in case Rb is true under *some* b-variant of $\mathcal{I}^*$. One b-variant of $\mathcal{I}^*$ assigns Uranus to b. If Uranus has rings, Rb is true under that interpretation, hence $\exists yRy$ is true under $\mathcal{I}^*$. Now, taking the sentence as a whole, you can see that the sentence is true under $\mathcal{I}^*$

if and only if either it is not true that every planet has an atmosphere, or some planet has rings, or both. Given that it is true that Uranus has rings, the sentence is true.

This is a lot to take in, but, if you take the trouble to think through it all, you can see how the characterization of 'true under $\mathcal{I}$' applies to more complex sentences. Before pushing ahead to derivations of $\mathcal{L}_\mathcal{P}$, I propose to step back and reflect briefly on what the structure of $\mathcal{L}_\mathcal{P}$ implies about the structure of reality. Does the fact that $\mathcal{L}_\mathcal{P}$ can be used to describe our universe have any implications for the character of what is being described?

Vocabulary Expansion

$\mathcal{L}_\mathcal{P}$ is fitted out with a finite vocabulary: twenty individual constants, twenty-six predicates, and six variables.

You might, for some particular purpose, need to expand the vocabulary of $\mathcal{L}_\mathcal{P}$. The result would be an extended version of $\mathcal{L}_\mathcal{P}$, $\mathcal{L}_\mathcal{P}^*$. In addition to the individual constants $a, b, c, \ldots$, $\mathcal{L}_\mathcal{P}^*$ might include constants distinguished by subscripts: $a_0, a_1, a_2, \ldots a_n, \ldots b_0, \ldots b_n, \ldots t_n$. This would allow for the open-ended creation of individual names. The stock of individual variables and predicates could be extended in the same way.

One advantage of $\mathcal{L}_\mathcal{P}^*$ over $\mathcal{L}_\mathcal{P}$ is that $\mathcal{L}_\mathcal{P}^*$ captures an important feature of natural languages. A natural language cannot run out of words. Although English contains a finite number of terms, it provides mechanisms for the creation of endless new terms. The possibility enlarging the vocabulary of $\mathcal{L}_\mathcal{P}$ when the occasion arises dramatically increases its usefulness and expressive power, and brings it into closer alignment with natural languages.

4.17 Logic and Ontology

A tradition running deep in Western philosophy is encapsulated in the idea that language mirrors reality: linguistic categories reflect categories of being. Words name features of reality; features thus designated serve as the words' meanings. The sentence 'Socrates is wise', contains a pair of names, 'Socrates' and 'is wise'. The former names Socrates, the latter, the property of being wise. The meaning of 'Socrates' is the man, Socrates, and the meaning of 'is wise' is the property of being wise, *wisdom*.

How, on such a picture, could you account for falsehood? A false sentence is not meaningless. But the sentence 'Socrates is spherical' corresponds to nothing in reality, and so would seem to lack meaning. One way to circumvent this difficult would be to suppose that reality contains entities corresponding to 'Socrates' and 'sphericality', but that these entities are not configured in the way the sentence says they are configured. Socrates lacks the property.

What of 'Athena is wise'? Is the sentence false? No object answering to 'Athena' exists. 'Athena', then, lacks a meaning. Sentences containing meaningless terms are meaningless, so the sentence must be meaningless. The sentence is certainly meaningful, however, so there must be some *other* entity fit to serve as the meaning of 'Athena'.

Some philosophers have thought that, in such cases, words are being used not to designate things, but to designate concepts or ideas. So, although there is no Athena, there is the *idea* of Athena, a

stand-in for a real Athena. Problem solved? If the meaning 'Athena' is not the goddess Athena but the idea of Athena, then 'Athena is wise' would mean that an idea, the idea of Athena, is wise, and that is not what the sentence means.

Such thoughts have led philosophers at various times to propose fundamental linguistic reforms. If we could clean up language, our talk about reality would avoid nonsense and confusion. We would still utter the occasional falsehood, but falsehood is curable in a way that confusion is not.

Suppose names were reserved for existing individuals, 'Socrates', for instance. Merely apparent names such as 'Athena' and 'Euterpe' could be placed in a distinct grammatical category that marks them as designating not ordinary objects but extraordinary objects. The dream is to construct an ideal language, one in which the structure of reality is an open book. Perhaps $\mathcal{L}_\mathcal{P}$ represents a step along the road toward the fulfilment of this dream.

More likely, the dream is untenable. Language is used to express beliefs about reality among other things. You can discover what you believe by listening to yourself. Other things equal, true beliefs, true theories, are preferred to false beliefs and theories. Your view of reality, if not reality as it is, is revealed in what you say about it when you are sincere.

The idea that the structure of our language mirrors, or ought to mirror, the structure of reality could be turned around and transformed into the idea that our language mirrors *us*. Our beliefs and theories about the world are reflected in what we choose to say. There is reason to regard $\mathcal{L}_\mathcal{P}$ as particularly revealing in this respect. An unanticipated advantage of pressing $\mathcal{L}_\mathcal{P}$ into service in the formulation of your ideas is that, in so doing, you can come to see starkly what those ideas do and do not commit you to. You might, as a result, revise your ideas. Or you might decide that $\mathcal{L}_\mathcal{P}$ is an inadequate medium for their expression.

'To be is to be the value of a bound variable'. So says W. V. O. Quine (1908–2000; see *From a Logical Point of View* [New York: Harper & Row, 1963], 1–19). You can ascertain the character of your 'ontological commitments' by noting the sorts of object you countenance as values of variables in a language like $\mathcal{L}_\mathcal{P}$.

In setting up $\mathcal{L}_\mathcal{P}$, the character of objects was left open. This is as it should be. What there is is a matter for science, not logic, to discover. Do atoms and molecules exist? What of medium-sized items: tables, chairs, planets? A more manageable question from the perspective of logic is whether our conception of the world includes a commitment to the existence of these things. Insofar as you find it necessary to 'quantify over' variables that take atoms, molecules, and planets as values, then you are so committed.

Occasionally we discover that it is possible and, for the sake of overall plausibility, desirable to paraphrase objects away. Is there an average American? We sometimes talk as though there were. The average American, we are told, is right-handed. Thus: 'There is exactly one x such that x is an average American . . .'. If this strikes you as silly, you will find ways of paraphrasing away talk about the average American.

One question is, how far could we, and ought we, go in this direction? Might talk of tables and trees be paraphrased away, for instance, replaced by talk of clouds of particles? In such matters we are guided largely by pragmatic criteria. We accept paraphrases that tend to make our overall conception of reality simpler and more coherent. In this case our conception is simplified if we do not suppose that, in addition to individual Americans, the world contains another entity, the average American.

Might a comparable economy and coherence be achieved by eliminating reference to tables, trees, mountains? What of moral or aesthetic values? Minds? Properties? Numbers? Is our view of reality improved or muddied by supposing it to contain such entities? These questions go beyond the scope of this discussion, the aim of which is simply to sketch an application of logic not obvious to casual examination, an application the significance of which should not be underestimated.

Put to work, $\mathcal{L}_p$ affords a mirror not of reality but of your considered opinions about reality. As with any mirror, what it reveals might or might not meet with your approval. What matters is not that you like what you find, but that you can see your way to something better.

5. Derivations in $\mathcal{L}_p$

5.00 Preliminaries

Derivations in $\mathcal{L}_p$ closely resemble derivations in $\mathcal{L}_S$. Consider, for instance, the sequence

If Socrates is wise, then he is happy.

Socrates is wise.

Socrates is happy.

Letting Wx = x is wise and Hx = x is happy, this sequence can be represented in $\mathcal{L}_p$ as

1. $+ Ws \supset Hs$
2. $+ Ws$
3. $? Hs$

and the conclusion derived in a single step via *MP*:

1. $+ Ws \supset Hs$
2. $+ Ws$
3. $? Hs$
4. Hs 1, 2 *MP*

The occurrence of quantifiers in sentences can complicate derivations, but that is not always the case. Consider the English sequence below and its counterpart in $\mathcal{L}_p$:

If anything is at rest, then everything is at rest.

Not everything is at rest.

Therefore, nothing is at rest.

Letting Rx = x is at rest:

1. $+ \exists xRx \supset \forall xRx$
2. $+ \neg\forall xRx$
3. $? \neg\exists xRx$
4. $\neg\exists xRx$ 1, 2 *MT*

The application of transformation rules to sentences in $\mathcal{L}_p$ can be equally straightforward. Consider the sentence of $\mathcal{L}_p$ in the first line of the sequence above:

$$\exists xRx \supset \forall xRx$$

You could apply *Cond* to this sentence, converting the ⊃ to a ∨, and changing the valence of the antecedent:

$$\neg\exists x Rx \vee \forall x Rx$$

An application of *DeM* to this sentence would yield

$$\neg(\exists x Rx \wedge \neg\forall x Rx)$$

The disjunction has been changed to a conjunction, and the valence of each conjunct has been reversed, along with the valence of the whole expression.

CP and *IP* also apply in $\mathcal{L}_{\mathcal{P}}$ just as they do in $\mathcal{L}_{\mathcal{S}}$, as the derivation below illustrates:

1.	$+\ \forall x Fx \supset \exists x Hx$	
2.	$?\ \forall x Fx \supset (\forall x Fx \wedge \exists x Hx)$	
3.	$\forall x Fx$	
4.	$?\ \forall x Fx \wedge \exists x Hx$	
5.	$\exists x Hx$	1, 3 *MP*
6.	$\forall x Fx \wedge \exists x Hx$	3, 5 $\wedge I$
7.	$\forall x Fx \supset (\forall x Fx \wedge \exists x Hx)$	3–6 *CP*

Although application of transformation rules to quantified expressions introduces nothing new, care must be exercised in the manipulation of negation signs. When *DeM* is applied to the quantified disjunction below, for instance,

$$\exists x(Fx \vee \neg Gx)$$

the *disjunction* changes its valence, so the negation sign is inserted *inside* the existential quantifier.

$$\exists x\neg(\neg Fx \wedge Gx)$$

Compare this with a case in which *DeM* is applied to a sentence containing a quantifier that is itself a part of a disjunction:

$$\forall y\neg(\exists x Rx \vee Gy)$$

DeM yields

$$\forall y(\neg\exists x Rx \wedge \neg Gy)$$

Mastering the application of transformation rules to sentences in $\mathcal{L}_{\mathcal{P}}$ requires understanding when the rules apply to a quantified expression and when they apply to an expression that *includes* a quantifier as a part. Consider the sentence

$$\forall x(Fx \supset Gx)$$

Were you to apply *Cond* to this sentence, you would be applying the rule to a quantified expression.

$$\forall x(\neg Fx \vee Gx)$$

In contrast, when you apply *Cond* to an expression that includes a quantifier as a part:

$$\forall y(\exists x Fx \supset Gy)$$

the contained quantifier is affected:

$$\forall y(\neg\exists x Fx \vee Gy)$$

All this makes perfect sense when you think about it. As is often the case in logic, it is easier done than said.

Exercises 5.00

Construct derivations for the $\mathcal{L}_\mathcal{P}$ sequences below.

1. $+ \forall x(Fx \supset Gx)$
   $+ \forall x(Gx \vee \neg Fx) \supset \exists y Fy$
   $? \exists y Fy$

2. $+ \exists x Fx \wedge \exists x Gx$
   $+ \exists x Hx \supset \neg\exists x Gx$
   $? \neg\exists x Hx$

3. $+ \exists y Hy$
   $+ (\forall x Fx \vee \exists y Hy) \supset Fa$
   $? Fa$

4. $+ \exists x Fx \supset \exists x Gx$
   $+ \neg\exists x Gx \vee \forall x Fx$
   $? \exists x Fx \supset \forall x Fx$

5. $+ \forall x(Fx \wedge (Gx \vee Hx))$
   $? \forall x((Fx \wedge Hx) \vee (Fx \wedge Gx))$

6. $+ \forall x(Fx \supset Gx)$
   $? \forall x\neg(Fx \wedge \neg Gx)$

7. $+ \forall x(Fx \supset Gx) \equiv \exists x Fx$
   $? \forall x(Fx \supset Gx) \supset \exists x Fx$

8. $+ \exists x((Fx \wedge \neg Gx) \vee \neg Gx)$
   $? \exists x((Fx \vee \neg Gx) \wedge \neg Gx)$

9. $+ \exists x Fx \supset \exists x(Fx \wedge Gx)$
   $+ \neg\exists x(Fx \wedge Gx) \supset \exists x Fx$
   $? \exists x(Fx \wedge Gx)$

10. $+ \forall x Fx \supset \neg\exists y Gy$
    $+ \neg\exists x Hx \supset \exists y Gy$
    $? \forall x Fx \supset \exists x Hx$

11. $+ \neg\forall x Hx$
    $+ (\exists x Fx \vee \exists x Gx) \supset \forall x Hx$
    $? \neg\exists x Fx$

12. $+ \forall x Fx \supset (Ga \wedge Ha)$
    $+ (\forall x Fx \supset Ha) \supset \exists x Jx$
    $? \exists x Jx$

13. $+ \forall x(\exists y(Fy \wedge Gy) \vee (Hx \wedge Jx))$
    $? \forall x((\exists y(Fy \wedge Gy) \vee Hx) \wedge (Jx \vee \exists y(Fy \wedge Gy)))$

14. $+ \forall x(Fx \supset Gx))$
    $? \forall x(\neg Gx \supset \neg Fx) \wedge \forall x(Gx \vee \neg Fx)$

15. $+ \neg((\forall x Fx \wedge \neg Ga) \vee \exists y(Gy \wedge Hy))$
    $? \forall x Fx \supset Ga$

5.01 Quantifier Transformation

English sentences that require quantifiers when they are translated into $\mathcal{L}_\mathcal{P}$ can often be translated in more than one way. Consider the sentence

Every philosopher is wise.

It would be natural to translate this sentence into $\mathcal{L}_\mathcal{P}$ (assuming Px = x is a philosopher and Wx = x is wise) as

$$\forall x(Px \supset Wx)$$

But consider: the original English sentence could be paraphrased, albeit awkwardly, as

It's not the case that there is a philosopher who isn't wise.

The most natural way to translate *this* sentence into $\mathcal{L}_\mathcal{P}$ would be

$$\neg\exists x(Px \wedge \neg Wx)$$

How is this sentence related to the original conditional sentence above?

Before answering this question, I invite you to look at some simpler sentences couched in quasi-$\mathcal{L}_\mathcal{P}$. (In these sentences, α and ϕ [*alpha* and *phi*] represent arbitrary terms.) Consider the sentence

All αs are ϕs.

and a companion sentence

It's not the case that some αs are not ϕ.

Do these sentences have the same truth conditions? If so, they are paraphrases of one another. If the quantificational structure of $\mathcal{L}_\mathcal{P}$ corresponds to that of English, then, the proto-$\mathcal{L}_\mathcal{P}$ sentence

$$\forall\alpha\phi$$

must be logically equivalent to

$$\neg\exists\alpha\neg\phi$$

Pairs of sentences corresponding to these expressions are equivalent. (A proof of this is provided below.)

A rule for *quantifier transformation* reflects the equivalence:

$$(QT)\ \forall\alpha\phi \dashv\vdash \neg\exists\alpha\neg\phi$$

The rule licenses the replacement of a universal quantifier with an existential quantifier flanked by negation signs.

Now return to the sentences considered earlier. Recall that, although it would be natural to translate the English sentence

Every philosopher is wise.

as

$$\forall x(Px \supset Wx)$$

a case could be made for translating it using an existential quantifier:

$$\neg\exists x(Px \wedge \neg Wx)$$

When you apply (QT) to the universally quantified original, you would replace the universal quantifier with an existential quantifier flanked by negation signs.

$$\neg\exists x\neg(Px \supset Wx)$$

The result is an odd-looking existentially quantified conditional. You can, however, get from this sentence to the existentially quantified alternative sentence by applying *Cond*:

$$\neg\exists x\neg(\neg Px \vee Wx)$$

then *DeM*:

$$\neg\exists x(Px \wedge \neg Wx)$$

Armed with the quantifier transformation rule, you can derive either sentence from the other. Given that the derivation rules are truth-preserving, this amounts to a proof that the sentences are logically equivalent in $\mathcal{L}_p$, just as they are in English.

Rule QT was described as permitting the replacement of a universal quantifier by an existential quantifier flanked by negation signs. Although this description is perfectly correct, it is potentially misleading. Chapter 3 introduced the convention of treating negation signs appearing in rules as instructions to *reverse the valence* of sentences introduced by applications of the rule.

In the case at hand, this means that when you replace a universal quantifier with an existential quantifier, you (i) reverse the valence of the quantifier, and (ii) reverse the valence of the expression to the immediate right of the quantifier. The rule is a two-way rule, licensing the substitution of a universal quantifier for an existential quantifier provided the same two conditions are satisfied.

Suppose you set out to replace the universal quantifier in

$$\neg\forall x(Fx \supset Gx)$$

with an existential quantifier. The rule permits the replacement, provided (i) the valence of the quantifier is reversed (changing it in this case from negative to positive), and (ii) the valence of the expression to its immediate right is reversed.

$$\exists x\neg(Fx \supset Gx)$$

Again, the rule can be applied in the opposite direction as well. Take the existentially quantified $\mathcal{L}_p$ sentence

$$\exists x\neg(Fx \wedge \neg Gx)$$

The existential quantifier could be replaced by a universal quantifier so long as its valence and the valence of the expression to its immediate right are both reversed.

$$\neg\forall x(Fx \wedge \neg Gx)$$

Rule *QT*, then, licenses transformations of the following sorts:

$$\forall\alpha\phi \dashv\vdash \neg\exists\alpha\neg\phi$$
$$\forall\alpha\neg\phi \dashv\vdash \neg\exists\alpha\phi$$
$$\neg\forall\alpha\phi \dashv\vdash \exists\alpha\neg\phi$$
$$\neg\forall\alpha\neg\phi \dashv\vdash \exists\alpha\phi$$

These reflect a comparable pattern in English.

All αs are ϕ $\dashv\vdash$ It's not the case that some αs are not ϕ.
All αs are not ϕ $\dashv\vdash$ It's not the case that some αs are ϕ.
Not all αs are ϕ $\dashv\vdash$ Some αs are not ϕ.
Not all αs are not ϕ $\dashv\vdash$ Some αs are ϕ.

Assuming these English equivalences hold, *QT* ensures that quantifier transformation in $\mathcal{L}_{\mathcal{P}}$ corresponds to *some/all* transformations in English.

What of sentences containing more than one quantifier? Consider the sentence

$$\forall x\exists y((Fx \wedge Gy) \supset Hxy)$$

Suppose you wanted to change both quantifiers to their opposites. *QT* is subject to a general restriction common to all rules: only a single application is permitted at a time. This means that, although it does not matter *which* quantifier is transformed first, you can transform *only one quantifier at a time*.

You might begin by substituting an existential quantifier for the universal quantifier, changing the quantifier, its valence, and the valence of the expression to its immediate right. In this instance, the expression to the immediate right is itself a quantifier.

$$\neg\exists x\neg\exists y((Fx \wedge Gy) \supset Hxy)$$

Now rule *QT* can be applied to the second existential quantifier, converting it to a universal quantifier. That quantifier, its valence, and the valence of the expression to its immediate right are all changed.

$$\neg\exists x\forall y\neg((Fx \wedge Gy) \supset Hxy)$$

If you so choose, you go on to apply various other transformation rules to the remainder of the expression to yield additional logically equivalent sentences.

Exercises 5.01

Construct derivations for the $\mathcal{L}_\mathcal{P}$ sequences below using rule *QT*.

1. $+ \exists x(Fx \wedge Gx)$
   $? \neg\forall x(Fx \supset \neg Gx)$

2. $+ \forall x(Fx \supset Gx)$
   $? \neg\exists x(Fx \wedge \neg Gx)$

3. $+ \forall x((Fx \wedge Gx) \supset Hx)$
   $? \neg\exists x(Fx \wedge (Gx \wedge \neg Hx))$

4. $+ \exists x((Fx \wedge Gx) \vee \neg Hx)$
   $? \neg\forall x(Hx \wedge (Fx \supset \neg Gx))$

5. $+ \forall x(Fx \wedge Gx)$
   $? \neg\exists x((Fx \wedge Gx) \supset (Fx \supset \neg Gx))$

6. $+ \forall x((Fx \supset Gx) \wedge Hx)$
   $? \neg\exists x((\neg Hx \vee Fx) \wedge (Hx \vee \neg Gx)$

7. $+ \exists xFx$
   $? \forall x\neg Fx \supset \exists xFx$

8. $+ \forall xFx$
   $? \neg(\forall xFx \supset \neg\forall xFx)$

9. $+ \exists xFx \supset \forall x\neg Fx$
   $? \forall x\neg Fx$

10. $+ \exists x\neg Fx$
    $? \neg(\neg\forall xFx \supset \forall xFx)$

11. $+ \forall x((Fx \wedge Gx) \supset Hx)$
    $? \neg\exists x(Fx \wedge (Gx \wedge \neg Hx))$

12. $+ \forall x(Fx \supset Gx)$
    $? \neg\exists x(\neg Gx \wedge Fx)$

13. $+ \neg\forall x(Fx \supset Gx)$
    $? \exists x(Fx \wedge \neg Gx)$

14. $+ \forall x(Fx \supset (Gx \supset Hx))$
    $? \neg\exists x((Fx \wedge Gx) \wedge \neg Hx)$

15. $+ \neg\exists x((Fx \vee Gx) \wedge Hx)$
    $? \forall x((\neg Fx \wedge \neg Gx) \vee \neg Hx)$

5.02 Universal Instantiation: *UI*

Recall that a sequence $\langle \Gamma, \phi \rangle$ is valid just in case its conclusion, ϕ, is not false if its premises, Γ, are true. The familiar syllogism below is plainly valid:

All people are mortal.

Socrates is a person.

Socrates is mortal.

Were you to translate the sequence into $\mathcal{L}_\mathcal{P}$, however, given only familiar $\mathcal{L}_\mathcal{S}$ derivation rules supplemented by rule *QT*, you would not be able to derive the conclusion from the premises.

1. $+ \forall x(Px \supset Mx)$
2. $+ Ps$
3. $? Ms$

The sequence hints at an application of *MP*, but *MP* permits the derivation of the consequent of a conditional given its antecedent. *Ps* is not the antecedent of $Px \supset Mx$, however, nor, for that matter, is *Ms* its consequent. Further, the conditional expression is bound by a quantifier. An application of *MP* would violate the general restriction on rules of inference: rules of inference apply only to *whole sentences*.

Still, there is something right about the thought that a proof for the validity of the sequence above involves an application of *MP*. Consider the first sentence in the sequence, 'All people are mortal'.

$$\forall x(Px \supset Mx)$$

If all people are mortal, then it surely follows that if Socrates is a person, then Socrates is mortal. It follows as well that if Euterpe is a person, Euterpe is mortal; and if Phar Lap is a person, then Phar Lap is mortal. So, if the sentence above is true, then the following sentences must be true:

$$Ps \supset Ms$$

$$Pe \supset Me$$

$$Pp \supset Mp$$

These sentences are *instances* of the original universally quantified sentence produced by (i) dropping the quantifier and (ii) replacing the variables it bound (in this case the *x*) with an individual constant (*s*, *e*, and *p* in the sentences above).

A rule for extracting instances from universally quantified sentences, *universal instantiation* (*UI*), captures this pattern of reasoning.

$$(UI) \qquad \forall \alpha \phi \vdash \phi\alpha/\beta$$

In (*UI*), $\forall \alpha \phi$ represents any universally quantified sentence, and ϕ stands for the sentence minus the quantifier, $\forall \alpha$. $\phi\alpha/\beta$ represents the sentence that results when you

(i) drop the universal quantifier, $\forall \alpha$, and

(ii) replace each instance of the variable it bound, each α, with some individual constant, β.

The idea becomes clear once you work through a few examples.

In applying *UI* to the sentence

$$\forall x(Fx \supset Gx)$$

you drop the universal quantifier, $\forall x$, and replace the variables it bound, the *x*s, with some individual constant. *Which* individual constant you choose depends on the circumstances. Each of these sentences results from an application of *UI* to the sentence above:

$$Fs \supset Gs$$

$$Fe \supset Ge$$

$$Fp \supset Gp$$

The derivation omitted earlier can now be completed.

1.	+ $\forall x(Px \supset Mx)$	
2.	+ Ps	
3.	? Ms	
4.	$Ps \supset Ms$	1 *UI*
5.	Ms	2, 4 *MP*

In this case, the universal quantifier was dropped and each occurrence of the variable it bound, x, changed to an s. The resulting sentence can be used with the sentence in line 2 and *MP* to derive the conclusion.

This application of *UI* captures the kind of informal reasoning about the original syllogism anyone would deploy in thinking it through. Suppose all people are mortal. In that case, 'If Socrates is a person, Socrates is mortal' would be an instance of the generalization. Socrates *is* a person, so Socrates is mortal.

In applying *UI* to the sentence

$$\forall x(\exists yFy \supset Gx)$$

the universal quantifier, $\forall x$, can be dropped and the variable it bound, the x in Gx, replaced with a constant:

$$\exists yFy \supset Ga$$

in this case, an a. The existential quantifier, $\exists y$, and the variable it binds, y in Fy, remain unaffected.

When the quantifier is dropped, the variable it bound can be replaced by *any* individual constant. In this instance, the constant a was used, but any other constant would have served. The constant you choose depends on the context. If you applied *UI* to the sentence below

$$\forall y(Fy \supset Gh)$$

you might want to change the variable bound by the quantifier to an h, yielding

$$Fh \supset Gh$$

although that would depend in part on other sentences in the sequence in which the quantifier is dropped.

Arbitrary Individuals

Rule *UI* introduces names for arbitrary individuals. An arbitrary individual is not an unreliable person, but a member of a class of individuals selected at random.

Take the class of red things. Every member of the class is red, so any individual selected at random from the class is red. If tomatoes are red, then any randomly selected tomato, any arbitrary tomato, is red. In allowing a constant to be substituted for variables bound by universal quantifiers, *UI* provides a way of assigning a name to an arbitrary individual member of a class in a way that reflects a familiar pattern of reasoning.

Rule *UI* does not permit the replacement of variables with other variables. Were you to do so, derivations would include strings of symbols that were not sentences because they contained free variables. Every line of a derivation must be a sentence. In dropping the $\forall y$ quantifier in the sentence above, then, the variable it bound, y, must be replaced by an individual constant, one of the lowercase letters a, b, c, . . ., t.

Rule *UI* requires that when a constant replaces a variable, it does so consistently. Suppose you dropped the universal quantifier in the sentence

$$\forall z(((Fz \vee Gz) \wedge Hz) \supset Jz)$$

In dropping the $\forall z$ every occurrence of z must be replaced by the *same* constant.

$$((Fa \vee Ga) \wedge Ha) \supset Ja$$

Rule *UI* yields the sentence above from the original sentence, but not the sentence below.

$$((Fa \vee Gb) \wedge Ha) \supset Ja$$

As in the case of rules of inference generally, *UI* applies only to *whole sentences*. A universal quantifier can be dropped only when its scope includes the entire sentence in which the quantifier occurs. If the scope of a universal quantifier is limited to a part of a sentence or to a sentence that is itself part of a larger sentence, rule *UI* does not apply. These restrictions are illustrated in the sentences below:

$$\neg\forall x(Fx \supset Gx)$$

$$\forall x(Fx \supset Gx) \supset \exists y(Fy \wedge Gy)$$

In the first sentence, the quantifier is negated. Part of the sentence in which the quantifier occurs is outside its scope, namely, the $\neg$. In the second sentence, the universal quantifier includes within its scope only the antecedent of a more inclusive sentence. In neither case does *UI* permit the dropping of the universal quantifier, $\forall$x.

Exercises 5.02

Construct derivations for the $\mathcal{L}_{\mathcal{P}}$ sequences below, making use of rule *UI* where necessary.

1. $+ \forall x(Fx \supset Gx)$
   $+ \neg Ga$
   $? \neg Fa$

2. $+ \forall x(Fx \supset Gx)$
   $+ \forall x(Gx \supset Hx)$
   $? Fa \supset Ha$

3. $+ \forall x(Fx \supset Gx)$
   $+ \forall x\neg(Gx \wedge \neg Hx)$
   $? Ha \vee \neg Fa$

4. $+ \forall x(Fx \supset Gx)$
   $+ \neg\exists x(Gx \wedge \neg Hx)$
   $? \neg(Fa \wedge \neg Ha)$

5. $+ \forall xFx \supset \forall xGx$
   $+ \neg\exists x\neg Fx$
   $? Ga$

6. $+ \forall xFx \supset \forall xGx$
   $+ \forall x\neg Gx$
   $? \exists x\neg Fx$

7. $+ \forall x(\neg Fx \supset Hx)$
   $+ \forall x(Hx \supset Gx)$
   $? \exists x(Fx \vee Gx)$

8. $+ \forall xFx$
   $+ \forall x(Fx \supset Gx)$
   $? \exists x(Fx \wedge Gx)$

9. $+ \forall x(Fx \supset Gx)$
   $+ \forall x(Fx \vee \exists y Jy)$
   $+ \exists y Jy \supset \forall x Hx$
   $? \exists x(Gx \vee Hx)$

10. $+ \neg\exists x(\neg Fx \vee Hx)$
    $+ \forall x(Jx \supset Gx)$
    $+ \forall x(Fx \supset Jx)$
    $? \exists x(Fx \wedge Gx)$

11. $+ \forall x(Fax \supset Ga)$
    $+ \neg\exists x(Gx \wedge Fxc)$
    $? \neg\forall x Fxc$

12. $+ \forall x(Fx \supset Gx)$
    $+ \neg\exists x(Gx \wedge \neg Hx)$
    $? \forall x Fx \supset Hb$

13. $+ \forall x(Fx \supset Gx)$
    $+ \neg\exists x Gx$
    $? \neg\forall x Fx$

14. $+ \forall x(\neg Fxa \supset Gax)$
    $+ \neg\exists x Gxb$
    $? \exists x Fxa$

15. $+ \forall x \forall y(Fxy \vee Fyx)$
    $+ \forall x(\neg\exists y(Fyx \wedge \neg Gx) \wedge \neg\exists y Fxy)$
    $? \exists x Gx$

5.03 Existential Generalization: *EG*

Rule *UI* permits the dropping of universal quantifiers in a way that mimics ordinary patterns of inference of the form

All horses are quadrupeds.

Therefore, if Phar Lap is a horse, Phar Lap is a quadruped.

Representing this sequence in $\mathcal{L}_p$ (and assuming that Hx = *x* is a horse and Qx = *x* is a quadruped):

1. $+ \forall x(Hx \supset Qx)$
2. $? Hp \supset Qp$
3. $Hp \supset Qp$ 1 *UI*

UI is an instantiation rule licensing inferences from universal generalizations to *instances* of those generalizations. Some inferences move in the other direction—from sentences about instances to sentences expressible in $\mathcal{L}_p$ via quantifiers.

Phar Lap is a horse.

Therefore, there is at least one horse.

This sequence would be translated into $\mathcal{L}_p$ as

1. $+ Hp$
2. $? \exists x Hx$

The principle at work in these sequences can be expressed in a rule, *existential generalization* (*EG*), that permits inferences of this kind.

$$(EG) \qquad \phi\alpha/\beta \vdash \exists\alpha\phi$$

Again, $\phi\alpha/\beta$ represents a sentence containing one or more occurrences of an individual constant, β. $\exists\alpha\phi$ is the sentence that results from

(i) the replacement of *one or more* occurrences of β by occurrences of a variable, α, and

(ii) the addition of an existential quantifier that captures every occurrence of α in ϕ.

Rule *EG* permits the derivation of the conclusion in the sequence above.

1.	$+ Hb$	
2.	$? \exists xHx$	
3.	$\exists xHx$	1 *EG*

The rule permits, as well, derivations that reflect the pattern of inference exhibited in the English sequence below:

All horses are quadrupeds.

Phar Lap is a horse.

There is at least one quadruped.

The sequence can be translated into $\mathcal{L}_{\mathcal{P}}$ and proven valid using *EG*:

1.	$+ \forall x(Hx \supset Qx)$	
2.	$+ Hp$	
3.	$? \exists xQx$	
4.	$Hp \supset Qp$	1 *UI*
5.	Qp	2, 4 *MP*
6.	$\exists xQx$	5 *EG*

Rule *EG*, like *UI*, applies only to whole sentences, not to sentences that are themselves parts of other sentences. When an existential quantifier is added to a sentence, then, the quantifier's scope must include the whole sentence. The sequence below illustrates both correct and incorrect applications of rule *EG*.

1.	$+ Fa \wedge Ga$	
2.	$+ \neg(Fa \wedge Ga)$	
3.	$\exists x(Fx \wedge Gx)$	1 *EG*
4.	$\exists x\neg(Fx \wedge Gx)$	2 *EG*
5.	$\exists xFx \wedge Ga$	1 *EG* (*not permitted*)
6.	$\neg\exists x(Fx \wedge Gx)$	2 *EG* (*not permitted*)

In lines 5 and 6, *EG* has been applied to sentences that are themselves parts of larger sentences. In neither case does the quantifier include within its scope the entire sentence to which it is added.

EG requires that when an individual constant is converted to a variable, the variable be bound (picked up) by the added existential quantifier. Given the sentence

$$\forall x(Fx \supset Ga)$$

EG would not permit the addition of an $\exists x$ and the conversion of a to x.

$$\exists x \forall x(Fx \supset Gx)$$

In this case the x in Gx is picked up not by the existential quantifier but by a universal quantifier, $\forall x$, already present in the sentence. *EG would* permit the following sentence to be inferred:

$$\exists y \forall x(Fx \supset Gy)$$

Here the y in Gy is bound by the newly introduced existential quantifier.

Although *EG* does not permit an inference from

$$Fa \wedge Gb$$

to the sentence

$$\exists x(Fx \wedge Gx)$$

(distinct individual constants, a and b, are converted to a single variable), *EG does* permit an inference from the sentence

$$Fa \wedge Ga$$

to the sentence

$$\exists x(Fx \wedge Ga)$$

In this case only one of the constants, a, is converted to a variable and bound by the new quantifier. Is this an oversight?

Reflect on the English sequence

Euterpe admires herself.

Therefore, Euterpe has an admirer.

The sequence is valid, and its validity can be captured in $\mathcal{L}_\mathcal{D}$ provided *EG* permits the addition of an existential quantifier to a sentence, ϕ, containing instances of an individual constant, β, without converting every instance of β in ϕ to a variable, α, picked up by the added quantifier. The sequence could be represented in $\mathcal{L}_\mathcal{D}$ (letting $A①② =$ ① admires ②).

1. $+ Aee$
2. $? \exists xAxe$
3. $\exists xAxe$ 1 *EG*

EG allows an existential quantifier to be added without requiring that every instance of the individual constant, e, is converted to a variable, x.

Exercises 5.03

Construct derivations for the $\mathcal{L}_\mathcal{P}$ sequences below, making use of rule *EG* where necessary.

1. $+ \forall x(Gx \supset Fx)$
   $+ Ga$
   $+ \exists xFx \supset \forall x(Gx \supset Hx)$
   $? \exists xHx$

2. $+ Fa$
   $+ \exists xFx \supset \forall x(Gx \vee Hx)$
   $+ \exists x(Gx \vee Hx) \supset Ha$
   $? \exists xHx$

3. $+ \forall x(Fx \supset Gx)$
   $+ \forall x(Gx \supset (Hx \wedge Jx))$
   $? \forall xFx \supset \exists x(Gx \wedge Hx)$

4. $+ \exists xFx \supset \forall x(Gx \supset Hx)$
   $+ \forall x(Fx \supset Gx)$
   $? Fa \supset (Ga \wedge Ha)$

5. $+ \neg\exists x(\neg Fx \vee Hx)$
   $+ \forall x(\exists y(Gy \wedge \neg Hy) \supset Jx)$
   $? \forall x(Fx \supset Gx) \supset \exists x(Fx \wedge Jx)$

6. $+ \forall x(\neg Gx \supset Hx)$
   $+ \forall x\neg(Fx \supset Hx)$
   $? \exists x(Fx \wedge Gx)$

7. $+ \forall x(Hx \supset Fx)$
   $+ \exists x(Hx \vee Kx) \supset \forall xGx$
   $+ \forall x(Hx \wedge Jx)$
   $? \exists x(Fx \wedge Gx)$

8. $+ \forall x(Fx \wedge Hx)$
   $+ \exists x(Gx \vee Ix) \supset Ja$
   $+ Ja \supset \forall x(Gx \supset \neg Fx)$
   $? \neg\forall x(Fx \supset Gx)$

9. $+ \exists x(\neg Fx \vee \neg Gx) \supset \forall x(Hx \supset Jx)$
   $+ \neg\exists x(Hx \wedge \neg Jx) \supset (Fa \wedge Ga)$
   $? \exists x(Fx \wedge Gx) \supset Fa$

10. $+ \forall x(Gx \supset \neg Hx)$
    $+ \forall x(Fx \vee Gx)$
    $? \forall xHx \supset \exists xFx$

11. $+ \forall x(Fx \supset \neg Gx)$
    $+ \neg\exists x\neg Fx$
    $? \exists x(Fx \wedge \neg Gx)$

12. $+ \forall x(Fx \supset Gx)$
    $+ \neg Gc$
    $? \exists x\neg Fx$

13. $+ \forall x\forall y(Fxy \supset Gx)$
    $+ \forall x\forall y(Gx \supset Hxy)$
    $? \forall xFxa \supset \exists xHxc$

14. $+ \forall x(Fx \supset Gx)$
    $+ \neg\exists x(\neg Gx \wedge Fx) \supset Hac$
    $? \exists x\exists yHxy$

15. $+ \forall x\forall y(Fxy \vee Fyx)$
    $+\exists x\exists yFxy \supset \forall xGxb$
    $? \exists x\exists yGxy$

5.04 Existential Instantiation: *EI*

Rule *EG* permits the addition of existential quantifiers to sentences and the conversion of constants to variables bound by those quantifiers. In so doing, it licenses inferences of the form

Euterpe is admirable.

Therefore, someone is admirable.

Consider a sequence that moves in the opposite direction:

Someone is admirable.

Therefore, Euterpe is admirable.

This sequence is clearly invalid. From the fact that *someone* is admirable, you cannot infer that Euterpe, a particular individual, is admirable.

What *could* be inferred from a sentence of the form 'Something is *F*'? If the sentence is true, *F* must be true of *some* individual (or other). What the sentence leaves unspecified is *which* individual is *F*. Suppose you reasoned as follows: if something is *F*, then *F* is true of something; call that something 'Jane Doe'. In so reasoning you would be introducing an *arbitrary name*, 'Jane Doe'. that serves solely as a designation for an individual, whichever individual it is, of which *F* is true. (If there is more than one object of which *F* is true, 'Jane Doe' designates an arbitrary individual in this collection.)

This point will become less mysterious once you work through applications of the rule for *existential instantiation* (*EI*) set out below:

(*EI*) $\quad \exists\alpha\phi \vdash \phi\alpha/\beta$

Restriction: β does not occur in
(i) an earlier sentence
(ii) in the conclusion
(iii) in $\exists\alpha\phi$

Rule *EI* is founded on the idea that, from the fact that something holds of at least one thing, it follows that there is some particular thing of which it holds. The rule allows you to drop an existential quantifier, and in that respect *EI* resembles *UI*. Both rules move from generalizations to instances. *EI* differs from *UI* in placing a restriction on the constant that replaces the variable bound by the existential quantifier. The constant must function, *in the context of the derivation*, as an arbitrary name. The restriction on the rule ensures that the constant introduced does so function.

Consider the English sequence

All quarks have charm.

Something is a quark.

Something has charm.

The sequence is valid. Were you to spell it out, you might do so as follows: if something is a quark, then there is at least one object that is a quark; call that object 'Gus'. (Here, 'Gus' serves as an arbitrary name.) Now, if all quarks have charm, then, if Gus is a quark, Gus has charm, and, if Gus has charm, something has charm.

This line of reasoning could be spelled out in $\mathcal{L}_\mathcal{P}$ as follows:

1.	$+ \forall x(Qx \supset Cx)$	
2.	$+ \exists xQx$	
3.	$? \exists xCx$	
4.	Qg	2 *EI*
5.	$Qg \supset Cg$	1 *UI*
6.	Cg	4, 5 *MP*
7.	$\exists xCx$	6 *EG*

Dropping the existential quantifier in line 4 satisfies the restrictions on *EI* and ensures that *g* functions as an arbitrary name, the name of something or other of which it is true that it is a quark. The universally quantified sentence in line 1 holds of everything, so it holds of *g*.

This derivation illustrates the importance of the *order* in which rules for dropping quantifiers are applied. Suppose the order of lines 4 and 5 had been reversed, suppose the universal quantifier had been dropped in line 4, and the variables it bound replaced with *g*. You could drop the existential quantifier in line 5, but replacing the variable it bound with *g* would have violated restriction (i) on *EI*.

Why should the order in which quantifiers are dropped matter? Rule *UI* permits inferences from universal generalizations—'everything is *F*'—to instances—'*Fa*'. If something is true of everything, it is true of *any* individual thing. In dropping an existential quantifier, a generalization true of *something*—'something is *F*'—the something is given a made-up arbitrary name. An individual constant that appears in a premise, in a sentence that occurs earlier in the derivation, or in the conclusion, could not then be introduced as an arbitrary name. Were the order of lines 4 and 5 reversed, the constant *b* would not function as an arbitrary name.

The invalid sequence below illustrates the point of the restriction:

Something is red all over.

Something is green all over.

Therefore, something is red and green all over.

Suppose you translated this sequence into $\mathcal{L}_\mathcal{P}$ (assuming Rx = *x* is red all over and Gx = *x* is green all over).

1.	$+ \exists xRx$
2.	$+ \exists xGx$
3.	$? \exists x(Rx \wedge Gx)$

You might then attempt to construct a derivation along the following lines:

1.	$+ \exists x Rx$	
2.	$+ \exists x Gx$	
3.	$? \exists x(Rx \wedge Gx)$	
4.	Ra	1 *EI*
5.	Ga	2 *EI violates* (i)
6.	$Ra \wedge Ga$	4, 5 $\wedge I$
7.	$\exists x(Rx \wedge Gx))$	6 *EG*

The derivation goes off the rails in line 5.

This invalid sequence nicely illustrates the thinking behind restriction (i) that *EI* places on the dropping of existential quantifiers. It is as though you reasoned as follows.

(1) Something is red all over; call that something 'Alfie'. (So far so good; 'Alfie' serves in this context as an arbitrary name.)

(2) Something is green all over. Call that something 'Alfie' too. (Whoops! 'Alfie' in step (2) would designate whatever 'Alfie' already designates in step (1), so the second 'Alfie' would not be an arbitrary name.)

(3) Alfie is red all over and green all over.

(4) So something is red all over and green all over.

The restrictions on *EI* provide simple syntactic devices that achieve the desired semantic result. Constructing derivations in accord with the restrictions ensures that names introduced when existential quantifiers are dropped function as arbitrary, made-up names.

Consider another invalid English sequence:

Someone admires Euterpe.

Therefore, Euterpe admires herself.

The restrictions on rule *EI* block the derivation of such sequences. Letting $A①②$ = ① admires ② and restricting the domain to persons

1.	$+ \exists x Axe$	
2.	$? Aee$	
3.	Aee	1 *EI violates* (ii) *and* (iii)

The application of *EI* in line 3 violates clauses (ii) and (iii) in the restriction. First, e occurs in the conclusion of the derivation, Aee. Second, e occurs in $\exists \alpha \phi$, the sentence from which the quantifier is dropped: $\exists x Axe$.

Remember that rules governing the adding and dropping of quantifiers apply only to sentences occupying whole lines of derivations. A quantifier can be dropped from or added to a sentence only if the entire sentence in which the quantifier occurs falls within its scope. The sequence below illustrates successive violations of this principle.

1.	$+\exists x(Fx \wedge Gx) \supset \exists x Hx$	
2.	$?(Fa \wedge Ga) \supset Hb$	
3.	$(Fa \wedge Ga) \supset \exists x Hx$	1 *EI* (*violates the principle*)
4.	$(Fa \wedge Ga) \supset Hb$	3 *EI* (*violates the principle*)

The quantifiers occurring in the sentence in line 1 include within their scope expressions that are themselves parts of a more inclusive sentence.

Derivation Heuristics

In constructing derivations that call for applications of *EI*, drop the existential quantifier *first*, replacing the variables it binds with some eligible constant, *then* check the resulting sentence to see whether any part of the restriction has been violated.

Exercises 5.04

Construct derivations for the $\mathcal{L}_{\mathcal{P}}$ sequences below, making use of rule *EI* where necessary.

1. $+ \exists x(Fx \wedge Gx)$
   $+ \forall x(Gx \supset Hx)$
   $? \exists x(Fx \wedge Hx)$

2. $+ \exists x \exists y Fxy$
   $+ \forall x \forall y(Fxy \supset Gx)$
   $? \exists x Gx$

3. $+ \forall x(Fx \supset Gx)$
   $+ \exists x Gx \supset \forall x(Fx \supset Hx)$
   $? \exists x(Fx \wedge Hx) \supset \forall x(Fx \supset Hx)$

4. $+ \neg \forall x Gx \supset \forall x Hx$
   $+ \exists x Hx \supset \forall x \neg Fx$
   $? \forall x(Fx \supset Gx)$

5. $+ \forall x(Fx \supset Hx)$
   $+ \exists x Hx \supset \neg \exists x Gx$
   $? \neg \exists x(Fx \wedge Gx)$

6. $+ \forall x(Fx \supset \exists y Gy)$
   $+ \exists y Gy \supset Ha$
   $? \exists x Fx \supset \exists x Hx$

7. $+ \forall x(Fx \supset Hx)$
   $+ \forall x \neg Gx$
   $? \exists x(Fx \vee Gx) \supset \exists x Hx$

8. $+ \exists x(Fx \vee Gx)$
   $+ \forall x(Fx \supset Hx)$
   $+ \forall x(Gx \supset Hx)$
   $? \exists x Hx$

9. $+ \exists x(Fx \wedge Gx)$
$+ \exists x(Hx \wedge Jx)$
$+ \exists x(Fx \wedge Hx) \supset \forall x Kx$
$? \forall x Fx \supset \exists x Kx$

10. $+ \exists x(\neg Fx \vee Hx)$
$+ \forall x(Fx \supset (Gx \supset Hx))$
$? \exists x(Fx \wedge Gx) \supset \exists x Hx$

11. $+ \forall x((Fx \vee Gx) \supset Hx)$
$+ \exists x(\neg Fx \vee Jx) \supset \forall x Kx$
$? \forall x Hx \vee \forall x Kx$

12. $+ \exists x(Fxa \wedge Gax)$
$+ \forall x(Fxa \supset \forall y Hxy)$
$? \exists x Hxc$

13. $+ \exists x(Fx \vee Gx)$
$+ \exists x Hx \supset \forall y \neg Gy$
$? \exists x(Fx \vee \neg Hx)$

14. $+ \exists x(Fx \equiv Gx)$
$+ \forall x(Fx \supset (Gx \supset Hx))$
$+ \forall x Fx \vee \forall y Gy$
$? \exists x Hx$

15. $+ \exists x(Fx \vee Hx)$
$+ \exists x Gx$
$+ \exists x Fx \supset \forall y(Gy \supset Hy)$
$? \exists x Hx$

5.05 Universal Generalization: *UG*

Applications of rule *UI* move from generalizations to instances. *Universal generalization* (*UG*) moves in the opposite direction, from instances to generalizations. The rule includes a restriction designed to block inferences of the form

Hesperus is bright.

Everything is bright.

Plainly, the premise does not imply the conclusion. From Hesperus's being bright, it does not follow that *everything* is bright. A universal conclusion cannot be derived from a premise that concerns a particular named individual.

The restriction blocks, as well, inferences of the form

Something is purple.

Everything is purple.

From the fact that F is true of something, it does not follow that F is true of everything.

So when *would* it be permissible to go from the particular to the general? The best way to answer this question is by first setting out rule *UG*, together with its associated restriction, and then noting how it applies in particular cases.

(*UG*) $\phi\alpha/\beta \vdash \forall\alpha\phi$

Restrictions: β does not occur
(i) in a premise
(ii) in an *undischarged* supposition
(iii) in $\forall\alpha\phi$
(iv) in a sentence previously obtained by means of an application of *EI*

Rule *UG* is founded on the idea that if you can show that something holds of *any* object belonging to a collection of objects, you would have thereby shown that it holds for *every* object in the collection. You move, in effect, from *any* to *every*.

Probably without realizing it, you have engaged in this kind of reasoning many times long before you were introduced to $\mathcal{L}_\mathcal{P}$ (or, for that matter, $\mathcal{L}_S$). Remember all those proofs in Euclidian geometry? In proving the Pythagorean theorem (the square of the hypotenuse of a right triangle = the sum of the squares of the remaining sides), you start with an arbitrary right triangle with sides *a*, *b*, and *c*. In proving that $a^2 = b^2 + c^2$, you establish that this holds for *any* right triangle.

The restriction associated with *UG* is designed to ensure that when a universal quantifier is added to a sentence, the variable it picks up is converted from an individual constant that is functioning as the name of an *arbitrary individual*. This semantic aim can be satisfied via a syntactic restriction, one the application of which depends only on purely formal characteristics of derivations.

The first part of the restriction is obvious: if an individual constant occurs in a premise of a derivation, it cannot also function in the derivation as a designation for an arbitrary object. Consider with the sequence

All horses are quadrupeds.
All quadrupeds have spleens.

All horses have spleens.

Arbitrary Names and Names for Arbitrary Individuals

The sense in which individual constants function as names of arbitrary *objects* should not be confused with the sense in which constants function as arbitrary *names* in applications of rule *EI*.

When you know that something is *F* ($\exists x Fx$), you can designate an arbitrary name to stand for that something.

In applying rule *UG*, in contrast, you are reasoning that, if *F* is true of an arbitrary object in the class of *F*s, *F* is true of every object in the class. If something holds true of an arbitrary right triangle, or an arbitrary tiger, it holds true of *any* triangle or tiger.

The restriction on *UG* ensures that a sentence containing a name can be converted to a universal generalization *only* in cases in which the name is functioning as the name of an arbitrary individual.

The sequence is valid. A proof that it is valid in $\mathcal{L}_p$ includes a representative application of *UG*. Suppose that Hx = *x* is a horse, Qx = *x* is a quadruped, and Sx = *x* has a spleen.

1.	+ $\forall x(Hx \supset Qx)$	
2.	+ $\forall x(Qx \supset Sx)$	
3.	? $\forall x(Hx \supset Sx)$	
4.	$Ha \supset Qa$	1 *UI*
5.	$Qa \supset Sa$	2 *UI*
6.	$Ha \supset Sa$	4, 5 *HS*
7.	$\forall x(Hx \supset Sx)$	6 *UG*

The sentence in line 4 results from reasoning as follows: if all horses are quadrupeds, then if an arbitrary individual, *a*, is a horse, *a* is a quadruped. Similarly, the sentence in line 5 is obtained by reasoning that if all quadrupeds have spleens, then if an arbitrary individual, *a*, is a quadruped, *a* has a spleen. The sentence in line 6 results from reasoning in accord with *HS*: if it is true that if *a* is a horse, *a* is a quadruped, and true as well that, if *a* is a quadruped, *a* has a spleen, then it is true that if *a* is a horse, *a* has a spleen. So long as *a* represents a genuinely arbitrary member of the class of horses, what holds for *any* horse holds for *every* horse, hence the universally quantified sentence in line 7.

The application of *UG* in the derivation accords with the restriction. The individual constant *a* does not occur in a premise, nor does *a* occur in an undischarged supposition: there are no suppositions, hence no undischarged suppositions. Finally, *a* does not occur in $\forall x(Hx \supset Sx)$. There are no applications of rule *EI*, so the last clause in the restriction is inapplicable.

You might worry that *a*, the constant introduced in line 5 of the derivation, does not name an arbitrary object. After all, *a* was introduced in previous lines via applications of *UI*. The use of constants as arbitrary *names* in applications of *EI*, however, differs from their use as names of arbitrary *individuals*. The quantifier rules are designed to work as a package. They ensure (in the case of *EI*) that existentially quantified sentences are replaced only by sentences containing arbitrary names. For *UG*, only names that designate arbitrary individuals are replaceable by variables that are picked up by a universal quantifier.

The discussion of *UG* began with examples of English sequences that were clearly invalid. Return to the first of those sequences:

Hesperus is bright.

Everything is bright.

The sequence can be translated into $\mathcal{L}_p$ (letting Bx = *x* is bright).

1.	+ Bh	
2.	? $\forall xBx$	
3.	$\forall xBx$	1 *UG violates restriction* (i)

The application of *UG* in line 3 violates the first restriction. You cannot convert an individual constant to a variable that is picked up by a universal quantifier when that constant occurs in a premise. In this case, *h*, the pertinent individual constant, appears in the premise in line 1.

The second example introduced earlier violates the restriction in a different way.

Something is purple.

Everything is purple.

In $\mathcal{L}_p$

1.	+ $\exists xPx$	
2.	? $\forall xPx$	
3.	Pa	1 *EI*
4.	$\forall xPx$	3 *UG violates restriction* (iv)

The restriction on *UG* is expressed syntactically, but its aim is to block invalid inferences of the kind illustrated by this sequence. In this case, the inference is blocked by barring the addition of a universal quantifier when the individual constant converted to a variable and picked up by the new quantifier—in this case *a*—occurs in a sentence introduced by an application of rule *EI*. In the example above, *a* occurs in a sentence introduced in line 3 by an application of *EI*, so the inference is blocked.

Notice that the restriction does *not* say that β, the constant occurring on a line obtained by an application of *EI*, was *introduced* by the application of *EI*, but that β cannot occur *on a line* arrived at by way of an application of *EI*. Consider the following sequence:

1.	+ $\forall x\exists yLxy$	
2.	? $\exists y\forall xLxy$	
3.	$\exists yLay$	I *UI*
4.	Lab	3 *EI*
5.	$\forall xLxb$	4 *UG violates restriction* (iv)
6.	$\exists y\forall xLxy$	5 *EG*

Although the *a* in line 4 was not obtained by way of an application of *EI*, *a* occurs on a line obtained by an application of *EI*.

The restriction blocks invalid sequences of the form

Everyone loves someone.

Someone is loved by everyone.

The premise says that everyone loves someone (or other). The conclusion says that there is someone—some *x*—that everyone loves.

Another way in which the restrictions on *UG* can be violated is illustrated by the invalid English sequence

Euterpe admires herself.

Everyone admires Euterpe.

Consider a derivation for this sequence in $\mathcal{L}_p$, again $A①②$ = ① admires ② and restricting the domain to persons.

1.	+ *Aee*	
2.	? ∀*xAxe*	
3.	∀*xAxe*	1 *UG violates restriction* (iii)

The third restriction on *UG* blocks inferences of this sort by requiring that the sentence resulting from the addition of a universal quantifier, $\forall\alpha\phi$, not include an instance of β, the constant being converted to a variable and picked up by the quantifier. The restriction amounts to the requirement that in an application of *UG*, you can convert an instance of an individual constant to a variable picked up by an added universal quantifier only if you convert *every* instance of that individual constant to the variable.

You might have noticed that the application of rule *UG* in the derivation above violates restriction (i) as well: the constant e occurs in a premise. As a result, in the context of this derivation, *e* is not functioning as the name of arbitrary object.

An illustration of the second clause in the restriction on *UG* requires a derivation that includes a supposition.

Everything with a mane is a quadruped.

Horses with manes are quadrupeds.

Using the interpretation of *Hx* and *Qx* set out above, and letting *Mx* = *x* has a mane, a derivation for this sequence can be constructed using *CP*.

1.	+ ∀*x*(*Mx* ⊃ *Qx*)	
2.	? ∀*x*((*Hx* ∧ *Mx*) ⊃ *Qx*)	
3.	*Ma* ⊃ *Qa*	1 *UI*
4.	*Ha* ∧ *Ma*	
5.	? *Qa*	
6.	*Ma*	4 ∧*E*
7.	*Qa*	3, 6 *MP*
8.	(*Ha* ∧ *Ma*) ⊃ *Qa*	4–7 *CP*
9.	∀*x*((*Hx* ∧ *Mx*) ⊃ *Qx*)	8 *UG*

In adding a universal quantifier in line 9, an individual constant, *a*, is converted to a variable. Does this violate the restriction on *UG*? The premise in line 1 does not contain an occurrence of *a*, and, in adding the quantifier in line 9, every instance of *a* has been converted to an *x* that is picked up by the quantifier. There are no applications of rule *EI*, so no violation of the clause in the restriction that blocks the conversion of constants in a sentence resulting from an application of *EI*.

What about restriction (ii)? Does *a* occur in an *undischarged* supposition? Although *a* occurs in a supposition (line 4), the supposition has been discharged. A supposition is discharged once ⌞ has been entered and the suppositional portion of a derivation is 'closed off'. Sentences included within the matched brackets become inactive, unavailable for further inferences. The universal quantifier is added to the sentence in line 8, a sentence that occurs *outside* the scope of the supposition: the supposition has been discharged.

You can see the point of the restriction by reflecting on a patently invalid sequence:

All horses are quadrupeds.

All quadrupeds have spleens.

If Old Regret is a horse, everything has a spleen.

The invalid inference is blocked in $\mathcal{L}_p$ because it violates the restriction on *UG* concerning the occurrence of individual constants in sentences introduced as suppositions.

1.	+ $\forall x(Hx \supset Qx)$	
2.	+ $\forall x(Qx \supset Sx)$	
3.	? $Ho \supset \forall xSx$	
4.	⌜ Ho	
5.	? $\forall xSx$	
6.	$Ho \supset Qo$	1 *UI*
7.	$Qo \supset So$	2 *UI*
8.	$Ho \supset So$	6, 7 *HS*
9.	So	4, 8 *MP*
10.	⌞ $\forall xSx$	9 *UG violates restriction* (ii)
11.	$Ho \supset \forall xSx$	4–10 *CP*

In the sentence in line 10, an individual constant, *o*, has been converted to a variable, *x*, and a universal quantifier has been added to capture that variable. This violates the restriction. The universal quantifier has been added *within* the scope of a supposition—the sentence introduced in line 4—that contains *o*, the constant converted and picked up by the universal quantifier added in line 10.

The restriction on *UG* does not prohibit the addition of a universal quantifier within the scope of a supposition in every case. The application of *UG* in line 10 is incorrect, not because it occurs within the scope of a supposition but because it occurs within the scope of a supposition that includes *o*, the individual constant that is converted to a variable and picked up by the quantifier.

Consider the sequence

Wombats have spleens.

Wombats have spleens or livers.

It is easy to prove the sequence valid using *CP*, but note, first, that the derivation below is incorrect.

1.	+ $\forall x(Wx \supset Sx)$	
2.	? $\forall x(Wx \supset (Sx \vee Lx))$	
3.	Wx	(*not a sentence*)
4.	? $Sx \vee Lx$	(*not a sentence*)
5.	$Wx \supset Sx$	1 *UI* (*not a sentence*)
6.	Sx	3, 5 *MP* (*not a sentence*)
7.	$Sx \vee Lx$	6 $\vee I$ (*not a sentence*)
8.	$Wx \supset (Sx \vee Lx)$	3–7 *CP* (*not a sentence*)
9.	$\forall x(Wx \supset (Sx \vee Lx))$	8 *UG*

Every line of a derivation must be a sentence. The expressions on lines 3–8 all contain free variables, and, as a result, do not qualify as sentences of $\mathcal{L}_p$.

The error is avoidable, as the derivation below illustrates:

1.	+ $\forall x(Wx \supset Sx)$	
2.	? $\forall x(Wx \supset (Sx \vee Lx))$	
3.	Wa	
4.	? $Sa \vee La$	
5.	$Wa \supset Sa$	1 *UI*
6.	Sa	3, 5 *MP*
7.	$Sa \vee La$	6 $\vee I$
8.	$Wa \supset (Sa \vee La)$	3–7 *CP*
9.	$\forall x(Wx \supset (Sx \vee Lx))$	8 *UG*

In this derivation, every line consists of a sentence of $\mathcal{L}_p$.

5.06 Quantifier Rules Summary

Before moving on to exercises incorporating applications of *UG*, you might find it helpful to pause and reflect on applications of all the quantifier rules in one place. If you are confident that you have a grip on the rules already, you should feel free to proceed directly to exercises 5.06.

Universal Instantiation

(*UI*) $\forall\alpha\varphi \vdash \varphi\alpha/\beta$

Kangaroos are marsupials.	1. + $\forall x(Kx \supset Mx)$	
Skippy is a kangaroo.	2. + Ks	
Skippy is a marsupial.	3. ? Ms	
	4. $Ks \supset Ms$	1 *UI*
	5. Ms	2, 4 *MP*

Existential Generalization

(*EG*) $\varphi\alpha/\beta \vdash \exists\alpha\varphi$

Kangaroos are marsupials.	1. + $\forall x(Kx \supset Mx)$	
Skippy is a kangaroo.	2. + Ks	
There is at least one marsupial.	3. ? $(\exists x)Mx$	
	4. $Ks \supset Ms$	1 *UI*
	5. Ms	2, 4 *MP*
	6. $\exists xMx$	5 *EG*

Existential Instantiation

(*EI*) $\exists\alpha\varphi \vdash \varphi\alpha/\beta$ *Restriction*: β does not occur

(i) in an earlier sentence

(ii) in the conclusion

(iii) in $\exists\alpha\varphi$

The point of the restriction is to block inferences such as the following (* indicates a violation of the restriction):

There are kangaroos.	1. + $\exists xKx$	
Socrates is a kangaroo.	2. ? Ks	
	3. Ks	1 *EI* *

. . . while allowing

Kangaroos are marsupials.	1. + $\forall x(Kx \supset Mx)$	
There is at least one kangaroo.	2. + $\exists x Kx$	
There is at least one marsupial.	3. ? $\exists x Mx$	
	4. Ka	2 *EI*
	5. $Ka \supset Ma$	1 *UI*
	6. Ma	4, 5 *MP*
	7. $\exists x Mx$	6 *EG*

The restriction on *EI* has the effect of ensuring that β functions as an arbitrary name ('Jack the Ripper', 'Jane Doe'), an invented dummy name used to designate an unknown something.

Compare this restriction on *EI* with the restriction on *UG*, which is designed to ensure that β names an arbitrary member of the domain: *any* member of the domain could have been β.

Universal Generalization

(*UG*) $\varphi\alpha/\beta \vdash \forall\alpha\varphi$ *Restriction*: β does not occur

(i) in a premise

(ii) in an *undischarged* supposition

(iii) in $\forall\alpha\phi$

(iv) in a sentence previously obtained by means of an application of *EI*

The restriction *allows* . . .

Kangaroos are marsupials.	1. + $\forall x(Kx \supset Mx)$	
Marsupials are warm-blooded.	2. + $\forall x(Mx \supset Wx)$	
Kangaroos are warm-blooded.	3. ? $\forall x(Kx \supset Wx)$	
	4. $Ka \supset Ma$	1 *UI*
	5. $Ma \supset Wa$	2 *UI*
	6. $Ka \supset Wa$	4, 5 *HS*
	7. $\forall x(Kx \supset Wx)$	6 *UG*

. . . and . . .

Anything with a pouch is a marsupial.	1. + $\forall x(Px \supset Mx)$	
Kangaroos with pouches are marsupials.	2. ? $\forall x((Kx \wedge Px) \supset Mx)$	
	3. $\vert\overline{Ka \wedge Pa}$	
	4. $\vert$? Ma	
	5. $\vert$ Pa	3 $\wedge E$
	6. $\vert$ $Pa \supset Ma$	1 *UI*
	7. $\vert$ Ma	5, 6 *MP*
(the *supposition is discharged*)	8. $(Ka \wedge Pa) \supset Ma$	3–7 *CP*
	9. $\forall x((Kx \wedge Px) \supset Mx)$	8 *UG*

. . . and . . .

Kangaroos have pouches.	1. + $\forall x(Kx \supset Px)$	
Kangaroos have pouches or tails.	2. ? $\forall x(Kx \supset (Px \vee Tx))$	
	3. $\vert\overline{Ka}$	
	4. $\vert$? $Pa \vee Ta$	
	5. $\vert$ $Ka \supset Pa$	1 *UI*
	6. $\vert$ Pa	3, 5 *MP*
	7. $\vert$ $Pa \vee Ta$	6 $\vee I$
(the *supposition is discharged*)	8. $Ka \supset (Pa \vee Ta)$	3–7 *CP*
	9. $\forall x(Kx \supset (Px \vee Tx))$	8 *UG*

. . . while blocking . . .

Skippy is a kangaroo.	1. + Ks	
Everything is a kangaroo.	2. ? $\forall xKx$	
(s occurs in a *premise*)	3. $\forall xKx$	1 *UG* *

. . . and

Something is a kangaroo.	1. + $\exists xKx$	
Everything is a kangaroo.	2. ? $\forall xKx$	
	3. Ka	1 *EI*
(a occurs in a line obtained by *EI*)	4. $\forall xKx$	3 *UG* *

The same restriction blocks

Everything resembles something.	1. + $\forall x \exists y Rxy$	
Something resembles everything.	2. ? $\exists y \forall x Rxy$	
	3. $\exists y Ray$	1 *UI*
	4. Rab	2 *EI*
(again, *a* occurs in a line obtained by *EI*)	5. $\forall x Rxb$	3 *UG* *
	6. $\exists y \forall x Rxy$	5 *EG*

Derivation Heuristics

Suppose you are in doubt as to whether dropping or adding a quantifier would be illegal and land you in trouble with the quantifier police. Rather than poring over the rules, it is easier to simply drop (or add) the quantifier, *then* check to see whether you have violated a restriction. If you have, look for a workaround.

Exercises 5.06

Construct derivations for the $\mathcal{L}_p$ sequences below, making use of rule *UG* where necessary.

1. + $\forall x(Fx \supset Gx)$
   + $\neg \exists x(Gx \wedge Hx)$
   ? $\forall x(Fx \supset \neg Hx)$

2. + $\forall x(Fx \supset Gx)$
   + $\exists x Gx \supset \forall x(Fx \supset Hx)$
   ? $\forall x Fx \supset \forall x Hx$

3. + $Fa \supset \forall x Gax$
   ? $\forall x(Fa \supset Gax)$

4. + $\forall x(Fx \wedge Gx)$
   ? $\forall x Fx \wedge \forall x Gx$

5. + $\forall x(Fx \supset Gx)$
   + $\exists x Fx \supset \neg \exists y Hy$
   ? $\forall x(\exists y Fy \supset \neg Hx)$

6. + $\exists x(Fx \vee Hx)$
   + $\forall x(Hx \supset Fx)$
   + $\exists x Gx \supset \forall x(Gx \supset Hx)$
   ? $\forall x(Gx \supset Fx)$

7. + $\forall x(Fx \supset Gx)$
   + $\forall x(Fx \supset Hx)$
   ? $\forall x(Fx \supset (Gx \wedge Hx))$

8. + $\forall x(Fx \supset Gx)$
   + $\forall x \exists y(Fy \wedge Hxy)$
   ? $\forall x \exists y(Gy \wedge Hxy)$

9. + $\forall x Fx$
   + $\forall x \neg Gx$
   ? $\neg \exists x(Fx \equiv Gx)$

10. + $\forall x((Fx \wedge \neg \exists y Hxy) \supset Gx)$
    + $\forall x(Jx \supset (Fx \wedge \neg(Kx \vee Gx)))$
    ? $\forall x(Jx \supset \exists y Hxy)$

11. $+ \forall x \forall y(Fxy \supset \exists z Gzxy)$
$+ \forall x \forall y \forall z(Gzxy \supset (Hzx \wedge Hzy))$
$+ \forall x Fxa$
$? \forall x \exists y(Gyxa \supset Hya)$

12. $+ \exists x(Fx \wedge \forall y(Gy \supset Hxy))$
$+ \forall x(Fx \supset \forall y(Jy \supset \neg Hxy))$
$? \forall x(Gx \supset \neg Jx)$

13. $+ \exists x \exists y Fxy$
$+ \forall x(\exists y Fxy \supset \exists y(\forall z Fyz \wedge Fxy))$
$? \exists x \forall y Fxy$

14. $+ \forall x(\exists y Gyx \supset Gxx)$
$+ \forall x(Fx \supset (\exists y Gxy \supset \exists y Gyx))$
$+ \neg \exists x Gxx$
$? \forall x(Fx \supset \forall y \neg Gxy)$

15. $+ \forall x \forall y((Fax \wedge Fya) \supset Fxy)$
$+ \forall x(Gx \supset Fxa)$
$+ \exists x(Gx \wedge Fax)$
$? \exists x(Gx \wedge \forall y(Gy \supset Fxy))$

5.07 Identity: *ID*

In chapter 4, you were introduced to the identity relation, a relation every individual bears to itself and to nothing else. Some, but not all, true identity statements are trivial. 'O. Henry is O. Henry'. for instance, is trivial. Individual objects can be designated by more than one singular term, however. The sentence

O. Henry is (is identical with) William Sydney Porter.

is true if and only if 'O. Henry' and 'William Sydney Porter' are names for one and the same individual. Similarly,

The author of *The Cop and the Anthem* is Porter.

is true just in case 'the author of *The Cop and the Anthem*' and 'Porter' co-designate.

Reflect on an English sequence that includes an identity statement of identity.

O. Henry is an author and a Tar Heel.

O. Henry is Porter.

Porter is an author and a Tar Heel.

The sequence is valid and can be represented in $\mathcal{L}_{\mathcal{P}}$ (assuming that $Ax = x$ is an author and $Tx = x$ is a Tar Heel).

1. $+ Ah \wedge Th$
2. $+ h = p$
3. $? Ap \wedge Tp$

A derivation of the conclusion from the premises in lines 1 and 2 requires a rule that exploits the nature of the identity relation. If O. Henry and Porter are the selfsame individual, whatever is true

of one must be true of the other. This is captured by rule *ID*. The rule includes two parts, each capturing an aspect of the concept of identity.

(*ID*) (i) $\vdash \beta = \beta$

(ii) $\phi, \beta = \gamma \vdash \phi\beta/\gamma$

In the formulation of the rule, β and γ stand for individual constants (a, b, c, . . ., t), ϕ is a sentence, and $\phi\beta/\gamma$ represents a sentence obtained from ϕ by replacing one or more occurrences of β in ϕ with γ.

The first clause of *ID* permits the introduction of a sentence consisting of an identity sign flanked by occurrences of a single individual constant at any point in a derivation. This reflects the logical truth that every object is identical with itself.

ID's second clause licenses an inference from a sentence containing an individual name, β, together with a statement of identity, $\beta = \gamma$, to another sentence that differs from the first only in its substitution of γ for β. This reflects our conviction that if a and b are identical, then whatever is true of a is true of b, and vice versa. If O. Henry is left-handed and has a scar on his forearm, then, if O. Henry is identical with Porter, Porter is left-handed and has a scar on his forearm.

This pattern of reasoning is evident in the sequence introduced above. If O. Henry is an author and a Tar Heel, then if O. Henry and Porter are one and the same person, Porter is an author and a Tar Heel.

1.	$+ Ah \wedge Th$	
2.	$+ h = p$	
3.	$? Ap \wedge Tp$	
4.	$Ap \wedge Tp$	1, 2 *ID*

Suppose you discovered that Porter had brown hair and a limp, and that Buchan had neither. It would follow that Porter and Buchan are not one and the same individual. This too can be proved in $\mathcal{L}_\mathcal{P}$ (letting $Bx = x$ has brown hair and $Lx = x$ limps).

1.	$+ Bp \wedge Lp$	
2.	$+ \neg(Bb \vee Lb)$	
3.	$? p \neq b$	
4.	$\boxed{p = b}$	
5.	$?\times$	
6.	$Bb \wedge Lb$	1, 4 *ID*
7.	Bb	6 $\wedge E$
8.	$\neg Bb \wedge \neg Lb$	2 *DeM*
9.	$\neg Bb$	8 $\wedge E$
10.	$Bb \wedge \neg Bb$	7, 9 $\wedge I$
11.	$p \neq b$	4–10 *IP*

When might you put the first clause of rule *ID* to work? Consider the sequence

Buchan is an author.

Something is an author and identical with Buchan.

This sequence can be represented in $\mathcal{L}_{\mathcal{P}}$ and proved valid as follows:

1.	$+ Ab$	
2.	$? \exists x(Ax \wedge x = b)$	
3.	$\neg\exists x(Ax \wedge x = b)$	
4.	$? \times$	
5.	$\forall x\neg(Ax \wedge x = b)$	4 *QT*
6.	$\neg(Ab \wedge b = b)$	6 *UI*
7.	$\neg Ab \vee b \neq b$	7 *DeM*
8.	$b \neq b$	1, 8 $\vee E$
9.	$b = b$	*ID*
10.	$b = b \wedge b \neq b$	8, 9 $\wedge I$
11.	$\exists x(Ax \wedge x = b)$	3–10 *IP*

The sentence introduced via an application of *ID* in line 9 is not derived from any earlier sentence, so no line numbers appear in its justification.

Finally, consider the English sequence

Someone loud booed Elvis.

Only Chet and Joe are loud.

Chet or Joe booed Elvis.

Suppose the domain is restricted to persons and Lx = x is loud and $B①②$ = ① booed ②. How might this sequence be captured in $\mathcal{L}_{\mathcal{P}}$? The first sentence and the conclusion are unproblematic. The second premise is more challenging. What happens if the second premise is expressed in quasi-$\mathcal{L}_{\mathcal{P}}$?

Chet and Joe are loud, and, for all x, if x is loud, then x is Chet or x is Joe.

This facilitates the translation of the second premise into $\mathcal{L}_{\mathcal{P}}$ (bearing in mind that the 'is' in 'x is Chet' and 'x is Joe' is the 'is' of identity).

1.	$+ \exists x(Lx \wedge Bxe)$
2.	$+ (Lc \wedge Lj) \wedge \forall x(Lx \supset (x = c \vee x = j)$
3.	$? Bce \vee Bje$

Now you can derive the conclusion from the premises.

1.	$+\ \exists x(Lx \wedge Bxe)$	
2.	$+\ (Lc \wedge Lj) \wedge \forall x(Lx \supset (x = c \vee x = j))$	
3.	$?\ Bce \vee Bje$	
4.	$\neg(Bce \vee Bje)$	
5.	$?\ \times$	
6.	$La \wedge Bae$	1 *EI*
7.	La	6 $\wedge E$
8.	Bae	6 $\wedge E$
9.	$\forall x(Lx \supset (x = c \vee x = j))$	2 $\wedge E$
10.	$La \supset (a = c \vee a = j)$	9 *UI*
11.	$a = c \vee a = j$	7, 10 *MP*
12.	$a = c$	
13.	$?\ Bce$	
14.	Bce	8, 12 *ID*
15.	$a = c \supset Bce$	12–14 *CP*
16.	$a = j$	
17.	$?\ Bje$	
18.	Bje	8, 16 *ID*
19.	$a = j \supset Bje$	16–18 *CP*
20.	$Bce \vee Bje$	11, 15, 19 *CD*
21.	$(Bce \vee Bje) \wedge \neg(Bce \vee Bje)$	4, 20 $\wedge I$
22.	$Bce \vee Bje$	4–21 *IP*

The derivation is worth working through. It includes an application of rule *CD* (in line 19) set up by a pair of embedded applications of *CP* (in lines 11–14 and 15–18).

Exercises 5.07

Construct derivations for the $\mathcal{L}_{\mathcal{P}}$ sequences below, making use of rule *ID* where necessary.

1. $+ \forall x(x = a \supset Fx)$
   $+ \forall x(Fx \supset Fb)$
   $? Fb$

2. $+ \exists x(Fx \wedge Gx)$
   $+ \exists x(Fx \wedge \neg Gx)$
   $? \exists x \exists y((Fx \wedge Fy) \wedge x \neq y)$

3. $+ \forall x \exists y Fxy$
   $+ \forall x \neg Fxx$
   $? \exists x \exists y(x \neq y)$

4. $+ \forall x(Fx \supset (\forall y)(Fy \supset x = y))$
   $+ \exists x(Fx \wedge Gx)$
   $? \forall x(Fx \supset Gx)$

5. $+ \forall x \forall y(Fy \equiv x = y)$
   $+ Fa \wedge Fb$
   $? a = b$

6. $+ \exists x(Fx \wedge \forall y(Fy \supset x = y))$
   $+ \neg Fb$
   $? \exists x(x \neq b)$

7. $+ \forall x(Fx \supset Gx)$
   $+ \forall x(Gx \supset Hx)$
   $+ Fa \wedge \neg Hb$
   $? a \neq b$

8. $+ \exists x((Fx \wedge Gax) \wedge Hx)$
   $+ Fb \wedge Gab$
   $+ \forall x((Fx \wedge Gax) \supset x = b)$
   $? Hb$

9 $+ \exists x((Fx \wedge \forall y(Fy \supset x = y)) \wedge Gx)$
   $+ \neg Ga$
   $? \neg Fa$

10. $+ \forall x \forall y((Fxy \wedge x \neq y) \supset Gxy)$
    $+ \exists x \forall y(x \neq y \supset Fxy)$
    $? \exists x \forall y(x \neq y \supset Gxy)$

11. $+ \neg \exists x(Fx \wedge \forall y(Fy \supset x = y))$
    $+ \exists x Fx$
    $? \exists x \exists y((Fx \wedge Fy) \supset x \neq y)$

12. $+ \exists x(Fx \wedge \forall y(Fy \supset x = y))$
    $+ \exists x(Fx \wedge Gx)$
    $? \forall x(Fx \supset Gx)$

13. $+ \forall x \neg Fxx$
    $+ \forall x \forall y \forall z((Fxy \wedge Fyz) \supset Fxz)$
    $? \forall x \forall y(Fxy \supset \neg Fyx)$

14. $+ \forall x \exists y Fxy$
    $+ \neg \exists x Fxx$
    $? \forall x(Fxa \supset a \neq x)$

15. $+ \exists x(x = a \wedge x = b)$
    $+ Fa$
    $? Fb$

5.08 Theorems in $\mathcal{L}_P$

Theorems in $\mathcal{L}_S$ make up the set of logical truths expressible in $\mathcal{L}_S$. These are $\mathcal{L}_S$ sentences true under any interpretation, sentences true no matter what. $\mathcal{L}_P$ expands the set of logical truths by adding logical truths expressible in $\mathcal{L}_P$ not expressible in $\mathcal{L}_S$. Just as in $\mathcal{L}_S$, a sentence is a theorem of $\mathcal{L}_P$ if it can be derived from the empty set of sentences. The derivation of theorems in $\mathcal{L}_P$ resembles the technique employed to derive theorems in $\mathcal{L}_S$; the only difference is that now quantifiers are in play.

Consider the sentence below:

$$\vdash \exists x(Fx \wedge Gx) \supset (\exists xFx \wedge \exists xGx)$$

As the $\vdash$ indicates, this sentence is a theorem. Indeed, the sentence is so derivable, via *CP*, from the empty set of sentences.

1.	$\exists x(Fx \wedge Gx)$	
2.	$? \exists xFx \wedge \exists xGx$	
3.	$Fa \wedge Ga$	1 *EI*
4.	Fa	3 $\wedge E$
5.	$\exists xFx$	4 *EG*
6.	Ga	3 $\wedge E$
7.	$\exists xGx$	6 *EG*
8.	$\exists xFx \wedge \exists xGx$	5, 7 $\wedge I$
9.	$\exists x(Fx \wedge Gx) \supset (\exists xFx \wedge \exists xGx)$	1–8 *CP*

The proof is straightforward, but proofs of other theorems can be more challenging. Consider the theorem

$$\vdash \forall x\exists y(Fx \wedge Gy) \supset \exists x(Fx \wedge Gx)$$

Here too, a derivation of this theorem can be constructed by deploying *CP*.

1.	$\forall x\exists y(Fx \wedge Gy)$	
2.	$? \exists x(Fx \wedge Gx)$	
3.	$\exists y(Fa \wedge Gy)$	1 *UI*
4.	$Fa \wedge Gb$	3 *EI*
5.	Gb	4 $\wedge E$
6.	$\exists y(Fb \wedge Gy)$	1 *UI*
7.	$Fb \wedge Gc$	6 *EI*
8.	Fb	7 $\wedge E$
9.	$Fb \wedge Gb$	5, 8 $\wedge I$
10.	$\exists x(Fx \wedge Gx)$	9 *EG*
11.	$\forall x\exists y(Fx \wedge Gy) \supset \exists x(Fx \wedge Gx)$	1–10 *CP*

Derivations of theorems often incorporate moves of the sort illustrated by lines 3 and 6. In those lines *UI* is applied to the same sentence twice, and *EI* is applied two times to the same sentence, once in line 3 and again in line 6. Moves of this kind are not difficult per se, but they can easily escape notice.

The derivation of the theorem below illustrates another feature common to derivations of theorems.

$$\vdash \exists x(Fx \supset Ga) \equiv (\forall xFx \supset Ga)$$

This theorem consists of a biconditional sentence. Bearing in mind that a biconditional is a two-way conditional, a derivation can be produced by deploying a pair of conditional proofs, one of which incorporates an embedded conditional proof, and the other an embedded indirect proof.

1.	$\exists x(Fx \supset Ga)$	
2.	$? \forall xFx \supset Ga$	
3.	$\forall xFx$	
4.	$? Ga$	
5.	$Fb \supset Ga$	1 *EI*
6.	Fb	3 *UI*
7.	Ga	5, 6 *MP*
8.	$\forall xFx \supset Ga$	3–7 *CP*
9.	$\exists x(Fx \supset Ga) \supset (\forall xFx \supset Ga)$	1–8 *CP*
10.	$\forall xFx \supset Ga$	
11.	$? \exists x(Fx \supset Ga)$	
12.	$\neg\exists x(Fx \supset Ga)$	
13.	$? \times$	
14.	$\forall x\neg(Fx \supset Ga)$	12 *QT*
15.	$\neg(Fc \supset Ga)$	14 *UI*
16.	$\neg(\neg Fc \vee Ga)$	15 *Cond*
17.	$Fc \wedge \neg Ga$	16 *DeM*
18.	Fc	17 $\wedge E$
19.	$\forall xFx$	18 *UG*
20.	Ga	10, 19 *MP*
21.	$\neg Ga$	17 $\wedge E$
22.	$Ga \wedge \neg Ga$	20, 21 $\wedge I$
23.	$\exists x(Fx \supset Ga)$	12–22 *IP*
24.	$(\forall xFx \supset Ga) \supset \exists x(Fx \supset Ga)$	10–23 *CP*
25.	$(\exists x(Fx \supset Ga) \supset (\forall xFx \supset Ga)) \wedge ((\forall xFx \supset Ga) \supset \exists x(Fx \supset Ga))$	9, 24 $\wedge I$
26.	$\exists x(Fx \supset Ga) \equiv (\forall xFx \supset Ga)$	25 *Bicond*

In line 25, the conditionals appearing in lines 9 and 24 are conjoined so that in line 26, *Bicond* can be applied yielding the theorem.

Notice that when the existential quantifier is dropped in line 5, the variable it bound cannot be replaced by *a*: *a* occurs in an earlier active sentence, the sentence in line 1. The application of *UG* in line 19 might look suspicious, but it accords with the restriction. The constant, *b*, that figures in the generalization does not occur in an *active* supposition or in the conclusion, nor does it result from an application of rule *EI*.

Exercises 5.08

Provide derivations for each of the theorems of $\mathcal{L}_\mathcal{P}$ below.

1. $\vdash \forall x \exists y(Fy \supset Fx)$
2. $\vdash \forall x(Fx \vee \neg Fx)$
3. $\vdash \neg \exists x(Fx \wedge \neg Fx)$
4. $\vdash \forall x \forall y((Fx \wedge x = y) \supset Fy)$
5. $\vdash \forall x \forall y(x = y \supset y = x)$
6. $\vdash \forall x \forall y(x = y \supset (Fx \equiv Fy))$
7. $\vdash \exists x(Fx \supset \forall y Fy)$
8. $\vdash \exists x \forall y Fxy \supset \forall y \exists x Fxy$
9. $\vdash \exists x Fx \supset (\forall x \neg Fx \supset \forall x Gx)$
10. $\vdash \forall x(Fx \supset Gx) \supset (\neg \exists x Gx \supset \neg \exists x Fx)$
11. $\vdash \exists x Fxx \supset \exists x \exists y Fxy$
12. $\vdash \forall x \forall y Fxy \supset \forall x Fxx$
13. $\vdash (\exists x Fx \wedge \exists x Gx) \supset \exists x \exists y(Fx \wedge Gy)$
14. $\vdash \forall x \forall y(Fxy \supset \neg Fxy) \supset \forall x \neg Fxx$
15. $\vdash \exists x \forall y(Fx \supset Gy) \supset (\forall x Fx \supset \forall x Gx)$

5.09 Invalidity in $\mathcal{L}_\mathcal{P}$

Assuming the rules included in $\mathcal{L}_\mathcal{P}$ have been properly formulated, they will yield only valid inferences (in which case $\mathcal{L}_\mathcal{P}$ would be *sound*). Imagine that you have been unsuccessful in discovering a derivation that proves a sequence valid. There are three possibilities.

First, you might have miscopied the sequence in the course of writing it out, a surprisingly common occurrence. Second, a derivation might be possible but temporarily escape your ingenuity. Third, the sequence might be *in*valid. Were that so, a derivation aimed at proving its validity would be not simply elusive but impossible. You could settle the matter decisively by discovering a derivation

showing that the sequence *is* valid. Failure to discover a derivation, however, would not entitle you to conclude that the sequence is not valid.

Might there be a *test* for invalidity, a procedure that would enable you to prove particular sequences invalid? In $\mathcal{L}_S$, the matter was simple. You could always construct a truth table—or a diagram—that demonstrated invalidity conclusively. Truth tables do not extend smoothly to $\mathcal{L}_P$, however. $\mathcal{L}_P$ sequences lacking quantifiers are, in effect, $\mathcal{L}_S$ sequences amenable to truth table analyses. Once quantifiers enter the picture, everything changes. Truth tables are finite devices, but the quantificational structure of $\mathcal{L}_P$ is open-ended.

Completeness

This discussion omits a fourth possibility: $\mathcal{L}_P$, as it stands, might be *incomplete*. In that case, there could be valid sequences that are not provable given only the rules introduced thus far. Completeness, as well as soundness, will be addressed in due course. Meanwhile, you have no choice but to trust that the author has done the right thing.

Skeptics are welcome to skip ahead to § 5.11 before proceeding further.

Reflect on a simple universally quantified sentence of the form $\forall xFx$: everything is *F*. This sentence is true under an interpretation if and only if every object in the domain is *F*. Given a finite domain—the set of planets, for instance—a universally quantified sentence is reducible to a conjunction. If $\forall xFx$, then

$$a \text{ is } F,\ b \text{ is } F,\ c \text{ is } F,\ \ldots,$$

and so on for every individual in the domain. Sentences, so reduced, could be evaluated using truth tables: *a* is *F* is true if *a is F*, false otherwise.

Suppose the domain is not finite, however. In that case $\forall xFx$ reduces to an open-ended conjunction:

$$Fa \wedge Fb \wedge Fc \wedge \ \ldots$$

Truth tables are not open-ended, however, so no such conjunction could be represented, much less tested, by means of a truth table. Indeed, it is precisely the open-ended character of quantification that makes it so significant. The truth conditions of the sentence

All horses are mammals.

concern an open-ended collection of individuals to which new members could be added indefinitely.

Strictly speaking, then, universally quantified sentences are not reducible to conjunctions. Sentences must be finite in length. If domains are infinite or open-ended, however, conjunctive reductions of universally quantified sentences would either be infinitely long or have no last member. Because such conjunctions would not count as sentences, they would not count as sentences reductively equivalent to the original universally quantified sentence.

Still, there is something right about thinking of universally quantified sentences as conjunctions, even if these conjunctions would not count as sentences of $\mathcal{L}_P$. 'All' and 'Every' mean this *and* this *and* this, *and* . . .

What of existentially quantified sentences? If a universally quantified sentence could be thought of as an open-ended conjunction, an existentially quantified sentence of the form $\exists xFx$ could be regarded as an open-ended *disjunction*. If *something* is *F*, then *a* is *F*, or *b* is *F*, *or c* is *F*, *or* . . .

Hold on to that thought, and recall truth table tests for validity of sequences in $\mathcal{L}_S$. Truth tables provide a systematic representation of the truth conditions of sentences in the sequence. The goal

is to discover whether, given those truth conditions, the sequence's premises could be true when its conclusion is false. In truth table lingo, that would mean discovering whether a truth table representation of the sequence contains one or more rows in which the value of each premise is *true* and the value of the conclusion is *false*.

Consider the $\mathcal{L}_S$ sequence below:

1. $+ P \supset Q$
2. $+ R \supset S$
3. $+ R \vee Q$
4. $? P \vee S$

You could construct a sixteen-row truth table for the whole sequence. The sequence would be invalid if the resulting truth table included a row in which the premises were true and the conclusion false.

In § 3.18 you were introduced to a less onerous method for proving invalidity. First the sequence was represented horizontally. You then looked for an interpretation—an assignment of truth values—to individual atomic sentences under which the premises were true and the conclusion false. As it happens, an assignment of truth values in which Q is true and P, R, and S are false establishes that there is an interpretation under which the premises are true and the conclusion is false, so the sequence is invalid.

$P \supset Q$	$R \supset S$	$R \vee Q$	$P \vee S$
F T	F F	F T	F F
T	T	T	F

Might this technique somehow be extended to $\mathcal{L}_P$?

The first step requires replacing universally quantified sentences with conjunctions and existentially quantified sentences with disjunctions. How many conjuncts and disjuncts? At this point simple comparisons between $\mathcal{L}_S$ and $\mathcal{L}_P$ break down. The number of conjuncts and disjuncts depends on the size of the domain. If the domain is open-ended, quantified sentences would require open-ended counterparts, and the truth table method of checking for validity would go by the board.

Think back on the requirements for validity in $\mathcal{L}_S$. A sequence, $\langle \Gamma, \phi \rangle$, is valid just in case Γ logically implies ϕ, ($\Gamma \vDash \phi$). Γ logically implies ϕ just in case there is no interpretation under which Γ is true and ϕ is false. In $\mathcal{L}_S$, an interpretation is an assignment of truth values to each distinct atomic sentence in Γ and ϕ. Might something like this work in $\mathcal{L}_P$?

In $\mathcal{L}_P$, an interpretation includes the specification of a domain of objects and an assignment of constants to objects in the domain (see § 4.14 for a more detailed discussion). Were these domains open-ended, the truth table method of proving invalidity of sequences containing quantified sentences could not be deployed. Because there is no reason to restrict the choice of domains, however, truth tables fail to provide foolproof tests for validity in $\mathcal{L}_P$. Even so, truth tables can be used as limited, but useful, aids—heuristics—for establishing the invalidity of sequences in $\mathcal{L}_P$.

A sequence is invalid if there is an interpretation—*any* interpretation—under which its premises are true and its conclusion false. Suppose you started with an interpretation with a finite domain for

Fermat's Last Theorem

If you have ever struggled with a sequence for which you can find no derivation, you might have a sense of the frustration felt by mathematicians who, for more than three hundred years, could not find a proof for the following theorem:

> The equation $x^n + y^n = z^n$, where n is an integer greater than 2, has no solution in the positive integers.

The theorem, commonly known as 'Fermat's Last Theorem', is named after French mathematician Pierre de Fermat (1601–1665), who wrote next to the theorem in the margin of a book, 'I have found a truly wonderful proof which this margin is too small to contain'.

In 1993, Andrew Wiles, a British mathematician working in the United States, announced that he had discovered a proof of the theorem. The proof turned out to contain errors, but in 1994, Wiles corrected his proof, and the corrected version was published in 1995.

Although you might suspect that some of the sequences for which you are asked to provide derivations are the logical counterparts of Fermat's Last Theorem, rest assured that derivations required for the exercises in this book are a breeze by comparison.

a sequence, $\langle \Gamma, \phi \rangle$. And suppose, under this interpretation, Γ is true and ϕ is false. Bingo! You would have established that the sequence is invalid.

By assuming a finite domain consisting of a single individual, a, for instance, or a domain comprising two individuals, a and b, you could represent quantified sentences as conjunctions or disjunctions, and set out the whole affair in a truth table. How might this work in practice?

Consider the sequence

$$+ \forall x(Fx \supset Gx)$$
$$+ \neg\exists x Fx$$
$$? \neg\exists x Gx$$

Start with an interpretation of the sentences in the sequence assuming a domain consisting of a single object, a. Given this domain, the sentence in line 1 is reducible to

$$Fa \supset Ga$$

If everything in the domain is such that if it is an F, then it is a G, and if the domain consists of just one individual, a, then it is true of a that if a is F then a is G. The sentence in line 2 says that it is not the case that something is F. There is only a single object in the domain to consider, namely, a, so this sentence is reducible to

$$\neg Fa$$

Finally, the conclusion can be rewritten as

$$\neg Ga$$

If it is not the case that something is G, and if there is only one object to consider, a, then it is not the case that a is G.

The sequence can now be represented in a truth table.

Fa *Ga*	*Fa* ⊃ *Ga*	¬*Fa*	¬*Ga*
T T	T	F	F
T F	F	F	T
F T	T	T	F
F F	T	T	T

In the third row of the truth table, the premises are true and the conclusion is false: the sequence is invalid.

Each row of the truth table above represents one member of a class of interpretations for that domain. So long as the domain is finite, the list of the possible interpretations can be set out exhaustively. So long as the domain includes only a single individual, differences between quantifiers are invisible. In a domain consisting of a single object, *a*, the sentence

$$\forall x Fx$$

and the sentence

$$\exists x Fx$$

are reducible to identical sentences. If every object is *F* and there is only one object, *a*, then *a* is *F*:

$$Fa$$

Similarly, if at least one object is *F* and there is only a single object, *a*, then *a* is *F*

$$Fa$$

Although interpretations that make use of domains consisting of a single individual cannot distinguish universally and existentially quantified sentences, matters change once you move to interpretations incorporating domains consisting of two or more individuals. Consider the quantified sentences above construed for a domain consisting of two objects, *a* and *b*. The universally quantified sentence is reducible to a conjunction:

$$Fa \wedge Fb$$

If every object in the domain is *F* and the domain contains exactly two objects, *a* and *b*, then *a* is *F* *and* *b* is *F*. In contrast, if, as in the existentially quantified version of the sentence, *some* object is *F*, then *a* is *F* *or* *b* is *F*.

$$Fa \vee Fb$$

Now suppose you are faced with the task of testing the sequence below for invalidity.

1. $+ \forall x(Fx \supset Gx)$
2. $+ \exists x Fx$
3. $? \forall x Gx$

You might start with a domain consisting of a single object, *a*. Given that domain, the sequence is reducible to the sequence

1. $+ Fa \supset Ga$
2. $+ Fa$
3. $? Ga$

A truth table reveals that no interpretation incorporating a domain consisting of a single object is such that the premises of the sequence are true and its conclusion false.

Fa	*Ga*	*Fa* ⊃ *Ga*	*Fa*	*Ga*
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	F	F

Is the original sequence valid? Intuitively, it seems not to be. From 'All *F*s are *G*s' and 'Something is an *F*'. it does not appear to follow that 'Everything is a *G*'. The sequence's invalidity has something to do with the quantifiers in play. Recalling that quantifier differences appear only in domains consisting of two or more individuals, you might turn to a domain that includes two individuals, *a* and *b*. In such a domain, the original sequence is reducible to

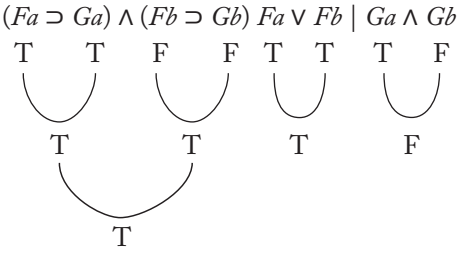
1. $+ (Fa \supset Ga) \wedge (Fb \supset Gb)$
2. $+ Fa \vee Fb$
3. $? Ga \wedge Gb$

Now the distinction between universally and existentially quantified sentences is signaled by the occurrence of ∧s and ∨s, respectively. A truth table for this sequence provides a systematic representation of the truth conditions of each sentence for every interpretation that makes use of the domain.

Fa *Fb* *Ga* *Gb*	*Fa* ⊃ *Ga*	*Fb* ⊃ *Gb*	(*Fa* ⊃ *Ga*) ∧ (*Fb* ⊃ *Gb*)	*Fa* ∨ *Fb*	*Ga* ∧ *Gb*
T T T T	T	T	T	T	T
T T T F	T	F	F	T	F
T T F T	F	T	F	T	F
T T F F	F	F	F	T	F
T F T T	T	T	T	T	T
T F T F	T	T	T	T	F
T F F T	F	T	F	T	F
T F F F	F	T	F	T	F
F T T T	T	T	T	T	T
F T T F	T	F	F	T	F
F T F T	T	T	T	T	F
F T F F	T	F	F	T	F
F F T T	T	T	T	F	T
F F T F	T	T	T	F	F
F F F T	T	T	T	F	F
F F F F	T	T	T	F	F

In both the sixth and eleventh rows of the truth table, the premises are true and the conclusion is false. The sequence is invalid. A sequence is invalid if there is *any* interpretation under which its premises are true and its conclusion is false, and, as the truth table makes clear, there are interpretations with domains consisting of two individuals, *a* and *b*, in which the premises are true and the conclusion is false.

Now the way is open to deploy the less cumbersome technique for the evaluation of sequences in $\mathcal{L}_S$ discussed earlier, to sequences in $\mathcal{L}_P$. After writing out the sequence horizontally, try to find an assignment of truth values that would make the premises true and the conclusion false. Applying this method to the sequence under consideration yields



The diagram provides a representation of the sixth row of the original truth table. There is, then, an interpretation, *I*, under which the premises are true and the conclusion false; I : {Fa = T, Fb = F, Ga = T, Gb = F}.

As in $\mathcal{L}_\mathcal{S}$, you might start by finding an assignment of truth values that makes the conclusion false, and carry these assignments over to the premises. You would then assign truth values to any remaining elements of the premises so as to make them true. These assignments correspond to distinct rows of the original truth table.

In some of the rows in which the conclusion is false, one or more of the premises is false. Remember that, in a given sequence, if there is more than one way to make the conclusion false or the premises true, you cannot conclude that the sequence is probably valid solely on the grounds that *one* of these ways does not result in an interpretation of the sequence that demonstrates its invalidity. You must first make certain that there is no *alternative* assignment that will make the premises of the sequence true and its conclusion false.

Suppose you find no interpretation under which the premises of a sequence are true and its conclusion is false, assuming a domain of two objects. Would this show that the sequence is valid? No. There might be an interpretation incorporating a larger domain under which its premises are true and its conclusion false. The truth table technique does not guarantee success. It can be used to prove invalidity, but not every invalid sequence could be demonstrated invalid using the technique. In general, any invalid sequence whose invalidity would only emerge under interpretations incorporating very large or perhaps open-ended domains would escape detection.

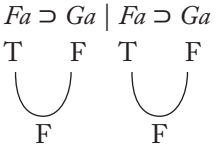
Despite these limitations, the truth table technique can be useful in practice. Consider the sequence

1. $+ \forall x Fx \supset \forall x Gx$
2. $? \forall x(Fx \supset Gx)$

Is this sequence valid? Maybe, maybe not. You might try showing that it is invalid under an interpretation that assumes a domain consisting of a single object, *a*, in which case the sequence would be represented as

1. $+ Fa \supset Ga$
2. $? Fa \supset Ga$

Assuming this domain, there is no interpretation under which the conclusion is false and the premise is true: the premise and conclusion are identical, so if the conclusion is false, the premise must be false as well as the diagram below illustrates.



There is no interpretation incorporating a domain consisting of a single individual under which the premise is true and the conclusion false. What of interpretations incorporating more populous

domains? What of an interpretation incorporating a domain consisting of a pair of individuals, *a* and *b*? Assuming such a domain, you could represent the original sequence as follows:

1. $+ (Fa \wedge Fb) \supset (Ga \wedge Gb)$
2. $? (Fa \supset Ga) \wedge (Fb \supset Gb)$

This makes differences between the premise and the conclusion obvious. The premise says

If everything is *F*, then everything is *G*.

If the 'everything' in question is *a* and *b*, then the sentence is reducible to

If *a* and *b* are *F*, then *a* and *b* are *G*.

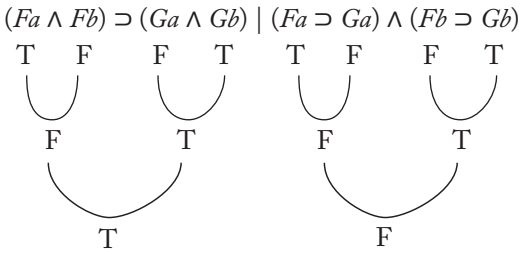
The conclusion is

Everything is such that, if it is *F*, then it is *G*.

that is,

If *a* is *F*, then *a* is *G*, and if *b* is *F*, then *b* is *G*.

Assuming this domain, it is easy to find an interpretation under which the premise is true and the conclusion is false:



The sequence is shown to be invalid under *I*: {Fa = T; Fb = F; Ga = F; Gb = T}.

A complication remains. Given finite domains, universally quantified sentences are reducible to conjunctions and existentially quantified sentences are reducible to disjunctions. Many sentences in $\mathcal{L}_P$, however, contain *mixed* quantification. How are these to be reduced assuming domains of more than one individual? Consider the sentence

$$\forall x \exists y Mxy$$

The sentence says, in effect, that *everything* bears relation *M* to *something*. Assuming a domain consisting of a single object, *a*, the reduction is simple:

$$Maa$$

If the domain includes only a single object, then to say that every object in the domain bears relation *M* to some object is to say that *a* bears *M* to itself, a reflection of the fact that quantifier differences emerge only in domains consisting of more than one individual. In a two-object domain, the reduction looks like this:

$$(Maa \vee Mab) \wedge (Mba \vee Mbb)$$

Given this domain, the sentence says that *a* bears *M* to *a* or to *b*, and *b* bears *M* to *a* or to *b*. The ∨s carry the sense of the existential quantifier and the ∧s reflect universal quantification. Similarly, assuming a two-individual domain, the sentence

$$\forall x Fx \supset \exists y Gy$$

is reducible to

$$(Fa \wedge Fb) \supset (Ga \vee Gb)$$

and the sentence

$$\exists x(Fx \supset \forall y Mxy)$$

reduces to

$$(Fa \supset (Maa \wedge Mab)) \vee (Fb \supset (Mba \wedge Mbb))$$

If this seems confusing, think of the original sentence in quasi-$\mathcal{L}_\mathcal{P}$:

There is some *x* such that if *x* is *F*, then, for all *y*, *x* bears *M* to *y*.

If the domain consists of just *a* and *b*, then this amounts to

If *a* is *F* then *a* bears *M* to both *a* and *b*, or if *b* is *F* then *b* bears *M* to both *a* and *b*.

Although the truth table test for invalidity falls short of perfection, it provides an important heuristic for uncovering instances of invalidity. In most cases, when you suspect that a sequence is invalid, you can find an interpretation under which its conclusion is false and its premises true by assuming a domain restricted to one, or, when the invalidity of a sequence turns on differences in quantifiers, two individuals. If such an interpretation emerges, you will have proved the sequence invalid.

Sequences Containing Quantifiers and Constants

What happens when you are faced with a sequence that includes both quantifiers and individual constants? Consider the sequence below.

1. $+ \forall x(x = c \supset Fx)$
2. $+ \forall x(Fx \supset Fd)$
3. $? \neg Fd$

The sequence includes a pair of individual constants, *c* and *d*. This means that your choice of a domain must include at least two individuals, *c* and *d*. In derivations containing constants, then, any interpretation must include (at least) the individuals corresponding to those constants.

Sequences that include three or more individual constants require a domain consisting of three (or more) individuals. This complicates reductions, but it does not alter the procedure. Universally quantified sentences reduce to conjunctions, and existentially quantified sentences reduce to disjunctions.

Failure to find an interpretation that establishes the invalidity of a sequence could be due to the fact that (i) the sequence is, after all, valid or (ii) its premises are true and its conclusion false only under interpretations involving larger domains. In practice, however, if you cannot prove invalidity with a domain consisting of two individuals, you would be safe in looking for a proof of validity.

Exercises 5.09

Provide interpretations that demonstrate the invalidity of the sequences below. Use truth tables or diagrams to support your answer.

1. $+ \exists x(Fx \wedge \neg Gx)$
   $+ \forall x(Hx \supset Fx)$
   $? \exists x(Hx \wedge \neg Fx)$

2. $+ \forall x(Fx \supset Gx)$
   $+ \forall x(\neg Fx \supset Hx)$
   $? \forall x(\neg Gx \supset \neg Hx)$

3. $+ \neg\exists x(Fx \wedge \neg Gx)$
   $+ \neg\exists x(Fx \wedge \neg Hx)$
   $+ \exists x(\neg Gx \wedge Jx)$
   $? \exists x((Fx \wedge Gx) \wedge Jx)$

4. $+ \forall x(Fx \supset \exists y Gxy)$
   $+ \exists x Fx$
   $+ \exists x \exists y Gxy$
   $? \forall x \exists y Gxy$

5. $+ \forall x(Fx \supset Gx)$
   $+ \forall x Gx \supset \exists x Hx$
   $? \forall x(Fx \supset Hx)$

6. $+ \exists y(Fy \supset \forall x Gxy)$
   $+ \forall y(Fy \wedge \exists x Hyx)$
   $? \exists x \forall y(Gxy \wedge Hyx)$

7. $+ \forall x Fx \supset \forall x Gx$
   $+ \exists x(Fx \vee Gx)$
   $? \neg\forall x(Fx \supset Gx)$

8. $+ \forall x(Fx \supset (Gx \wedge Hx))$
   $+ Fa$
   $? \exists x \neg Hx$

9. $+ \forall x(Fx \supset Gxa)$
   $+ \exists x Gxa \supset \exists x Hxa$
   $? \forall x \exists y(Fx \supset Hxy)$

10. $+ \exists x(Fx \supset \forall y(Fy \supset Jy))$
    $+ \exists x(Fx \wedge (Gx \vee Hx))$
    $? \forall x(Fx \supset Hx)$

11. $+ \forall x \exists y Fxy$
    $? \exists y \forall x Fxy$

12. $+ \forall x Fxx$
    $? \forall x \forall y Fxy$

13. $+ \exists x \exists y Fxy$
    $? \exists x Fxx$

14. $+ \exists x(Fx \wedge Gx)$
    $+ \forall x(Gx \supset Hx)$
    $? \forall x(Gx \vee Fx)$

15. $+ \forall x \exists y Fxy$
    $+ \exists x Fxx$
    $? \forall x Fxx$

5.10 Prenex Normal Form

Logicians prize elegance. If there are two ways of writing a sentence or constructing a derivation, a card-carrying logician prefers the more elegant of the two. In logic and mathematics, elegance amounts to formal simplicity. If there are two ways to derive a sentence, for instance, the simpler of the two, the derivation with the fewest steps, receives the nod. Both derivations do the job, but one does it more cleanly than the other.

Elegance is favored no less in the sciences than in logic. Quine captured the ethos when he observed that logicians (and mathematicians, and scientists) have a taste for desert landscapes. Even if you do not share Quine's aesthetic sensibilities, however, you might at least agree that once you have mastered a technique, a certain satisfaction attends your performing it well, as against just performing it.

You have probably already encountered examples of what I am referring to in the course of constructing derivations. Sometimes you embark on a derivation and, partway through it, you realize that some steps you have taken are superfluous: you could have reached your goal without them. There is no harm in leaving the unneeded steps in, but you recognize that, if you were to do the derivation over, you would omit them.

When it comes to sentences, matters are different. There, elegance is not a matter of leaving out superfluous symbols but a matter of form. In most sentences containing quantifiers, the quantifiers are sprinkled throughout the sentence, as in the sentence below, a sentence discussed in § 4.11.

The detective is never happy.

$$\exists x((Dx \wedge \forall y(Dy \supset x = y)) \wedge \neg\exists z(Tz \wedge Lxz))$$

The sentence is perfectly satisfactory as it stands. How could it possibly be improved?

As it happens, once you have grown comfortable with translations, you can fine-tune the sentence in a way that makes it more elegant formally by locating all of the quantifiers at the front of the sentence. Once you move beyond this book, you might discover that, in addition to affording a measure of aesthetic gratification, locating quantifiers at the front of sentences can also be useful.

Quantifiers cannot be moved around in sentences willy-nilly, however. Doing so would risk changing the sentences' meaning. Moving quantifiers requires compensating changes elsewhere in the sentence that can be spelled out in a simple procedure. A sentence that is either quantifier-free or consists of a string of quantifiers followed by a quantifier-free expression is said to be in *prenex normal form* (PNF). Any sentence containing quantifiers can be converted to PNF by following the steps below:

(1) Remove all occurrences of $\supset$ and $\equiv$ by applying appropriate transformation rules.

(2) 'Bring in' negation signs until they stand only in front of predicates by applying *QT* and, if necessary, *DeM*.

(3) Rewrite quantifiers until no two quantifiers contain the same variable, and adjust the bound variables accordingly.

(4) Place all the quantifiers at the beginning of the sentence *in the order in which they occur* in the original sentence. The scope of each quantifier must include the entire expression to its right.

You can see how this works in the case of a sentence discussed earlier.

1.	$\exists x((Dx \wedge \forall y(Dy \supset x = y)) \wedge \neg\exists z(Tz \wedge Lxz))$	(*original sentence*)
2.	$\exists x((Dx \wedge \forall y(\neg Dy \vee x = y)) \wedge \neg\exists z(Tz \wedge Lxz))$	1 *Cond*
3.	$\exists x((Dx \wedge \forall y(\neg Dy \vee x = y)) \wedge \forall z\neg(Tz \wedge Lxz))$	2 *QT*
4.	$\exists x((Dx \wedge \forall y(\neg Dy \vee x = y)) \wedge \forall z(\neg Tz \vee \neg Lxz))$	3 *DeM*
5.	$\exists x\forall y\forall z((Dx \wedge (\neg Dy \vee x = y)) \wedge (\neg Tz \vee \neg Lxz))$	4 *PNF*

Although there is no rule for moving quantifiers to the front of a sentence, the move made in line 5 above is valid. (This could be proved by deriving the PNF sentence from the original sentence using *IP*.)

1.	$+\ \exists x((Dx \wedge \forall y(Dy \supset x = y)) \wedge \neg\exists z(Tz \wedge Lxz))$	(*original sentence*)
2.	$?\ \exists x\forall y\forall z((Dx \wedge (\neg Dy \vee x = y)) \wedge (\neg Tz \vee \neg Lxz))$	(*PNF sentence*)
3.	$\neg\exists x\forall y\forall z((Dx \wedge (\neg Dy \vee x = y)) \wedge (\neg Tz \vee \neg Lxz))$	
4.	$? \times$	
5.	⋮	

You might never find a reason to regiment sentences by putting them into PNF, but you never know! Just in case, you might try converting the sentences in exercises 5.09.

Exercises 5.10

Put each of the sentences below into PNF.

1. $\forall xFx \supset Ga$
2. $\forall x(Cx \supset \forall y(Fy \supset Lxy))$
3. $\exists x\neg\forall yTxy$
4. $\neg\exists x\forall y(Ly \supset Txy)$
5. $\forall x(Px \supset \exists y(Ty \wedge Hxy))$
6. $\neg\exists x(Hx \wedge Bx) \supset \exists x(Hx \wedge Dx)$
7. $\exists x(Px \wedge Fxs) \supset \forall x(Px \supset Fxs)$
8. $\exists x(Px \wedge \exists y(Py \wedge x \neq y))$
9. $\forall x(Fx \supset Gx) \supset \exists x(Fx \wedge Gx)$
10. $Me \wedge \forall x((Mx \wedge x \neq e) \supset Wex)$

5.11 Soundness and Completeness of $\mathcal{L}_p$

The metalogical notions of soundness and completeness were introduced in § 3.17. $\mathcal{L}_p$ is *sound* just in case every derivation expressible in $\mathcal{L}_p$ is valid: where $\langle\Gamma,\phi\rangle$ is a sequence, if $\Gamma \vdash \phi$, then $\Gamma \vDash \phi$ (if a set of sentences, Γ, *deductively yields* a sentence, ϕ, then Γ *logically implies* ϕ). $\mathcal{L}_p$ is *complete* just in case, if a sequence is valid, a derivation could be given for it in $\mathcal{L}_p$: if $\Gamma \vDash \phi$, then $\Gamma \vdash \phi$ (if a set of sentences, Γ, logically implies a sentence, ϕ, then Γ deductively yields ϕ).

In discussing soundness and completeness in $\mathcal{L}_S$, no attempt was made to provide exhaustive proofs of either. I observed only what would be involved in such proofs. I shall adopt the same informal mode of exposition here.

Suppose that the soundness of the rules used in $\mathcal{L}_S$ derivations has been established. If $\mathcal{L}_S$ is sound, then its rules are *truth-preserving*: if you start with true sentences, any sentences derived from these using the rules yields true sentences: true in, true out. An $\mathcal{L}_P$ derivation that uses only rules used in $\mathcal{L}_S$, then, must be sound.

Extending the proof of soundness to the quantifier rules (*QT*, *UI*, *UG*, *EI*, and *EG*) together with the rule for identity (*ID*) would require showing that these rules too are truth-preserving: applying them to true sentences yields true sentences. This amounts to showing that for any sentence, ϕ, if ϕ occurs as the result of the application of one of these rules, ϕ is logically implied by the sentence or sentences from which ϕ was derived.

Consider rule *UI*. Suppose that Γ is a sequence in which ϕ occurs as the result of an application of *UI*, and that Γ includes the universally quantified sentence

$$\forall x Fx$$

and ϕ is the sentence

$$Fa$$

If rule *UI* is truth-preserving, there is no interpretation under which the first sentence is true and the second sentence is false. I hope it is clear that, given the account of what it means for a universally quantified sentence such as $\forall x Fx$ to be true under an interpretation (see § 4.16), if the first sentence is true, the second must be true as well.

This will be so for any pair of sentences consisting of a universally quantified sentence, $\forall x \phi$, and a sentence, $\phi\alpha/\beta$, derived from that sentence via an application of *UI*. This line of reasoning extends to the remaining rules, resulting in a proof of the soundness of $\mathcal{L}_P$: ϕ is derivable from Γ only if Γ implies ϕ.

A proof for the completeness of $\mathcal{L}_P$ is more complicated, just as it was in $\mathcal{L}_S$. A proof for completeness would establish that every valid sequence expressible in $\mathcal{L}_P$ is derivable. The truth table technique mentioned in § 3.20 cannot be extended to cover sequences containing quantified sentences, however. How, then, might a proof for completeness go?

Start with the idea that a set of sentences, Γ, is *consistent with respect to derivability* or, for short, *d*-consistent, just in case the sentence $\gamma \wedge \neg\gamma$ is not derivable from Γ. (In English, a set of sentences is *d*-consistent if you cannot derive a contradiction from it.)

The concept of *d*-consistency differs from consistency *tout court*, the ordinary semantic notion of consistency. If a set of sentences, Γ, is consistent (in the semantic sense), it does not *imply* a contradiction. If Γ is *d*-consistent, a contradiction cannot be *derived from* Γ. A pivotal move in the proof of completeness is a proof that *d*-consistent sets of sentences are consistent in the semantic sense.

The task now would be to show that for any sentence, ϕ, and any set of sentences, Γ, ϕ is derivable from Γ if and only if $\{\Gamma,\neg\phi\}$ is not *d*-consistent.

Suppose, first, that ϕ is derivable from Γ, and that you have constructed a derivation of ϕ from Γ in which ϕ appears on the last line. Next, suppose $\neg\phi$ is added to Γ. Now $\gamma \wedge \neg\gamma$ is derivable from Γ,

so $\{\Gamma,\neg\phi\}$ is not *d*-consistent. In general, once you show that a sequence yields some sentence *and* its contradiction, you can show that it yields *any* sentence:

⋮		
i.	ϕ	
j.	$\neg\phi$	
k.	$\phi \vee (\gamma \wedge \neg\gamma)$	i $\vee I$
l.	$\gamma \wedge \neg\gamma$	j,k $\vee E$

Now suppose that $\{\Gamma,\neg\phi\}$ is not *d*-consistent. In that case, $\gamma \wedge \neg\gamma$ is derivable from $\{\Gamma,\neg\phi\}$. If that is so, the sentence

$$\neg\phi \supset (\gamma \wedge \neg\gamma)$$

can be derived from Γ. (In general, for any set of sentences, Γ, and any sentences ϕ and γ, if $\{\Gamma, \phi\} \vdash \gamma$, then $\Gamma \vdash (\phi \supset \gamma)$. Having derived this sentence, ϕ can be derived from Γ by deriving $\neg(\gamma \wedge \neg\gamma)$ by means of *IP*, and then applying *MT*.

⋮		
i.	$\neg\phi \supset (\gamma \wedge \neg\gamma)$	
j.	$\gamma \wedge \neg\gamma$	
k.	? ×	
l.	γ	j $\wedge E$
m.	$\neg\gamma$	j $\wedge E$
n.	$\gamma \wedge \neg\gamma$	l, m $\wedge I$
o.	$\neg(\gamma \wedge \neg\gamma)$	j–o *IP*
p.	ϕ	l,o *MT*

Given the concept of *d*-consistency, you can define a related notion of *maximal d-consistency*:

> a set of sentences, Γ, is maximally *d*-consistent if and only if
>
> (i) Γ is *d*-consistent, and
>
> (ii) Γ is not a proper subset of any *d*-consistent set of sentences.

If Γ is maximally *d*-consistent, then, if ϕ is a sentence not in Γ, $\{\Gamma,\phi\}$ is not *d*-consistent.

Finally, suppose that Γ is maximally *d*-consistent and that ϕ is a sentence of $\mathcal{L}_P$. It follows that ϕ is a member of Γ if and only if $\neg\phi$ is *not* a member of Γ. It follows, as well, that ϕ is a member of Γ if and only if ϕ is derivable from Γ. (The derivation in this case will be trivial: if ϕ is in Γ, then *f* can be derived from Γ simply by deriving $\phi \vee \phi$ from ϕ and then applying *Taut*.) Given these results, you can show that

1. $\phi \wedge \varphi$ is in Γ if and only if ϕ is in Γ and φ is in Γ.
2. $\phi \vee \varphi$ is in Γ if and only if ϕ is in Γ, or φ is in Γ, or both.
3. $\phi \supset \varphi$ is in Γ if and only if ϕ is not in Γ, or φ is in Γ, or both.
4. $\phi \equiv \varphi$ is in Γ if and only if ϕ is in Γ and φ is in Γ, or neither ϕ nor φ is in Γ.

Γ can be said to be ω-complete (*omega*-complete) just in case, for every expression ϕ and every variable, α, if $\exists\alpha\phi$ is in Γ, then $\phi\alpha/\beta$ is in Γ (where $\phi\alpha/\beta$ is the sentence that results from dropping the quantifier in $\exists a\phi$ and replacing each instance of the variable α bound by that quantifier with some individual constant, β). If Γ is ω-complete, then, Γ contains an existentially quantified sentence if and only if it contains, as well, a sentence from which that existentially quantified sentence can be derived by an application of rule *EG*. If Γ is both maximally *d*-consistent and ω-complete, then the following will hold:

5. $\forall\alpha\phi$ is in Γ if and only if for every individual constant, β, $\phi\alpha/\beta$ is in Γ.
6. $\exists\alpha\phi$ is in Γ if and only if there is an individual constant, β, such that $\phi\alpha/\beta$ is in Γ.

A proof for the completeness of $\mathcal{L}_\mathcal{P}$ is now within range. Such a proof would begin by establishing that

Every *d*-consistent set of sentences is consistent.

Once this link between the semantic notion of consistency and the syntactic notion of *d*-consistency is forged, completeness could be proved as follows.

Suppose that Γ is a set of sentences and ϕ is logically implied by Γ ($\Gamma \vDash \phi$). In that case, $\{\Gamma,\neg\phi\}$ is not consistent, so not *d*-consistent. If $\{\Gamma, \neg\phi\}$ is not *d*-consistent, ϕ is derivable from Γ ($\Gamma \vdash \phi$). Thus $\mathcal{L}_\mathcal{P}$ is complete.

The foregoing constitutes only the barest outline of a full proof of completeness. A more detailed account can be found in B. Mates, *Elementary Logic*, 2nd ed. (New York: Oxford University Press, 1972), 142–47.

The discussion here is intended only to provide a hint of a deeper and much richer territory. $\mathcal{L}_\mathcal{S}$ and $\mathcal{L}_\mathcal{P}$ are the tip of an iceberg. This book will have achieved its purpose if it has provided you with a feel for the formal languages and a sense of what lies beneath the surface. And now, gentle reader,

Go well!

Solutions to Even-Numbered Exercises[1]

Chapter 2

Exercises 2.01 (p. 9)

(2) S

(4) Q

Exercises 2.05 (p. 15)

(2) $H \wedge G$

(4) $\neg(F \wedge \neg H)$

(6) $\neg(G \wedge H)$

(8) $\neg H \wedge \neg G$

(10) $\neg((H \wedge G) \wedge \neg F)$ or $\neg(H \wedge G) \wedge \neg F$

Exercises 2.06 (p. 19)

(2) $H \vee \neg G$

(4) $\neg(F \vee \neg H)$

(6) $(H \vee G) \wedge \neg(H \wedge G)$

(8) $\neg H \vee \neg G$

(10) $E \vee (G \vee H)$

Exercises 2.08 (p. 27)

(2) $E \supset H$

(4) $(G \vee H) \supset F$

(6) $E \supset (G \supset F)$

(8) $E \supset (H \vee G)$

(10) $F \supset (E \vee (G \vee H))$

Exercises 2.09 (p. 29)

(2) $H \supset E$

(4) $H \equiv (E \vee F)$

(6) $E \equiv (G \wedge H)$

(8) $(H \vee G) \equiv E$

(10) $F \supset (E \equiv \neg G)$

1. Instructors may request a full answer key online at www.hackettpublishing.com/heil-answer-key.

Exercises 2.10 (p. 35)

(2)

P	Q	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

(4)

P	Q	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$	$\neg(\neg P \vee \neg Q)$
T	T	F	F	F	T
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	F

(6)

P	Q	$\neg Q$	$P \supset \neg Q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	T

(8)

P	Q	R	$\neg R$	$Q \wedge \neg R$	$\neg(Q \wedge \neg R)$	$P \supset \neg(Q \wedge \neg R)$
T	T	T	F	F	T	T
T	T	F	T	T	F	F
T	F	T	F	F	T	T
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	T	F	T	T	F	T
F	F	T	F	F	T	T
F	F	F	T	F	T	T

(10)

P	Q	R	$\neg P$	$\neg Q$	$\neg R$	$\neg Q \wedge \neg R$	$\neg P \vee (\neg Q \wedge \neg R)$	$\neg(\neg P \vee (\neg Q \wedge \neg R))$
T	T	T	F	F	F	F	F	T
T	T	F	F	F	T	F	F	T
T	F	T	F	T	F	F	F	T
T	F	F	F	T	T	T	T	F
F	T	T	T	F	F	F	T	F
F	T	F	T	F	T	F	T	F
F	F	T	T	T	F	F	T	F
F	F	F	T	T	T	T	T	F

Exercises 2.11 (p. 37)

(2)

P Q	$P \supset Q$	$\neg Q$	$P \wedge \neg Q$	$\neg(P \wedge \neg Q)$
T T	T	F	F	T
T F	F	T	T	F
F T	T	F	F	T
F F	T	T	F	T

(4)

P Q	$P \wedge Q$	$P \mid Q$	$(P \mid Q) \mid (P \mid Q)$
T T	T	F	T
T F	F	T	F
F T	F	T	F
F F	F	T	F

(6)

P Q	$P \vee Q$	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$	$\neg(\neg P \wedge \neg Q)$
T T	T	F	F	F	T
T F	T	F	T	F	T
F T	T	T	F	F	T
F F	F	T	T	T	F

(8)

P Q	$P \supset Q$	$\neg P$	$\neg P \vee Q$
T T	T	F	T
T F	F	F	F
F T	T	T	T
F F	T	T	T

(10)

P Q	$P \mid Q$	$\neg Q$	$P \supset \neg Q$
T T	F	F	F
T F	T	T	T
F T	T	F	T
F F	T	T	T

Exercises 2.13 (p. 41)

(2) $J \wedge C$

(4) $\neg(F \wedge G)$

(6) $(F \equiv J) \wedge \neg G$

(8) $(J \vee \neg C) \supset G$

(10) $(G \wedge \neg F) \wedge C$ or $G \wedge (\neg F \wedge C)$

Exercises 2.14 (p. 43)

(2) $\neg(I \vee G)$ or $\neg I \wedge \neg G$

(4) $\neg(I \vee G) \supset F$ or $(\neg I \wedge \neg G) \supset F$

(6) $\neg(I \wedge G)$

(8) $\neg C \supset (I \vee G)$

(10) $(\neg C \wedge J) \supset (I \vee G)$

Exercises 2.15 (p. 46)

(2) $\neg G \supset \neg J$

(4) $J \supset G$

(6) $J \supset G$

(8) $F \equiv (C \wedge \neg J)$

(10) $J \supset (F \equiv \neg G)$

Exercises 2.16 (p. 50)

(2) $I \vee G$ or $\neg G \supset I$

(4) $\neg F \vee C$ or $\neg C \supset \neg F$

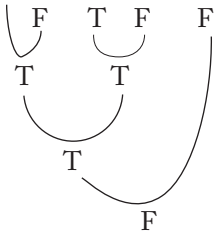
(6) $I \supset (J \vee G)$ or $I \supset (\neg G \supset J)$

(8) $(I \wedge \neg G) \wedge J$

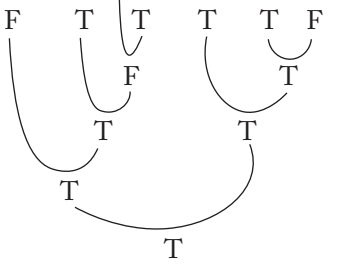
(10) $(C \supset F) \wedge J$

Exercises 2.17 (p. 53)

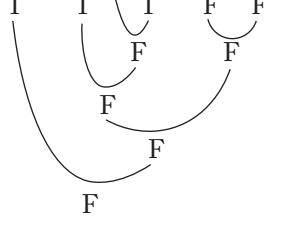
(2) $(\neg Q \wedge (B \vee S)) \supset P$



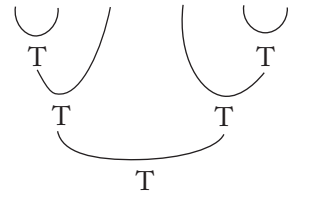
(8) $P \supset ((A \vee \neg B) \wedge (B \supset (C \vee Q)))$



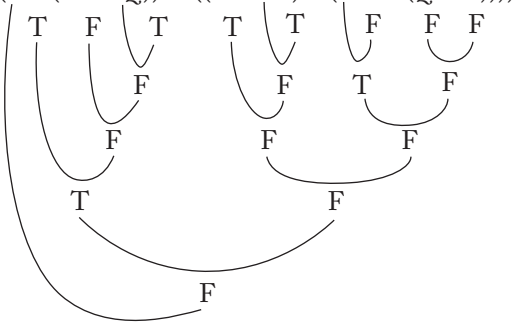
(4) $A \supset ((B \wedge \neg C) \vee (P \vee Q))$



(6) $((A \wedge B) \wedge P) \equiv (B \supset (C \vee S))$



(10) $\neg((B \vee (P \wedge \neg Q)) \supset ((A \wedge \neg B) \vee (\neg P \supset (Q \vee R))))$



Exercises 2.18 (p. 56)

logical truth (tautology)

(2)

$A\ Q$	$\neg A$	$A \wedge \neg A$	$(A \wedge \neg A) \supset Q$
TT	F	F	T
TF	F	F	T
FT	T	F	T
FF	T	F	T

contradiction

(4)

P	$\neg P$	$\neg P \supset P$	$P \supset (\neg P \supset P)$	$\neg(P \supset (\neg P \supset P))$
T	F	T	T	F
F	T	F	T	F

logical truth (tautology)

(6)

PQR	$P \supset Q$	$Q \supset R$	$(P \supset Q) \wedge (Q \supset R)$	$P \supset R$	$((P \supset Q) \wedge (Q \supset R)) \supset (P \supset R)$
TTT	T	T	T	T	T
TTF	T	F	F	F	T
TFT	F	T	F	T	T
TFF	F	T	F	F	T
FTT	T	T	T	T	T
FTF	T	F	F	T	T
FFT	T	T	T	T	T
FFF	T	T	T	T	T

contradiction

(8)

$A\ B$	$\neg A$	$\neg B$	$B \vee A$	$(B \vee A) \wedge \neg B$	$((B \vee A) \wedge \neg B) \wedge \neg A$
T T	F	F	T	F	F
T F	F	T	T	T	F
F T	T	F	T	F	F
F F	T	T	F	F	F

logical truth (tautology)

(10)

PST	$\neg P$	$S \wedge \neg P$	$\neg(S \wedge \neg P)$	$T \supset \neg(S \wedge \neg P)$	$\neg(T \supset \neg(S \wedge \neg P))$	$T \wedge S$	$(T \wedge S) \vee P$	$\neg(T \supset \neg(S \wedge \neg P)) \supset ((T \wedge S) \vee P)$
TTT	F	F	T	T	F	T	T	T
TTF	F	F	T	T	F	F	T	T
TFT	F	F	T	T	F	F	T	T
TFF	F	F	T	T	F	F	T	T
FTT	T	T	F	F	T	T	T	T
FTF	T	T	F	T	F	F	F	T
FFT	T	F	T	T	F	F	F	T
FFF	T	F	T	T	F	F	F	T

Chapter 3

Exercises 3.01 (p. 65)

(2) 'One plus one' is not identical with 'two'.

(4) 'Sincerity' involves 'sin'.

(6) 'One' is not identical with 'one'.

(8) (Sentences (8) and (9) cannot both be true.)

(10) I love the sound of 'a cellar door'.

Exercises 3.02 (p. 70)

invalid (sixth row)

(2)

P Q R	$P \supset Q$	$P \supset R$	$Q \supset R$
TTT	T	T	T
TTF	T	F	F
TFT	F	T	T
TFF	F	F	T
FTT	T	T	T
FTF	T	T	F
FFT	T	T	T
FFF	T	T	T

valid

(4)

PQRS	¬*P*	¬*Q*	¬*R*	¬*S*	*Q* ∧ ¬*R*	¬(*Q* ∧ ¬*R*)	*P* ∨ ¬*Q*	¬(*Q* ∧ ¬*R*) ⊃ ¬*S*	¬*P*	¬*Q*
TTTT	F	F	F	F	F	T	T	F	F	F
TTTF	F	F	F	T	F	T	T	T	F	F
TTFT	F	F	T	F	T	F	T	T	F	F
TTFF	F	F	T	T	T	F	T	T	F	F
TFTT	F	T	F	F	F	T	T	F	F	T
TFTF	F	T	F	T	F	T	T	T	F	T
TFFT	F	T	T	F	F	T	T	F	F	T
TFFF	F	T	T	T	F	T	T	T	F	T
FTTT	T	F	F	F	F	T	F	F	T	F
FTTF	T	F	F	T	F	T	F	T	T	F
FTFT	T	F	T	F	T	F	F	T	T	F
FTFF	T	F	T	T	T	F	F	T	T	F
FFTT	T	T	F	F	F	T	T	F	T	T
FFTF	T	T	F	T	F	T	T	T	T	T
FFFT	T	T	T	F	F	T	T	F	T	T
FFFF	T	T	T	T	F	T	T	T	T	T

invalid (third and fourth row)

(6)

P Q	Q ⊃ *P*	*P* ⊃ (*Q* ⊃ *P*)
T T	T	T
T F	T	T
F T	F	T
F F	T	T

invalid (fifth row)

(8)

P Q R	*P* ⊃ *Q*	*R* ⊃ *Q*
T T T	T	T
T T F	T	T
T F T	F	F
T F F	F	T
F T T	T	T
F T F	T	T
F F T	T	F
F F F	T	T

invalid (first row)

(10)

P Q	$P \vee Q$	$\neg Q$
T T	T	F
T F	T	T
F T	T	F
F F	F	T

valid

(12)

P Q	$P \supset Q$	$\neg Q$	$\neg P$
T T	T	F	F
T F	F	T	F
F T	T	F	T
F F	T	T	T

valid

(14)

P Q R	$P \supset Q$	$\neg Q$	$P \supset R$
T T T	T	F	T
T T F	T	F	F
T F T	F	T	T
T F F	F	T	F
F T T	T	F	T
F T F	T	F	T
F F T	T	T	T
F F F	T	T	T

Exercises 3.06 (p. 78)

(2)

1. $+ \neg P \supset \neg Q$
2. $+ \neg P$
3. $? \neg Q$
4. $\neg Q$ 1, 2 *MP*

(4)

1. $+ P \supset Q$
2. $+ \neg S$
3. $+ \neg(Q \supset R) \supset S$
4. $? P \supset R$
5. $Q \supset R$ 2, 3 *MT*
6. $P \supset R$ 1, 5 *HS*

(6)

1.	$+ P \supset (Q \supset R)$	
2.	$+ P$	
3.	$+ Q$	
4.	$? R$	
5.	$Q \supset R$	1, 2 *MP*
6.	R	3, 5 *MP*

(8)

1.	$+ P \supset Q$	
2.	$+ Q \supset R$	
3.	$+ P$	
4.	$? R$	
5.	$P \supset R$	1, 2 HS
6.	R	3, 5 *MP*

(10)

1.	$+ \neg(P \supset R) \supset \neg Q$	
2.	$+ P$	
3.	$+ Q$	
4.	$? R$	
5.	$P \supset R$	1, 3 *MT*
6.	R	2, 5 *MP*

(12)

1.	$+ P \supset Q$	
2.	$+ Q \supset \neg R$	
3.	$+ R$	
4.	$? \neg P$	
5.	$P \supset \neg R$	1, 2 *HS*
6.	$\neg P$	3, 5 *MT*

(14)

1.	$+ \neg(P \wedge \neg S) \supset (Q \vee R)$	
2.	$+ (Q \vee R) \supset \neg T$	
3.	$+ T$	
4.	$? P \wedge \neg S$	
5.	$\neg(Q \vee R)$	2, 3 *MT*
6.	$P \wedge \neg S$	1, 5 *MT*

Exercises 3.07 (p. 81)

(2)

1.	$+ P \wedge (\neg Q \wedge \neg R)$	
2.	$? \neg R$	
3.	$\neg Q \wedge \neg R$	1 $\wedge E$
4.	$\neg R$	3 $\wedge E$

(4)

1.	$+ \neg P$	
2.	$+ Q$	
3.	$+ (\neg P \wedge Q) \supset R$	
4.	$? R$	
5.	$\neg P \wedge Q$	1, 2 $\wedge I$
6.	R	3, 5 *MP*

(6)

1.	$+ P \supset (Q \wedge \neg R)$	
2.	$+ P$	
3.	$? \neg R$	
4.	$Q \wedge \neg R$	1, 2 *MP*
5.	$\neg R$	4 $\wedge E$

(8)

1.	$+ (P \wedge Q) \supset (R \wedge S)$	
2.	$+ Q$	
3.	$+ P$	
4.	$? R$	
5.	$P \wedge Q$	2, 3 $\wedge I$
6.	$R \wedge S$	1, 5 *MP*
7.	R	6 $\wedge E$

(10)

1.	$+ S \wedge ((P \equiv Q) \supset R)$	
2.	$+ P \equiv Q$	
3.	$? R$	
4.	$(P \equiv Q) \supset R$	1 $\wedge E$
7.	R	2, 4 *MP*

(12)

1.	$+ P \supset Q$	
2.	$+ Q \supset (R \wedge S)$	
3.	$+ P \wedge T$	
4.	$? S \wedge T$	
5.	$P \supset (R \wedge S)$	1, 2 *HS*
6.	P	3 $\wedge E$
7.	$R \wedge S$	5, 6 *MP*
8.	S	7 $\wedge E$
9.	T	3 $\wedge E$
10.	$S \wedge T$	8, 9 $\wedge I$

(14)

1.	$+ P \supset (Q \supset \neg R)$	
2.	$+ P \wedge Q$	
3.	$? \neg R$	
4.	P	2 $\wedge E$
5.	$Q \supset \neg R$	1, 4 *MP*
6.	Q	2 $\wedge E$
7.	$\neg R$	5, 6 *MP*

Exercises 3.08 (p. 85)

(2)

1.	$+ (P \vee Q) \supset (R \wedge S)$	
2.	$+ P$	
3.	$? S$	
4.	$P \vee Q$	2 $\vee I$
5.	$R \wedge S$	1, 4 *MP*
6.	S	5 $\wedge E$

(4)

1.	$+ P \supset (Q \vee R)$	
2.	$+ \neg(Q \vee R) \vee S$	
3.	$+ \neg S$	
4.	$? \neg P$	
5.	$\neg(Q \vee R)$	2, 3 $\vee E$
6.	$\neg P$	1, 5 *MT*

(6)

1.	$+ P \supset \neg(Q \wedge R)$	
2.	$+ (Q \wedge R) \vee S$	
3.	$+ \neg S$	
4.	$? \neg P$	
5.	$Q \wedge R$	2, 3 $\vee E$
6.	$\neg P$	1, 5 *MT*

(8)

1.	$+ P \supset (Q \vee \neg S)$	
2.	$+ P \wedge S$	
3.	$? Q$	
4.	P	2 $\wedge E$
5.	$Q \vee \neg S$	1, 4 *MP*
6.	S	2 $\wedge E$
7.	Q	5, 6 $\vee E$

(10)

1.	$+ P \supset (Q \wedge R)$	
2.	$+ S \vee \neg T$	
3.	$+ S \supset P$	
4.	$+ T$	
5.	$? Q$	
6.	S	2, 4 $\vee E$
7.	P	3, 6 *MP*
8.	$Q \wedge R$	1, 7 *MP*
9.	Q	8 $\wedge E$

(12)

1.	$+ P \supset Q$	
2.	$+ (Q \vee (R \supset S)) \supset (S \vee T)$	
3.	$+ \neg S \wedge P$	
4.	$? T$	
5.	P	3 $\wedge E$
6.	Q	1, 5 *MP*
7.	$Q \vee (R \supset S)$	6 $\vee I$
8.	$S \vee T$	2, 7 *MP*
9.	$\neg S$	3 $\wedge E$
10.	T	8, 9 $\vee E$

(14)

1.	$+ P \supset Q$	
2.	$+ (Q \vee R) \supset (R \vee \neg S)$	
3.	$+ P \wedge \neg R$	
4.	$? \neg S$	
5.	P	3 $\wedge E$
6.	Q	1, 5 *MP*
7.	$Q \vee R$	6 $\vee I$
8.	$R \vee \neg S$	2, 7 *MP*
9.	$\neg R$	3 $\wedge E$
10.	$\neg S$	8, 9 $\vee E$

Exercises 3.09 (p. 88)

(2)

1.	$+ P \supset (S \supset (Q \wedge R))$	
2.	$+ (Q \wedge R) \supset \neg P$	
3.	$+ T \supset S$	
4.	$? P \supset \neg T$	
5.	$\lceil P$	
6.	$\quad ? \neg T$	
7.	$\quad S \supset (Q \wedge R)$	1, 5 *MP*
8.	$\quad \neg(Q \wedge R)$	2, 5 *MT*
9.	$\quad \neg S$	7, 8 *MT*
10.	$\quad \neg T \rfloor$	3, 9 *MT*
11.	$P \supset \neg T$	5–10 *CP*

(4)

1.	$+ (P \wedge Q) \supset R$	
2.	$+ P$	
3.	$? Q \supset R$	
4.	$\lceil Q$	
5.	$\quad ? R$	
6.	$\quad P \wedge Q$	2, 4 $\wedge I$
7.	$\quad R \rfloor$	1, 6 *MP*
8.	$Q \supset R$	4–7 *CP*

(6)

1.	$+ P \supset (Q \vee R)$	
2.	$+ P \supset \neg Q$	
3.	$? P \supset R$	
4.	$\quad P$	
5.	$\quad ? R$	
6.	$\quad \neg Q$	2, 4 *MP*
7.	$\quad Q \vee R$	1, 4 *MP*
8.	$\quad R$	6, 7 $\vee E$
9.	$P \supset R$	4–8 *CP*

(8)

1.	$+ P \supset S$	
2.	$+ R \supset S$	
3.	$? P \supset (R \supset S)$	
4.	$\quad P$	
5.	$\quad ? R \supset S$	
6.	$\quad\quad R$	
7.	$\quad\quad ? S$	
8.	$\quad\quad S$	1, 4 *MP*
9.	$\quad R \supset S$	6–8 *CP*
10.	$P \supset (R \supset S)$	4–9 *CP*

(10)

1.	$+ Q \supset (T \vee S)$	
2.	$+ \neg R \wedge \neg T$	
3.	$+ P$	
4.	$? P \wedge (Q \supset S)$	
5.	$\quad Q$	
6.	$\quad ? S$	
7.	$\quad T \vee S$	1, 5 *MP*
8.	$\quad \neg T$	2 $\wedge E$
9.	$\quad S$	7, 8 $\vee E$
10.	$Q \supset S$	5–9 *CP*
11.	$P \wedge (Q \supset S)$	3, 10 $\wedge I$

(12)

1.	$+ (P \vee \neg T) \supset ((S \vee T) \supset Q)$	
2.	$+ \neg P \vee S$	
3.	$? (P \supset Q) \vee (S \supset T)$	
4.	$\quad P$	
5.	$\quad ? Q$	
6.	$\quad S$	2, 4 $\vee E$
7.	$\quad P \vee \neg T$	4 $\vee I$
8.	$\quad (S \vee T) \supset Q$	1, 7 *MP*
9.	$\quad S \vee T$	6 $\vee I$
10.	$\quad Q$	8, 9 *MP*
11.	$P \supset Q$	4–11 *CP*
12.	$(P \supset Q) \vee (S \supset T)$	11 $\vee I$

(14)

1.	$+ P \supset (S \vee T)$	
2.	$+ (S \vee T) \supset (Q \supset (R \vee \neg S))$	
3.	$+ S$	
4.	$? P \supset (Q \supset R)$	
5.	$\quad P$	
6.	$\quad ? Q \supset R$	
7.	$\quad\quad Q$	
8.	$\quad\quad ? R$	
9.	$\quad\quad S \vee T$	1, 5 *MP*
10.	$\quad\quad Q \supset (R \vee \neg S)$	2, 9 *MP*
11.	$\quad\quad R \vee \neg S$	7, 10 *MP*
12.	$\quad\quad R$	3, 11 $\vee E$
13.	$\quad Q \supset R$	7–12 *CP*
14.	$P \supset (Q \supset R)$	5–13 *CP*

Exercises 3.10 (p. 92)

(2)

1.	$+ P \vee Q$	
2.	$+ P \supset (R \wedge S)$	
3.	$+ (R \wedge S) \supset Q$	
4.	$? Q$	
5.	$\neg Q$	
6.	$? \times$	
7.	P	1, 5 $\vee E$
8.	$R \wedge S$	2, 7 *MP*
9.	Q	3, 8 *MP*
10.	$Q \wedge \neg Q$	5, 9 $\wedge I$
11.	Q	5–10 *IP*

(4)

1.	$+ \neg R \supset \neg(\neg P \vee Q)$	
2.	$+ \neg R$	
3.	$? P$	
4.	$\neg P$	
5.	$? \times$	
6.	$\neg(\neg P \vee Q)$	1, 2 *MP*
7.	$\neg P \vee Q$	4 $\vee I$
8.	$(\neg P \vee Q) \wedge \neg(\neg P \vee Q)$	6, 7 $\wedge I$
9.	P	4–8 *IP*

(6)

1.	$+ P \supset Q$	
2.	$+ S \supset T$	
3.	$? (P \vee S) \supset \neg(\neg Q \wedge \neg T)$	
4.	$P \vee S$	
5.	$? \neg(\neg Q \wedge \neg T)$	
6.	$\neg Q \wedge \neg T$	
7.	$? \times$	
8.	$\neg Q$	6 $\wedge E$
9.	$\neg P$	1, 8 *MT*
10.	S	4, 9 $\vee E$
11.	$\neg T$	6 $\wedge E$
12.	$\neg S$	2, 11 *MT*
13.	$S \wedge \neg S$	10, 12 $\wedge I$
14.	$\neg(\neg Q \wedge \neg T)$	6-13 *IP*
15.	$(P \vee S) \supset \neg(\neg Q \wedge \neg T)$	4–14 *CP*

(8)

1.	$+ (P \vee Q) \supset (R \supset S)$	
2.	$+ \neg P \supset T$	
3.	$+ R \wedge \neg S$	
4.	$? T$	
5.	$\neg T$	
6.	$? \times$	
7.	P	2, 5 *MT*
8.	$P \vee Q$	7 $\vee I$
9.	$R \supset S$	1, 8 *MP*
10.	R	3 $\wedge E$
11.	S	9, 10 *MP*
12.	$\neg S$	3 $\wedge E$
13.	$S \wedge \neg S$	11, 12 $\wedge I$
14.	T	5–13 *IP*

(10)

1.	$+ P \supset (\neg Q \wedge R)$	
2.	$+ S \vee \neg T$	
3.	$+ P \vee T$	
4.	$? Q \supset S$	
5.	$\quad Q$	
6.	$\quad ? S$	
7.	$\quad \neg S$	
8.	$\quad ? \times$	
9.	$\quad \neg T$	2, 7 $\vee E$
10.	$\quad P$	3, 9 $\vee E$
11.	$\quad \neg Q \wedge R$	1, 10 *MP*
12.	$\quad \neg Q$	11 $\wedge E$
13.	$\quad Q \wedge \neg Q$	5, 12 $\wedge I$
14.	S	7–13 *IP*
15.	$Q \supset S$	5–14 *CP*

(12)

1.	$+ \neg(S \wedge T) \supset (Q \supset R)$	
2.	$+ P \supset \neg T$	
3.	$? P \supset (Q \supset R)$	
4.	$\quad P$	
5.	$\quad ? Q \supset R$	
6.	$\quad \neg(Q \supset R)$	
7.	$\quad ? \times$	
8.	$\quad S \wedge T$	1, 6 *MT*
9.	$\quad T$	8 $\wedge E$
10.	$\quad \neg T$	2, 4 *MP*
11.	$\quad T \wedge \neg T$	9, 10 $\wedge I$
12.	$Q \supset R$	6–11 *IP*
13.	$P \supset (Q \supset R)$	4–12 *CP*

(14)

1.	$+ P \vee S$	
2.	$+ S \supset (R \supset T)$	
3.	$+ R \wedge (T \supset P)$	
4.	$? P \vee Q$	
5.	$\quad \neg P$	
6.	$\quad ? \times$	
7.	$\quad S$	1, 5 $\vee E$
8.	$\quad R \supset T$	2, 7 *MP*
9.	$\quad R$	3 $\wedge E$
10.	$\quad T$	8, 9 *MP*
11.	$\quad T \supset P$	3 $\wedge E$
12.	$\quad \neg T$	5, 11 *MT*
13.	$\quad T \wedge \neg T$	10, 12 $\wedge I$
14.	P	5–13 *IP*
15.	$P \vee Q$	14 $\vee I$

Exercises 3.11 (p. 98)

(2)

1.	$+ (P \supset Q) \vee (R \vee S)$	
2.	$? (S \vee R) \vee (P \supset Q)$	
3.	$(R \vee S) \vee (P \supset Q)$	1 *Com*
4.	$(S \vee R) \vee (P \supset Q)$	3 *Com*

(4)

1.	$+ (P \vee Q) \wedge (R \wedge S)$	
2.	$? (R \wedge (P \vee Q)) \wedge S$	
3.	$((P \vee Q) \wedge R) \wedge S$	1 *Assoc*
4.	$(R \wedge (P \vee Q)) \wedge S$	3 *Com*

(6)

1.	$+ P \supset ((R \vee Q) \supset S)$	
2.	$+ (T \vee S) \supset W$	
3.	$? P \supset (Q \supset W)$	
4.	$\boxed{P}$	
5.	$? Q \supset W$	
6.	$\quad\boxed{Q}$	
7.	$\quad? W$	
8.	$\quad(R \vee Q) \supset S$	1, 4 *MP*
9.	$\quad Q \vee R$	6 $\vee I$
10.	$\quad R \vee Q$	9 *Com*
11.	$\quad S$	8, 10 *MP*
12.	$\quad S \vee T$	11 $\vee I$
13.	$\quad T \vee S$	12 *Com*
14.	$\quad W$	2, 13 *MP*
15.	$Q \supset W$	6–14 *CP*
16.	$P \supset (Q \supset W)$	4–15 *CP*

(8)

1.	$+ (P \vee (Q \vee R)) \supset T$	
2.	$+ (S \vee \neg T) \supset R$	
3.	$? T$	
4.	$\boxed{\neg T}$	
5.	$? \times$	
6.	$\neg T \vee S$	4 $\vee I$
7.	$S \vee \neg T$	6 *Com*
8.	R	2, 7 *MP*
9.	$R \vee (P \vee Q)$	8 $\vee I$
10.	$(P \vee Q) \vee R$	9 *Com*
11.	$P \vee (Q \vee R)$	10 *Assoc*
12.	T	1, 11 *MP*
13.	$T \wedge \neg T$	4, 12 $\wedge I$
14.	T	4–13 *IP*

(10)

1.	$+ P \supset ((Q \wedge R) \vee S)$	
2.	$+ (R \wedge Q) \supset \neg P$	
3.	$+ T \supset \neg S$	
4.	$? P \supset \neg T$	
5.	$\boxed{P}$	
6.	$? \neg T$	
7.	$(Q \wedge R) \vee S$	1, 5 *MP*
8.	$\boxed{T}$	
9.	$? \times$	
10.	$\neg S$	3, 8 *MP*
11.	$Q \wedge R$	7, 10 $\vee E$
12.	$\neg(R \wedge Q)$	2, 5 *MT*
13.	$\neg(Q \wedge R)$	12 *Com*
14.	$(Q \wedge R) \wedge \neg(Q \wedge R)$	11, 13 $\wedge I$
15.	$\neg T$	8–14 *IP*
16.	$P \supset \neg T$	5–15 *CP*

(12)

1.	$+ P \supset R$	
2.	$+ P \vee (R \wedge S)$	
3.	$? Q \vee R$	
4.	$\boxed{\neg R}$	
5.	$? \times$	
6.	$\neg P$	1, 4 *MT*
7.	$R \wedge S$	2, 6 $\vee E$
8.	R	7 $\wedge E$
9.	$R \wedge \neg R$	4, 8 $\wedge I$
10.	R	4–9 *IP*
11.	$R \vee Q$	10 $\vee I$
12.	$Q \vee R$	11 *Com*

(14)

1.	$+ P \vee S$	
2.	$+ S \supset (T \supset P)$	
3.	$+ S \supset T$	
4.	$? R \vee (Q \vee P)$	
5.	$\boxed{\neg P}$	
6.	$? \times$	
7.	S	1, 5 $\vee E$
8.	$T \supset P$	2, 7 *MP*
9.	T	3, 7 *MP*
10.	P	8, 9 *MP*
11.	$P \wedge \neg P$	5, 10 $\wedge I$
12.	P	5–11 *IP*
13.	$P \vee (R \vee Q)$	12 $\vee I$
14.	$(R \vee Q) \vee P$	13 *Com*
15.	$R \vee (Q \vee P)$	14 *Assoc*

Exercises 3.12 (p. 100)

(2)

1.	$+ \neg(\neg P \vee (\neg Q \wedge \neg R))$	
2.	$? P \wedge (Q \vee R)$	
3.	$P \wedge \neg(\neg Q \wedge \neg R)$	1 *DeM*
4.	$P \wedge (Q \vee R)$	3 *DeM*

(4)

1.	$+ P \supset \neg(Q \wedge (R \vee \neg S))$	
2.	$? P \supset (\neg Q \vee (\neg R \wedge S))$	
3.	$P \supset (\neg Q \vee \neg(R \vee \neg S))$	1 *DeM*
4.	$P \supset (\neg Q \vee (\neg R \wedge S))$	3 *DeM*

(6)

1.	$+ Q \supset S$	
2.	$+ S \supset P$	
3.	$? P \vee \neg Q$	
4.	$\neg(P \vee \neg Q)$	
5.	$? \times$	
6.	$\neg P \wedge Q$	4 *DeM*
7.	Q	6 $\wedge E$
8.	S	1, 7 *MP*
9.	P	2, 8 *MP*
10.	$\neg P$	6 $\wedge E$
11.	$P \wedge \neg P$	9, 10 $\wedge I$
12.	$P \vee \neg Q$	4–11 *IP*

(8)

1.	$+ P \supset (Q \vee S)$	
2.	$+ \neg S$	
3.	$? \neg P \vee Q$	
4.	$\neg(\neg P \vee Q)$	
5.	$? \times$	
6.	$P \wedge \neg Q$	4 *DeM*
7.	P	6 $\wedge E$
8.	$\neg Q$	6 $\wedge E$
9.	$Q \vee S$	1, 7 *MP*
10.	S	8, 9 $\vee E$
11.	$S \wedge \neg S$	2, 10 $\wedge I$
12.	$\neg P \vee Q$	4–11 *IP*

(10)

1.	$+ P \vee (Q \vee R)$	
2.	$+ Q \supset (R \wedge S)$	
3.	$? P \vee R$	
4.	$\neg(P \vee R)$	
5.	$? \times$	
6.	$\neg P \wedge \neg R$	4 *DeM*
7.	$\neg P$	6 $\wedge E$
8.	$Q \vee R$	1, 7 $\vee E$
9.	$\neg R$	6 $\wedge E$
10.	Q	8, 9 $\vee E$
11.	$R \wedge S$	2, 10 *MP*
12.	R	11 $\wedge E$
13.	$R \wedge \neg R$	9, 12 $\wedge I$
14.	$P \vee R$	4–13 *IP*

(12)

1.	$+ S \vee (Q \supset R)$	
2.	$+ P \vee (Q \wedge (T \vee \neg R))$	
3.	$? S \vee (T \vee P)$	
4.	$\neg(S \vee (T \vee P))$	
5.	$? \times$	
6.	$\neg S \wedge \neg(T \vee P)$	4 *DeM*
7.	$\neg S$	6 $\wedge E$
8.	$Q \supset R$	1, 7 $\vee E$
9.	$\neg(T \vee P)$	6 $\wedge E$
10.	$\neg T \wedge \neg P$	9 *DeM*
11.	$\neg P$	10 $\wedge E$
12.	$Q \wedge (T \vee \neg R)$	2, 11 $\vee E$
13.	$T \vee \neg R$	12 $\wedge E$
14.	$\neg T$	10 $\wedge E$
15.	Q	12 $\wedge E$
16.	R	8, 15 *MP*
17.	$\neg R$	13, 14 $\vee E$
18.	$R \wedge \neg R$	16, 17 $\wedge I$
19.	$S \vee (T \vee P)$	4–18 *IP*

(14)

1.	$+ S \vee (P \supset (R \supset Q))$	
2.	$+ S \vee P$	
3.	$+ R$	
4.	$? Q \vee S$	
5.	$\neg(Q \vee S)$	
6.	$? \times$	
7.	$\neg Q \wedge \neg S$	5 *DeM*
8.	$\neg S$	7 $\wedge E$
9.	P	2, 8 $\vee E$
10.	$P \supset (R \supset Q)$	1, 8 $\vee E$
11.	$R \supset Q$	9, 10 *MP*
12.	Q	3, 11 *MP*
13.	$\neg Q$	7 $\wedge E$
14.	$Q \wedge \neg Q$	12, 13 $\wedge I$
15.	$Q \vee S$	5–14 *IP*

Exercises 3.13 (p. 103)

(2)

1.	$+ \neg(P \vee Q) \supset R$	
2.	$? \neg P \supset (\neg Q \supset R)$	
3.	$(\neg P \wedge \neg Q) \supset R$	1 *DeM*
4.	$\neg P \supset (\neg Q \supset R)$	3 *Exp*

(4)

1.	$+ P \vee (Q \wedge \neg R)$	
2.	$? (P \vee Q) \wedge \neg(\neg P \wedge R)$	
3.	$(P \vee Q) \wedge (P \vee \neg R)$	1 *Dist*
4.	$(P \vee Q) \wedge \neg(\neg P \wedge R)$	3 *DeM*

(6)

1.	$+ P \supset (Q \wedge R)$	
2.	$+ R \supset (Q \supset S)$	
3.	$? P \supset S$	
4.	$(R \wedge Q) \supset S$	2 *Exp*
5.	$(Q \wedge R) \supset S$	4 *Com*
6.	$P \supset S$	1, 5 *HS*

(8)

1.	$+ (P \wedge R) \supset Q$	
2.	$+ P \supset R$	
3.	$? P \supset Q$	
4.	$(R \wedge P) \supset Q$	1 *Com*
5.	$R \supset (P \supset Q)$	4 *Exp*
6.	$P \supset (P \supset Q)$	1, 5 *HS*
7.	$(P \wedge P) \supset Q$	6 *Exp*
8.	$P \supset Q$	7 *Taut*

(10)

1.	$+ (S \vee T) \supset (\neg P \vee \neg R)$	
2.	$+ S \vee (Q \wedge T)$	
3.	$? P \supset \neg R$	
4.	$\boxed{P}$	
5.	$\quad ? \neg R$	
6.	$\quad (S \vee Q) \wedge (S \vee T)$	2 *Dist*
7.	$\quad S \vee T$	6 $\wedge E$
8.	$\quad \neg P \vee \neg R$	1, 7 *MP*
9.	$\quad \neg R$	4, 8 $\vee E$
10.	$P \supset \neg R$	4–9 *CP*

(12)

1.	$+ S$	
2.	$+ \neg R \supset T$	
3.	$? (R \vee S) \wedge (R \vee T)$	
4.	$\boxed{\neg(R \vee T)}$	
5.	$\quad ? \times$	
6.	$\quad \neg R \wedge \neg T$	4 *DeM*
7.	$\quad \neg R$	6 $\wedge E$
8.	$\quad T$	2, 7 *MP*
9.	$\quad \neg T$	6 $\wedge E$
10.	$\quad T \wedge \neg T$	8, 9 $\wedge I$
11.	$R \vee T$	4–10 *IP*
12.	$S \vee R$	1 $\vee I$
13.	$R \vee S$	12 *Com*
14.	$(R \vee S) \wedge (R \vee T)$	11, 13 $\wedge I$

(14)

1.	$+ P \supset (Q \supset R)$	
2.	$+ S \supset (P \wedge Q)$	
3.	$+ (S \supset R) \supset T$	
4.	$? P \supset T$	
5.	$\boxed{P}$	
6.	$\quad ? T$	
7.	$\quad (P \wedge Q) \supset R$	1 *Exp*
8.	$\quad S \supset R$	2, 7 *HS*
9.	$\quad T$	3, 8 *MP*
10.	$P \supset T$	5–9 *CP*

Exercises 3.14 (p. 105)

(2)

1.	$+ \neg P \supset (Q \wedge R)$	
2.	$? \neg(Q \wedge R) \supset P$	
3.	$\quad \neg(Q \wedge R) \supset P$	1 *Contra*

(4)

1.	$+ \neg P \supset (\neg Q \supset R)$	
2.	$? (P \vee Q) \vee R$	
3.	$\quad P \vee (\neg Q \supset R)$	1 *Cond*
4.	$\quad P \vee (Q \vee R)$	3 *Cond*
5.	$\quad (P \vee Q) \vee R$	4 *Assoc*

(6)

1.	$+ (\neg S \vee R) \supset (T \supset P)$	
2.	$+ S \supset R$	
3.	$? \neg T \vee P$	
4.	$\neg S \vee R$	2 *Cond*
5.	$T \supset P$	1, 4 *MP*
6.	$\neg T \vee P$	5 *Cond*

(8)

1.	$+ ((P \supset Q) \supset R) \supset S$	
2.	$+ R$	
3.	$? S$	
4.	$\neg S$	
5.	$? \times$	
6.	$\neg((P \supset Q) \supset R)$	1, 4 *MT*
7.	$\neg(\neg(P \supset Q) \vee R)$	6 *Cond*
8.	$(P \supset Q) \wedge \neg R$	7 *DeM*
9.	$\neg R$	8 $\wedge E$
10.	$R \wedge \neg R$	2, 9 $\wedge I$
11.	S	4–10 *IP*

(10)

1.	$+ \neg P \vee R$	
2.	$+ (P \wedge \neg R) \vee S$	
3.	$+ (R \wedge S) \supset Q$	
4.	$? P \supset Q$	
5.	P	
6.	$? Q$	
7.	$\neg(P \wedge \neg R)$	1 *DeM*
8.	S	2, 7 $\vee E$
9.	R	1, 5 $\vee E$
10	$R \wedge S$	8, 9 $\wedge I$
11.	Q	3, 10 *MP*
12.	$P \supset Q$	5–11 *CP*

(12)

1.	$+ S \supset R$	
2.	$+ R \supset \neg(T \supset Q)$	
3.	$? S \supset T$	
4.	S	
5.	$? T$	
6.	R	1, 4 *MP*
7.	$\neg(T \supset Q)$	2, 6 *MP*
8.	$\neg(\neg T \vee Q)$	7 *Cond*
9.	$T \wedge \neg Q$	8 *DeM*
10.	T	9 $\wedge E$
11.	$S \supset T$	4–10 *CP*

(14)

1.	$+ \neg P \supset Q$	
2.	$+ S \supset \neg(P \vee Q)$	
3.	$+ R \supset S$	
4.	$? R \supset T$	
5.	R	
6.	$? \times$	
7.	S	3, 5 *MP*
8.	$\neg(P \vee Q)$	2, 7 *MP*
9.	$\neg(\neg P \supset Q)$	8 *Cond*
10.	$(\neg P \supset Q) \wedge \neg(\neg P \supset Q)$	1, 9 $\wedge I$
11.	$\neg R$	5–10 *IP*
12.	$\neg R \vee T$	11 $\vee I$
13.	$R \supset T$	12 *Cond*

Exercises 3.15 (p. 107)

(2)

1.	$+ P \equiv Q$	
2.	$? (P \vee \neg Q) \wedge (\neg P \vee Q)$	
3.	$(P \supset Q) \wedge (Q \supset P)$	1 *Bicond*
4.	$(\neg P \vee Q) \wedge (Q \supset P)$	3 *Cond*
5.	$(\neg P \vee Q) \wedge (\neg Q \vee P)$	4 *Cond*
6.	$(\neg Q \vee P) \wedge (\neg P \vee Q)$	5 *Com*
7.	$(P \vee \neg Q) \wedge (\neg P \vee Q)$	6 *Com*

(4)

1.	$+ (P \supset Q) \wedge \neg(\neg P \wedge Q)$	
2.	$? P \equiv Q$	
3.	$(P \supset Q) \wedge (P \vee \neg Q)$	1 *DeM*
4.	$(P \supset Q) \wedge (\neg Q \vee P)$	3 *Com*
5.	$(P \supset Q) \wedge (Q \supset P)$	4 *Cond*
6.	$P \equiv Q$	5 *Bicond*

(6)

1.	$+ (P \vee Q) \supset (R \equiv \neg S)$	
2.	$+ (S \vee T) \supset (P \wedge R)$	
3.	$? \neg S$	
4.	$\boxed{S}$	
5.	$? \times$	
6.	$S \vee T$	4 $\vee I$
7.	$P \wedge R$	2, 6 *MP*
8.	P	7 $\wedge E$
9.	$P \vee Q$	8 $\vee I$
10.	$R \equiv \neg S$	1, 9 *MP*
11.	$(R \supset \neg S) \wedge (\neg S \supset R)$	10 *Bicond*
12.	$R \supset \neg S$	11 $\wedge E$
13.	R	7 $\wedge E$
14.	$\neg S$	12, 13 *MP*
15.	$S \wedge \neg S$	4, 14 $\wedge I$
16.	$\neg S$	4–14 *IP*

(8)

1.	$+ P \supset (Q \equiv R)$	
2.	$+ \neg S \supset (P \vee R)$	
3.	$+ P \equiv Q$	
4.	$? S \vee R$	
5.	$\boxed{\neg(S \vee R)}$	
6.	$? \times$	
7.	$\neg S \wedge \neg R$	5 *DeM*
8.	$\neg S$	7 $\wedge E$
9.	$P \vee R$	12, 8 *MP*
10.	$\neg R$	7 $\wedge E$
11.	P	9, 10 $\vee E$
12.	$Q \equiv R$	1, 11 *MP*
13.	$(Q \supset R) \wedge (R \supset Q)$	12 *Bicond*
14.	$(P \supset Q) \wedge (Q \supset P)$	3 *Bicond*
15.	$P \supset Q$	14 $\wedge E$
16.	$Q \supset R$	13 $\wedge E$
17.	$P \supset R$	15, 16 *HS*
18.	R	11, 17 *MP*
19.	$R \wedge \neg R$	10, 18 $\wedge I$
20.	$S \vee R$	5–19 *IP*

(10)

1.	$+ S \equiv T$	
2.	$+ S \supset (P \vee Q)$	
3.	$? \neg Q \supset (T \supset P)$	
4.	$(S \supset T) \wedge (T \supset S)$	1 *Bicond*
5.	$T \supset S$	4 $\wedge E$
6.	$T \supset (P \vee Q)$	2, 5 *HS*
7.	$\neg T \vee (P \vee Q)$	6 *Cond*
8.	$(\neg T \vee P) \vee Q$	7 *Assoc*
9.	$Q \vee (\neg T \vee P)$	8 *Com*
10.	$\neg Q \supset (\neg T \vee P)$	9 *Cond*
11.	$\neg Q \supset (T \supset P)$	10 *Cond*

(12)

1.	$+ P \supset (Q \equiv R)$	
2.	$+ (\neg Q \vee R) \supset T$	
3.	$? P \supset T$	
4.	$\boxed{P}$	
5.	$? T$	
6.	$Q \equiv R$	1, 4 *MP*
7.	$(Q \supset R) \wedge (R \supset Q)$	6 *Bicond*
8.	$Q \supset R$	7 $\wedge E$
9.	$(Q \supset R) \supset T$	2 *Cond*
10.	T	8, 9 *MP*
11.	$P \supset T$	4–10 *CP*

(14)

1.	$+ P \equiv (\neg Q \vee R)$	
2.	$+ (Q \supset R) \supset S$	
3.	$+ S \supset \neg P$	
4.	$? \neg P$	
5.	$\quad P$	
6.	$\quad ? \times$	
7.	$\quad \neg S$	3, 5 *MT*
8.	$\quad \neg(Q \supset R)$	2, 7 *MT*
9.	$\quad (P \supset (\neg Q \vee R)) \wedge ((\neg Q \vee R) \supset P)$	1 *Bicond*
10.	$\quad P \supset (\neg Q \vee R)$	9 $\wedge E$
11.	$\quad \neg Q \vee R$	5, 10 *MP*
12.	$\quad Q \supset R$	11 *Cond*
13.	$\quad (Q \supset R) \wedge \neg(Q \supset R)$	8, 12 $\wedge I$
14.	$\neg P$	5–13 *IP*

Exercises 3.17 (p. 112)

(2)

1.	$+ S \supset P$	
2.	$+ Q \supset P$	
3.	$+ \neg Q \supset S$	
4.	$? P$	
5.	$\quad Q \vee S$	3 *Cond*
6.	$\quad P \vee P$	1, 2, 5 *CD*
7.	$\quad P$	6 *Taut*

(4)

1.	$+ P \supset (R \wedge T)$	
2.	$+ Q \supset (S \wedge T)$	
3.	$? (P \vee Q) \supset (R \vee S)$	
4.	$\quad P \vee Q$	
5.	$\quad ? R \vee S$	
6.	$\quad (R \wedge T) \vee (S \wedge T)$	1, 2, 4 *CD*
7.	$\quad (T \wedge R) \vee (S \wedge T)$	6 *Com*
8.	$\quad (T \wedge R) \vee (T \wedge S)$	7 *Com*
9.	$\quad T \wedge (R \vee S)$	8 *Dist*
10.	$\quad R \vee S$	9 $\wedge E$
11.	$(P \vee Q) \supset (R \vee S)$	4–10 *CP*

(6)

1.	$+ P \vee R$	
2.	$+ P \supset (Q \wedge \neg S)$	
3.	$+ (\neg R \vee T) \wedge \neg S$	
4.	$? Q \vee T$	
5.	$\quad \neg P \vee (Q \wedge \neg S)$	2 *Cond*
6.	$\quad (\neg P \vee Q) \wedge (\neg P \vee \neg S)$	5 *Dist*
7.	$\quad \neg P \vee Q$	6 $\wedge E$
8.	$\quad P \supset Q$	7 *Cond*
9.	$\quad \neg R \vee T$	3 $\wedge E$
10.	$\quad R \supset T$	9 *Cond*
11.	$\quad Q \vee T$	1, 8, 10 *CD*

(8)

1.	$+ \neg P$	
2.	$+ \neg Q$	
3.	$? (P \vee Q) \supset (R \vee S)$	
4.	$\quad P \vee Q$	
5.	$\quad ? R \vee S$	
6.	$\quad \neg P \vee R$	1 $\vee I$
7.	$\quad \neg Q \vee S$	2 $\vee I$
8.	$\quad P \supset R$	6 *Cond*
9.	$\quad Q \supset S$	7 *Cond*
10.	$\quad R \vee S$	4, 8, 9 *CD*
11.	$(P \vee Q) \supset (R \vee S)$	4–10 *CP*

(10)

1.	$+ P \supset (Q \vee R)$	
2.	$+ S \supset (R \vee T)$	
3.	$+ \neg R$	
4.	$? (P \vee S) \supset (Q \vee T)$	
5.	$\quad P \vee S$	
6.	$\quad ? Q \vee T$	
7.	$\quad (Q \vee R) \vee (R \vee T)$	1, 2, 5 *CD*
8.	$\quad Q \vee (R \vee (R \vee T))$	7 *Assoc*
9.	$\quad Q \vee ((R \vee R) \vee T)$	8 *Assoc*
10.	$\quad Q \vee (R \vee T)$	9 *Taut*
11.	$\quad Q \vee (T \vee R)$	10 *Com*
12.	$\quad (Q \vee T) \vee R$	11 *Assoc*
13.	$\quad Q \vee T$	3, 12 $\vee E$
14.	$(P \vee S) \supset (Q \vee T)$	4–10 *CP*

(12)

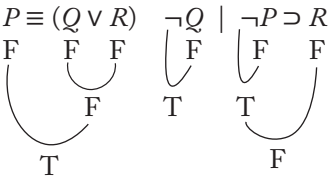
1.	$+ P \vee Q$	
2.	$+ R \vee S$	
3.	$? \neg(Q \wedge S) \supset (P \vee R)$	
4.	$\quad \neg(Q \wedge S)$	
5.	$\quad ? P \vee R$	
6.	$\quad \neg Q \vee \neg S$	4 *DeM*
7.	$\quad Q \vee P$	1 *Com*
8.	$\quad \neg Q \supset P$	7 *Cond*
9.	$\quad S \vee R$	2 *Com*
10.	$\quad \neg S \supset R$	9 *Cond*
11.	$\quad P \vee R$	6, 8, 10 *CD*
12.	$\neg(Q \wedge S) \supset (P \vee R)$	4–11 *CP*

(14)

1.	$+ S \supset T$	
2.	$+ R \supset (T \vee Q)$	
3.	$+ (T \vee Q) \supset P$	
4.	$? (S \vee R) \supset (T \vee P)$	
5.	$\quad S \vee R$	
6.	$\quad ? T \vee P$	
7.	$\quad R \supset P$	2, 3 *HS*
8.	$\quad T \vee P$	1, 5, 7 *CD*
9.	$(S \vee R) \supset (T \vee P)$	5–8 *CP*

Exercises 3.18 (p. 117)

(2)



Invalid under *I*: {*P* = F; *Q* = F; *R* = F}

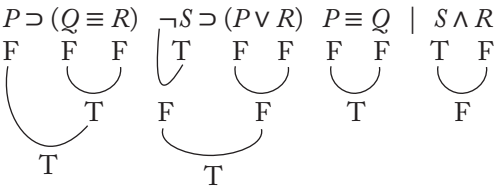
(4)

1.	+ (P ∨ Q) ⊃ (R ≡ S)	
2.	+ ¬(¬*S* ∧ *P*)	
3.	+ *R* ⊃ *T*	
4.	? *P* ⊃ (*T* ∧ *R*)	
5.	*P*	
6.	? *T* ∧ *R*	
7.	*P* ∨ *Q*	5 ∨*I*
8.	*R* ≡ *S*	1, 7 *MP*
9.	*S* ∨ ¬*P*	2 *DeM*
10.	*S*	5, 9 ∨*E*
11.	(*R* ⊃ *S*) ∧ (*S* ⊃ *R*)	8 *Bicond*
12.	*S* ⊃ *R*	11 ∧*E*
13.	*R*	10, 12 *MP*
14.	*T*	3, 13 *MP*
15.	*T* ∧ *R*	13, 14 ∧*I*
16.	*P* ⊃ (*T* ∧ *R*)	5–15 *CP*

(6)

1.	+ ¬*P* ⊃ (*Q* ⊃ *R*)	
2.	+ (*P* ∨ *S*) ⊃ *T*	
3.	+ *R* ⊃ (*P* ∨ *S*)	
4.	+ ¬*T*	
5.	? ¬*Q*	
6.	¬(*P* ∨ *S*)	2, 4 *MT*
7.	¬*P* ∧ ¬*S*	6 *DeM*
8.	¬*P*	7 ∧*E*
9.	*Q* ⊃ *R*	1, 8 *MP*
10.	¬*R*	3, 6 *MT*
11.	¬*Q*	9, 10 *MT*

(8)



Invalid under I: {*P* = F; *Q* = F; *R* = F; *S* = T}

(10)

1.	$+ P \supset (Q \supset R)$	
2.	$+ \neg R$	
3.	$? P \supset \neg Q$	
4.	$\lceil P$	
5.	$\quad ? \neg Q$	
6.	$\quad Q \supset R$	1, 4 *MP*
7.	$\quad \neg Q \rfloor$	2, 6 *MT*
8.	$P \supset \neg Q$	4–7 *CP*

(12)

P	$\supset$	$(Q$	$\vee$	$R)$	S	$\supset$	$(T$	$\vee$	$R)$	$\mid$	$(P$	$\vee$	$S)$	$\supset$	R
T		T		F	T		T		F		T	T		F	
			T					T					T		
	T					T						F			

Invalid under I: {P = T; Q = T; R = F; S = T; T = T}

(14)

1.	$+ (P \vee Q) \supset R$	
2.	$+ (P \vee Q) \supset S$	
3.	$+ \neg S$	
4.	$? \neg P$	
5.	$\quad \neg(P \vee Q)$	2, 3 *MT*
6.	$\quad \neg P \wedge \neg Q$	5 *DeM*
7.	$\quad \neg P$	6 $\wedge E$

Exercises 3.19 (p. 120)

(2) $\vdash P \supset (\neg P \supset P)$

1.	$\lceil P$	
2.	$\quad ? \neg P \supset P$	
3.	$\quad P \vee P$	1 $\vee I$
4.	$\quad \neg P \supset P \rfloor$	3 *Cond*
5.	$P \supset (\neg P \supset P)$	1–4 *CP*

(4) $\vdash ((P \supset Q) \wedge \neg Q) \supset \neg P$

1.	$\lceil (P \supset Q) \wedge \neg Q$	
2.	$\quad ? \neg P$	
3.	$\quad P \supset Q$	1 $\wedge E$
4.	$\quad \neg Q$	1 $\wedge E$
5.	$\quad \neg P \rfloor$	3, 4 *MT*
6.	$((P \supset Q) \wedge \neg Q) \supset \neg P$	1–5 *CP*

(6) $\vdash (P \vee Q) \equiv \neg(\neg P \wedge \neg Q)$

1.	$P \vee Q$	
2.	$? \neg(\neg P \wedge \neg Q)$	
3.	$\neg(\neg P \wedge \neg Q)$	1 *DeM*
4.	$(P \vee Q) \supset \neg(\neg P \wedge \neg Q)$	1–3 *CP*
5.	$\neg(\neg P \wedge \neg Q)$	
6.	$? P \vee Q$	
7.	$P \vee Q$	5 *DeM*
8.	$\neg(\neg P \wedge \neg Q \supset (P \vee Q)$	5–7 *CP*
9.	$((P \vee Q) \supset \neg(\neg P \wedge \neg Q)) \wedge (\neg(\neg P \wedge \neg Q) \supset (P \vee Q))$	4, 8 $\wedge I$
10.	$(P \vee Q) \equiv \neg(\neg P \wedge \neg Q)$	9 *Bicond*

(8) $\vdash (P \supset Q) \vee (Q \supset P)$

1.	$\neg(P \supset Q)$	
2.	$? Q \supset P$	
3.	$\neg(\neg P \vee Q)$	1 *Cond*
4.	$P \wedge \neg Q$	3 *DeM*
5.	$\neg Q$	4 $\wedge E$
6.	$\neg Q \vee P$	5 $\vee I$
7.	$Q \supset P$	6 *Cond*
8.	$\neg(P \supset Q) \supset (Q \supset P)$	1–7 *CP*
9.	$(P \supset Q) \vee (Q \supset P)$	8 *Cond*

(10) $\vdash \neg(P \supset Q) \equiv (P \wedge \neg Q)$

1.	$\neg(P \supset Q)$	
2.	$? P \wedge \neg Q$	
3.	$\neg(\neg P \vee Q)$	1 *Cond*
4.	$P \wedge \neg Q$	3 *DeM*
5.	$\neg(P \supset Q) \supset (P \wedge \neg Q)$	1–4 *CP*
6.	$P \wedge \neg Q$	
7.	$? \neg(P \supset Q)$	
8.	$\neg(\neg P \vee Q)$	6 *DeM*
9.	$\neg(P \supset Q)$	8 *Cond*
10.	$(P \wedge \neg Q) \supset \neg(P \supset Q)$	6–10 *CP*
11.	$(\neg(P \supset Q) \supset (P \wedge \neg Q)) \wedge ((P \wedge \neg Q) \supset \neg(P \supset Q))$	5, 10 $\wedge I$
12.	$\neg(P \supset Q) \equiv (P \wedge \neg Q)$	11 *Bicond*

(12) $\vdash \neg P \supset (P \supset Q)$

1.	$\neg P$	
2.	? $P \supset Q$	
3.	$\neg P \vee Q$	1 $\vee I$
4.	$P \supset Q$	3 *Cond*
5.	$\neg P \supset (P \supset Q)$	1–4 *CP*

(14) $\vdash (P \supset Q) \supset (P \supset (Q \equiv P))$

1.	$P \supset Q$	
2.	? $P \supset (Q \equiv P)$	
3.	P	
4.	? $Q \equiv P$	
5.	$\neg(Q \equiv P)$	
6.	? ×	
7.	$\neg((Q \supset P) \wedge (P \supset Q))$	5 *Bicond*
8.	$\neg(Q \supset P) \vee \neg(P \supset Q)$	7 *DeM*
9.	$\neg(Q \supset P)$	1, 8 $\vee E$
10.	$\neg(\neg Q \vee P)$	9 *Cond*
11.	$Q \wedge \neg P$	10 *DeM*
12.	$\neg P$	11 $\wedge E$
13.	$P \wedge \neg P$	3, 12 $\wedge I$
14.	$Q \equiv P$	5–13 *IP*
15.	$P \supset (Q \equiv P)$	3–14 *CP*
16.	$(P \supset Q) \supset (P \supset (Q \equiv P))$	1–15 *CP*

Chapter 4

Exercises 4.01 (p. 130) [All general terms are one-place unless otherwise noted]

(2) Callie is taller than Joe. (two-place relation)

(4) If Callie is taller than Joe and Joe is taller than Iola, then Callie is taller than Iola (all are two-place relations)

(6) Gertrude sits between Frank and Joe. (three-place relation)

(8) Fenton admires himself. (two-place relation)

(10) Iola is shorter than Callie or Joe, but taller than Fenton. (both are two-place relations)

(12) Callie and Iola live in Bayport. (two-place relation)

(14) If Gertrude is a detective, she admires Frank and Joe. (two-place relation)

Exercises 4.02 (p. 134)

(2) *Tji*

(4) $(Tcj \wedge Tji) \supset Tci$

(6) *Bgfj*

(8) *Aff*

(10) $(Sic \wedge Sij) \wedge Tif$

(12) $Lcb \wedge Lib$

(14) $Dg \supset (Agf \wedge Agj)$

Exercises 4.03 (p. 140)

(2) $\neg Sg \wedge Ag$

(4) $\exists x(Ax \wedge Sx) \supset \exists x(Sx \wedge Ax)$

(6) $Cf \wedge Sf$

(8) $\forall x(Ax \supset Cx)$

(10) $\forall x(Ax \supset Cx) \supset Cg$

(12) $\exists x(Sx \wedge (Cx \wedge Ax))$

(14) $\forall x(Ax \supset (Cx \supset Sx))$

Exercises 4.04 (p. 142)

(2) $\exists x(Fx \wedge Gx)$

(4) $\exists y(Fy \wedge Gy) \wedge \forall x(Fx \supset Gx)$

(6) $\exists y((Fy \wedge Gy) \wedge (Hy \wedge Iy))$

(8) $((\exists y Fy \wedge Gy) \wedge (Hy \wedge Iy))$

(10) $\forall x(Fx \supset Gx) \supset Ha$

Exercises 4.05 (p. 144)

(2) $Kc \supset \exists x(Sx \wedge \neg Wx)$

(4) $\forall x(Sx \supset Wx) \supset (\neg Wf \supset \neg Sf)$

(6) $\neg\exists x(Sx \wedge \neg Wx)$

(8) $\exists x(Sx \wedge Kx) \supset \neg\forall x(Sx \supset Wx)$

(10) $(\neg Kg \wedge \neg Kc) \supset Wi$ or $\neg(Kg \vee Kc) \supset Wi$

(12) $\exists x(Sx \wedge Kx) \wedge \forall x(Sx \supset Wx)$

(14) $Kg \supset \exists x(Sx \wedge \neg Wx)$

Exercises 4.06 (p. 146)

(2) $\exists x(Sx \wedge Cxf)$

(4) $\forall x(Sx \supset Cxc)$

(6) $\neg Eg \supset \neg\exists x((Sx \wedge Cx) \wedge Ex)$

(8) $\neg\forall x((Sx \wedge Cx) \supset Ex)$

(10) $\exists x((Sx \wedge Cx) \wedge Ex) \supset \exists x((Sx \wedge Cx) \wedge Cxg)$

(12) $Ef \supset \exists x((Sx \wedge Cx) \wedge Cxg)$

(14) $\exists x((Sx \wedge Cx) \wedge Ex) \supset \forall x((Sx \wedge Cx) \supset Ex)$

Exercises 4.07 (p. 150)

(2) $\neg\exists x(Sx \wedge \exists y(Cy \wedge Axy))$

(4) $\exists x(Sx \wedge \forall y(Cy \supset Axy))$

(6) $\forall x((Cx \wedge \neg Mx) \supset Afx)$

(8) $\neg\forall x(Sx \supset \exists y(Cy \wedge Axy))$

(10) $\neg\forall x((Sx \wedge Mx) \supset \exists y((Cy \wedge \neg My) \wedge \neg Axy))$

(12) $\forall x(Sx \supset \exists y((Cy \wedge \neg My) \wedge \neg Axy))$

(14) $\forall x(Cx \supset Afx) \supset \forall x((Cx \wedge Mx) \supset Afx))$

Exercises 4.08 (p. 154)

(2) $\exists x(Px \wedge \forall y(Sy \supset Ayx))$

(4) $\neg\exists x(Sx \wedge \forall y(Py \supset Ayx))$

(6) $\neg(Sf \wedge Wf) \supset \neg\exists x(Px \wedge Axf)$

(8) $Acf \supset \exists x((Sx \wedge Wx) \wedge Acx)$

(10) $\forall x((Sx \wedge Wx) \supset \neg\exists y Axy)$

(12) $\neg\forall x((Sx \wedge Wx) \supset Afx)$

(14) $\exists x((Sx \wedge Wx) \wedge \forall y(Py \supset Ayx))$

Exercises 4.09 (p. 159)

(2) $g = h$

(4) $\exists x((Axj \wedge \forall y(Ayj \supset x = y)) \wedge \neg Sx)$

(6) $j \neq f$

(8) $g = h \supset (Sg \supset Sh)$

(10) $Ahj \vee h \neq g$

(12) $\exists x Sx \supset (Sf \wedge \forall x(Sx \supset x = f))$

(14) $Agj \supset (\neg Sg \supset g \neq h)$

Exercises 4.11 (p. 166)

(2) $\exists x(Sx \wedge Wx)$

(4) $\exists x(((Sx \wedge Wx) \wedge \forall y((Sy \wedge Wy) \supset x = y)) \wedge (\exists z)(Pz \wedge Axz))$

(6) $\exists x(((Sx \wedge Wx) \wedge \forall y((Sy \wedge Wy) \supset x = y)) \wedge Agx)$

(8) $\forall x \forall y(((Sx \wedge Wx) \wedge (Sy \wedge Wy)) \supset \forall z((Sz \wedge Wz) \supset (z = x \vee z = y)))$ or

$\forall x((Sx \wedge Wx) \supset \forall y(((Sy \wedge Wy) \wedge x \neq y) \supset \forall z((Sz \wedge Wz) \supset (x = z \vee y = z))))$

(10) $\exists x \exists y(((Sx \wedge Sy) \wedge x \neq y) \wedge \forall z(Sz \supset (z = x \vee z = y))) \wedge (Afx \wedge Afy))$ or

$\exists x((Sx \wedge \exists y((Sy \wedge x \neq y) \wedge \forall z(Sz \supset (x = z \vee y = z)))) \wedge (Afx \wedge Afy))$

(12) $\exists x(((Sx \wedge Wx) \wedge \forall y((Sy \wedge Wy) \supset x = y)) \wedge \exists z(Pz \wedge Azx))$

(14) $Wf \supset \exists x((Sx \wedge \forall y(Sy \supset x = y)) \wedge Axf)$

Exercises 4.12 (p. 169)

(2) $\exists x(Sx \wedge \neg Tfx)$

(4) $\neg \exists x(Sx \wedge Txc)$

(6) $(Sf \wedge Sc) \wedge \forall x((Sx \wedge (x \neq f \wedge x \neq c)) \supset Tfx)$

(8) $\exists x((Sx \wedge \forall y((Sy \wedge x \neq y) \supset Txy)) \wedge Txc)$

(10) $Sf \supset \neg \forall x((Sx \wedge x \neq f) \supset Tfx)$

(12) $((Sc \wedge Bc) \wedge (Sf \wedge Bf)) \wedge \forall x((Sx \wedge Bx) \supset (x = c \vee x = f))$

(14) $\exists x(Sx \wedge Bx) \supset \exists x((Sx \wedge \forall y((Sy \wedge x \neq y) \supset Txy)) \wedge Bx)$

Exercises 4.14 (p. 173)

(2) $\forall x(Sx \supset \exists y((Sy \wedge By) \wedge Axy)) \quad \Rightarrow \quad \forall x \exists y(By \wedge Axy)$

(4) $\exists x((Sx \wedge Bx) \wedge \forall y(Sy \supset Axy)) \quad \Rightarrow \quad \exists x(Bx \wedge \forall y Axy)$

(6) $\forall x((Sx \wedge Bx) \supset \forall y((Sy \wedge By) \supset x = y)) \quad \Rightarrow \quad \forall x(Bx \supset \forall y(By \supset x = y))$

(8) $\forall x \forall y(((Sx \wedge Sy) \wedge x \neq y) \supset \forall z(Sz \supset (z = x \vee z = y))) \quad \Rightarrow \quad \forall x \forall y(x \neq y \supset \forall z(z = x \vee z = y))$

(10) $\neg \exists x((Sx \wedge \exists y(Sy \wedge \neg By)) \wedge Axy) \quad \Rightarrow \quad \neg \exists x \exists y(\neg By \wedge Axy)$

Chapter 5

Exercises 5.00 (p. 186)

(2)
1. + $\exists xFx \wedge \exists xGx$
2. + $\exists xHx \supset \neg\exists xGx$
3. ? $\neg\exists xHx$
4. $\exists xGx$ 1 $\wedge E$
5. $\neg\exists xHx$ 2, 4 *MT*

(4)
1. + $\exists xFx \supset \exists xGx$
2. + $\neg\exists xGx \vee \forall xFx$
3. ? $\exists xFx \supset \forall xFx$
4. $\exists xGx \supset \forall xFx$ 2 *Cond*
5. $\exists xFx \supset \forall xFx$ 1, 4 *HS*

(6)
1. + $\forall x(Fx \supset Gx)$
2. ? $\forall x\neg(Fx \wedge \neg Gx)$
3. $\forall x(\neg Fx \vee Gx)$ 1 *Cond*
4. $\forall x\neg(Fx \wedge \neg Gx)$ 3 *DeM*

(8)
1. + $\exists x((Fx \wedge \neg Gx) \vee \neg Gx)$
2. ? $\exists x((Fx \vee \neg Gx) \wedge \neg Gx)$
3. $\exists x(\neg Gx \vee (Fx \wedge \neg Gx))$ 1 *Com*
4. $\exists x((\neg Gx \vee Fx) \wedge (\neg Gx \vee \neg Gx))$ 3 *Dist*
5. $\exists x((\neg Gx \vee Fx) \wedge \neg Gx)$ 4 *Taut*
6. $\exists x((Fx \vee \neg Gx) \wedge \neg Gx)$ 5 *Com*

(10)
1. + $\forall xFx \supset \neg\exists yGy$
2. + $\neg\exists xHx \supset \exists yGy$
3. ? $\forall xFx \supset \exists xHx$
4. $\neg\exists yGy \supset \exists xHx$ 2 *Contra*
5. $\forall xFx \supset \exists xHx$ 1, 4 *HS*

(12)
1. + $\forall xFx \supset (Ga \wedge Ha)$
2. + $(\forall xFx \supset Ha) \supset \exists xJx$
3. ? $\exists xJx$
4. | $\neg\exists xJx$
5. | ? $\times$
6. | $\neg(\forall xFx \supset Ha)$ 2, 4 *MT*
7. | $\neg(\neg\forall xFx \vee Ha)$ 6 *Cond*
8. | $\forall xFx \wedge \neg Ha$ 7 *DeM*
9. | $\forall xFx$ 8 $\wedge E$
10. | $Ga \wedge Ha$ 1, 9 *MP*
11. | Ha 10 $\wedge E$
12. | $\neg Ha$ 8 $\wedge E$
13. | $Ha \wedge \neg Ha$ 11, 12 $\wedge I$
14. $\exists xJx$ 4–13 *IP*

(14)
1. + $\forall x(Fx \supset Gx))$
2. ? $\forall x(\neg Gx \supset \neg Fx) \wedge \forall x(Gx \vee \neg Fx)$
3. $\forall x(\neg Gx \supset \neg Fx)$ 1 *Contra*
4. $\forall x(Gx \vee \neg Fx)$ 3 *Cond*
5. $\forall x(\neg Gx \supset \neg Fx) \wedge \forall x(Gx \vee \neg Fx)$ 3, 4 $\wedge I$

Exercises 5.01 (p. 190)

(2)

1.	+ $\forall x(Fx \supset Gx)$	
2.	? $\neg\exists x(Fx \wedge \neg Gx)$	
3.	$\neg\exists x\neg(Fx \supset Gx)$	1 *QT*
4.	$\neg\exists x\neg(\neg Fx \vee Gx)$	3 *Cond*
5.	$\neg\exists x(Fx \wedge \neg Gx))$	4 *DeM*

(4)

1.	+ $\exists x((Fx \wedge Gx) \vee \neg Hx)$	
2.	? $\neg\forall x(Hx \wedge (Fx \supset \neg Gx))$	
3.	$\neg\forall x\neg((Fx \wedge Gx) \vee \neg Hx)$	1 *QT*
4.	$\neg\forall x(\neg(Fx \wedge Gx) \wedge Hx)$	3 *DeM*
5.	$\neg\forall x((\neg Fx \vee \neg Gx) \wedge Hx)$	4 *DeM*
6.	$\neg\forall x((Fx \supset \neg Gx) \wedge Hx)$	5 *Cond*
7.	$\neg\forall x(Hx \wedge (Fx \supset \neg Gx))$	6 *Com*

(6)

1.	+ $\forall x((Fx \supset Gx) \wedge Hx)$	
2.	? $\neg\exists x((\neg Hx \vee Fx) \wedge (Hx \vee \neg Gx)$	
3.	$\neg\exists x\neg((Fx \supset Gx) \wedge Hx)$	1 *QT*
4.	$\neg\exists x(\neg(Fx \supset Gx) \vee \neg Hx)$	3 *DeM*
5.	$\neg\exists x(\neg(\neg Fx \vee Gx) \vee \neg Hx)$	4 *Cond*
6.	$\neg\exists x((Fx \wedge \neg Gx) \vee \neg Hx)$	5 *DeM*
7.	$\neg\exists x(\neg Hx \vee (Fx \wedge \neg Gx))$	6 *Com*
8.	$\neg\exists x((\neg Hx \vee Fx) \wedge (\neg Hx \vee \neg Gx)$	7 *Dist*

(8)

1.	+ $\forall xFx$	
2.	? $\neg(\forall xFx \supset \neg\forall xFx)$	
3.	$\forall xFx \wedge \forall xFx$	1 *Taut*
4.	$\neg(\neg\forall xFx \vee \neg\forall xFx)$	3 *DeM*
5.	$\neg(\forall xFx \supset \neg\forall xFx)$	4 *Cond*

(10)

1.	+ $\exists x\neg Fx$	
2.	? $\neg(\neg\forall xFx \supset \forall xFx)$	
3.	$\neg\forall xFx$	1 *QT*
4.	$\neg\forall xFx \wedge \neg\forall xFx$	3 *Taut*
5.	$\neg(\forall xFx \vee \forall xFx)$	4 *DeM*
6.	$\neg(\neg\forall xFx \supset \forall xFx)$	5 *Cond*

(12)

1.	+ $\forall x(Fx \supset Gx)$	
2.	? $\neg\exists x(\neg Gx \wedge Fx)$	
3.	$\neg\exists x\neg(Fx \supset Gx)$	1 *QT*
4.	$\neg\exists x\neg(\neg Fx \vee Gx)$	3 *Cond*
5.	$\neg\exists x(Fx \wedge \neg Gx)$	4 *DeM*
6.	$\neg\exists x(\neg Gx \wedge Fx)$	5 *Com*

(14)

1.	+ $\forall x(Fx \supset (Gx \supset Hx))$	
2.	? $\neg\exists x((Fx \wedge Gx) \wedge \neg Hx)$	
3.	$\forall x((Fx \wedge Gx) \supset Hx)$	1 *Exp*
4.	$\neg\exists x\neg((Fx \wedge Gx) \supset Hx)$	3 *QT*
5.	$\neg\exists x\neg(\neg(Fx \wedge Gx) \vee Hx)$	4 *Cond*
6.	$\neg\exists x((Fx \wedge Gx) \wedge \neg Hx)$	5 *DeM*

Exercises 5.02 (p. 193)

(2)

1.	+ $\forall x(Fx \supset Gx)$	
2.	+ $\forall x(Gx \supset Hx)$	
3.	? $Fa \supset Ha$	
4.	$Fa \supset Ga$	1 *UI*
5.	$Ga \supset Ha$	2 *UI*
6.	$Fa \supset Ha$	4, 5 *HS*

(4)

1.	+ $\forall x(Fx \supset Gx)$	
2.	+ $\neg\exists x(Gx \wedge \neg Hx)$	
3.	? $\neg(Fa \wedge \neg Ha)$	
4.	$Fa \supset Ga$	1 *UI*
5.	$\forall x\neg(Gx \wedge \neg Hx)$	2 *QT*
6.	$\neg(Ga \wedge \neg Ha)$	5 *UI*
7.	$\neg Ga \vee Ha$	6 *DeM*
8.	$Ga \supset Ha$	7 *Cond*
9.	$Fa \supset Ha$	4, 8 *HS*
10.	$\neg Fa \vee Ha$	9 *Cond*
11.	$\neg(Fa \wedge \neg Ha)$	10 *DeM*

(6)

1.	+ $\forall xFx \supset \forall xGx$	
2.	+ $\forall x\neg Gx$	
3.	? $\exists x\neg Fx$	
4.	$\neg\exists x\neg Fx$	
5.	? ×	
6.	$\forall xFx$	4 *QT*
7.	$\forall xGx$	1, 5 *MP*
8.	Ga	7 *UI*
9.	$\neg Ga$	2 *UI*
10.	$Ga \wedge \neg Ga$	8, 9 $\wedge I$
11.	$\exists x\neg Fx$	4–10 *IP*

(8)

1.	+ $\forall xFx$	
2.	+ $\forall x(Fx \supset Gx)$	
3.	? $\exists x(Fx \wedge Gx)$	
4.	$\neg\exists x(Fx \wedge Gx)$	
5.	? ×	
6.	$\forall x\neg(Fx \wedge Gx)$	4 *QT*
7.	$\neg(Fa \wedge Ga)$	6 *UI*
8.	$\neg Fa \vee \neg Ga$	7 *DeM*
9.	Fa	1 *UI*
10.	$Fa \supset Ga$	2 *UI*
11.	Ga	9, 10 *MP*
12.	$\neg Ga$	8, 9 $\vee E$
13.	$Ga \wedge \neg Ga$	11, 12 $\wedge I$
14.	$\exists x(Fx \wedge Gx)$	4–13 *IP*

(10)

1.	+ $\neg\exists x(\neg Fx \vee Hx)$	
2.	+ $\forall x(Jx \supset Gx)$	
3.	+ $\forall x(Fx \supset Jx)$	
4.	? $\exists x(Fx \wedge Gx)$	
5.	$\neg\exists x(Fx \wedge Gx)$	
6.	? ×	
7.	$\forall x\neg(\neg Fx \vee Hx)$	1 *QT*
8.	$\neg(\neg Fa \vee Ha)$	7 *UI*
9.	$Fa \wedge \neg Ha$	8 *DeM*
10.	$\forall x\neg(Fx \wedge Gx)$	5 *QT*
11.	$\neg(Fa \wedge Ga)$	10 *UI*
12.	$\neg Fa \vee \neg Ga$	11 *DeM*
13.	$Ja \supset Ga$	2 *UI*
14.	$Fa \supset Ja$	3 *UI*
15.	$Fa \supset Ga$	13, 14 *HS*
16.	Fa	9 ∧*E*
17.	Ga	15, 16 *MP*
18.	$\neg Ga$	12, 16 ∨*E*
19.	$Ga \wedge \neg Ga$	17, 18 ∧*I*
20.	$\exists x(Fx \wedge Gx)$	5–19 *IP*

(12)

1.	+ $\forall x(Fx \supset Gx)$	
2.	+ $\neg\exists x(Gx \wedge \neg Hx)$	
3.	? $\forall xFx \supset Hb$	
4.	$\forall xFx$	
5.	? Hb	
6.	$Fb \supset Gb$	1 *UI*
7.	Fb	4 *UI*
8.	Gb	6, 7 *MP*
9.	$\forall x\neg(Gx \wedge \neg Hx)$	2 *QT*
10.	$\neg(Gb \wedge \neg Hb)$	9 *UI*
11.	$\neg Gb \vee Hb$	10 *DeM*
12.	Hb	8, 11 ∨*E*
13.	$\forall xFx \supset Hb$	4–12 *CP*

(14)

1.	+ $\forall x(\neg Fxa \supset Gax)$	
2.	+ $\neg\exists xGxb$	
3.	? $\exists xFxa$	
4.	$\neg\exists xFxa$	
5.	? ×	
6.	$\neg Fba \supset Gab$	1 *UI*
7.	$\forall x\neg Gxb$	2 *QT*
8.	$\neg Gab$	7 *UI*
9.	$\forall x\neg Fxa$	4 *QT*
10.	$\neg Fba$	9 *UI*
11.	Gab	6, 10 *MP*
12.	$Gab \wedge \neg Gab$	8, 11 ∧*I*
13.	$\exists xFxa$	4–12 *IP*

Exercises 5.03 (p. 197)

(2)

1.	+ Fa	
2.	+ $\exists xFx \supset \forall x(Gx \vee Hx)$	
3.	+ $\exists x(Gx \vee Hx) \supset Ha$	
4.	? $\exists xHx$	
5.	$\exists xFx$	1 *EG*
6.	$\forall x(Gx \vee Hx)$	2, 5 *MP*
7.	$Ga \vee Ha$	6 *UI*
8.	$\exists x(Gx \vee Hx)$	7 *EG*
9.	Ha	3, 8 *MP*
10.	$\exists xHx$	9 *EG*

(4)

1.	+ $\exists xFx \supset \forall x(Gx \supset Hx)$	
2.	+ $\forall x(Fx \supset Gx)$	
3.	? $Fa \supset (Ga \wedge Ha)$	
4.	$\vert$ Fa	
5.	$\vert$? $Ga \wedge Ha$	
6.	$\vert$ $Fa \supset Ga$	2 *UI*
7.	$\vert$ $\exists xFx$	4 *EG*
8.	$\vert$ $\forall x(Gx \supset Hx)$	1, 7 *MP*
9.	$\vert$ $Ga \supset Ha$	8 *UI*
10.	$\vert$ $Fa \supset Ha$	6, 9 *HS*
11.	$\vert$ Ga	4, 6 *MP*
12.	$\vert$ Ha	4, 10 *MP*
13.	$\vert$ $Ga \wedge Ha$	11, 12 $\wedge I$
14.	$Fa \supset (Ga \wedge Ha)$	4–13 *CP*

(6)

1.	+ $\forall x(\neg Gx \supset Hx)$	
2.	+ $\forall x\neg(Fx \supset Hx)$	
3.	? $\exists x(Fx \wedge Gx)$	
4.	$\neg Ga \supset Ha$	1 *UI*
5.	$\neg(Fa \supset Ha)$	2 *UI*
6.	$\neg(\neg Fa \vee Ha)$	5 *Cond*
7.	$Fa \wedge \neg Ha$	6 *DeM*
8.	$\neg Ha$	7 $\wedge E$
9.	Ga	4, 8 *MT*
10.	Fa	7 $\wedge E$
11.	$Fa \wedge Ga$	9, 10 $\wedge I$
12.	$\exists x(Fx \wedge Gx)$	11 *EG*

(8)

1.	+ $\forall x(Fx \wedge Hx)$	
2.	+ $\exists x(Gx \vee Ix) \supset Ja$	
3.	+ $Ja \supset \forall x(Gx \supset \neg Fx)$	
4.	? $\neg\forall x(Fx \supset Gx)$	
5.	$\vert$ $\forall x(Fx \supset Gx)$	
6.	$\vert$? ×	
7.	$\vert$ $Fa \wedge Ha$	1 *UI*
8.	$\vert$ $Fa \supset Ga$	5 *UI*
9.	$\vert$ Fa	7 $\wedge E$
10.	$\vert$ Ga	8, 9 *MP*
11.	$\vert$ $Ga \vee Ia$	10 $\vee I$
12.	$\vert$ $\exists x(Gx \vee Ix)$	11 *EG*
13.	$\vert$ Ja	2, 12 *MP*
14.	$\vert$ $\forall x(Gx \supset \neg Fx)$	3, 13 *MP*
15.	$\vert$ $Ga \supset \neg Fa$	14 *UI*
16.	$\vert$ $\neg Fa$	10, 15 *MP*
17.	$\vert$ $Fa \wedge \neg Fa$	9, 16 $\wedge I$
18.	$\neg\forall x(Fx \supset Gx)$	5–17 *IP*

(10)

1.	$+ \forall x(Gx \supset \neg Hx)$	
2.	$+ \forall x(Fx \vee Gx)$	
3.	$? \forall xHx \supset \exists xFx$	
4.	$\forall xHx$	
5.	$? \exists xFx$	
6.	$Ga \supset \neg Ha$	1 *UI*
7.	$Fa \vee Ga$	2 *UI*
8.	Ha	4 *UI*
9.	$\neg Ga$	6, 8 *MT*
10.	Fa	7, 9 $\vee E$
11.	$\exists xFx$	10 *EG*
12.	$\forall xHx \supset \exists xFx$	4–11 *CP*

(12)

1.	$+ \forall x(Fx \supset Gx)$	
2.	$+ \neg Gc$	
3.	$? \exists x\neg Fx$	
4.	$Fc \supset Gc$	1 *UI*
5.	$\neg Fc$	2, 4 *MT*
6.	$\exists x\neg Fx$	5 *EG*

(14)

1.	$+ \forall x(Fx \supset Gx)$	
2.	$+ \neg\exists x(\neg Gx \wedge Fx) \supset Hac$	
3.	$? \exists x\exists yHxy$	
4.	$\neg Hac$	
5.	$? \times$	
6.	$\exists x(\neg Gx \wedge Fx)$	2, 4 *MT*
7.	$\neg\forall x\neg(\neg Gx \wedge Fx)$	6 *QT*
8.	$\neg\forall x(Gx \vee \neg Fx)$	7 *DeM*
9.	$\neg\forall x(\neg Fx \vee Gx)$	8 *Com*
10.	$\neg\forall x(Fx \supset Gx)$	9 *Cond*
11.	$\forall x(Fx \supset Gx) \wedge \neg\forall x(Fx \supset Gx)$	1, 10 $\wedge I$
12.	Hac	4–11 *IP*
13.	$\exists yHay$	12 *EG*
14.	$\exists x\exists yHxy$	13 *EG*

Exercises 5.04 (p. 201)

(2)

1.	+ $\exists x \exists y Fxy$	
2.	+ $\forall x \forall y (Fxy \supset Gx)$	
3.	? $\exists x Gx$	
4.	$\exists y Fay$	1 *EI*
5.	Fab	4 *EI*
6.	$\forall y (Fay \supset Ga)$	2 *UI*
7.	$Fab \supset Ga$	6 *UI*
8.	Ga	5, 7 *MP*
9.	$\exists x Gx$	8 *EG*

(4)

1.	+ $\neg \forall x Gx \supset \forall x Hx$	
2.	+ $\exists x Hx \supset \forall x \neg Fx$	
3.	? $\forall x (Fx \supset Gx)$	
4.	$\neg \forall x (Fx \supset Gx)$	
5.	? ×	
6.	$\exists x \neg (Fx \supset Gx)$	4 *QT*
7.	$\neg (Fa \supset Ga)$	6 *EI*
8.	$\neg (\neg Fa \vee Ga)$	7 *Cond*
9.	$Fa \wedge \neg Ga$	8 *DeM*
10.	Fa	9 $\wedge E$
11.	$\exists x Fx$	10 *EG*
12.	$\neg \forall x \neg Fx$	11 *QT*
13.	$\neg \exists x Hx$	2, 12 *MT*
14.	$\forall x \neg Hx$	13 *QT*
15.	$\neg Ha$	14 *UI*
16.	$\exists x \neg Hx$	15 *EG*
17.	$\neg \forall x Hx$	16 *QT*
18.	$\forall x Gx$	1, 17 *MT*
19.	Ga	18 *UI*
20.	$\neg Ga$	9 $\wedge E$
21.	$Ga \wedge \neg Ga$	19, 20 $\wedge I$
22.	$\forall x (Fx \supset Gx)$	4–21 *IP*

(6)

1.	+ $\forall x (Fx \supset \exists y Gy)$	
2.	+ $\exists y Gy \supset Ha$	
3.	? $\exists x Fx \supset \exists x Hx$	
4.	$\exists x Fx$	
5.	? $\exists x Hx$	
6.	Fb	4 *EI*
7.	$Fb \supset \exists y Gy$	1 *UI*
8.	$\exists y Gy$	6, 7 *MP*
9.	Ha	2, 8 *MP*
10.	$\exists x Hx$	9 *EG*
11.	$\exists x Fx \supset \exists x Hx$	4–10 *CP*

(8)

1.	+ $\exists x (Fx \vee Gx)$	
2.	+ $\forall x (Fx \supset Hx)$	
3.	+ $\forall x (Gx \supset Hx)$	
4.	? $\exists x Hx$	
5.	$Fa \vee Ga$	1 *EI*
6.	$Fa \supset Ha$	2 *UI*
7.	$Ga \supset Ha$	3 *UI*
8.	$Ha \vee Ha$	5, 6, 7 *CD*
9.	Ha	8 *Taut*
10.	$\exists x Hx$	9 *EG*

(10)

1.	$+ \exists x(\neg Fx \vee Hx)$	
2.	$+ \forall x(Fx \supset (Gx \supset Hx))$	
3.	$? \exists x(Fx \wedge Gx) \supset \exists x Hx$	
4.	$\exists x(Fx \wedge Gx)$	
5.	$? \exists x Hx$	
6.	$Fa \wedge Ga$	4 *EI*
7.	$Fa \supset (Ga \supset Ha)$	1 *UI*
8.	$(Fa \wedge Ga) \supset Ha$	7 *Exp*
9.	Ha	6, 8 *MP*
10.	$\exists x Hx$	9 *EG*
11.	$\exists x Fx \supset \exists x Hx$	4–10 *CP*

(12)

1.	$+ \exists x(Fxa \wedge Gax)$	
2.	$+ \forall x(Fxa \supset \forall y Hxy)$	
3.	$? \exists x Hxc$	
4.	$Fba \wedge Gab$	1 *EI*
5.	$Fba \supset \forall y Hby$	2 *UI*
6.	Fba	4 $\wedge E$
7.	$\forall y Hby$	5, 6 *MP*
8.	Hbc	7 *UI*
9.	$\exists x Hxc$	8 *EG*

(14)

1.	$+ \exists x(Fx \equiv Gx)$	
2.	$+ \forall x(Fx \supset (Gx \supset Hx))$	
3.	$+ \forall x Fx \vee \forall y Gy$	
4.	$? \exists x Hx$	
5.	$\neg \exists x Hx$	
6.	$? \times$	
7.	$Fa \equiv Ga$	1 *EI*
8.	$\forall x \neg Hx$	5 *QT*
9.	$\neg Ha$	8 *UI*
10.	$Fa \supset (Ga \supset Ha)$	2 *UI*
11.	$(Fa \supset Ga) \wedge (Ga \supset Fa)$	7 *Bicond*
12.	$Ga \supset Fa$	11 $\wedge E$
13.	$Ga \supset (Ga \supset Ha)$	10, 12 *HS*
14.	$(Ga \wedge Ga) \supset Ha$	13 *Exp*
15.	$Ga \supset Ha$	14 *Taut*
16.	$\neg Ga$	9, 15 *MT*
17.	$\exists y \neg Gy$	16 *EG*
18.	$\neg \forall y Gy$	17 *QT*
19.	$\forall x Fx$	3, 18 $\vee E$
20.	Fa	19 *UI*
21.	$Fa \supset Ga$	11 $\wedge E$
22.	Ga	20, 21 *MP*
23.	$Ga \wedge \neg Ga$	16, 22 $\wedge I$
24.	$\exists x Hx$	5–23 *IP*

Exercises 5.06 (p. 212)

(2)

1.	$+ \forall x(Fx \supset Gx)$	
2.	$+ \exists xGx \supset \forall x(Fx \supset Hx)$	
3.	$? \forall xFx \supset \forall xHx$	
4.	$\forall xFx$	
5.	$? \forall xHx$	
6.	$Fa \supset Ga$	1 *UI*
7.	Fa	4 *UI*
8.	Ga	6, 7 *MP*
9.	$\exists xGx$	8 *EG*
10.	$\forall x(Fx \supset Hx)$	2, 9 *MP*
11.	$Fa \supset Ha$	10 *UI*
12.	Ha	7, 11 *MP*
13.	$\forall xHx$	12 *UG*
14.	$\forall xFx \supset \forall xHx$	4–13 *CP*

(4)

1.	$+ \forall x(Fx \wedge Gx)$	
2.	$? \forall xFx \wedge \forall xGx$	
3.	$Fa \wedge Ga$	1 *UI*
4.	Fa	3 $\wedge E$
5.	Ga	3 $\wedge E$
6.	$\forall xFx$	4 *UI*
7.	$\forall xGx$	5 *UI*
8.	$\forall xFx \wedge \forall xGx$	6, 7 $\wedge I$

(6)

1.	$+ \exists x(Fx \vee Hx)$	
2.	$+ \forall x(Hx \supset Fx)$	
3.	$+ \exists xGx \supset \forall x(Gx \supset Hx)$	
4.	$? \forall x(Gx \supset Fx)$	
5.	Ga	
6.	$? Fa$	
7.	$\exists xGx$	5 *EG*
8.	$\forall x(Gx \supset Hx)$	3, 7 *MP*
9.	$Ga \supset Ha$	8 *UI*
10.	Ha	5, 9 *MP*
11.	$Ha \supset Fa$	2 *UI*
12.	Fa	10, 11 *MP*
13.	$Ga \supset Fa$	5–12 *CP*
14.	$\forall x(Gx \supset Fx)$	13 *UG*

(8)

1.	$+ \forall x(Fx \supset Gx)$	
2.	$+ \forall x\exists y(Fy \wedge Hxy)$	
3.	$? \forall x\exists y(Gy \wedge Hxy)$	
4.	$\neg\forall x\exists y(Gy \wedge Hxy)$	
5.	$? \times$	
6.	$\exists x\neg\exists y(Gy \wedge Hxy)$	4 *QT*
7.	$\exists x\forall y\neg(Gy \wedge Hxy)$	6 *QT*
8.	$\forall y\neg(Gy \wedge Hay)$	7 *EI*
9.	$\exists y(Fy \wedge Hay)$	2 *UI*
10.	$Fb \wedge Hab$	9 *EI*
11.	$\neg(Gb \wedge Hab)$	8 *UI*
12.	$\neg Gb \vee \neg Hab$	11 *DeM*
13.	$Fb \supset Gb$	1 *UI*
14.	Fb	10 $\wedge E$
15.	Gb	13, 14 *MP*
16.	$\neg Hab$	12, 15 $\vee E$
17.	Hab	10 $\wedge E$
18.	$Hab \wedge \neg Hab$	16, 17 $\wedge I$
19.	$\forall x\exists y(Gy \wedge Hxy)$	4–18 *IP*

(10)

1.	$+ \forall x((Fx \wedge \neg\exists yHxy) \supset Gx)$	
2.	$+ \forall x(Jx \supset (Fx \wedge \neg(Kx \vee Gx)))$	
3.	$? \forall x(Jx \supset \exists yHxy)$	
4.	Ja	
5.	$? \exists yHay$	
6.	$Ja \supset (Fa \wedge \neg(Ka \vee Ga))$	2 *UI*
7.	$Fa \wedge \neg(Ka \vee Ga)$	4, 6 *MP*
8.	$\neg(Ka \vee Ga)$	7 $\wedge$*E*
9.	$\neg Ka \wedge \neg Ga$	8 *DeM*
10.	$\neg Ga$	9 $\wedge$*E*
11.	$(Fa \wedge \neg\exists yHay) \supset Ga$	1 *UI*
12.	$\neg(Fa \wedge \neg\exists yHay)$	10, 11 *MT*
13.	$\neg Fa \vee \exists yHay$	12 *DeM*
14.	Fa	7 $\wedge$*E*
15.	$\exists yHay$	13, 14 $\vee$*E*
16.	$Ja \supset \exists yHay$	4–15 *CP*
17.	$\forall x(Jx \supset \exists yHxy)$	16 *UG*

(12)

1.	$+ \exists x(Fx \wedge \forall y(Gy \supset Hxy))$	
2.	$+ \forall x(Fx \supset \forall y(Jy \supset \neg Hxy))$	
3.	$? \forall x(Gx \supset \neg Jx)$	
4.	$Fa \wedge \forall y(Gy \supset Hay)$	1 *EI*
5.	Gb	
6.	$? \neg Jb$	
7.	$\forall y(Gy \supset Hay)$	4 $\wedge$*E*
8.	$Gb \supset Hab$	7 *UI*
9.	$Fa \supset \forall y(Jy \supset \neg Hay)$	2 *UI*
10.	Fa	4 $\wedge$*E*
11.	$\forall y(Jy \supset \neg Hay)$	9, 10 *MP*
12.	$Jb \supset \neg Hab$	11 *UI*
13.	Hab	5, 8 *MP*
14.	$\neg Jb$	12, 13 *MT*
15.	$Gb \supset \neg Jb$	5–14 *CP*
16.	$\forall x(Gx \supset \neg Jx)$	15 *UG*

(14)

1.	$+ \forall x(\exists yGyx \supset Gxx)$	
2.	$+ \forall x(Fx \supset (\exists yGxy \supset \exists yGyx))$	
3.	$+ \neg\exists xGxx$	
4.	$? \forall x(Fx \supset \neg Gxy)$	
5.	Fa	
6.	$? \forall y\neg Gay$	
7.	$Fa \supset (\exists yGay \supset \exists yGya)$	2 *UI*
8.	$\exists yGay \supset \exists yGya$	5, 7 *MP*
9.	$\exists yGya \supset Gaa$	1 *UI*
10.	$\exists yGay \supset Gaa$	8, 9 *HS*
11.	$\forall x\neg Gxx$	3 *QT*
12.	$\neg Gaa$	11 *UI*
13.	$\neg\exists yGay$	10, 12 MT
14.	$\forall y\neg Gay$	13 *QT*
15.	$Fa \supset \forall y\neg Gay$	5–14 *CP*
16.	$\forall x(Fx \supset \forall y\neg Gxy)$	15 *UG*

Exercises 5.07 (p. 217)

(2)

1.	$+\ \exists x(Fx \wedge Gx)$	
2.	$+\ \exists x(Fx \wedge \neg Gx)$	
3.	$?\ \exists x\exists y((Fx \wedge Fy) \wedge x \neq y)$	
4.	$Fa \wedge Ga$	1 *EI*
5.	$Fb \wedge \neg Gb$	2 *EI*
6.	$a = b$	
7.	$?\ \times$	
8.	Ga	4 $\wedge E$
9.	$\neg Gb$	5 $\wedge E$
10.	$\neg Ga$	6, 9 *ID*
11.	$Ga \wedge \neg Ga$	8, 10 $\wedge I$
12.	$a \neq b$	6–11 *IP*
13.	Fa	4 $\wedge E$
14.	Fb	5 $\wedge E$
15.	$Fa \wedge Fb$	13, 14 $\wedge I$
16.	$(Fa \wedge Fb) \wedge a \neq b$	12, 15 $\wedge I$
17.	$\exists y((Fa \wedge Fy) \wedge a \neq y)$	16 *EG*
18.	$\exists x\exists y((Fx \wedge Fy) \wedge x \neq y)$	17 *EG*

(4)

1.	$+\ \forall x(Fx \supset \forall y(Fy \supset x = y))$	
2.	$+\ \exists x(Fx \wedge Gx)$	
3.	$?\ \forall x(Fx \supset Gx)$	
4.	Fa	
5.	$?\ Ga$	
6.	$Fb \wedge Gb$	2 *EI*
7.	$Fa \supset \forall y(Fy \supset a = y)$	1 *UI*
8.	$\forall y(Fy \supset a = y)$	4, 7 *MP*
9.	$Fb \supset a = b$	8 *UI*
10.	Fb	6 $\wedge E$
11.	$a = b$	9, 10 *MP*
12.	Gb	6 $\wedge E$
13.	Ga	11, 12 *ID*
14.	$Fa \supset Ga$	4–13 *CP*
15.	$\forall x(Fx \supset Gx)$	14 *UG*

(6)

1.	$+\ \exists x(Fx \wedge \forall y(Fy \supset x = y))$	
2.	$+\ \neg Fb$	
3.	$?\ \exists x(x \neq b)$	
4.	$\neg\exists x(x \neq b)$	
5.	$?\ \times$	
6.	$Fa \wedge \forall y(Fy \supset a = y)$	1 *EI*
7.	$\forall y(Fy \supset a = y)$	6 $\wedge E$
8.	$Fb \supset a = b$	7 *UI*
9.	$\forall x(x = b)$	4 *QT*
10.	$a = b$	9 *UI*
11.	Fa	6 $\wedge E$
12.	Fb	10, 11 *ID*
13.	$Fb \wedge \neg Fb$	2, 12 $\wedge E$
14.	$\exists x(x \neq b)$	4–13 *IP*

(8)

1.	$+\ \exists x((Fx \wedge Gax) \wedge Hx)$	
2.	$+\ Fb \wedge Gab$	
3.	$+\ \forall x((Fx \wedge Gax) \supset x = b)$	
4.	$?\ Hb$	
5.	$(Fc \wedge Gac) \wedge Hc$	1 *EI*
6.	$(Fc \wedge Gac) \supset c = b$	3 *UI*
7.	$Fc \wedge Gac$	5 $\wedge E$
8.	$c = b$	6, 7 *MP*
9.	Hc	5 $\wedge E$
10.	Hb	8, 9 *ID*

(10)

1.	$+ \forall x \forall y((Fxy \wedge x \neq y) \supset Gxy)$	
2.	$+ \exists x \forall y(x \neq y \supset Fxy)$	
3.	$? \exists x \forall y(x \neq y \supset Gxy)$	
4.	$\forall y(a \neq y \supset Fay)$	2 *EI*
5.	$\neg \exists x \forall y(x \neq y \supset Gxy)$	
6.	$? \times$	
7.	$\forall x \neg \forall y(x \neq y \supset Gxy)$	5 *QT*
8.	$\forall x \exists y \neg(x \neq y \supset Gxy)$	7 *QT*
9.	$\exists y \neg(a \neq y \supset Gay)$	8 *UI*
10.	$\neg(a \neq b \supset Gab)$	9 *EI*
11.	$\neg(a = b \vee Gab)$	10 *Cond*
12.	$a \neq b \wedge \neg Gab$	11 *DeM*
13.	$\forall y((Fay \wedge a \neq y) \supset Gay)$	1 *UI*
14.	$(Fab \wedge a \neq b) \supset Gab$	13 *UI*
15.	$a \neq b \supset Fab$	4 *UI*
16.	$a \neq b$	12 $\wedge E$
17.	Fab	15, 16 *MP*
18.	$Fab \wedge a \neq b$	16, 17 $\wedge I$
19.	Gab	14, 18 *MP*
20.	$\neg Gab$	12 $\wedge E$
21.	$Gab \wedge \neg Gab$	19, 20 $\wedge I$
22.	$\exists x \forall y(x \neq y \supset Gxy)$	5–21 *IP*

(12)

1.	$+ \exists x(Fx \wedge \forall y(Fy \supset x = y))$	
2.	$+ \exists x(Fx \wedge Gx)$	
3.	$? \forall x(Fx \supset Gx)$	
4.	$\neg \forall x(Fx \supset Gx)$	
5.	$? \times$	
6.	$Fa \wedge \forall y(Fy \supset a = y)$	1 *EI*
7.	$\exists x \neg(Fx \supset Gx)$	4 *QT*
8.	$\neg(Fb \supset Gb)$	7 *EI*
9.	$\neg(\neg Fb \vee Gb)$	8 *Cond*
10.	$Fb \wedge \neg Gb$	9 *DeM*
11.	$\forall y(Fy \supset a = y)$	6 $\wedge E$
12.	$Fb \supset a = b$	11 *UI*
13.	Fb	10 $\wedge E$
14.	$a = b$	12, 13 *MP*
15.	$Fc \wedge Gc$	2 *EI*
16.	$Fc \supset a = c$	11 *UI*
17.	Fc	15 $\wedge E$
18.	$a = c$	16, 17 *MP*
19.	Gc	15 $\wedge E$
20.	$\neg Gb$	0 $\wedge E$
21.	Ga	18, 19 *ID*
22.	$\neg Ga$	14, 20 *ID*
23.	$Ga \wedge \neg Ga$	21, 22 $\wedge I$
24.	$\forall x(Fx \supset Gx)$	4–23 *IP*

(14)

1.	$+ \forall x \exists y Fxy$	
2.	$+ \neg \exists x Fxx$	
3.	$? \forall x(Fxa \supset a \neq x)$	
4.	$\neg \forall x(Fxa \supset a \neq x)$	
5.	$? \times$	
6.	$\exists x \neg(Fxa \supset a \neq x)$	4 *QT*
7.	$\neg(Fba \supset a \neq b)$	6 *EI*
8.	$\neg(\neg Fba \vee a \neq b)$	7 *Cond*
9.	$Fba \wedge a = b$	8 *DeM*
10.	$\forall x \neg Fxx$	2 *QT*
11.	$\neg Faa$	10 *UI*
12.	Fba	9 $\wedge E$
13.	$a = b$	9 $\wedge E$
14.	Faa	12, 13 ID
15.	$Faa \wedge \neg Faa$	11, 14 $\wedge I$
16.	$\forall x(Fxa \supset a \neq x)$	4–15 *IP*

Exercises 5.08 (p. 220)

(2) $\vdash \forall x(Fx \vee \neg Fx)$

1.	$\neg \forall x(Fx \vee \neg Fx)$	
2.	$? \times$	
3.	$\exists x \neg \exists y(Fy \supset Fx)$	1 *QT*
4.	$\neg \exists y(Fy \supset Fa)$	3 *EI*
5.	$\forall y \neg(Fy \supset Fa)$	4 *QT*
6.	$\neg(Fa \supset Fa)$	5 *UI*
7.	$\neg(\neg Fa \vee Fa)$	6 *Cond*
8.	$Fa \wedge \neg Fa$	7 *DeM*
9.	$\forall x \exists y(Fy \supset Fx)$	1–8 *IP*

(4) $\vdash \forall x \forall y((Fx \wedge x = y) \supset Fy)$

1.	$Fa \wedge a = b$	
2.	$? Fb$	
3.	Fa	1 $\wedge E$
4.	$a = b$	1 $\wedge E$
5.	Fb	3, 4 *ID*
6.	$Fa \wedge a = b \supset Fb$	1–5 *CP*
7.	$\forall y((Fa \wedge a = y) \supset Fy)$	6 *UG*
8.	$\forall x \forall y((Fx \wedge x = y) \supset Fy)$	7 *UG*

(6) $\vdash \forall x \forall y(x = y \supset (Fx \equiv Fy))$

1.	$\neg\forall x\forall y(x = y \supset (Fx \equiv Fy))$	
2.	? $\times$	
3.	$\exists x\neg\forall y(x = y \supset (Fx \equiv Fy))$	1 *QT*
4.	$\exists x\exists y\neg(x = y \supset (Fx \equiv Fy))$	3 *QT*
5.	$\exists y\neg(a = y \supset (Fa \equiv Fy))$	4 *EI*
6.	$\neg(a = b \supset (Fa \equiv Fb))$	5 *EI*
7.	$\neg(a \neq b \vee (Fa \equiv Fb))$	6 *Cond*
8.	$a = b \wedge \neg(Fa \equiv Fb)$	7 *DeM*
9.	$a = b$	8 $\wedge E$
10.	$\neg(Fa \equiv Fb)$	8 $\wedge E$
11.	$\neg((Fa \supset Fb) \wedge (Fb \supset Fa))$	10 *Bicond*
12.	$\neg(Fa \supset Fb) \vee \neg(Fb \supset Fa)$	11 *DeM*
13.	$\neg(Fa \supset Fb) \vee \neg(Fa \supset Fa)$	9, 12 *ID*
14.	$\neg(Fa \supset Fa) \vee \neg(Fa \supset Fa)$	9, 13 *ID*
15.	$\neg(Fa \supset Fa)$	14 *Taut*
16.	$\neg(\neg Fa \vee Fa)$	15 *Cond*
17.	$Fa \wedge \neg Fa$	16 *DeM*
18.	$\exists x\exists y((Fx \wedge Fy) \wedge x \neq y)$	1–17 *IP*

(8) $\vdash \exists x\forall y Fxy \supset \forall y\exists x Fxy$

1.	$\exists x\forall y Fxy$	
2.	? $\forall y\exists x Fxy$	
3.	$\forall y Fay$	1 *EI*
4.	Fab	3 *UI*
5.	$\exists x Fxb$	4 *EG*
6.	$\forall y\exists x Fxy$	5 *UG*
7.	$\exists x\forall y Fxy \supset \forall y\exists x Fxy$	1–6 *CP*

(10) $\vdash \forall x(Fx \supset Gx) \supset (\neg\exists x Gx \supset \neg\exists x Fx)$

1.	$\forall x(Fx \supset Gx)$	
2.	? $\neg\exists x Gx \supset \neg\exists x Fx$	
3.	$\neg\exists x Gx$	
4.	? $\neg\exists x Fx$	
5.	$\forall x\neg Gx$	3 *QT*
6.	$\neg Ga$	5 *UI*
7.	$Fa \supset Ga$	1 *UI*
8.	$\neg Fa$	6, 7 *MT*
9.	$\forall x\neg Fx$	8 *UI*
10.	$\neg\exists x Fx$	9 *QT*
11.	$\neg\exists x Gx \supset \neg\exists x Fx$	3–10 *CP*
12.	$\forall x(Fx \supset Gx) \supset (\neg\exists x Gx \supset \neg\exists x Fx)$	1–11 *CP*

(12) $\vdash \forall x\forall y Fxy \supset \forall x Fxx$

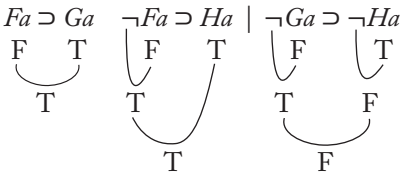
1.	$\forall x\forall y Fxy$	
2.	? $\forall x Fxx$	
3.	$\forall y Fay$	1 *UI*
4.	Faa	3 *UI*
5.	$\forall x Fxx$	4 *UG*
6.	$\forall x\forall y Fxy \supset \forall x Fxx$	1–5 *CP*

(14) $\vdash \forall x\forall y(Fxy \supset \neg Fxy) \supset \forall x\neg Fxx$

1.	$\forall x\forall y(Fxy \supset \neg Fxy)$	
2.	$? \forall x\neg Fxx$	
3.	$\forall y(Fay \supset \neg Fay)$	1 *UI*
4.	$Faa \supset \neg Faa$	3 *UI*
5.	$\neg Faa \vee \neg Faa$	4 *Cond*
6.	$\neg Faa$	5 *Taut*
7.	$\forall x\neg Fxx$	6 *UG*
8.	$\forall x\forall y(Fxy \supset \neg Fxy) \supset \forall x\neg Fxx$	1–7 *CP*

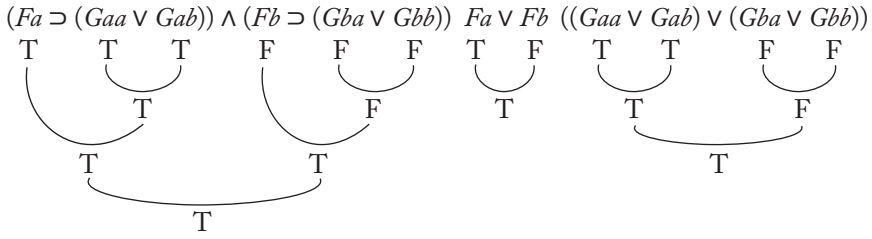
Exercises 5.09 (p. 230)

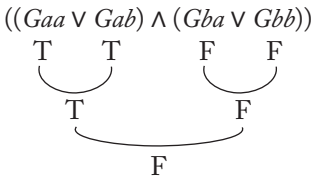
(2)



Invalid under *I*: {*Fa* = F; *Ga* = F; *Ha* = T}

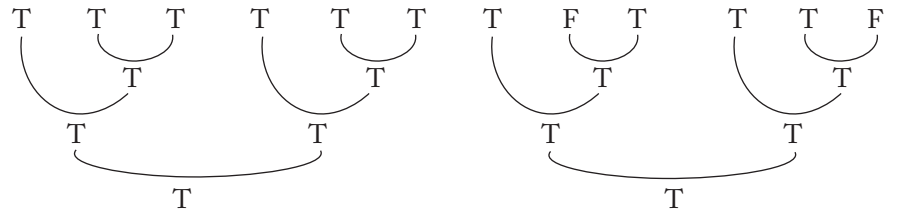
(4)



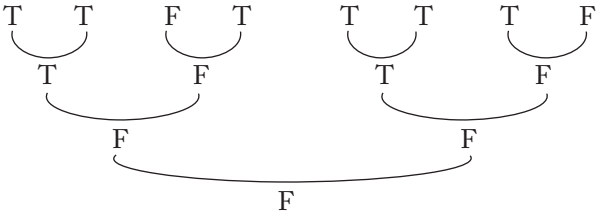


Invalid under *I*:
{*Fa* = T; *Fb* = F; *Gaa* = T; *Gab* = T; *Gba* = F; *Gbb* = F}

(6) $Fa \supset (Gaa \wedge Gba)) \vee (Fb \supset (Gab \wedge Gbb))$ $(Fa \wedge (Haa \vee Hab)) \wedge (Fb \wedge (Hba \vee Hbb))$

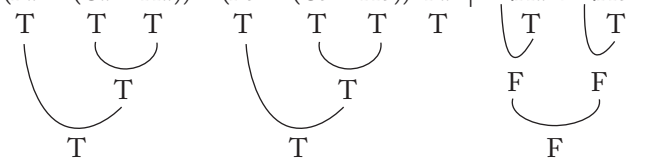


$((Gaa \wedge Gab) \wedge (Haa \wedge Hba)) \vee ((Gba \wedge Gbb) \wedge (Hab \wedge Hbb))$



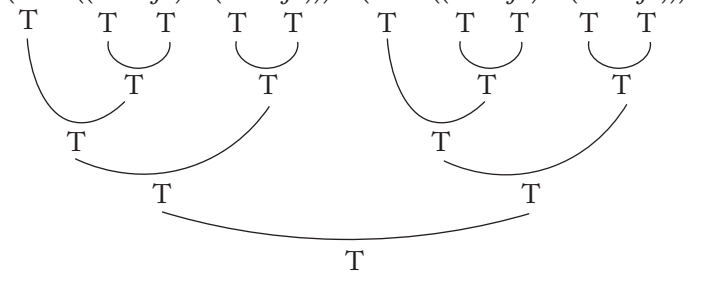
Invalid under *I*:
{*Fa* = T; *Fb* = T; *Gaa* = T; *Gab* = T; *Gba* = T; *Gbb* = T; *Haa* = F; *Hab* = T; *Hba* = T; *Hbb* = F}

(8) $(Fa \supset (Ga \wedge Ha)) \wedge (Fb \supset (Gb \wedge Hb))$ Fa | $\neg Ha \vee \neg Hb$

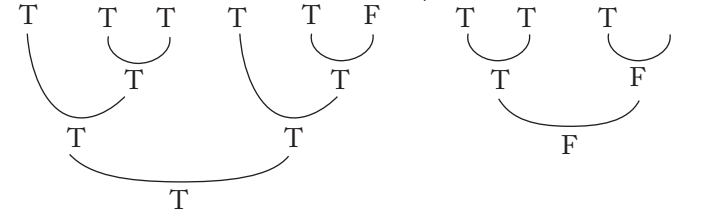


Invalid under *I*: {*Fa* = T; *Fb* = T; *Ga* = T; *Gb* = T; *Ha* = T; *Hb* = T}

(10) $(Fa \supset ((Fa \supset Ja) \wedge (Fb \supset Jb))) \vee (Fb \supset ((Fa \supset Ja) \wedge (Fb \supset Jb)))$

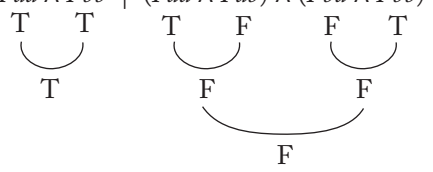


$(Fa \wedge (Ga \vee Ha)) \vee (Fb \wedge (Gb \vee Hb))$ | $(Fa \supset Ha) \wedge (Fb \supset Hb)$



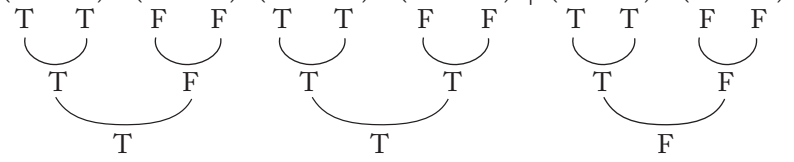
Invalid under *I*: {*Fa* = T; *Fb* = T; *Ga* = T; *Gb* = T; *Ha* = T; *Hb* = F; *Ja* = T; *Jb* = T}

(12) $Faa \wedge Fbb \mid (Faa \wedge Fab) \wedge (Fba \wedge Fbb)$



Invalid under I: {Faa = T; Fab = F; Fba = F; Fbb = T}

(14) $(Fa \wedge Ga) \vee (Fb \wedge Gb) \quad (Ga \supset Ha) \wedge (Gb \supset Hb) \mid (Ga \vee Fa) \wedge (Gb \vee Fb)$



Invalid under I: {Fa = T; Fb = F; Ga = T; Gb = F; Ha = T; Hb = F}

Exercises 5.10 (p. 232)

(2) $\forall x(Cx \supset \forall y(Fy \supset Lxy))$

$\forall x(\neg Cx \vee \forall y(Fy \supset Lxy))$	*Cond*
$\forall x(\neg Cx \vee \forall y(\neg Fy \vee Lxy))$	*Cond*
$\forall x\forall y(\neg Cx \vee (\neg Fy \vee Lxy))$	*PNF*

(4) $\neg\exists x\forall y(Ly \supset Txy)$

$\forall x\neg\forall y(Ly \supset Txy)$	*QT*
$\forall x\exists y\neg(Ly \supset Txy)$	*QT*
$\forall x\exists y\neg(\neg Ly \vee Txy)$	*Cond*
$\forall x\exists y(Ly \wedge \neg Txy)$	*DeM*

(6) $\neg\exists x(Hx \wedge Bx) \supset \exists x(Hx \wedge Dx)$

$\exists x(Hx \wedge Bx) \vee \exists x(Hx \wedge Dx)$	*Cond*
$\exists x(Hx \wedge Bx) \vee \exists y(Hy \wedge Dy)$	(y for x)
$\exists x\exists y((Hx \wedge Bx) \vee (Hy \wedge Dy))$	*PNF*

(8) $\exists x(Px \wedge \exists y(Py \wedge x \neq y))$

$\exists x\exists y(Px \wedge (Py \wedge x \neq y))$	*PNF*

(10) $Me \wedge \forall x((Mx \wedge x \neq e) \supset Wex)$

$Me \wedge \forall x(\neg(Mx \wedge x \neq e) \vee Wex)$	*Cond*
$Me \wedge \forall x((\neg Mx \vee x = e) \vee Wex)$	*DeM*
$\forall x(Me \wedge ((\neg Mx \vee x = e) \vee Wex))$	*PNF*

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QUANTIFIER RULES

Quantifier Transformation (*QT*)

$\forall\alpha\phi \dashv\vdash \neg\exists\alpha\neg\phi$

Identity (*ID*)

(i) $\vdash \beta = \beta$

(ii) $\phi, \beta = \gamma \vdash \phi\beta/\gamma$

In the formulation of rule ID, β and γ stand for individual constants, ϕ is a sentence, and $\phi\beta/\gamma$ represents a sentence obtained from ϕ by replacing one or more occurrences of β in ϕ with γ.

Instantiation Rules

Universal Instantiation (*UI*)

$\forall\alpha\phi \vdash \phi\alpha/\beta$

Existential Instantiation (*EI*)

$\exists\alpha\phi \vdash \phi\alpha/\beta$

Restriction: β does not occur

(i) in an earlier sentence

(ii) in the conclusion

(iii) in $\exists\alpha\phi$

In rules *UI* and *EI*, $\forall\alpha\phi$ and $\exists\alpha\phi$ each represent some quantified sentence, and ϕ stands for that sentence less the quantifier; $\phi\alpha/\beta$ represents a sentence that results from (1) dropping the quantifier and (2) replacing each instance of the variable it bound, α, by an individual constant, β.

Generalization Rules

Universal Generalization (*UG*)

$\phi\alpha/\beta \vdash \forall\alpha\phi$

Restriction: β does not occur

(i) in a premise

(ii) in an undischarged supposition

(iii) in $\forall\alpha\phi$

(iv) in a sentence previously obtained by means of an application of *EI*

Existential Generalization (*EG*)

$\phi\alpha/\beta \vdash \exists\alpha\phi$

In rules *UG* and *EG*, $\phi\alpha/\beta$ represents a sentence containing one or more occurrences of an individual constant, β; $\forall\alpha\phi$ is a sentence that results from (1) replacing *every* occurrence of β with a variable α free in ϕ and (2) adding a universal quantifier $\forall\alpha$ that captures every occurrence of α in ϕ; $\exists\alpha\phi$ results from (1) replacing *one or more* occurrences of β with a variable α free in ϕ and (2) adding an existential quantifier $\exists\alpha$ that captures every occurrence of α in ϕ.

"In his introduction to this most welcome second edition of his logic text, Heil clarifies his aim in writing and revising this book: 'I believe that anyone unfamiliar with the subject who set out to learn formal logic could do so relying solely on [this] book. That, in any case, is what I set out to create in writing *An Introduction to First-Order Logic*.' Heil has certainly accomplished this with perhaps **the most explanatorily thorough and pedagogically rich text I've personally come across**. . . .

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—Jonah Schupbach, University of Utah

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